

NMAGIF Research Dossiers Supplementary Research Material

Companion to the NMAGIF Monograph - Stage 7.1 “Mind 1.1” Edition

Available at: <https://nmagif.org>

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About This Document

This document is a supplementary companion to the NMAGIF Monograph (Neuro-Monolithic Architecture Global Intelligence Framework: The Open Science of Topological Cognition).

It contains the complete, unabridged corpus of research dossiers, literature surveys, architectural catalogs, and third-party analyses that informed the theoretical foundations of the NMAGIF project. These materials were assembled during the research phase (2024-2026) and are provided here for reference and reproducibility.

This document is NOT the NMAGIF Monograph itself. The core theory, proofs, and architecture are presented in the main monograph, available at:

<https://nmagif.org>

Contents Overview

Part I - Analytical Research Dossiers (11 chapters)

Deep analytical reports covering: theoretical limits of LLMs, frontier mathematical frameworks, ETFE validation, AGI paradigm history, heterodox AI archaeology, collective cognition mechanisms, and NMAGIF-specific audits.

Part II - Star Chapters & Theoretical Syntheses (4 chapters)

Extended theoretical essays on formal models of thinking, thermodynamics of meaning, and the “Edge of the Fractal” synthesis/verification chapter.

Part III - AGI Architecture Survey 2024-2026 (9 chapters)

Comprehensive survey of AGI architectures covering foundational paradigms, multi-agent systems, memory/metacognition, evaluation benchmarks, heterodox paradigms, analog-discrete transformations, entropic physics, world models, and synthetic subconscious architectures.

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Chapter 1

Catalog of Theoretical Limits of Large Language Models and the Deep Learning Paradigm (2020-2026)

1.1 Abstract

This catalog assembles the mathematically substantive critiques of autoregressive Transformer-based deep learning that hold under strict scrutiny of assumptions, excluding all ethical, societal, and policy considerations. The critiques are organized around twelve fatal-or-near-fatal theoretical walls: (1) circuit-complexity ceilings on Transformer expressivity (Merrill & Sabharwal, 2023); (2) the disputed status of "emergent" reasoning as a metric artifact (Schaeffer, Miranda & Koyejo, 2023); (3) formal impossibility results for hallucination elimination (Xu, Jain & Kankanhalli, 2024; Karpowicz, 2025); (4) rank/attention degeneracy with depth (Dong, Cordonnier & Loukas, 2021); (5) thermodynamic lower bounds on training energy (Tkachenko, 2025); (6) No-Free-Lunch/Kolmogorov-complexity constraints on inductive bias (Goldblum, Finzi, Rowan & Wilson, 2024); (7) non-identifiability and singular geometry of loss landscapes (Lakapragada, 2025); (8) provable and statistically characterized model collapse under recursive synthetic-data training (Shumailov et al., 2024; Suresh, Thangaraj & Khandavally, 2024); (9) provable limits on length and compositional generalization (Ahuja & Mansouri, 2024); (10) formal hallucination/generalization equivalence under the open-world assumption (Xu, 2025); (11) scaling-law ceilings and information-theoretic bounds on irreducible loss (Jeon & Van Roy, 2024; Caballero, Gupta, Rish & Krueger, 2023); and (12) chain-of-thought and in-context learning's provable computational ceiling at bounded-depth uniform threshold circuits absent unbounded intermediate tokens (Amiri, Huang, Rofin & Hahn, 2025; Zubić, Soldá, Sulser & Scaramuzza, 2025). Each wall is stated with its precise mathematical scope, distinguishing rigorously proved theorems from empirically suggestive but unproven claims, and cross-referenced against known loopholes. The synthesis concludes that the paradigm faces several genuinely load-bearing mathematical ceilings - chiefly circuit-complexity bounds on constant/log-depth inference and provable hallucination non-elimination under open-world assumptions - while many other claims (model collapse severity, emergence-as-mirage, topology blindness) remain conditional on specific architectural or distributional assumptions rather than universal impossibility results.

1.2 Notation and Definitions

TC^0

Constant-depth, polynomial-size threshold-circuit complexity class with unbounded fan-in majority/threshold gates

NC^1

Problems solvable by circuits of $O(\log n)$ depth, polynomial size, bounded fan-in

AC^0

Constant-depth polynomial-size circuits with AND/OR/NOT gates, unbounded fan-in

$PTIME$

Polynomial-time decidable problems

Log-precision transformer

Transformer whose internal numeric representations use $O(\log n)$ bits for input length n (Merrill &

Sabharwal, 2023)

Kolmogorov complexity $K(x)$

Length of the shortest program on a fixed universal Turing machine that outputs string x ; uncomputable in general (Vítányi, 2020)

MDL

Minimum Description Length principle: model selection by minimizing $L(\text{model}) + L(\text{data} \mid \text{model})$

No Free Lunch (NFL) Theorem

For supervised learning, all learning algorithms achieve identical average performance across a uniform distribution over all possible target functions (Goldblum et al., 2024)

Rank collapse

Convergence of all token representations in a self-attention stack toward a common rank-1 subspace with increasing depth (Dong, Cordonnier & Loukas, 2021)

Model collapse

Progressive divergence of a model's learned distribution from the true data distribution under recursive training on model-generated (synthetic) data (Shumailov et al., 2024, discussed in Borji, 2024)

Singular Learning Theory (SLT)

Watanabe's algebraic-geometric framework for statistical models whose parameter-to-function map is non-injective (singular), replacing classical asymptotic normality with a real log-canonical threshold ("learning coefficient") λ (Lakapragada, 2025)

Weisfeiler-Lehman (WL) test

Graph-isomorphism heuristic; the 1-WL test bounds the distinguishing power of standard message-passing GNNs (Franks, Morris, Velingker & Geerts, 2024)

Over-squashing

Exponential compression of information from distant nodes into a fixed-size vector as message-passing depth/graph diameter grows

Landauer bound

Thermodynamic lower bound of $kT \ln 2$ per irreversibly erased bit at temperature T

Broken Neural Scaling Law (BNSL)

Smoothly-broken power law $y = a + (bx^{-c_0}) \prod_i [1 + (x/d_i)^{1/f_i}]^{-c_i f_i}$ fitting empirical scaling curves across regimes (Caballero et al., 2023)

1.3 Taxonomy Table

- 1 Autoregressive next-token prediction limits - Teacher-forcing vs. autoregressive divergence; universal-learner theorem - Proved (narrow scope) + disputed critique - 2 - Malach, 2024; Bachmann & Nagarajan, 2024
- 2 Emergence as metric artifact - Nonlinear-metric argument, statistical re-analysis - Strong empirical/statistical, not a theorem - 3 - Schaeffer et al., 2023
- 3 Circuit-complexity bounds on Transformers - TC^0 -simulation theorem - Rigorously proved (log-precision, fixed depth) - 5 - Merrill & Sabharwal, 2023
- 4 Topology/geometry blindness in attention/MPNN - WL-bound, over-squashing, rank collapse theorems - Proved for specific architectures - 4 - Dong et al., 2021; Franks et al., 2024
- 5 Thermodynamic/energetic bounds - Free-energy functional lower bound $E < 2NDkT$ - Proved under quasi-static/analog assumptions - 2 - Tkachenko, 2025
- 6 NFL/Kolmogorov/MDL/uncomputability - NFL theorem, Kolmogorov incomputability, MDL bounds - Proved (classical) + reframed for DL - 3 - Goldblum et al., 2024; Vítányi, 2020

- 7 Loss-landscape degeneracy/non-identifiability - Singular Learning Theory, rank-collapse spectra - Proved (SLT); active empirical extension - 3 - Lakkapragada, 2025
- 8 Model collapse under recursive synthetic training - Statistical divergence theorems, collapse-rate bounds - Proved for specific statistical families - 4 - Shumailov et al./Borji, 2024; Suresh et al., 2024
- 9 OOD/compositional/length generalization impossibility - Provable length-generalization theorems (conditional) - Proved under explicit support/architecture conditions - 4 - Ahuja & Mansouri, 2024; Wang & Sun, 2025
- 10 Hallucination as formal impossibility - Impossibility theorem (knowledge aggregation) - Proved under stated axioms; scope-limited - 4 - Xu, Jain & Kankanhalli, 2024; Karpowicz, 2025
- 11 Scaling-law ceilings/irreducible error - Information-theoretic scaling bound; BNSL - Proved for specific data-generating models - 3 - Jeon & Van Roy, 2024; Maloney, Roberts & Sully, 2022
- 12 In-context learning / CoT computational ceiling - CoT lower bounds, $TC^0 \rightarrow PTIME$ extension theorem - Proved (conditional on hard-attention, precision) - 4 - Amiri et al., 2025; Zubić et al., 2025

Fatality is scored 1 (minor, easily circumvented) to 5 (structural, no known architectural fix within the paradigm) under the stated assumptions only - not as an unconditional ranking.

1.4 Wall 1 - Fundamental Limits of Autoregressive Next-Token Prediction

Claim. Training via teacher-forced next-token prediction can produce models that are formally powerful (universal, in a specific sense) yet the *inference-time* autoregressive procedure - repeatedly conditioning on self-generated tokens - is a distinct stochastic process from training, and this training/inference mismatch is the actual source of many observed failure modes, not an inherent limitation of the objective itself.

Formal framework. Malach (2024) proves an informal Theorem 1.1: for any function f efficiently computable by a Turing machine, there exists a dataset D such that training a linear next-token predictor with Chain-of-Thought data on D yields a predictor approximating f , introducing "length complexity" as the number of intermediate CoT tokens required. This is a *positive* universality result - next-token prediction is not intrinsically weak; expressivity is bought with intermediate-token length. Conversely, Bachmann & Nagarajan (2024) ("The pitfalls of next-token prediction") argue the two phases - autoregressive inference and teacher-forced training - are conflated in popular criticism, and identify concrete failure constructions (e.g., the "Clever Hans" cheat where teacher-forced training allows shortcut solutions that fail catastrophically at autoregressive generation time on specific graph-planning tasks) that are architecture-and-task-specific rather than universal.

What is proved vs. suggested. Proved: existence of Turing-complete simulation via CoT-augmented linear predictors (conditional on unbounded CoT length) (Malach, 2024); existence of specific tasks where teacher-forcing masks an error class invisible during training but fatal at inference (Bachmann & Nagarajan, 2024). Not proved: that all autoregressive LMs suffer catastrophic inference-time divergence in general - this remains task-dependent.

Why it threatens the paradigm. If the dominant training objective (teacher-forced cross-entropy) is provably decoupled from the deployment-time generative process, then benchmark performance under teacher forcing systematically overstates deployment reliability, and no amount of scale under the *same objective* guarantees closing this gap.

Loopholes/counterarguments. Multi-token prediction objectives, diffusion-style parallel decoding, and self-consistency/verification loops directly target the train/inference mismatch and are proposed as mitigations, not refutations, of the underlying gap (Bachmann & Nagarajan, 2024). A Law of Next-Token Prediction and universality results suggest the objective per se is not the bottleneck. Fractal/self-similarity analyses argue the statistical structure of language partially explains why next-token prediction succeeds empirically (Alcalá et al., 2024).

Key researchers. Eran Malach (Harvard/Kempner), Gregor Bachmann & Vaishnavh Nagarajan (Google/ETH).

1.5 Wall 2 - Emergent Reasoning as Metric/Scale Artifact

Claim. Apparent "emergent abilities" - capabilities absent at small scale that appear discontinuously at larger scale - are, for many reported cases, an artifact of nonlinear or discontinuous evaluation metrics (e.g., exact-match accuracy) applied to continuously and predictably improving underlying model likelihoods, rather than genuine phase transitions in capability.

Formal framework. Schaeffer, Miranda & Koyejo (2023) show that for a fixed model family, when a smoothly, continuously, and predictably improving per-token error rate is passed through a nonlinear or discontinuous metric (e.g., accuracy requiring an entire multi-token sequence to be exactly correct), the resulting metric can appear as a sharp, unpredictable jump at a threshold scale - even though no discontinuity exists in the underlying per-token loss. They further show that changing the metric to a continuous one (e.g., token-edit-distance or per-token log-likelihood) or increasing measurement resolution frequently causes the "emergent" jump to disappear, and that this effect can be reproduced by construction on non-LLM models.

What is proved vs. suggested. Proved: a mathematical mechanism (metric nonlinearity composed with smooth underlying improvement) *sufficient* to produce apparent emergence, plus empirical re-analysis showing several published "emergent" capabilities vanish under continuous metrics. Not proved: that *all* claimed emergent abilities are explained this way, nor that no genuine phase transitions occur in any LLM capability - the paper is explicitly a sufficiency/mirage argument, not an exhaustive impossibility proof. Later work complicates the picture: Berti, Giorgi & Kasneci (2025) survey the debate and note inconsistent definitions of "emergent," while Du et al. (2024) analyze emergent abilities via the loss landscape and find some emergent behaviors are tied to genuine, non-metric-artifact transitions in specific loss regimes.

Why it threatens the paradigm. If widely cited "emergent reasoning" claims are largely metric artifacts, the narrative that scale alone induces qualitatively new cognitive capacities is undermined, weakening the theoretical basis for expecting further scale to produce reasoning improvements beyond smooth interpolation of existing capability curves.

Loopholes/counterarguments. This is explicitly *not* a theorem that emergence never occurs; some tasks (e.g., certain arithmetic and symbolic tasks) show genuine sharp transitions even under continuous metrics, and grokking phenomena (Section 8) constitute independently verified sharp generalization transitions under fixed loss functions, suggesting real discontinuities exist in some training dynamics even if many benchmark-level "emergent abilities" are artifacts.

Key researchers. Rylan Schaeffer, Brando Miranda, Sanmi Koyejo (Stanford).

1.6 Wall 3 - Expressivity and Circuit-Complexity Bounds for Transformers

Claim. Fixed-depth Transformers operating with realistic (logarithmic) numerical precision cannot compute functions outside the complexity class TC^0 , and therefore provably cannot solve certain problems believed to require more computational depth (e.g., certain graph connectivity, some arithmetic problems, general Boolean-circuit evaluation) without an unbounded external scratchpad.

Formal framework - the central theorem. Merrill & Sabharwal (2023), building on Merrill & Sabharwal (2022), "Log-Precision Transformers are Uniform Threshold Circuits", prove: any log-precision, depth- d transformer (arbitrary attention function, no saturation assumption required) can be simulated by a non-uniform threshold-circuit family of depth $3 + (9 + 2d_{\oplus})d$ and polynomial size, where $d_{\oplus}$ is the depth needed for approximate floating-point addition - hence log-precision transformers lie in non-uniform TC^0 . The proof is inductive over layers: position embeddings are computed by constant-depth gates, similarity scores and softmax-normalized sums are computed by depth-3 polynomial circuits, and floating-point summation is realized in TC^0 . Related results: Chiang, Cholak & Pillay (2023), "Tighter Bounds on the Expressivity of Transformer Encoders"; Hao, Angluin & Frank (2022), "Saturated Transformers are Constant-Depth Threshold Circuits"; Angluin, Chiang & Yang (2023), "Masked Hard-Attention Transformers Recognize Exactly the Star-Free Languages"; and the survey Strobl, Merrill, Weiss, Chiang & Angluin, "What Formal Languages Can Transformers Express?".

Hard-attention variants. Feng, Merrill et al. and Yao et al. formally characterize unique hard-attention

Transformers (UHAT), generalized unique hard-attention Transformers (GUHAT), and averaging hard-attention Transformers (AHAT) against AC^0 ; AHAT can implement MAJORITY (which is outside AC^0), establishing that even hard-attention variants exceed AC^0 but remain circuit-complexity-bounded.

What is proved vs. suggested. Rigorously proved, under explicit assumptions: (i) fixed number of layers d independent of input length n ; (ii) numerical precision of internal representations is $O(\log n)$ bits; (iii) the simulating circuit family may be non-uniform (later work strengthens to uniform TC^0). The result does *not* apply to transformers with per-input adaptive/unbounded depth (e.g., unbounded Chain-of-Thought - see Section 13) or unbounded/high-precision arithmetic, and does not claim that no problems in TC^0 are practically hard for transformers to *learn* (learnability and expressivity are distinct).

Why it threatens the paradigm. $TC^0 \subsetneq P$ is widely believed (though unproven, akin to $P \neq NP$), so if this separation holds, fixed-depth transformers without external scratchpads are provably unable to solve certain polynomial-time problems (e.g., some formulations of graph connectivity, general boolean formula evaluation) regardless of parameter count or training data - an architecture-intrinsic computational ceiling that no amount of scaling within the same architectural template can remove.

Loopholes/counterarguments. (a) The bound is conditional on log-precision; models with higher (e.g., linear) precision escape the proof mechanism, though this is not how practical transformers are typically implemented (bf16/fp16/int8). (b) Chain-of-Thought with polynomially many intermediate tokens is proven to extend expressivity from TC^0 up to $PTIME$ (Merrill & Sabharwal; Amiri et al., 2025) - the wall is about fixed-depth, single-pass inference specifically. (c) $TC^0 \neq PTIME$ is not proven; the "wall" rests on an unresolved complexity-theoretic conjecture, so it is a strong conditional statement, not an unconditional impossibility.

Key researchers. William Merrill (NYU/AI2), Ashish Sabharwal (AI2), David Chiang (Notre Dame), Yiding Hao, Dana Angluin, Lena Strobl.

1.7 Wall 4 - Topology, Geometry, and Message-Passing Limitations

Claim. (a) Attention stacks provably lose representational rank exponentially with depth absent skip connections/MLPs; (b) message-passing GNNs are bounded in distinguishing power by the 1-dimensional Weisfeiler-Lehman graph-isomorphism test; (c) message passing over long-range graph dependencies suffers exponential "over-squashing" of information into fixed-size vectors.

Formal framework - rank collapse. Dong, Cordonnier & Loukas (2021), "Attention is Not All You Need: Pure Attention Loses Rank Doubly Exponentially with Depth", prove that for a pure self-attention network (no skip connections, no MLPs) of depth L with H heads, the residual (deviation from a rank-1 matrix) of the token-representation matrix decays at a rate governed by $\|\cdot\|_{1,\infty}$ with a doubly-exponential dependence on depth L , formally establishing that output tokens converge toward a single shared representation as depth increases - "rank collapse." Skip connections and MLPs are shown (theoretically and empirically) to counteract this collapse, which is why practical Transformers avoid catastrophic collapse. Follow-up work qualifies the universality: Dovonon, Bronstein & Kusner (2024), "Setting the Record Straight on Transformer Oversmoothing" prove that for a simplified Transformer layer (fixed attention, fixed weights, residual connection), convergence to a common vector is *possible but not inevitable* - it depends on the eigenspectrum of value/projection weights and the sign of LayerNorm weights, and they empirically exhibit pretrained models that do *not* oversmooth. Bao, Hataya & Karakida (2024) similarly show localization (the inverse of collapse) is governed by the query-key eigenspectrum mean/-variance.

Formal framework - WL-boundedness of GNNs. The Weisfeiler-Lehman correspondence (foundational, pre-2020, established by Xu et al. 2019 and formalized further since) shows standard message-passing GNNs cannot distinguish graphs that the 1-WL test cannot distinguish (e.g., regular graphs of the same degree sequence), bounding expressivity below full graph-isomorphism power. Kanatsoulis & Ribeiro (2023) show this bound is *not tight for all GNN variants*: standard GNNs with anonymous inputs, analyzed via graph-operator eigendecomposition, can in some regimes exceed 1-WL by distinguishing graphs with different eigenvalues while preserving permutation equivariance - nuancing the popular claim that "GNNs = 1-WL" as an absolute ceiling. Franks, Morris, Velingker & Geerts (2024) further show, using margin theory, that increased WL-style expressivity does *not* straightforwardly translate into better generalization - expressivity and generalization are shown to be only loosely coupled, empirically

confirmed.

Formal framework - over-squashing. Shao, Shi, Han, Guo, Zhao & Gao (2023) unify over-smoothing and over-squashing via a graph heat-equation/physics-informed formalism, showing both pathologies emerge from the same diffusion-operator spectral properties, and propose reversed-time high-pass filtering (MHKG) as a partial architectural remedy - implying the pathology is not architecture-independent but tied to specific message-passing diffusion operators.

What is proved vs. suggested. Proved: rank collapse for *pure* attention without skip/MLP (a narrow architectural regime rarely deployed as-is); the 1-WL upper bound for a broad but specific class of message-passing GNNs; spectral characterizations of over-squashing/over-smoothing for diffusion-type GNNs. Not proved: that all practical, skip-connected, LayerNorm-equipped Transformers or attention-augmented GNNs suffer these pathologies in the limit - subsequent work explicitly shows these are avoidable, architecture-dependent effects, not universal laws of attention.

Why it threatens the paradigm. If depth-scaling drives representational collapse or requires increasingly delicate spectral tuning to avoid it, then "just add more layers" is not a free-lunch scaling axis - it interacts destructively with representational diversity, and long-range/topological reasoning (e.g., multi-hop graph reasoning, global structural inference) is bounded by provable information-theoretic bottlenecks in the underlying message-passing/attention substrate.

Loopholes/counterarguments. Modern architectures (residual streams, LayerNorm, multi-head diversity, graph rewiring, curvature-based edge sparsification) are explicit, empirically effective countermeasures; the rank-collapse and over-squashing theorems apply to idealized/simplified variants and are best read as design constraints rather than as proofs that scaled, engineered Transformers/GNNs necessarily fail in deployment.

Key researchers. Yihe Dong, Andreas Loukas (attention rank collapse); Christopher Morris, Floris Geerts (WL expressivity); Michael Bronstein, Matt Kusner (oversmoothing); Junbin Gao (over-squashing unification).

1.8 Wall 5 - Thermodynamic and Energetic Bounds on Training

Claim. Analog/physical implementations of deep-network training are subject to a provable thermodynamic lower bound on total dissipated energy, scaling with the number of trainable parameters and dataset size, derived from mapping feedforward architectures onto a physical free-energy functional.

Formal framework. Tkachenko (2025), "Thermodynamic Bounds on Energy Use in Quasi-Static Deep Neural Networks" establishes a universal lower bound $E < 2NDkT$ on training energy for quasi-static analog DNNs, where N is the number of trainable parameters, D is dataset size, k is Boltzmann's constant, and T is temperature - derived by mapping the network's parameter update dynamics onto a physical free-energy functional under a quasi-static (near-equilibrium) assumption. Complementary generalized-Landauer-bound work (e.g., work formalizing Landauer's principle for general information-processing systems, MDPI Entropy 2022) extends the classical $kT \ln 2$ -per-bit-erasure bound to broader computational settings, though full access to that paper's specific theorem text was not available (page access denied at fetch time); the classical Landauer bound itself is well established in the physics literature predating this survey window.

What is proved vs. suggested. Proved, under explicit and restrictive assumptions: quasi-static (slow, near-equilibrium) parameter updates and an *analog* physical substrate for computation. Digital GPU/TPU training - the actual deployed regime - operates far from thermodynamic equilibrium and dissipates orders of magnitude more energy than the Landauer-type bound, for engineering reasons (switching losses, memory movement, cooling) unrelated to the abstract computational lower bound; the theorem constrains an idealized physical lower bound, not the observed energy cost of digital LLM training, which is dominated by non-fundamental inefficiencies. It also does not directly bound the *statistical* efficiency of the learning algorithm - only the physical energy cost of updates in the specified analog regime.

Why it threatens the paradigm. It formalizes the intuitive claim that training cost scales at least linearly in $N \times D$ even under an idealized reversible-computation limit, meaning aggressive continued parameter and data scaling faces an energy floor that cannot be engineered away by better silicon alone - relevant

to sustained scaling-law extrapolation (Section 12) independent of algorithmic efficiency gains.

Loopholes/counterarguments. The bound is specific to quasi-static analog systems; current large-scale training is digital and operates in a regime dominated by non-thermodynamic (architectural, algorithmic, cooling) losses, so the practical relevance of the *exact* bound to today's GPU clusters is limited - it is best read as a physics lower bound establishing an ultimate floor rather than a binding constraint on current hardware. Reversible/adiabatic computing research (not covered in the surveyed window with LLM-specific results) could in principle approach the bound, but no LLM-scale reversible training system has been demonstrated as of 2026.

Key researchers. Alexei V. Tkachenko.

1.9 Wall 6 - Information Closure, No-Free-Lunch, Kolmogorov Complexity, and MDL/Bayesian Limits

Claim. (a) No-Free-Lunch theorems formally state that averaged over a uniform distribution on all possible target functions, no learning algorithm outperforms any other, including random guessing; (b) real-world learnability derives entirely from a match between inductive bias and the (typically low) Kolmogorov complexity of real tasks, not from architecture-independent universality; (c) Kolmogorov complexity itself is uncomputable, imposing a fundamental identifiability ceiling on any complexity-based justification of generalization.

Formal framework. The classical NFL theorems (Wolpert & Macready, pre-2020) are reframed by Goldblum, Finzi, Rowan & Wilson (2024), "The No Free Lunch Theorem, Kolmogorov Complexity, and the Role of Inductive Biases in Machine Learning": they formally argue that while NFL holds exactly under a uniform distribution over target functions, virtually all uniformly sampled datasets have near-maximal Kolmogorov complexity, whereas real-world tasks disproportionately have low Kolmogorov complexity; they show (empirically and via compression arguments) that neural architectures - including randomly initialized language models - exhibit an inductive bias toward generating and compressing low-complexity sequences, and that domain-specific architectures (e.g., vision CNNs) can compress datasets from ostensibly unrelated domains, evidencing a broad, not narrowly domain-specific, low-complexity bias. Vitányi (2020), "How Incomputable Is Kolmogorov Complexity?" formally reaffirms the classical uncomputability of $K(x)$ (no algorithm computes $K(x)$ exactly for all x ; it is only upper-semicomputable), which underlies why any MDL-style justification of a specific learned model as "optimal" is asymptotic and non-constructive rather than verifiable in finite time. Shaw, Cohan, Eisenstein & Toutanova (2026), "Bridging Kolmogorov Complexity and Deep Learning" introduce "asymptotically optimal description length objectives" for Transformers, proving a minimizer of such an objective achieves optimal compression for any dataset up to an additive constant - an attempt to operationalize MDL for practical Transformer training, explicitly built atop the acknowledged uncomputability of exact Kolmogorov complexity (approximated, not computed exactly).

What is proved vs. suggested. Proved (classical, foundational): the NFL theorem under a uniform-function-distribution assumption; uncomputability of exact Kolmogorov complexity. Proved (2024, Goldblum et al.): the near-certainty that uniformly sampled data has near-maximal complexity, and a compression-based inductive-bias-matching argument. Empirically suggested, not proved as a theorem: that *all* practical LLM inductive biases are well matched to real-world task complexity in a quantitatively tight sense - the paper offers compression experiments as evidence, not a closed-form bound relating LLM architecture bias to task complexity distributions.

Why it threatens the paradigm. NFL implies architectural universality claims ("this one model class is broadly best") are automatically false outside the low-complexity-task regime; if scaling merely exploits an ever-more-refined match to internet-text complexity statistics rather than acquiring general intelligence, then out-of-distribution or genuinely novel-complexity tasks are structurally disadvantaged by the very inductive bias that makes in-distribution scaling work - connecting directly to OOD limits (Section 11) and hallucination (Section 12).

Loopholes/counterarguments. NFL is a statement about *uniform* distributions over all functions, a distribution that essentially never arises in practice; critics correctly note NFL alone says nothing damaging about learning on structured, low-complexity real-world data - the informative content is entirely in the low-complexity concentration argument, which is empirical/statistical rather than a hard theorem

bounding LLM performance. MDL-based Transformer objectives (Shaw et al., 2026) show constructive progress is possible despite uncomputability, via approximation rather than exact computation.

Key researchers. Micah Goldblum, Andrew Gordon Wilson (NYU); Paul Vitányi (CWI, foundational Kolmogorov complexity); Peter Shaw, Kristina Toutanova (Google DeepMind).

1.10 Wall 7 - Loss-Landscape Degeneracy, Non-Identifiability, and Optimization Pathologies

Claim. Deep networks are statistically singular models - their parameter-to-function map is non-injective - which invalidates classical asymptotic statistics (Fisher-information-based generalization bounds, BIC) and requires a fundamentally different (algebraic-geometric) theory to explain training dynamics such as grokking and phase transitions.

Formal framework. Lakkapragada (2025), "Using Physics-Inspired Singular Learning Theory to Understand Grokking & Other Phase Transitions" applies Watanabe's Singular Learning Theory (SLT), explicitly noting that "classical statistical inference and learning theory often fail to explain the success of modern neural networks" because "these models are non-identifiable (singular), violating core assumptions behind PAC bounds and asymptotic normality." The paper empirically tests an Arrhenius-style rate hypothesis for the SLT free energy $\mathcal{F}_n$ using a grokking modulo-arithmetic model and measures how the local learning coefficient λ_α scales with problem difficulty across polynomial regressors, low-rank linear networks, and low-rank autoencoders, finding partial agreement with and partial deviation from theoretical SLT predictions. Complementary rank-collapse spectral theory (Section 6) shows optimization dynamics interact with representational degeneracy directly: the eigenspectrum of value/query/key weight matrices governs whether training converges to degenerate (rank-collapsed or oversmoothed) solutions (Dovonon, Bronstein & Kusner, 2024; Bao, Hataya & Karakida, 2024).

What is proved vs. suggested. Proved (foundational SLT, Watanabe, pre-2020): for singular statistical models, the generalization error scales with a real log-canonical threshold λ rather than half the parameter count (as in the regular/non-singular case), formally replacing BIC-style asymptotics. Empirically explored, not fully proved as a general theorem for Transformer-scale LLMs: the specific quantitative Arrhenius-rate behavior of the free energy during grokking-type phase transitions in realistic (not toy) LLMs - the 2025 paper explicitly reports "meaningful deviations from theoretical expectations" in some tested regimes, an epistemically important caveat against overclaiming SLT as a complete explanation.

Why it threatens the paradigm. If neural network parameter spaces are irreducibly singular, then flatness/sharpness-based generalization arguments, information-criterion model-selection heuristics, and any claim of a well-defined "effective parameter count" for LLMs are not merely approximate but categorically inapplicable without the SLT correction - undermining a large body of classical generalization theory as applied naively to Transformers, and explaining otherwise-mysterious discontinuous training phenomena (grokking) as genuine phase transitions in the free-energy landscape rather than optimization noise.

Loopholes/counterarguments. SLT is itself an active, incompletely validated research program at LLM scale - most rigorous SLT results are established for toy/small models; extrapolation to billion-parameter Transformers is an open empirical question, and the surveyed paper itself reports partial disagreement with theory. Flatness-based generalization measures, while shown to be reparametrization-ambiguous in isolation, remain empirically predictive in restricted settings (e.g., Nagarajan & Kolter, "Sharp Minima Can Generalize for Deep Nets" shows sharp minima are not disqualifying), so the pathology is one of *naive* application of classical tools, not proof that no generalization theory can work.

Key researchers. Sumio Watanabe (SLT foundations, pre-2020), Anish Lakkapragada, Gbétondji J-S Dovonon, Michael Bronstein, Matt Kusner.

1.11 Wall 8 - Mode Collapse, Model Collapse, and Error Accumulation Under Recursive Synthetic-Data Training

Claim. Recursively training generative models on their own (or predecessor models') synthetic outputs causes a provable, statistically characterized divergence from the true data distribution - "model collapse"

- with formally derived collapse rates for specific model families, though the *severity and universality* of the effect (especially with mixed real/synthetic data) remains actively contested.

Formal framework. Shumailov et al. (2024), "AI Models Collapse When Trained on Recursively Generated Data," (original Nature paper, critiqued and extended below) established the empirical/theoretical phenomenon; Borji (2024), "A Note on Shumailov et al." develops a Kernel-Density-Estimation-based theoretical model of iteratively resampling from a fitted distribution, concluding the collapse is "a statistical phenomenon" that "may be unavoidable" under the studied resampling regime - an important epistemic distinction: the *mechanism* is unavoidable under the stated conditions (finite-sample re-estimation with resampling), but conditions such as accumulating (not replacing) real data can change the outcome. Suresh, Thangaraj & Khandavally (2024), "Rate of Model Collapse in Recursive Training" derive the *exact* asymptotic rate of collapse under maximum-likelihood estimation for specific well-studied distribution families (a rate not previously known even for Gaussian families), addressing an open quantitative question. Seddik, Chen, Hayou, Youssef & Debbah (2024), "How Bad Is Training on Synthetic Data?" provide a statistical model separating synthetic-only vs. mixed real/synthetic recursive training scenarios, formally characterizing how the tails of the original distribution disappear under recursive resampling - a specific, provable tail-truncation mechanism.

What is proved vs. suggested. Proved, for explicit statistical families (e.g., Gaussian, other well-studied parametric families) under maximum-likelihood or near-MLE recursive re-estimation: a specific asymptotic collapse rate (Suresh et al., 2024); a mechanism by which distributional tails vanish under pure-synthetic recursive training (Seddik et al., 2024). Not proved as universal: that model collapse is unavoidable for arbitrary neural generative models under arbitrary real/synthetic data-mixing strategies - subsequent work explicitly proposes and partially validates escape strategies (e.g., Feng et al., "A Theoretical Perspective: How to Prevent Model Collapse"; "Escaping Model Collapse via Synthetic Data Verification," 2025; "Escaping Collapse: The Strength of Weak Data," 2025), indicating the phenomenon is real and formally characterized for specific regimes but conditionally avoidable, not an absolute law for all training protocols.

Why it threatens the paradigm. As an increasing fraction of internet text used for future pretraining is itself LLM-generated, the formally derived collapse mechanisms suggest a self-inflicted data-quality ceiling on the entire scaling paradigm (Section 12): each generation of "internet-scale" data becomes progressively less representative of the true human-generated distribution unless real data provenance is actively tracked and preserved, connecting directly to the sustainability of continued pretraining-data scaling.

Why it is not fully fatal. Provenance-aware training (mixing verified real data, verification/filtering of synthetic data, and non-replacement accumulation strategies) is shown in multiple 2025 papers to mitigate or avoid collapse, so the wall is better characterized as a structural risk requiring active countermeasures than an unconditional endpoint for the paradigm.

Key researchers. Ilya Shumailov et al. (original phenomenon); Ali Borji (statistical critique); Ananda Theertha Suresh, Andrew Thangaraj (rate theorem); Mohamed El Amine Seddik, Soufiane Hayou, Merouane Debbah (statistical modeling).

1.12 Wall 9 - Out-of-Distribution, Domain-Generalization, and Compositional/Systematic Generalization Limits

Claim. Formal, provable conditions exist under which sequence models cannot generalize beyond the length or compositional structure seen during training - "provable" in the sense of explicit necessary/sufficient conditions on the data-generating process and hypothesis class, not an unconditional impossibility for all architectures and all tasks.

Formal framework. Ahuja & Mansouri (2024), "On Provable Length and Compositional Generalization" set up a formal framework with input sequences $x_{\leq t} \in \mathbb{R}^n$, labels $y_t = f(x_{\leq t})$, a family of input distributions and labeling functions $P = (P_{X_{\leq t=1}}^\infty, f)$, risk $R(h; T) = \sum_{t=1}^T \mathbb{E}[\ell(h(x_{\leq t}), y_t)]$, and a test distribution/labeling pair Q potentially at longer lengths or novel compositions than seen in training. They derive explicit conditions (on the support of Q relative to P , and on architecture classes including linear RNNs, some Transformer variants) under which a learner achieving zero training risk under P provably achieves zero risk under length/compositional extrapolation to Q - a *provable positive* result

under stated conditions, and conversely a characterization of when such extrapolation is not guaranteed absent those conditions. Wang & Sun (2025), "Is the Reversal Curse a Binding Problem?" empirically and theoretically link the well-documented "reversal curse" (models trained on "A is B" often fail to infer "B is A") to the cognitive-science "binding problem," identifying it as a basic generalization failure tied to how Transformer representations bind entities to relations, without full training-dynamics proofs of inevitability. Hupkes, Dankers, Mul & Bruni (2020), a foundational (pre-2020-effective, published JAIR 2020) empirical decomposition of "compositionality" into five testable aspects, explicitly disclaim providing "a new definition of compositionality" or any general theorem, cautioning their empirical findings "should not be taken as general" claims about neural network compositionality - an important epistemic anchor showing the compositional-generalization literature is largely empirical/taxonomic rather than proof-based prior to 2024's provable-generalization work.

What is proved vs. suggested. Proved, under explicit distributional/support and architectural conditions: length and compositional generalization guarantees (positive result) and corresponding failure conditions (negative result) in Ahuja & Mansouri (2024) - this is the closest the literature comes to a genuine formal OOD generalization theorem for sequence models, but it is explicitly conditional (support inclusion, architecture class), not a universal impossibility statement. Empirically documented, not proved as inevitable: the reversal curse and general "binding" failures - these remain robust empirical phenomena with proposed but not fully settled theoretical explanations.

Why it threatens the paradigm. If genuine compositional/length generalization requires support and architectural conditions that current large-scale pretraining does not guarantee or verify, then benchmark generalization successes may not transfer to genuinely novel compositional structures, and the reversal-curse-type binding failures suggest a structural encoding limitation in how relational information is stored, independent of scale.

Loopholes/counterarguments. The Ahuja & Mansouri framework is explicitly double-edged: it proves generalization *is* possible under identifiable conditions, meaning the correct reading is "generalization requires specific structural alignment between train support and test targets," not "generalization is impossible." Architectural mitigations for the reversal curse (e.g., bidirectional training objectives, explicit relation-symmetrization) are active, partially successful research directions.

Key researchers. Kartik Ahuja (Meta FAIR), Amin Mansouri (Mila); Boshi Wang, Huan Sun (Ohio State); Dieuwke Hupkes (Meta AI, formerly Amsterdam).

1.13 Wall 10 - Hallucination and Unfaithfulness as Formal Impossibility or Undecidability Results

Claim. Under explicit formalizations of "hallucination" as inconsistency between a computable LLM and a computable ground-truth function, or as a knowledge-aggregation process subject to information-theoretic conservation constraints, hallucination is provably impossible to eliminate - though the scope of "impossible to eliminate" is tightly bound to the specific formalization chosen, and different formalizations yield different (sometimes disputed) conclusions.

Formal framework 1 - computability-theoretic. Xu, Jain & Kankanhalli (2024), "Hallucination Is Inevitable: An Innate Limitation of Large Language Models" formalize a world in which hallucination is defined as inconsistency between a computable LLM and a computable ground-truth function, considering LLMs as general problem solvers; the abstract states they "formalize the problem of hallucination" and "show that it is impossible to eliminate hallucination in LLMs," employing results from computability theory (implicitly invoking diagonalization/halting-problem-style arguments, standard in such "inevitability" proofs, though the specific proof mechanics were not fully retrievable from the abstract page) and extending the claim to "real-world LLMs constrained by provable time complexity."

Formal framework 2 - information-aggregation impossibility. Karpowicz (2025), "On the Fundamental Impossibility of Hallucination Control in Large Language Models" establishes what is explicitly labeled a "Fundamental Impossibility Theorem": "no LLM performing non-trivial knowledge aggregation can simultaneously achieve truthful knowledge representation, semantic information conservation, complete revelation of relevant knowledge, and knowledge-constrained optimality." The paper explicitly states the impossibility "stems from the mathematical structure of information aggregation" and "does not stem from engineering limitations" - a structural, not merely practical, claim, though full access to the

formal definitions of the four properties and the proof mechanism was not obtained from the fetched page content.

Formal framework 3 - open-world generalization framing. Xu (2025), "Hallucination Is Inevitable for LLMs with the Open World Assumption" reframes hallucination as a manifestation of the generalization problem: under a Closed World assumption (train/test distributions consistent), hallucination may be mitigated; under an Open World assumption (unbounded environment), hallucination becomes inevitable - explicitly distinguishing correctable from unavoidable hallucination classes, an important refinement against blanket "hallucination is always unavoidable" claims.

What is proved vs. suggested. Proved, under the *stated* formalizations: hallucination-elimination is impossible in each of the three frameworks above, *given* their respective axioms (computable ground truth mismatch; four simultaneously-required knowledge-aggregation properties; open-world/unbounded environment assumption). Not proved: that hallucination is impossible to *reduce arbitrarily* in practice, nor that these three formalizations are equivalent or jointly necessary - they are three distinct, non-identical impossibility results with different scope, and the field has not converged on a single canonical formalization, meaning claims of "hallucination is mathematically inevitable" should be read as "inevitable under this specific formal model," not as a single unconditional theorem.

Why it threatens the paradigm. If credible under any of these formalizations, no amount of RLHF, retrieval augmentation, or scale can drive hallucination to zero in the open-world deployment regime that matters commercially - a structural ceiling on trustworthy factual reliability rather than an engineering deficiency awaiting a fix.

Loopholes/counterarguments. All three results depend on specific, contestable formal definitions (e.g., what counts as "non-trivial knowledge aggregation," what constitutes the "open world" boundary); critics can accept the theorem while disputing its practical bite - e.g., the Closed-World/Open-World distinction (Xu, 2025) implies most commercially deployed narrow-domain LLM applications may operate closer to the closed-world regime where mitigation, not elimination-impossibility, is the operative constraint. The Karpowicz theorem's four desiderata may not all be simultaneously desired in practice (e.g., "complete revelation of relevant knowledge" may be deliberately sacrificed for calibrated abstention).

Key researchers. Ziwei Xu, Sanjay Jain, Mohan Kankanhalli (NUS); Michał P. Karpowicz; Bowen Xu.

1.14 Wall 11 - Scaling-Law Ceilings, Data-Entropy Limits, and Irreducible Error

Claim. Empirical power-law scaling of loss with compute/parameters/data has an information-theoretic floor set by the entropy of the data-generating process, breaks down ("bends") outside the fitted regime, and is subject to a rigorously derivable compute-optimal parameter/data allocation for at least one analytically tractable model class.

Formal framework. Jeon & Van Roy (2024), "Information-Theoretic Foundations for Neural Scaling Laws" develop what they describe as "rigorous information-theoretic foundations for neural scaling laws," characterizing scaling for data generated by an infinite-width two-layer network, deriving an upper bound on the minimal information-theoretically achievable expected error as a function of parameter count p and token count T , and deriving the compute-optimal p - T relation, which is shown to be linear (up to logarithmic factors) for large compute budgets - consistent with, and providing a first rigorous mathematical derivation underlying, the empirically fitted Hoffmann et al. ("Chinchilla") 2022 law. Maloney, Roberts & Sully (2022), "A Solvable Model of Neural Scaling Laws" construct a joint generative-data/random-feature model, solvable in the dual large-training-set/large-feature limit, deriving necessary conditions (power-law spectral statistics of the data) for scaling laws to arise, along with predicted breakdown/-plateau behavior from finite spectral extent - an explicit, derived account of *why* and *when* scaling laws saturate. Caballero, Gupta, Rish & Krueger (2023), "Broken Neural Scaling Laws" propose the empirically fitted functional form $y = a + (bx^{-c_0}) \prod_{i=1}^n [1 + (x/d_i)^{1/f_i}]^{-c_i f_i}$, explicitly parameterizing multiple smooth power-law "breaks," with the constant a representing an irreducible floor as $x \rightarrow \infty$ - this is a fitted empirical model, not a first-principles derivation, but is widely used to characterize observed ceiling/plateau behavior across many domains (including LLM downstream task performance, adversarial robustness, and diffusion models). Porian, Wortsman, Jitsev, Schmidt & Carmon (2024), "Resolving Discrepancies in Compute-Optimal Scaling of Language Models" show that the well-known disagreement between the Kaplan et al. and Hoffmann et al. ("Chinchilla") scaling-law predictions is explained by

three specific confounds (last-layer computational cost, warmup duration, scale-dependent optimizer tuning), and that correcting these yields agreement with the Chinchilla law - an important methodological result showing that some "scaling law disagreements" are artifacts of experimental confounds rather than genuine theoretical disputes.

What is proved vs. suggested. Proved, for the specific analytically tractable model classes studied (infinite-width two-layer teacher networks; random-feature/power-law-spectrum generative models): the existence of an information-theoretic error floor and its scaling exponents, and necessary conditions on data spectral statistics for power-law scaling to emerge at all. Empirically fitted, not derived from first principles: the Broken Neural Scaling Law functional form itself, though it is grounded in and consistent with the more fundamental derivations. Not proved in general: that all LLM training regimes (with real, non-idealized internet text, not power-law-spectrum synthetic data) obey a single universal, first-principles-derived scaling exponent - the rigorous derivations are for tractable toy models, and their applicability to frontier LLM pretraining is an extrapolation, not a proven identity.

Why it threatens the paradigm. If loss has a provable, entropy-set floor and the compute-optimal parameter/data ratio is fixed by data spectral statistics rather than free to be pushed arbitrarily by more compute, continued brute-force scaling faces diminishing, eventually vanishing, marginal returns - directly bounding the "scale is all you need" strategic thesis, and combining multiplicatively with the data-availability/model-collapse concerns of Section 10.

Loopholes/counterarguments. The rigorous information-theoretic floor is specific to the fitted teacher-student model class; real-world data does not perfectly satisfy the model's generative assumptions, so the *exact* floor value for frontier LLMs is not known in closed form - only the *existence* of some floor and the qualitative scaling regime are established with rigor. Algorithmic/data-quality improvements (better tokenization, curriculum, data curation, mixture-of-experts sparsity) can improve the *effective* constant terms even if the asymptotic exponent is fixed, so scaling-law ceilings bound brute-force parameter/data scaling specifically, not all forms of capability improvement.

Key researchers. Hong Jun Jeon, Benjamin Van Roy (Stanford); Alexander Maloney, Daniel A. Roberts, James Sully (McGill/MIT/Anthropic); Ethan Caballero, David Krueger (Mila/Cambridge); Jared Kaplan; Jordan Hoffmann et al. (DeepMind, Chinchilla).

1.15 Wall 12 - Limitations of In-Context Learning and Chain-of-Thought as Computation/Learning Mechanisms

Claim. Chain-of-Thought (CoT) provably extends Transformer expressivity beyond the fixed-depth TC^0 ceiling (Section 5) toward P TIME, but only when the number of intermediate reasoning tokens is allowed to scale with problem size; explicit, tight *lower bounds* on the number of CoT steps required for specific algorithmic problems have now been derived, and specific architectural function classes (e.g., one-layer State Space Models) are proven unable to perform function composition efficiently even with CoT.

Formal framework. Merrill & Sabharwal (Section 5) establish that polynomial-length CoT scratchpads extend transformer expressivity from TC^0 to P TIME - a positive result contingent on unbounded (polynomial) intermediate token budgets. Amiri, Huang, Rofin & Hahn (2025), "Lower Bounds for Chain-of-Thought Reasoning in Hard-Attention Transformers" initiate systematic *lower bounds* on the number of CoT steps required across various algorithmic problems in the hard-attention regime, proving bounds "tight up to logarithmic factors," and note the empirically striking finding that transformers require scratchpads even for some problems provably inside TC^0 (e.g., Parity, Multiplication) - challenging the optimistic reading that all TC^0 -complete tasks are "easy" for transformers without CoT, since the *practical* CoT-length requirement can exceed naive circuit-complexity-based expectations. Zubić, Soldá, Sulser & Scaramuzza (2025), "Limits of Deep Learning: Sequence Modeling through the Lens of Complexity Theory" prove that one-layer Structured State Space Models (SSMs) cannot efficiently perform function composition over large domains without impractically large state sizes, and that even with CoT prompting, the number of required reasoning steps scales unfavorably with function-composition complexity; they further show the language recognized by a finite-precision SSM lies within the regular languages - a strict expressivity ceiling for that architecture class - and empirically corroborate compounding-error "shortcut" behavior on multi-digit multiplication, dynamic programming, and

constraint-satisfaction (Einstein’s puzzle) tasks even with advanced prompting.

What is proved vs. suggested. Proved: the $TC^0 \rightarrow PTIME$ CoT-extension theorem (conditional on polynomial CoT length); tight-up-to-log-factor lower bounds on required CoT steps for specific algorithmic problem classes in the hard-attention regime; strict regular-language containment for finite-precision one-layer SSMs, and provable inefficiency of one-layer SSM function composition. Empirically demonstrated, not fully proved as a general theorem for arbitrary multi-layer Transformer CoT: that models resort to "shortcuts" leading to compounding errors on compositional tasks - this is an experimental finding in Zubić et al. (2025), consistent with but not identical to a formal lower-bound proof for full multi-layer Transformer architectures (their strict theorems are stated for one-layer SSMs specifically).

In-context learning as implicit optimization. A distinct strand (Bao, Hataya et al.; induction-head mechanistic work, e.g., Olsson et al. and subsequent induction-head studies) characterizes in-context learning mechanistically as pattern-completion via induction-head circuits rather than genuine novel-task learning, though this remains a predominantly mechanistic/empirical research program rather than a closed formal impossibility result as of the survey window.

Why it threatens the paradigm. If genuine algorithmic reasoning via CoT requires a number of intermediate tokens that scales polynomially (or worse) with problem complexity - and specific architectures (one-layer SSMs) are proven to require *impractically* large state or step counts for basic composition - then "reasoning via more tokens" is not a free scaling axis either: it inherits its own complexity-theoretic cost structure, and shortcut/compounding-error behavior under advanced prompting suggests current training does not reliably induce the theoretically sufficient CoT-length behavior even when architecturally possible.

Loopholes/counterarguments. The strict SSM composition-inefficiency theorem is stated for one-layer SSMs, a narrow (though influential, e.g., early Mamba-style) architecture class - multi-layer, hybrid SSM-attention, or full Transformer architectures are not covered by the same strict proof, and the paper’s own multi-layer/Transformer results are described as complexity-theoretic bounds rather than the identical composition-inefficiency statement. The CoT lower-bound results are specific to the hard-attention regime, which is an idealized proxy for practical soft-attention transformers; soft attention may behave differently, and the authors explicitly frame their contribution as "initiating" (not concluding) systematic lower-bound study.

Key researchers. William Merrill, Ashish Sabharwal; Alireza Amiri, Michael Hahn (Saarland); Nikola Zubić, Davide Scaramuzza (University of Zurich).

1.16 Comparative Synthesis and Fatality Ranking

The critiques divide sharply along an axis of conditional rigor vs. unconditional bite. Ranking fatality requires separating (a) results that are mathematically airtight but narrow in scope, (b) results that are broad in claimed scope but rest on contested or unproven complexity-theoretic conjectures, and (c) results that are genuinely empirical/statistical rather than theorem-backed.

Tier 1 - Structural, architecture-specific, rigorously proved (Fatality 5). The TC^0 circuit-complexity ceiling for fixed-depth, log-precision Transformers (Merrill & Sabharwal) is the most mathematically decisive result in this catalog: it is a genuine theorem (not a conjecture) under explicit, realistic assumptions (bounded depth, bounded precision), and it correctly predicts practically observed failure classes (compositional arithmetic, certain graph problems) absent CoT. Its only escape hatch - unbounded CoT - is itself now subject to matching lower bounds (Amiri et al., 2025), so the wall and its stated loophole are both formally characterized, an unusually mature state for a theoretical AI critique. Its bite is nonetheless *conditional* on the unproven $TC^0 \subsetneq PTIME$ separation, so even this "fatality 5" result is one conjecture away from being reclassified.

Tier 2 - Broad-scope impossibility, but definitionally contingent (Fatality 4). Hallucination-impossibility results (Xu, Jain & Kankanhalli; Karpowicz; Xu, 2025), provable-generalization conditions (Ahuja & Mansouri), model-collapse rate theorems (Suresh et al.), and topology/rank-collapse results (Dong et al.; Franks et al.) are each rigorously proved *given their stated formalization*, but the formalizations themselves (what counts as "hallucination," "non-trivial knowledge aggregation," "collapse," "over-squashing") are contestable, non-unique, and in several cases explicitly shown to be avoidable under alter-

native architectural or data-provenance choices. These are genuine mathematical walls for the *specific formal model*, not universal walls for "LLMs" as commercially deployed.

Tier 3 - Statistically/empirically grounded, theorem-adjacent (Fatality 3). No-Free-Lunch/Kolmogorov-complexity arguments (Goldblum et al.), Singular Learning Theory applications to grokking (Lakkapragada), and information-theoretic scaling-law derivations (Jeon & Van Roy; Maloney et al.) combine rigorous classical theorems (NFL, SLT foundations, information theory) with a significant empirical/extrapolative leap to apply them to frontier-scale LLMs; the leap itself is not proved, only motivated and partially validated.

Tier 4 - Diagnostic/statistical, not theorem-backed (Fatality 2). The emergence-as-mirage critique (Schaeffer et al.) is a sufficiency argument and re-analysis, not a theorem ruling out genuine emergence; the thermodynamic bound (Tkachenko) is rigorous but for a physical regime (quasi-static analog) far removed from deployed digital training; teacher-forcing/autoregressive-mismatch critiques (Bachmann & Nagarajan) are constructive counterexamples on specific tasks, not general impossibility proofs.

Cross-cutting observation. The single most consequential *interaction effect* across walls is between Tier 1 (circuit-complexity ceiling), Tier 2 (model collapse, scaling-law data ceilings), and Tier 2 (hallucination impossibility): if fixed-depth inference is circuit-complexity-bounded (Section 5), CoT-length requirements grow with task complexity (Section 14), usable non-synthetic data is finite and subject to collapse dynamics under recursive self-training (Section 10), and hallucination is provably non-eliminable in open-world deployment (Section 12), then the three most durable levers of the current paradigm - more depth, more inference-time compute, and more/better data - each face an independently derived, only partially escapable mathematical ceiling, even though no single result proves the paradigm cannot progress further within these ceilings.

1.17 Explicit Epistemic Flags

Disputed / actively contested: the universality and severity of "model collapse" outside specific resampling protocols (Borji's re-analysis vs. multiple 2025 collapse-avoidance papers); whether "emergent abilities" are entirely metric artifacts or partly genuine phase transitions (Schaeffer et al. vs. Du et al.); whether GNN expressivity beyond 1-WL meaningfully improves generalization (Franks et al. shows it often does not).

Conditional, not universal, despite "impossibility" framing in source titles: all three hallucination-inevitability results (Section 12) depend on specific formal definitions that are not community-standardized; the SSM composition-inefficiency theorem (Zubić et al.) is proved for one-layer SSMs, not all sequence architectures.

Architecture-specific, not general limits on learning: rank collapse, over-squashing, and the TC^0 ceiling are properties of specific attention/message-passing/fixed-depth computational templates, not of "learning" or "intelligence" as such - alternative computational substrates (unbounded-depth recurrent computation, external memory, tool use) are not covered by these proofs.

Resting on unproven complexity conjectures: the practical bite of the TC^0 ceiling depends on $TC^0 \neq PTIME$, which is widely believed but unproven.

Chapter 2

Catalog of Frontier Research (2015-2026): Physical, Electromagnetic, and Quantum-Biological Mechanisms of Macroscopic Co

2.1 Scope and Evidentiary Grading

This catalog compiles mathematically grounded frameworks addressing whether and how spatially separated biological cognitive agents can be physically coupled through electromagnetic, bioelectric, or wave-mechanical fields. Every framework is graded on a three-tier evidential scale:

E1 - Established mechanism: Direct experimental demonstration, replicated, with quantitative agreement to theory (e.g., ephaptic coupling, Kuramoto-model fits to cortical data, radical-pair magnetoreception).

E2 - Plausible but incomplete biophysical hypothesis: Coherent mathematical formalism, some empirical correlates, but the causal chain from mechanism to macroscopic cognitive effect (especially cross-agent field mediation) is not closed.

E3 - Unsupported/speculative: No demonstrated bridge from the physical mechanism to inter-agent cognitive coupling; included only to explicitly rule it out or to show where popular claims exceed the data.

No framework surveyed here demonstrates direct field-mediated transfer of cognitive content between spatially separated brains at a distance greater than tens of centimeters under normal (unaided, non-technological) conditions. This constraint is stated up front because it recurs as the central negative result across every EM-field, quantum-biological, and morphogenetic-field framework examined below.

2.2 Coupled Oscillator Networks: The Kuramoto/Sakaguchi Family

2.2.1 Canonical model and order parameter

The baseline model for macroscopic phase synchronization is the Kuramoto model of N globally coupled phase oscillators:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N$$

where θ_i is the phase, ω_i the natural frequency (drawn from a distribution $g(\omega)$), and K the coupling strength. The macroscopic coherence is captured by the complex order parameter

$$r e^{i\psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}$$

with a synchronization transition at critical coupling $K_c = 2/[\pi g(0)]$ for symmetric unimodal $g(\omega)$ (Dynamics in the Sakaguchi-Kuramoto model with bimodal distribution).

2.2.2 Sakaguchi generalization (phase lag/frustration)

The Sakaguchi-Kuramoto model adds a phase-lag (frustration) parameter α , which is mathematically equivalent to a small transmission/synaptic delay:

$$\dot{\theta}_i = \omega_i - \frac{K}{N} \sum_{j=1}^N \sin(\theta_i - \theta_j + \alpha)$$

For $|\alpha| < \pi/2$ the system still synchronizes spontaneously; at $\alpha = \pi/2$ the system becomes conservative/integrable, and larger phase lags produce chimera states, traveling waves, or discordant (π -state) synchronization (saka18.eps, arXiv; Model Reduction for the Kuramoto-Sakaguchi Model). Two-population extensions yield an explicit inter-group phase-difference variable $\delta = \Theta_1 - \Theta_2$ governing zero-lag, π -state, or traveling-wave collective regimes (Two-Population Kuramoto-Sakaguchi Model) - this is the closest legitimate mathematical analogue to "coupling between two separated oscillator populations," but empirically it has been applied to intra-brain or ecological populations, not to EM-mediated inter-brain coupling.

2.2.3 Higher-order (hypergraph) interactions

Recent work (2024-2026) generalizes pairwise Kuramoto coupling to simplicial/hypergraph interactions, where triads or larger groups of oscillators interact jointly rather than only pairwise. This produces first-order (discontinuous) synchronization transitions and novel bifurcation structure not seen in pairwise models (When higher-order interactions enhance synchronization: the case of the Kuramoto model; Synchronization and Chaos in Adaptive Kuramoto Networks with Higher-Order Interactions: A Review; Global synchronization in generalized multilayer higher-order networks). These frameworks are mathematically rigorous (E1 for the pure dynamics) but their biological application is to *intra*-brain epileptic network stability (Synchronization stability of epileptic brain network with higher-order interactions), not inter-agent coupling - E2/E1 for intra-brain dynamics; no bridge demonstrated to inter-agent coupling (E3 boundary).

2.2.4 Empirical fit to cortical data

A 2025 Kuramoto-based reconstruction of cortical activity ("Around the brain in 80 Hz") demonstrates that resting-state and task EEG/MEG dynamics are reasonably captured by connectome-weighted Kuramoto oscillator ensembles (Around the brain in 80 Hz: A Kuramoto model perspective of cortical neuronal activity). Whole-brain modeling using hybrid structural connectomes (E1 for intra-brain fit quality) is well established (Enhanced simulations of whole-brain dynamics using hybrid resting-state structural connectomes). These models are strictly intra-cranial: the coupling matrix is the physical white-matter connectome, not any hypothesized inter-cranial field.

2.2.5 Phase Response Curves (PRCs)

PRCs quantify how a single perturbation shifts oscillator phase, and are the standard tool for reducing a full neural (Hodgkin-Huxley-class) model to a scalar phase equation, enabling Kuramoto-style analysis of real neurons (The Utility of Phase Models in Studying Neural Synchronization; Phase response approaches to neural activity models with distributed delay). PRC-based coupling functions $H(\theta_i - \theta_j)$ generalize the sinusoidal Kuramoto coupling term when the perturbation is weak and the oscillator is on a stable limit cycle - this is E1, a rigorous and experimentally validated reduction technique, but again bounded to physically connected (synaptic/ephaptic/electrical) neural coupling, not distance-independent field transmission.

2.3 Neural Mass Models, Criticality, and Scale-Free Dynamics

2.3.1 Neural mass / Wilson-Cowan framework

The Wilson-Cowan equations model coupled excitatory (E) and inhibitory (I) neural populations:

$$\tau_E \dot{E} = -E + (1 - rE)S_E(c_1 E - c_2 I + P)$$

$$\tau_I \dot{I} = -I + (1 - rI)S_I(c_3E - c_4I + Q)$$

where S_E, S_I are sigmoidal gain functions (The Wilson-Cowan model, 36 years later; Wilson-Cowan Equations for Neocortical Dynamics). Extending this to a spatial continuum yields neural field theory with nonlocal integral coupling kernels $w(x - x')$, producing traveling waves, bumps, and multistable cortical rhythms (Travelling waves in a neural field model with refractoriness; Biophysical Mechanisms of Multistability in Resting-State Cortical Rhythms). This is E1 as an intra-cortical continuum-field description, but the "field" here is a mathematical continuum limit of anatomically coupled neurons, not a physically propagating radiative EM field extending outside the skull.

2.3.2 Criticality and scale-free dynamics

Empirically, cortical dynamics show hallmarks of self-organized criticality: power-law neuronal avalanche size distributions, long-range temporal correlations, and near-critical branching ratios (Universal critical dynamics in high resolution neuronal avalanche data; Broadback Criticality of Human Brain Network Synchronization). Brain criticality measurably predicts individual variation in inter-areal synchronization strength (Brain criticality predicts individual levels of inter-areal synchronization in human electrophysiological data), and homeostatic plasticity rules can self-tune networks toward the synchronization edge (Optimal input reverberation and homeostatic self-organization towards the edge of synchronization). This is E1: robust, replicated, mathematically characterized via branching-process and mean-field theory. It explains scale-free *intra-brain* dynamics; it does not by itself imply or require any shared extracranial field, though criticality maximizes a system's dynamic range and susceptibility to weak external perturbation - a fact exploited (speculatively) by field-based consciousness theories (Section 4).

2.3.3 Chimera states and generalized synchronization manifolds

Chimera states - coexisting synchronized and desynchronized subpopulations in identically coupled oscillator networks - are now documented in multiplex/multilayer neuronal network models and even purportedly in "uncoupled" neurons mediated by multilayer structure (Chimera states in uncoupled neurons induced by a multilayer structure; Chimeras in Multiplex Networks: Interplay of Inter- and Intra-Layer Delays). These are E1 as rigorous nonlinear dynamics but describe within-network partial synchrony, not cross-agent field transmission.

2.4 Graph, Multilayer, and Hypergraph Network Models of Brain and Inter-Brain Structure

Multilayer network formalisms represent each cognitive/behavioral modality (EEG bands, fMRI, physiology) as a layer, with intra-layer and inter-layer coupling matrices $A_{ij}^{(\alpha)}$ and $C_i^{(\alpha\beta)}$ (Multilayer Networks in Neuroimaging; Multilayer Network Approach in EEG Motor Imagery with an Adaptive Threshold). Hypergraph extensions capture beyond-pairwise (triadic+) statistical dependencies in fMRI connectomes via tensor decompositions and hypergraph convolutional networks (Dynamic weighted hypergraph convolutional network for brain functional connectome analysis; Hypergraph representation in brain network analysis). A 2025 multilayer network approach was explicitly applied to *inter*-brain synchrony in hyperscanning data ("Is what I think what you think? Multilayer network-based inter-brain synchrony approach," PMC), representing each participant's brain as a layer and computing cross-layer coupling statistics. This is E1 as a statistical/graph-theoretic formalism for describing correlated activity across brains, but it is descriptive of *statistical coupling*, not proof of a physical field carrier connecting the two brains.

2.5 Electromagnetic Field Theories of Consciousness and Their Range Limits

2.5.1 CEMI field theory (McFadden)

The Conscious Electromagnetic Information (CEMI) field theory proposes that consciousness is instantiated in the brain's global electromagnetic field, generated by summed neuronal dipole currents and detected as EEG/MEG (Integrating information in the brain's EM field: the cemi field theory of consciousness). The theory's key mathematical claim is that fields perform spatial (simultaneous) information integration where discrete computation (logic gates, Turing machines, spiking networks) can only

integrate information temporally/causally - because a field state at any point encodes the superposed contribution of all sources. McFadden formalizes this with an EM-field "AND gate" model: two dipole transmitters generate overlapping fields; a receiver anywhere in the overlap region reads out the joint (AND) computation, illustrating field-distributed analog computation.

Critically, McFadden's own quantitative estimate constrains the coupling range: a single cortical neuron's electric-field perturbation extends only about 80 micrometres, affecting at most roughly 200 neighboring neurons; because the electric component follows an inverse-square law and the magnetic component an inverse-cube law, long-range integration within a brain requires relay chains of synchronously firing neurons, not raw field propagation (CEMI field theory). The theory explicitly does not claim or demonstrate brain-to-brain conscious-field transfer; its own text states only that EM fields could in principle be "downloaded" from any point in a field (analogized to Wi-Fi/television broadcast) and that photon light-cones from brain sources could reach external charged particles with "rapidly diminishing probability" - a statement of physical possibility in the Maxwellian sense, not a demonstrated biological mechanism or effect size. Grade: E2 for intra-brain field-binding hypothesis (biologically plausible mechanism, some correlative EEG/TMS support, contested causal role); E3 for any brain-to-brain shared-field cognition claim - explicitly not evidenced in the primary literature.

2.5.2 "Field resonance" binding/criticality hypothesis (Hunt, 2026)

A 2026 *Frontiers in Computational Neuroscience* theory paper by Tam Hunt (UC Santa Barbara) proposes that the neuroscientific "binding problem" and "criticality problem" are jointly resolved by treating cognition as continuous electromagnetic field resonance rather than discrete spike-based computation - a "gooey" versus "prickly" framing (The goo that binds us: how field resonance solves neuroscience's binding and criticality problems). The paper proposes "field computing" as an analog-computing paradigm in which binding is a natural consequence of unified field dynamics and criticality a consequence of multiscale field interactions. This is explicitly framed as a Hypothesis and Theory article (not an empirical study) - Grade: E2 (bordering E3): mathematically underspecified relative to the Kuramoto/neural-mass literature (no explicit coupled differential equations or falsifiable quantitative predictions are given in the abstracted content), and makes no claim about inter-brain field coupling; its scope is strictly intra-brain binding.

2.5.3 Related field/skepticism literature

A companion critical literature directly compares the "spike code" hypothesis against the EM-field hypothesis (Fields or firings? Comparing the spike code and the electromagnetic field hypothesis), and a Frontiers editorial catalogs the opportunities and obstacles facing all EM field theories of consciousness, including CEMI, Pockett's theory, and related proposals (Editorial: Electromagnetic field theories of consciousness: opportunities and obstacles). A cross-disciplinary neuroscience/physics collaborative paper argues EM theories offer explanatory bridges across the "explanatory gap" between physical description and subjective experience, while acknowledging this remains speculative regarding qualia and does not resolve or test inter-brain field transfer (Electromagnetism's Bridge Across the Explanatory Gap). A directly relevant (2021) explicitly hypothetical paper proposes brain-to-brain communication via EM fields as "a potential hypothesis," underscoring that even proponents label it hypothesis-stage, not demonstrated mechanism (Brain-to-brain communication: the possible role of brain electromagnetic fields (As a Potential Hypothesis)). Grade: E3 for the specific brain-to-brain claim; E2 for intra-brain EM correlational effects (TMS-evoked entrainment, alpha-band phase-locking) which are experimentally supported (Frequency- and Area-Specific Phase Entrainment... by Repetitive Transcranial Magnetic Stimulation).

2.5.4 Ephaptic coupling - the one demonstrated non-synaptic field mechanism

Ephaptic coupling - direct modulation of neuronal firing by the local extracellular electric field generated by neighboring active neurons, without any chemical synapse - is the most rigorously demonstrated non-wired electrical coupling mechanism in neuroscience (Ephaptic coupling of cortical neurons; Neuronal coupling by endogenous electric fields: cable theory and applications to coincidence detector neurons in the auditory brain stem). Cable-theory models extend integrate-and-fire neurons to include weak extracellular field terms as an added current $I_{ephaptic}(x,t) \propto -\partial V_{ext}/\partial x$ acting on the dendritic membrane (Extending Integrate-and-Fire Model Neurons to Account for the Effects of Weak Electric Fields). Grade:

E1 - but the effective range is sub-millimeter to at most a few millimeters (cortical column scale), several orders of magnitude below any macroscopic inter-agent distance, and requires direct tissue proximity (no case of trans-cranial or trans-body ephaptic coupling between two organisms has been demonstrated).

2.6 Bioelectric Morphogenetic Fields - the Strict Levin Biophysical Sense

2.6.1 Core framework: bioelectric networks as "cognitive glue"

Michael Levin (Tufts/Wyss Institute) has developed the most rigorously operationalized biophysical framework for scale-free collective cognition, grounded in measurable transmembrane voltage patterns (V_{mem}) propagated through gap junctions. In "Bioelectric networks: the cognitive glue enabling evolutionary scaling from physiology to mind" (Anim. Cogn. 2023), Levin argues that developmental bioelectric signaling long preceded neurons evolutionarily, using the same physical substrate - voltage gradients across cell membranes, propagated via gap-junctional electrical synapses - to coordinate multicellular anatomical outcomes (embryogenesis, regeneration, cancer suppression) as neurons later co-opted for behavior.

2.6.2 The "Computational Boundary of a Self" model

Levin's most mathematically explicit formalization is "The Computational Boundary of a 'Self': Developmental Bioelectricity Drives Multicellularity and Scale-Free Cognition" (Frontiers in Psychology 2019). Key formal constructs:

Individual/Self = defined operationally by the scale of goal states a control system can pursue, modeled in the cybernetic/control-theory sense as a dynamical system with attractors in state space.

Cognitive light cone: the spatio-temporal region within which a system can measure, model, predict, and act - formally analogous to (but not identical with, and explicitly not claimed to be physically identical with) a relativistic light cone; it bounds "what the system can care about."

Computational surface / cognitive boundary: the demarcation of information flow that separates a coherent "Self" from its environment, set by the balance between local information restriction (individuality) and shared information/influence (integration into a larger Self).

Scale-free cognition: cells with local bioelectric goal-states (e.g., target V_{mem} values in regenerating planaria) can integrate into tissue-level "Selves" with larger cognitive light cones when gap-junctional communication sufficiently synchronizes their states - and conversely fragment into smaller Selves (e.g., cancer) when communication drops.

This framework is explicitly a *biophysical and information-theoretic generalization of cognition* - not a claim of exotic field mediation. The physical carrier is the bioelectric potential propagated through gap junctions and ion channels, a well-characterized, purely classical electrochemical mechanism. Grade: E1/E2 - gap-junction-mediated bioelectric pattern control of anatomy is experimentally demonstrated with strong causal manipulation evidence (e.g., targeted V_{mem} shifts reprogramming planarian head/tail identity: Long-Term, Stochastic Editing of Regenerative Anatomy via Targeting Endogenous Bioelectric Gradients; The Role of Early Bioelectric Signals in the Regeneration of Planarian Anterior/Posterior Polarity); the "scale-free cognition" extrapolation to arbitrary macroscopic collective-cognition claims across unconnected agents is E2 theoretical generalization, since gap junctions require direct cell-cell contact - this mechanism categorically cannot mediate coupling between physically separated organisms/brains without an intervening physical (wired or radiative) carrier.

2.6.3 Bioelectric gene-regulatory network models

Bistable, hysteretic bioelectric circuit models formalize how stochastic ion-channel/gap-junction dynamics produce discrete, heritable "somatic pattern memories" in regenerating tissue, using coupled ODE/stochastic models of voltage-gated conductances and gene-regulatory feedback (Bistability of somatic pattern memories: stochastic outcomes in bioelectric circuits underlying regeneration; Bioelectric gene and reaction networks: computational modelling of genetic, biochemical and bioelectrical dynamics in pattern regulation). Grade: E1 for the intracellular/intercellular mechanism; strictly local (contact-mediated), no long-range field claim.

2.6.4 Collective intelligence as a cross-scale unifying framework

Levin and colleagues extend this to a general "collective intelligence" framework spanning molecular networks, cells, tissues, organisms, and swarms, arguing for common formal principles (goal-directedness, competency, multi-scale agency) rather than a single physical carrier (Collective intelligence: a unifying concept for integrating biology across scales and substrates, *Nature Communications Biology* 2024; *The Multiscale Wisdom of the Body*). Related recent work explores whether physically separated bioengineered constructs ("Xenobots") can influence each other's behavior/collective dynamics without direct contact, framed as an open experimental question, not an established phenomenon (*Bidirectional Communication Between Xenobot Cultures? Toward Collective Intelligence Across Diverse Biological Systems*, 2025). Grade: E2 for the unifying conceptual framework (well-argued, cross-validated across many biological substrates); E3 for any claim of non-contact/field-mediated coupling between separate biological systems - this remains an open experimental question explicitly flagged as unresolved in the primary source.

2.7 Reaction-Diffusion and Turing-Pattern Models - the Mathematical Backbone of Morphogenetic Fields

Turing's reaction-diffusion framework remains the canonical mathematical model underlying morphogen-based (as opposed to purely bioelectric) pattern formation:

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u + f(u, v), \quad \frac{\partial v}{\partial t} = D_v \nabla^2 v + g(u, v)$$

with instability toward periodic spatial patterns when the diffusion coefficients $D_u \neq D_v$ satisfy the classical Turing conditions (Modern perspectives on near-equilibrium analysis of Turing systems; A Selection Criterion for Patterns in Reaction-Diffusion Systems). Recent extensions show that many biochemical reaction networks generically support Turing instabilities without requiring fine-tuned feedback loops (Widespread biochemical reaction networks enable Turing patterns without imposed feedback), and that non-ideal fluid physical interactions (crowding, viscoelasticity) promote pattern robustness (Physical interactions in non-ideal fluids promote Turing patterns). Grade: E1 for the mathematical and in-vitro/in-vivo chemical mechanism; this framework describes chemical morphogen gradients (diffusible molecules), a distinct carrier from bioelectric V_{mem} fields (Section 5) and wholly distinct from electromagnetic radiative fields (Section 4). Coupling ranges are set by diffusion length scales ($L \sim \sqrt{D\tau}$), typically micrometers to millimeters over developmental timescales (hours-days) - irrelevant to real-time cognition or to any inter-organism signaling at macroscopic distance.

2.8 Quantum Biology and Open Quantum Systems: Decoherence Constraints

2.8.1 The decoherence bound (Tegmark)

The rigorous quantitative baseline for assessing any quantum-coherence-based claim about brain function is Tegmark's decoherence calculation (The importance of quantum decoherence in brain processes, *Phys. Rev. E* 2000 / arXiv:quant-ph/9907009). For a charge/dipole superposition of separation s interacting with an environmental ion at distance a , temperature T , the decoherence time scales as:

$$\tau \sim \frac{4\pi\epsilon_0 a^3 \sqrt{mk_B T}}{Nq_e s}$$

Tegmark's evaluation for both (a) superpositions of neuron firing/non-firing states and (b) microtubule kink-soliton superpositions yields decoherence times of 10^{-13} to 10^{-20} seconds, twelve to seventeen orders of magnitude shorter than relevant neural dynamical timescales (10^{-3} to 10^{-1} s, i.e., milliseconds). Grade: E1 - this calculation is widely accepted as a rigorous lower-bound argument against sustained macroscopic quantum coherence surviving in warm, wet, noisy neural tissue at the relevant scale and timescale.

2.8.2 Contested revisions (Hagan, Hameroff, Penrose)

Hagan, Hameroff, and Tepper disputed several of Tegmark's parameter choices (superposition separation, charge-vs-dipole treatment, dielectric screening), rederiving decoherence times of 10^{-5} to 10^{-4} seconds for microtubule-interior states, and arguing that metabolically-sustained ordered water layers could further extend coherence by an order of magnitude (arXiv:quant-ph/0005025). This remains an unresolved, unreplicated dispute in the literature - later syntheses treat the corrected estimate range (roughly 10^{-9} to 10^{-4} s depending on assumptions) as still far short of gamma-band (25 ms) or slower cognitive integration windows unless auxiliary discrete-event mechanisms (ion-channel tunnelling, Posner-molecule nuclear spins) are separately invoked. Grade: E2 (contested) at best for any microtubule-based quantum-coherence contribution to single-brain cognition; E3 for any extension to inter-brain quantum coupling, which no source in this literature attempts to formalize or test.

2.8.3 Orch-OR and its critical reception

The Penrose-Hameroff Orchestrated Objective Reduction (Orch-OR) theory proposes that quantum superpositions in tubulin dimers within microtubules undergo gravitationally-induced objective collapse, generating discrete conscious moments (Consciousness Is Quantum State Reduction Which Creates the Flow of Time; How quantum brain biology can rescue conscious free will). Recent papers explore experimental tests linking microtubule quantum computation to consciousness (Could there be an experimental way to link consciousness and quantum computations of brain microtubules?, 2024), but no controlled experiment has yet demonstrated the predicted quantum-gravitational collapse signature in living neural tissue. Grade: E3 for Orch-OR as a whole (unfalsified core mechanism, decoherence problem unresolved, no experimental confirmation of the gravitational self-energy collapse criterion in biological tissue) - and there is no extension of Orch-OR in the peer-reviewed literature to shared/entangled cognition between separate brains; any such claim would additionally require biologically implausible long-range entanglement preservation, which is not proposed or defended in the primary Orch-OR literature itself.

2.8.4 Legitimately demonstrated quantum biology (contrast case)

Two quantum-biological mechanisms *are* reasonably well-supported and provide the correct comparison baseline for what "quantum biology done right" looks like:

Radical-pair magnetoreception: cryptochrome-based radical pairs in migratory animals show microsecond-scale spin coherence sufficient for geomagnetic field sensing, modeled via open quantum system master equations for the singlet-triplet interconversion rate as a function of external field orientation (An open quantum system approach to the radical pair mechanism; Magnetosensitivity of tightly bound radical pairs in cryptochrome is enabled by the quantum Zeno effect). Grade: E1 - but this is a magnetic *sensing* mechanism operating on the ambient geomagnetic field (a passive receiver), not a mechanism for one organism to transmit cognitive state to another.

FMO photosynthetic complex coherence: quantum coherence in the Fenna-Matthews-Olson light-harvesting complex was long claimed to enhance energy-transfer efficiency; subsequent higher-fidelity analysis substantially walked this back - "Do photosynthetic complexes use quantum coherence to increase their efficiency? Probably not" (PMC 2021), with coherence times around 60 femtoseconds at room temperature. Grade: E1/E2 (mechanism real, functional significance now doubted) - again strictly intramolecular, no bearing on inter-agent coupling.

2.8.5 Verdict on quantum-biological inter-agent coupling

No peer-reviewed open-quantum-systems framework in the 2015-2026 literature proposes, let alone demonstrates, a decoherence-surviving quantum channel between spatially separated human (or animal) brains. The decoherence bound in Section 7.1, even under the most generous contested revision, constrains coherence to sub-millisecond intracellular scales - many orders of magnitude short of what would be required for macroscopic inter-brain quantum coupling, and no experimental claim of biological quantum entanglement between organisms exists in the citable literature. This entire category is graded E3 for the specific "shared quantum field between separated agents" question.

2.9 Topological Physics: Legitimate Bridges and Explicit Non-Bridges

2.9.1 Non-Hermitian and topological wave physics (legitimate physics, no cognition bridge)

Non-Hermitian topological physics - exceptional points, non-Abelian topological charge, topological Anderson insulators in acoustic/photonic metamaterials - is a mature, rapidly advancing area of condensed-matter and photonics physics (Scaling theory for non-Hermitian topological transitions; Non-Hermitian Topological Phononic Metamaterials; Observation of an acoustic non-Hermitian topological Anderson insulator). These systems describe topologically protected wave transport in engineered metamaterials (photonic crystals, acoustic lattices, electrical circuit lattices) governed by non-Hermitian Hamiltonians with complex eigenvalue spectra and winding-number invariants. Grade: E1 as physics - but there is no experimentally grounded biological analogue and no published bridge to neural or cognitive systems in the surveyed literature. Any claim that "resonant topological field interactions" mediate cognition or inter-brain coupling in the sense of these condensed-matter systems is unsupported (E3): the mathematical machinery (winding numbers, exceptional-point sensitivity, topological protection against disorder) has no demonstrated realization in neural or bioelectric media as of this literature search.

2.9.2 Topological data analysis (TDA) of brain states - a genuine, different, and correctly-scoped bridge

Topological Data Analysis (persistent homology, simplicial complexes) is legitimately and increasingly applied to characterize the *shape* of brain-state trajectories in fMRI/MEG/EEG data - this is a data-analysis/dynamical-systems tool, not a physical field-topology claim. Key results:

Persistent homology reveals a robust "hub-like transition state" at rest, replicated across precision datasets (Precision dynamical mapping using topological data analysis reveals a hub-like transition state at rest, Nature Communications 2022).

TDA distinguishes individual variability and correlates topological features with behavior (Topological signatures of brain dynamics: persistent homology reveals individuality and brain-behavior links, 2025).

TDA is used diagnostically in Alzheimer's, autism, and rumination-trait studies (Topological data analysis reveals altered brain connectivity in Alzheimer's disease, 2026; Topological data analysis reveals rigid brain-state dynamics during self-viewing in trait rumination, 2025).

Grade: E1 as a validated descriptive/statistical methodology for single-brain state-space geometry. This is explicitly not a claim about physical field topology or inter-brain topological coupling - it characterizes the shape of high-dimensional activity-correlation data, a mathematical abstraction, and the surveyed literature does not extend it to multi-brain shared-field claims.

2.9.3 Explicit statement on "resonant topological field interactions" and cognition

Across all three legitimate topological-physics domains searched (non-Hermitian wave systems, TDA, and topological soliton/pattern-formation field theory in reaction-diffusion contexts - Optical skyrmion and its "zipper-like" topological behavior in an energy flux field; Structure and Pattern Formation in Biological Liquid Crystals), no peer-reviewed source establishes a mechanistic bridge between topological wave/soliton physics and macroscopic inter-agent cognitive coupling. Where "resonant topological field interactions" is invoked in relation to shared cognition (as in some popular-science and non-peer-reviewed sources), this constitutes E3: unsupported extrapolation - the terminology borrows legitimate physics vocabulary without a demonstrated or even theoretically specified physical mapping onto neural or bioelectric media at macroscopic, inter-organism scale.

2.10 Hyperscanning and Inter-Brain Synchrony: Correlation versus Causal Coupling

2.10.1 The central methodological problem

Hyperscanning (simultaneous multi-brain EEG/fMRI/fNIRS recording during joint tasks) reliably detects statistical correlation between separated brains' signal time series ("inter-brain coupling") during social interaction (Brain-to-Brain Neural Synchrony During Social Interactions: A Systematic Review on Hyperscanning Studies). However, a 2024 review explicitly frames the field's central unresolved question: existing hyperscanning studies establish associations, not causal direction - it remains undetermined

whether inter-brain coupling drives behavioral synchrony, whether behavioral synchrony drives the coupling, or whether the relation is bidirectional (Hyperscanning: from inter-brain coupling to causality, *Front. Hum. Neurosci.* 2024).

2.10.2 Common-input confound

Two participants performing a joint task typically share correlated sensory input (the same visual/auditory stimulus stream) and predict each other's actions, either of which can produce inter-brain statistical correlation without any direct brain-to-brain causal coupling - this is the "common input" or "common environment" confound, structurally identical to the classic confound in any multivariate time-series causal-inference problem. The review notes both mechanisms (shared stimulus-driven correlated response, and reciprocal predictive adjustment) are sufficient to generate correlation absent direct coupling.

2.10.3 Isolating true coupling: methods and evidence

Several methodological advances attempt to separate true interpersonal coupling from common-input artifacts:

Between- vs. within-brain coupling decomposition: shows measurable direct contribution of *between-brain* coupling to movement synchronization beyond what within-brain dynamics alone predict (Differential contribution of between and within-brain coupling to movement synchronization).

Directed multi-brain network analysis: using directed coupling metrics (Granger-causality-style, transfer entropy) shows "generalized synchrony" during social exchange with directional asymmetries consistent with genuine interpersonal influence, not mere shared-stimulus artifact (Directed coupling in multi-brain networks underlies generalized synchrony during social exchange).

Information flow / transfer entropy between brains: identifies directional information flow between interacting brains correlated with social expertise, providing a quantitative information-theoretic (not merely correlational) measure of coupling asymmetry (Information flow between interacting human brains: Identification, validation, and relationship to social expertise).

Hyper-brain cell assembly hypothesis (Müller and colleagues): proposes that Hebbian-like cell-assembly formation principles operating within a single brain may extend across brains when two nervous systems are functionally coupled through sensorimotor interaction loops - explicitly a *hypothesis*, grounded in established within-brain Hebbian plasticity mathematics extended by analogy, not yet validated as an actual cross-brain physical mechanism (Neural Synchrony and Network Dynamics in Social Interaction: A Hyper-Brain Cell Assembly Hypothesis).

2.10.4 The definitive causal test: multibrain stimulation

The strongest methodological critique in the field states plainly that "Hyperscanning Alone Cannot Prove Causality. Multibrain Stimulation Can" (*Trends in Cognitive Sciences*, 2020; response and rejoinder at PMC). The argument: because hyperscanning is purely observational, no correlational inter-brain coupling metric - however statistically robust - can establish that coupling *causes* social/behavioral outcomes, or that one brain's state is physically transmitted to the other, versus both arising from a shared external driver. The proposed remedy is experimental multibrain stimulation: using TMS/tACS to artificially entrain or desynchronize two brains simultaneously and observing behavioral consequences - a true interventional design. As of the reviewed literature (through 2024-2025), multibrain stimulation studies remain nascent; the review paper itself frames this as the necessary next step rather than reporting completed causal demonstrations at scale.

2.10.5 Physical carrier for hyperscanning correlation - never a field between brains

It is essential to state explicitly what all hyperscanning literature agrees on: the observed inter-brain correlation is a statistical relationship between two independently-measured internal neural signals - sensory input (sound, sight, touch) is the only established physical channel connecting the two nervous systems. No hyperscanning paper in the literature claims or provides evidence for a non-sensory (i.e., direct electromagnetic or bioelectric field) transmission channel between the two skulls. This directly bounds any interpretation of hyperscanning results as evidence for a "shared signal field": Grade: E1 for the statistical correlation phenomenon itself (robustly replicated); E2 for whether *any* of it reflects

genuinely direct (non-sensory-mediated) coupling - an open, actively investigated question with partial positive evidence from directed-coupling/transfer-entropy studies; E3/ruled-out for any field-mediated (non-sensory, non-motor) channel, since no such channel has been proposed with a specified carrier and independently confirmed in this body of work.

2.10.6 Physiological (non-neural) interpersonal synchrony

A parallel and more physically transparent literature documents interpersonal *physiological* synchrony - heart-rate coupling, electrodermal activity - during social interaction, again mediated entirely through sensory channels (shared visual/auditory context, mutual gaze, synchronized movement), not through any hypothesized field (Heart rate synchrony as a marker of real-world social engagement; Interpersonal Physiological Synchrony During Dyadic Joint Action Is Increased by Task Novelty and Reduced by Social Anxiety). This category is useful as the clean baseline: it shows biologically real, physiologically important cross-agent synchrony with an entirely mundane, well-understood physical carrier (sound, sight, touch), reinforcing that where cross-agent coupling is unambiguously demonstrated, the carrier is always identifiable and conventional.

2.11 Active Inference / Free Energy Principle: Physical Instantiation Status

The Free Energy Principle (FEP) and active inference formalize biological self-organization as approximate Bayesian inference minimizing variational free energy F under a generative model, with the "Markov blanket" concept partitioning a system's internal states from external states via a statistical conditional-independence boundary (Active inference on discrete state-spaces: a synthesis; Generalised free energy and active inference). A rigorous experimental instantiation exists: in-vitro cortical neuron cultures interfaced with closed-loop stimulation validate free-energy-principle predictions in living neural tissue (Experimental validation of the free-energy principle with in vitro neural networks, 2022) - Grade: E1, a genuine physical (electrophysiological) instantiation, though at the scale of a single cultured network, not inter-organism coupling.

Multi-agent extensions are an active 2025 research frontier: "What the flock knows that the birds do not: exploring the emergence of joint agency in multi-agent active inference" formally examines whether a Markov blanket can emerge at the group level from individually-blanketed agents interacting through their environment (arXiv 2511.10835, 2025); a companion paper, "As One and Many: Relating Individual and Emergent Group-Level Generative Models in Active Inference" (Entropy 2025), formalizes the mathematical relationship between individual and group-level generative models. Levin's own "Morphogenesis as Bayesian inference" work (Prog. Biophys. Mol. Biol. 2020) applies active-inference formalism to bioelectric pattern control, with a critical commentary noting that Markov blankets are general physical interaction surfaces applicable broadly, not unique markers of cognition (Markov blankets are general physical interaction surfaces).

Grade: E1/E2 for the formal mathematics and single-system physical instantiation; E2 for multi-agent group-level active inference as a coherent, currently-being-formalized theoretical framework; the shared "field" in this framework is explicitly the ordinary environment (shared sensory/behavioral signals each agent can access), not any novel physical field - active inference does not posit or require any exotic field carrier, and none of the surveyed sources claim one.

2.12 Biophotons: Ultraweak Photon Emission

Cells emit measurable ultraweak photon emissions (biophotons), a well-documented but low-intensity phenomenon linked to oxidative metabolism (reactive oxygen species) (Physical properties of biophotons and their biological functions; First Experimental Measurements of Biophotons from Astrocytes and Glioblastoma Cell Cultures, 2025). Some cell-cell death communication studies suggest biophoton signals may traverse non-aqueous environments to affect neighboring cells (Cell cell death communication by signals passing through non-aqueous environments), and recent work explores biophotons as optical markers of brain activity (Exploring ultraweak photon emissions as optical markers of brain activity, 2025). Grade: E2 for biophoton-mediated cell-cell (contact/near-contact range, μm -mm) signaling as an intriguing but still-developing research area with modest replication; E3 for any macroscopic-distance or inter-organism biophoton signaling claim - emission intensities are extremely low (single-photon-per-second-per-cell order of magnitude), and no study demonstrates biologically meaningful photon flux

surviving propagation across meters of open air or tissue to a separate organism's sensory apparatus. Note also one search result returning a study on biophoton emission "exposed to a scalar field" during meditation - such sources sit outside the peer-reviewed physics/biology mainstream and are flagged here explicitly as outside this catalog's evidentiary standard (E3, excluded from further analysis).

2.13 Fröhlich Coherence: Historical Foundational Model

Fröhlich's 1968 theory proposed that metabolically pumped biological systems could sustain long-range coherent (Bose-Einstein-condensate-like) collective vibrational excitations at GHz-THz frequencies, potentially explaining long-range order in biomolecular assemblies (Long-range coherence and energy storage in biological systems, 1968; Coherent Excitations in Biological Systems, Springer 1983). This is included as essential foundational background (pre-2015) because later microtubule-coherence and CEMI-adjacent theories draw on its formalism. Direct experimental confirmation of Fröhlich condensation in living tissue at physiological temperature remains elusive and contested; more recent microtubule-coherent-energy-transfer feasibility studies remain theoretical (The Feasibility of Coherent Energy Transfer in Microtubules, arXiv 2014). Grade: E2/E3 - mathematically well-posed (non-equilibrium statistical mechanics of driven phonon modes) but empirically unconfirmed in vivo at the relevant biological temperature and noise level; no demonstrated role in inter-agent coupling.

2.14 Master Comparison Table

Kuramoto / Sakaguchi oscillator networks (arXiv:1702.03641, PMC7725404)

$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i + \alpha)$; order parameter $re^{i\psi}$ - Synaptic/anatomical connectome coupling - Intra-brain (mm-cm, whole cortex) - Strong fit to EEG/MEG/fMRI dynamics; critical coupling K_c matched empirically - Predicted synchronization transition at measurable K_c ; testable via connectome-weighted simulation vs. recorded phase-locking - Coupling matrix is physical (axonal), not a free field; no inter-cranial extension - E1 (intra-brain); not applicable inter-agent

Higher-order/hypergraph Kuramoto (Springer 2025)

Simplicial/triadic coupling terms added to Kuramoto - Same as above, generalized topology - Intra-brain, network-model - Simulation-validated; applied to epileptic network models - Predicts first-order synchronization transitions distinguishable from pairwise models - No biological hypergraph substrate directly measured; theoretical extension - E1 (math); E2 (bio-application)

Neural mass / Wilson-Cowan continuum (PMC2866289)

$\tau \dot{E} = -E + (1 - rE)S_E(\cdot)$ etc.; nonlocal field kernel $w(x - x')$ - Cortical population firing rates - mm (columns) to cm (cortex) - Reproduces EEG rhythms, traveling waves, multistability - Predicted wave speed/multistability boundaries testable via ECoG - Deterministic mean-field approximation; not a radiative field - E1

Self-organized criticality / scale-free avalanches (PRL 2012)

Branching process, power-law avalanche size $P(s) \sim s^{-\tau}$ - Cortical spiking/LFP networks - Whole-brain - Power laws replicated across species/recording modalities - Predicts specific critical exponents, deviations under pathology - Criticality describes dynamics, not a coupling carrier between organisms - E1

CEMI field theory (OUP 2020)

Field superposition/AND-gate analog computation; inverse-square/cube attenuation - Endogenous brain EM field (EEG/MEG-detectable) - 80 μm per neuron (≤ 200 neurons) without relay; whole-brain via synchronized relay chains - TMS-behavior effects; EEG/MEG correlates of perception; no direct test of field causal sufficiency - Predicts phase/frequency-specific external EM field effects on motor responses - Field strength decays rapidly; long-range integration requires neuronal relay, not raw field propagation; brain-to-brain transfer not evidenced - E2 (intra-brain); E3 (inter-brain)

Field-resonance binding/criticality hypothesis (Frontiers 2026)

Qualitative EM-resonance/binding framework; no explicit coupled ODEs given - Brain EM field - Intra-brain - None presented (theory article) - Not yet formalized into testable quantitative predictions - Explicitly a "Hypothesis and Theory" article; no inter-brain claim - E2 (intra-brain, theoretical only)

Ephaptic coupling (Nature Neuroscience 2011)

Cable-theory extension with extracellular field current term - Local extracellular electric field - Sub-mm to few mm (cortical column) - Directly measured in hippocampal/cortical slices - Predicts firing modulation as function of local field gradient; confirmed - Range far below any inter-organism distance - E1

Bioelectric "cognitive glue" / Computational Boundary of a Self (PMC6923654, PMC10770221)

Cybernetic control-theory attractor model; V_{mem} gap-junction propagation - Transmembrane bioelectric potential via gap junctions - Cell-cell contact scale (μm); tissue-scale via gap-junction networks (mm-cm) - Targeted V_{mem} manipulation reprograms planarian anatomy (causal, replicated) - Predicts specific anatomical outcomes from specific bioelectric perturbations; extensively validated - Requires direct cell contact (gap junctions); cannot mediate coupling between physically separated organisms - E1 (mechanism); E2 (scale-free cognition generalization)

Reaction-diffusion / Turing patterning (PMC10336379)

$\partial_t u = D_u \nabla^2 u + f(u, v)$, etc. - Diffusible morphogens - μm -mm, hours-days - Extensively validated in vitro/in vivo - Predicts pattern wavelength from D , reaction kinetics - Diffusion-limited; far too slow/short-range for real-time cognition or inter-organism signaling - E1

Open quantum systems / decoherence bound (arXiv:quant-ph/9907009)

$\tau \sim 4\pi\epsilon_0 a^3 \sqrt{mk_B T} / (N q_e^2 s)$ - Ionic/dipole environment in neural tissue - Sub-cellular (nm- μm) - Calculation-based; widely cited, not experimentally falsified - Predicts decoherence timescale as function of temperature/geometry; falsifiable via measured coherence times - Decoherence times (fs-ps, contested up to sub-ms) orders of magnitude below cognitive timescales - E1 (bound); constrains all quantum-cognition claims

Orch-OR (Brill 2023)

Gravitationally-induced objective state reduction criterion - Tubulin quantum superposition in microtubules - Intracellular (nm) - No confirmed experimental detection of predicted collapse signature - Predicts specific collapse time from gravitational self-energy; testable in principle, untested in vivo - Decoherence problem unresolved; no inter-brain extension proposed - E3

Radical-pair magnetoreception (PMC6200754)

Open-system master equation for singlet-triplet radical pair dynamics - Cryptochrome protein spin states - Molecular (nm); sensing whole-Earth field - Behavioral and biochemical confirmation in multiple species - Predicts field-orientation-dependent reaction yield; confirmed - Passive sensing of ambient geomagnetic field; not an inter-organism transmission channel - E1

Hyperscanning inter-brain coupling (PMC11599244)

Cross-correlation / coherence / transfer entropy between recorded signals - Behavior + shared sensory input; measured neural correlation - Meters (dyad distance); mediated by light/sound, not field - Extensively replicated correlational findings across EEG/fMRI/fNIRS - Predicts increased coupling during cooperative vs. non-interactive conditions; confirmed - Correlational; common-input confound; causal direction unresolved - E1 (correlation); E2 (causal coupling, partial)

Multibrain stimulation (Trends Cogn. Sci. 2020)

Interventional TMS/tACS entrainment across two brains simultaneously - External applied EM stimulation, not endogenous shared field - Meters (separate stimulators, separate skulls) - Proposed methodology; large-scale causal studies still emerging - Predicts behavioral change from artificially synchronized vs. desynchronized stimulation - Nascent evidence base as of 2025-2026 - E2

Active inference / Markov blanket multi-agent models (arXiv:2511.10835)

Variational free energy minimization; group-level generative model - Ordinary sensory/behavioral environment - Any distance within sensory/behavioral range - In-vitro single-network validation; multi-agent math still developing - Predicts emergent group-level Markov blanket under specified interaction conditions - Shared "field" = conventional environment, not a novel physical field - E1 (single-system); E2 (multi-agent formal theory)

Non-Hermitian topological wave physics (Adv. Mater. 2023)

Non-Hermitian Hamiltonian, topological winding number invariants - Engineered photonic/acoustic/electronic metamaterials - Metamaterial device scale (μm -cm) - Extensively validated in physics

labs - Predicts topologically protected transport robust to disorder; confirmed in metamaterials - No biological/neural realization demonstrated - E1 (physics); E3 (bio-cognition bridge, unestablished)

Topological data analysis of brain states (Nat. Commun. 2022)

Persistent homology on point-cloud/correlation data - Statistical/mathematical abstraction of neural activity - Whole-brain (single subject) - Replicated across large fMRI/MEG datasets - Predicts topological signatures distinguishing cognitive/clinical states; confirmed - Describes state-space geometry, not physical field topology; no inter-brain extension - E1 (methodology); not applicable to field-mediation claims

Fröhlich coherence (Wiley 1968)

Driven phonon-mode Bose-condensation-like model - Biomolecular vibrational modes - Molecular-cellular (nm- μ m) - Contested; no confirmed in-vivo detection at physiological temperature - Predicts coherent GHz-THz spectral lines; searched for, not confirmed - Foundational/historical; unconfirmed in living tissue - E2/E3

Biophoton emission (MDPI 2025)

Chemiluminescence from ROS-driven oxidative metabolism - Ultraweak photon flux - Cell-cell (μ m) to short range - Detected instrumentally; weak, low-replication field - Predicts photon flux correlated with metabolic/oxidative state; partially confirmed - Flux far too low for macroscopic-distance or inter-organism signaling - E2 (local); E3 (macroscopic/inter-organism)

2.15 Explicit Verdict on Key Claim Categories

Direct field-mediated cognitive content transfer between spatially separated brains: Not demonstrated by any peer-reviewed source in this catalog. The CEMI/field-resonance literature is confined to intra-brain binding hypotheses with acknowledged rapid field-strength decay (inverse-square/cube) over micrometer-to-millimeter native ranges (McFadden, CEMI); hyperscanning literature explicitly attributes observed inter-brain correlation to shared sensory input and reciprocal behavioral prediction rather than a field carrier (Hyperscanning: from inter-brain coupling to causality).

Morphogenetic fields (Levin) as a model for "shared signal fields" between separated agents: The bioelectric mechanism is real, causally validated, and mathematically formalized, but its physical carrier (gap-junction-propagated transmembrane voltage) is categorically contact-dependent - it cannot mediate coupling across a physical gap between separated organisms without an intervening classical carrier (light, sound, wired connection). The "scale-free cognition" and "cognitive light cone" framework is a rigorous generalization of *what counts as an agent's boundary*, not a claim about exotic non-local transmission (Computational Boundary of a Self).

Quantum-biological substrates for inter-agent coupling: Ruled out by the decoherence bound at any biologically relevant inter-agent distance and timescale; even the most generous contested revisions of Tegmark's calculation remain many orders of magnitude short of what would be needed, and no source proposes or tests inter-organism quantum entanglement (arXiv:quant-ph/9907009; Hagan et al. rebuttal).

Topological field interactions as a mechanism for shared cognition: No bridge exists between legitimate non-Hermitian/topological wave physics (an active, well-validated area of condensed matter and photonics) and any neural or bioelectric system; this is stated explicitly as an evidentiary gap, not merely an open question (Non-Hermitian Topological Phononic Metamaterials).

What is actually established: Robust, mathematically precise, and empirically validated mechanisms exist for (a) intra-brain macroscopic phase synchronization via Kuramoto/neural-mass/criticality dynamics operating over the physical connectome; (b) local (sub-millimeter) ephaptic field coupling between neurons; (c) contact-mediated bioelectric morphogenetic patterning across cells via gap junctions; (d) statistically robust but causally underdetermined inter-brain correlation during hyperscanning, mediated by conventional sensory channels; and (e) genuine quantum-biological sensing (radical-pair magnetoreception) operating on ambient fields, not inter-organism transmission. Collectively, these establish that biological systems achieve remarkable collective coordination through entirely conventional physical carriers (electrochemical, synaptic, sensory, diffusive) operating at well-characterized, short-to-moderate ranges - and that no currently validated mechanism supports unmediated field-based cognitive coupling

across macroscopic inter-agent distances.

2.16 Key Researchers Referenced

Michael Levin (Tufts University / Wyss Institute, Harvard) - bioelectric morphogenetic fields, scale-free cognition, collective intelligence, Xenobots (PMC10770221, PMC6923654, Nature Comm. Biology 2024).

Johnjoe McFadden (University of Surrey) - CEMI field theory of consciousness (OUP 2020).

Tam Hunt (UC Santa Barbara) - field-resonance binding/criticality hypothesis (Frontiers 2026).

Susan Pockett - cited in EM field theories of consciousness comparative literature (PMC9245352).

Max Tegmark (MIT) - foundational quantum decoherence bound for brain processes (arXiv:quant-ph/9907009).

Stuart Hameroff and Roger Penrose - Orchestrated Objective Reduction (Orch-OR) theory and contested decoherence revisions (Hagan, Hameroff, Tuszynski, arXiv:quant-ph/0005025).

Yoshiki Kuramoto and Hidetsugu Sakaguchi - foundational coupled-oscillator synchronization models underlying the entire phase-oscillator literature reviewed in Sections 1 and 13.

Herbert Fröhlich - foundational (1968) theory of long-range coherent excitation in biological systems (Wiley 1968).

Karl Friston - Free Energy Principle / active inference, Markov blankets (arXiv:2001.07203).

Ulman Lindenberger and Viktor Müller - hyperscanning, hyper-brain cell assembly hypothesis, inter-brain network dynamics (PMC9101304).

2.17 Bottom Line for "Shared Signal Fields Across Spatially Separated Agents"

The mathematically rigorous literature on coupled oscillators, neural mass models, criticality, and bioelectric morphogenesis provides powerful, validated tools for understanding how large numbers of physically connected biological units achieve coherent collective states. None of this literature, however, establishes a validated physical mechanism by which two or more spatially separated cognitive agents (without a conventional sensory, wired, or engineered EM-transmission channel) can be coupled through a shared ambient field. Where field-based hypotheses for consciousness or binding exist (CEMI, field-resonance), their own quantitative range estimates (tens to hundreds of micrometers per neuron without relay chains) place inter-brain field coupling at distances beyond any single-neuron field's reach, requiring either direct neural relay (found only within a connected nervous system) or an as-yet-unspecified amplification mechanism that no source in this survey provides. Quantum-biological and topological-physics avenues, while individually legitimate and active research areas, currently have no demonstrated or even fully specified bridge to macroscopic inter-agent cognitive coupling. The empirically demonstrated cross-agent coordination phenomena in this literature (hyperscanning correlations, physiological synchrony) are consistently and explicitly attributed to conventional sensory-behavioral channels, not to an exotic shared field.

Chapter 3

Catalog of Frontier Mathematical Frameworks, Theoretical Models, and Topological Concepts (2015-2026)

3.1 RICCI FLOWS (Non-Standard)

Terminology note: "Lacunae" is not standard terminology in Ricci flow theory. The nearest rigorous constructs are (a) *singular Ricci flow* (Ricci flow spacetimes with controlled, stratified singular sets while the flow itself continues), (b) *synthetic/weak Ricci flow* on metric-measure spaces (no smooth structure at all, so "gaps" are replaced by measure-theoretic degeneracy), and (c) *Ricci flow with rough/non-smooth initial data* (continuity of the flow persists even though the initial metric has lower regularity than the flow's interior smoothing would suggest). None of these literatures use the word "lacuna."

Singular Ricci flows (established framework). Bruce Kleiner and John Lott (2014-2017) constructed *singular Ricci flows*: Ricci flow spacetimes satisfying asymptotic conditions that give a canonical way to flow a 3-manifold "through" a singularity without surgery, resolving a long-standing existence problem stated by Perelman (Kleiner-Lott, *Singular Ricci Flows I*, arXiv:1408.2271; published in *Acta Mathematica* 219 (2017)). Richard Bamler and Kleiner then proved uniqueness and stability of this flow through singularities, i.e., that the singular Ricci flow is the unique such spacetime determined by initial data (Bamler-Kleiner, *Uniqueness and stability of Ricci flow through singularities*, 2017-2022). Mechanism: the manifold develops a stratified singular set of Type-I curvature blow-up; the flow spacetime continues on the complementary open regular part with controlled asymptotic geometry near the singular strata - a continuity-preserving, surgery-free continuation.

Compactness theory for super-Ricci flows (2023). Bamler's *Structure Theory of Non-Collapsed Limits of Ricci Flows* and the associated *compactness theory of the space of Super Ricci flows* (Springer, *Inventiones Mathematicae* 2023, DOI 10.1007/s00222-023-01196-3) establishes a partial-regularity theory: limits of Ricci flow sequences converge (in a synthetic/metric-measure sense) even when smooth convergence fails, producing singular strata of controlled Hausdorff codimension - an active, technically mature research program, not speculative.

Synthetic/weak notions of Ricci flow on metric measure spaces (2020-2025, active program). Multiple groups have proposed definitions of "Ricci flow" for time-dependent metric measure spaces using optimal transport and heat-flow monotonicity rather than smooth curvature tensors: (i) K. Sturm and collaborators' "super Ricci flows" framework via entropic curvature-dimension conditions (arXiv:2604.10007, *On weak formulations of (super) Ricci flows*); (ii) "Synthetic notions of Ricci flow for metric measure spaces" (arXiv:2501.07175, 2025); (iii) a review, "Synthetic approaches to Ricci flows" (arXiv:2511.11477, 2025); and (iv) "On Weak Super Ricci Flow through Neckpinch" (*Analysis and Geometry in Metric Spaces* 2021, DOI 10.1515/agms-2020-0123), which explicitly studies a *weak* super-Ricci-flow continuation through a neckpinch singularity - the closest published construct to a "continuity-preserving gap." Mechanism: monotonicity of entropy/heat-kernel functionals along the Wasserstein geodesic structure substitutes for the smooth PDE, allowing the "flow" to be defined even where the metric is merely a measurable, possibly degenerate object.

RCD(K,N) spaces and synthetic Ricci curvature lower bounds (mature but still actively extended, 2018-2025). The RCD (Riemannian Curvature-Dimension) theory of Ambrosio-Gigli-Savaré and Lott-Villani-

Sturm defines Ricci curvature lower bounds on general metric measure spaces via convexity of entropy along Wasserstein geodesics, with no smoothness required. Recent frontier results: stratified spaces satisfy $\text{RCD}(K,N)$ iff the Ricci tensor is bounded below on the regular stratum (Numdam, **Annales de l'Institut Fourier** 71 (2021), pp.123-173); the vanishing of the metric-measure boundary on $\text{RCD}(K,N)$ spaces, resolving a Kapovitch-Lytchak-Petrulin conjecture (**GAFA** 2023, DOI 10.1007/s00039-023-00626-x); stability of synthetic Ricci lower bounds for inverse-limit extended metric measure spaces (2020s, Polish extended mm-spaces as inverse limits); and a program toward a "great unification" of Riemannian and sub-Riemannian synthetic Ricci bounds (Barilari et al., published as AMS Memoir 1613). This is the most rigorously established branch of "non-smooth Ricci geometry" and underlies essentially all subsequent weak/singular Ricci flow work.

Ricci flow with rough (non-smooth) initial metrics (Peter Topping and collaborators, ongoing survey program 2020-2024). "Preserving curvature lower bounds when Ricci flowing non-smooth initial data" (arXiv:2411.13204, 2024) surveys results in which the flow is launched from initial data with only measure-theoretic or distributional regularity (e.g., metrics with unbounded curvature, orbifold points, or conical singularities), yet instantaneously produces a smooth flow for $t > 0$ - the "gap" is at $t = 0$ only, and disappears immediately, distinguishing this from surgery. Related German DFG project SPP2026 explicitly frames its goal as "a synthetic theory of Ricci flows for non-smooth spaces, monotonic quantities, and rigidity" (spp2026.de).

Ricci solitons, conical singularities, and non-uniqueness (2022). Continuations of Ricci flow through conical singularities can be **non-unique** in some regimes - a genuine mathematical surprise contrasting with the Bamler-Kleiner uniqueness result for the singular Ricci flow proper (**GAFA** 2022, DOI 10.1007/s00039-022-00601-y).

3.2 KALMAN FILTERS & STATE ESTIMATION (Absolute Edge)

Invariant Extended Kalman Filter (IEKF) on matrix Lie groups (established, actively extended 2020-2026). Barrau and Bonnabel's IEKF constructs filters whose estimation error is defined intrinsically via the Lie-group structure (left- or right-invariant error), giving convergence guarantees independent of the true trajectory (arXiv:1410.1465; overview at Wikipedia). Recent frontier work: equivalence of left- and right-invariant EKFs on matrix Lie groups (arXiv:2506.01514, 2025); invariant Kalman filtering for relative dynamics, revisiting state-trajectory independence on Lie groups (arXiv:2412.10519, 2024); distributed invariant Kalman filtering for cooperative multi-robot localization (arXiv:2405.04000, 2024); and neural-aided adaptive innovation-based invariant Kalman filters blending learned innovation covariances with the geometric IEKF structure (arXiv:2603.26709).

Topological Kalman Filtering on Cell Complexes (2026, genuinely novel - the literal "topological Kalman filter"). Liu, Money, Gao, Sabbagi, Beferull-Lozano, and Isufi introduce a Kalman filter whose state lives on a cell complex (simplicial/CW-complex domain), inferring latent dynamics for multivariate time series with higher-order (edge, face) interactions rather than only node-wise interactions (arXiv:2605.15955, 2026). This is the closest rigorous match to "topological Kalman filter"; the term itself is new and not yet a standardized subfield, but the underlying formalism (Hodge-decomposed state-transition operators over a cell complex) is mathematically explicit. Distinguish from separate uses of persistent-homology **features** as inputs to an ordinary Kalman/state-space filter (e.g., topological-data-analysis-derived features for nonstationary structural-dynamics state estimation, Laflamme et al., **Mechanical Systems and Signal Processing**), which is a looser, non-topological-domain usage.

Quantum filtering / quantum trajectory theory (established quantum stochastic calculus, actively refined 2015-2025). The quantum filtering equation (Belavkin; Bouten-van Handel-James formalization) computes the conditional (a posteriori) quantum state given continuous weak measurement records via a quantum stochastic master equation. Recent rigor: a fully measure-theoretic treatment, "On quantum stochastic master equations" (**Electronic Journal of Probability** 30, 2025, DOI 10.1214/25-EJP1322); the quantum Kushner-Stratonovich equation, an optimal nonlinear quantum-state estimator under continuous measurement using quantum Itô calculus (summarized via Emergent Mind); and quantum projection filters reformulated via truncated Stratonovich-Taylor expansions for tractable real-time implementation (**Automatica** 112, 2020, DOI in PolyU repository). Mechanism: the filter is a stochastic PDE on the space of density operators, non-commutative analogue of the classical nonlinear (Kushner-Stratonovich/Zakai) filtering equation.

Wasserstein Distributionally Robust Kalman Filtering (established 2018, extended through 2025). Shafieezadeh-Abadeh, Nguyen, Kuhn, and Mohajerin Esfahani formulate the minimax Kalman filter over a Wasserstein ambiguity ball around a nominal noise distribution, solving a zero-sum game against nature to obtain a distribution-free robust estimator (arXiv:1809.08830, NeurIPS 2018). Frontier extensions: a noise-centric distributionally robust Kalman filter (arXiv:2512.06286); distributionally robust filtering over finite and infinite horizons (arXiv:2407.18837); a marginal distributionally robust Kalman filter for multi-sensor fusion (2025, IEEE); and a minimum-error-entropy Wasserstein Kalman filter combining information-theoretic (entropy) criteria with distributional robustness under non-Gaussian noise (*Signal Processing*, ScienceDirect, DOI 10.1016/j.sigpro.2022.108716-adjacent).

Manifold-valued / Riemannian-geometric Kalman filters (2020s, active applied program). Filters that track state directly on Riemannian manifolds - SPD (symmetric positive-definite) matrix manifolds, Stiefel manifolds, or general Riemannian pose manifolds - using intrinsic metrics rather than local linearization: Log-Euclidean-metric covariance propagation on the SPD manifold for continuous-discrete EKF (ScienceDirect 2024); convergence of the EKF on Stiefel manifolds; and Riemannian-manifold Kalman filtering for indoor pose estimation fusing IMU and acoustics (arXiv:2409.01002, 2024).

Information-geometric (Fisher-Rao) estimation. Information geometry equips statistical manifolds with the Fisher-Rao metric, giving intrinsic geodesic distances between probability distributions; frontier work extends this to infinite-dimensional non-parametric Fisher-Rao metrics (arXiv:2512.21451) and to finite-energy signal manifolds for estimation under additive noise (PMC, *Fisher-Rao Distance for Finite-Energy Signal Manifolds*). While not itself a "Kalman filter," this framework underlies proposed entropy-maximizing/information-geometric filtering approaches that replace the Gaussian assumption with geodesic distance minimization on the statistical manifold.

Set-valued / possibilistic filtering. Older but relevant established construct: set-valued estimation for recursive robust H_∞ filtering (IET, DOI 10.1049/ip-cta:20041051) - bounds the state within a guaranteed set (zonotope/ellipsoid) rather than producing a probability distribution; this predates 2015 but remains the standard reference point for "possibilistic" (non-probabilistic) filtering, actively used as a robustness baseline in recent distributionally robust filter papers.

3.3 HYPERVECTORS (VSA / HDC)

Consolidated survey baseline (established, 2021-2022, still the reference texts). Kleyko, Rachkovskij, Osipov, and Rahimi's two-part survey, "A Survey on Hyperdimensional Computing aka Vector Symbolic Architectures" (Part I: Models and Data Transformations, arXiv:2111.06077; Part II: Applications, Cognitive Models, and Challenges, arXiv:2112.15424) is the authoritative modern taxonomy, unifying binary, bipolar, integer, real, complex, and sparse hypervector families under common bundling/binding/permutation operators.

Fractional binding (Furlong, Simone, and Voelker et al., 2021-2022, active). Fractional binding generalizes the discrete "role-filler" binding operator to a *continuous* exponentiation of a base vector ($\mathbf{v}^\alpha$ for real α), enabling continuous-valued symbol binding and spatial semantic pointers (SSPs) for continuous variables such as position (Furlong et al., *Fractional Binding in Vector Symbolic Architectures as Representations for Efficient Mobile Robot Task Learning*). This directly extends VSA/HDC into continuous vector symbolic architecture (CVSA) territory, treating bundled symbol sets as approximating a probability distribution over the underlying continuous space.

Generalized Holographic Reduced Representations (GHRR, 2024). GHRR extends the classical Fourier Holographic Reduced Representation (FHRR) with a unifying algebraic and kernel-theoretic framework, proving adherence to core HDC axioms and characterizing binding/bundling kernel geometry rigorously (arXiv:2405.09689, 2024). A closely related 2026 refinement, qFHRR, replaces continuous phase encoding with quantized/integer phase arithmetic for hardware efficiency while preserving the FHRR algebra (arXiv:2604.25939).

Tensor-product and geometric-algebra hypervectors. Frank Qiu's "Tensor Products and Hyperdimensional Computing" (arXiv:2305.10572, 2023) formalizes tensor-product binding as the "ground truth" binding operator from which circular-convolution-based HRR is a compressed approximation. Distinct geometric-algebra-based hypervector schemes ("geometric analogue of holographic reduced representation") use Clifford/geometric-algebra products (rather than Hadamard or circular convolution) as the

binding operator, giving richer multivector geometric structure (*Neural Networks* / ScienceDirect, DOI-linked paper) - this line is older (pre-2015 origin) but is actively cited and re-derived in 2020s HDC survey work as a rigorous alternative binding geometry.

Category-theoretic foundation for VSA (2025). "Developing a Foundation of Vector Symbolic Architectures Using Category Theory" (arXiv:2501.05368, 2025) proposes recasting binding/bundling/permutation as categorical constructions (functorial mappings), an explicitly non-standardized, actively proposed formal program aiming to unify disparate HDC variants under one abstract-algebraic umbrella.

Capacity theory and kernel-preserving encodings (2020-2023, mathematically rigorous). "A Theoretical Perspective on Hyperdimensional Computing" (arXiv:2010.07426) and "A Formal Perspective on Hyperdimensional Computing" establish capacity bounds (how many bound/bundled items can be reliably recovered as a function of dimension) and define kernel-preserving encodings - maps ϕ such that the inner product $\langle \phi(x), \phi(x') \rangle$ approximates a target kernel $k(x, x')$ up to bounded distortion $\Delta(d, n)$ (escholarship formal-perspective PDF). A dedicated 2023 paper, "Capacity Analysis of Vector Symbolic Architectures" (arXiv:2301.10352), formalizes this quantitatively across VSA variants - this is the rigorous mathematical core underlying informal claims about hypervector "capacity" and "kernel geometry."

Recursive/similarity-preserving binding for sequences (2022) and Kronecker-rotation clean-up (2025). "Recursive Binding for Similarity-Preserving Hypervector Representations" (arXiv:2201.11691) constructs sequence encodings preserving element-order similarity structure (non-trivial, since naive binding is a "hashing" operation that destroys similarity). "Linearithmic Clean-up for Vector-Symbolic Key-Value Memory with Kronecker Rotation Products" (arXiv:2506.15793, 2025) improves the computational complexity of the decoding ("clean-up") step from quadratic to $O(n \log n)$ via Kronecker-structured rotation operators - a genuine complexity-theoretic advance for VSA retrieval.

Learned binding operators (2020s, active). Multiple 2021-2024 papers explore learned, rather than fixed, binding operators - e.g., self-attention-based semantic decomposition within VSA (arXiv:2403.13218) and neuro-vector-symbolic architectures for relational reasoning (e.g., solving Raven's Progressive Matrices) - replacing the classical circular-convolution/Hadamard binding with differentiable, trainable analogues while retaining the algebraic bundling/similarity structure.

3.4 TOPOLOGY & FRACTALS (Thurston & Beyond)

Terminology note. "Thurston geometries applied to fractals" is not a single accepted framework. Thurston's geometrization program (eight model geometries, geometric structures on 3-manifolds, Teichmüller theory) is a smooth/hyperbolic-manifold theory; fractal geometry (Assouad/conformal dimension, resistance forms) operates on non-manifold, non-smooth metric spaces. The only genuinely rigorous bridge is the Sullivan dictionary, which maps Kleinian-group/hyperbolic-geometry concepts (Thurston's milieu) to complex dynamics/rational-map iteration, whose invariant Julia sets are fractals - this is a real, well-established correspondence, but it is a dictionary between *dynamical systems on fractal limit sets and hyperbolic 3-manifold theory*, not a direct transplant of Thurston's geometric structures onto fractals themselves.

Sullivan dictionary via Assouad-type dimensions and spectra (2024, direct evidence of the terminological gap). "A new perspective on the Sullivan dictionary via Assouad type dimensions and spectra" (*Bulletin of the AMS* 61(1), 2024, DOI 10.1090/bull/1796) explicitly states the correspondence between Kleinian groups acting on hyperbolic space (a Thurston-style setting) and rational maps of the complex plane, using Assouad-type dimension spectra as the quantitative bridge - the clearest rigorous instantiation of "Thurston-adjacent geometry meets fractal dimension theory" in the literature, though the authors do not claim a unified "Thurston geometry of fractals."

Conformal Assouad dimension (established, actively extended 2021-2025). The conformal Assouad dimension - the infimum of Assouad dimension over all quasisymmetric metric deformations - is proven to equal the critical exponent for combinatorial modulus (Mathav Murugan, arXiv:2209.01187), linking fractal dimension theory to discrete conformal/potential-theoretic invariants (a genuinely Teichmüller/Thurston-adjacent quantity, since combinatorial modulus is the discrete analogue of conformal modulus used in Thurston's circle-packing approach to uniformization). Further frontier work: Sobolev/quasiconformal distortion bounds relating intermediate dimension to conformal dimension (arXiv:2505.10525, 2025); Assouad-type dimension spectra for self-affine sponges and Apollonian gaskets, including a computer-

assisted proof that the infinitely-generated Apollonian gasket has full Hausdorff dimension spectrum (*Ergodic Theory and Dynamical Systems*, 2026, DOI-linked).

Resistance forms, Dirichlet forms, and spectral triples on fractals (established noncommutative-geometry program, 2013-2021, continuing). Cipriani, Guido, Isola, and collaborators constructed spectral triples (Connes-style noncommutative-geometric data: Hilbert space, Dirac-type operator, algebra representation) for self-similar fractals such as the Sierpinski gasket, recovering Hausdorff dimension and measure from purely operator-algebraic (Dixmier-trace) data (Cipriani-Guido-Isola, *Spectral triples for nested fractals*, arXiv:1601.08208; extension to solenoidal spaces via locally isometric ramified self-covering of the Sierpinski gasket, SIGMA 2021). A directly relevant unifying paper, "Metrics and spectral triples for Dirichlet and resistance forms", constructs the Dirac operator/spectral triple directly from a regular Dirichlet or resistance form, making the construction applicable to a broad class of fractal-like metric-measure spaces beyond specific self-similar examples (EMS Press, *Journal of Noncommutative Geometry*).

Spectral theory for fractal pseudodifferential operators (2024). A genuinely recent (2024) construction studies the eigenvalue distribution of a compact fractal pseudodifferential operator defined via an oscillatory-integral kernel with fractal-scaling symbol, extending classical pseudodifferential calculus into the fractal-geometric regime (arXiv:2405.15814).

Persistent homology and fractal dimension (2018-2020, still the reference results). Adams, Schweinhart, and collaborators proved the first rigorous results relating persistent homology of random geometric complexes to classical fractal (box-counting/Ahlfors-regularity) dimension, and defined a family of "persistent homology dimensions" $\dim_{PH}^i(\mu)$ for probability measures, offering a statistically robust, filtration-based fractal-dimension estimator distinct from Hausdorff or box-counting dimension (Schweinhart, *Fractal Dimension and the Persistent Homology of Random Geometric Complexes*, arXiv:1808.02196; Adams et al., *A fractal dimension for measures via persistent homology*).

Fractal Laplacians, energy forms on hybrid fractals, and self-similar Dirichlet forms (2020, active analysis-on-fractals program). "Convergence of energy forms on Sierpinski gaskets with added rotated triangle" (arXiv:2011.11149) and the "Analysis on hybrid fractals" program (fractals built by gluing distinct self-similar pieces) extend the classical Kigami theory of resistance forms/Laplacians on self-similar fractals to structurally heterogeneous, non-self-similar fractal constructions - a genuine 2020s extension beyond the classical (pre-2015) Kigami framework.

3.5 SOURCES, SINKS, AND NON-SINKS (Advanced Dynamical Flow Features)

Exceptional points as non-Hermitian saddle/degeneracy structures (mature research area, 2019-2026, still rapidly active). Exceptional points (EPs) are branch-point spectral degeneracies where both eigenvalues *and* eigenvectors of a non-Hermitian operator coalesce, producing fractional-power-law spectral response - qualitatively distinct from ordinary Hermitian degeneracies (diabolical points) and from simple hyperbolic saddles (review, *Applied Physics Letters* 124, 060502 (2024), "Exceptional points in non-Hermitian systems"). Frontier 2023-2026 results: a universal topological classification of exceptional points in nonlinear non-Hermitian systems (*Nature Communications*, 2026, DOI 10.1038/s41467-026-72854-2); higher-order exceptional points via non-Hermitian zero-index materials for ultrasensitive perturbation response (APS, DOI 10.1103/18gg-gvzc); disorder-driven exceptional points and concurrent topological phase transitions (APS, DOI 10.1103/x4vq-rlp5); and an explicit exceptional non-Hermitian topological edge mode applied to active matter (*Nature Communications* 11, 5745 (2020), DOI 10.1038/s41467-020-19488-0), linking EP topology directly to the active-matter domain in item 5 below.

Riddled and intermingled basins of attraction (established 1990s-2010s theory; 2022-2025 revival with new models). A basin is *riddled* when every point in it has, arbitrarily nearby, points belonging to a different basin, making basin membership practically unpredictable even though it remains deterministic. Recent frontier extensions: "The Basins Zoo" (arXiv:2504.01580, 2025) formally taxonomizes basin-boundary types (intertwined vs. intermingled basin boundaries) for modern nonlinear-dynamics classification; multifractal analysis of intermingled basins and blowout bifurcations in a two-parameter family of skew-product maps (arXiv:2502.04869, 2025); and riddled basins and noise-induced chaos in coupled optomechanical systems, a genuinely novel 2020s physical instantiation (*Physica Scripta*, IOP, DOI-linked).

Milnor attractors and chaotic itinerancy (established, actively used in neuroscience modeling 2020-2023). A Milnor attractor is an attractor in the weaker Milnor sense (attracts a positive-measure but not necessarily open neighborhood), central to chaotic itinerancy - successive, quasi-random transitions between "attractor ruins." Frontier application: heteroclinic networks for brain dynamics, explicitly framing metastable neural switching via heteroclinic-cycle theory (PMC, *heteroclinic networks for brain dynamics* review), and "chaotic itinerancy generated by coupling of Milnor attractors" as a mechanism for network-level metastability (Hokkaido University repository).

Ghost attractors / long ecological transients (established terminology, actively reviewed 2018-2021). "Ghost attractor" (sometimes "ghost of equilibrium past" or "bifurcation ghost") refers to the residual dynamical influence of a bifurcated-away fixed point or slow manifold, producing anomalously long transient dynamics before eventual - often abrupt - transition (Hastings et al., *Transient phenomena in ecology*, *Science* 361 (2018), DOI 10.1126/science.aat6412; Hastings et al., *Effects of stochasticity on the length and behaviour of ecological transients*, *J. R. Soc. Interface* 2021). This is the field's actual accepted term for what the task calls "pseudo-attractors" or "transient attractors" - the mechanism is a saddle-node bifurcation remnant (a "ghost" of the vanished fixed point) that still slows trajectories near its former location.

Pullback and random attractors for non-autonomous/stochastic systems (established framework, 2012 origin, continuously extended). The pullback attractor generalizes the omega-limit-set attractor concept to explicitly time-dependent (non-autonomous) or randomly forced systems by tracking the limit of trajectories pulled back from $t \rightarrow -\infty$ rather than forward-limiting - necessary because forward attractors need not exist for non-autonomous flows (Bixiang Wang, *Sufficient and Necessary Criteria for Existence of Pullback Attractors for Non-compact Random Dynamical Systems*, arXiv:1202.2390). Continuing applications include random pullback attractors for the stochastic Swift-Hohenberg equation with multiplicative noise (*Journal of Mathematical Physics* 61, 092703 (2020), DOI 10.1063/5.0007959-adjacent).

Edge modes, topological defects, and singularities in active matter (mature and rapidly growing field, 2020-2023). "Topological active matter" reviews chiral edge modes, half-integer/full-integer nematic and polar defects, and their non-equilibrium analogues of topological-insulator edge states (Shankar, Souslov, Bowick, Marchetti, Vitelli, *Nature Reviews Physics* 4 (2022), reviewed at arXiv:2010.00364). Frontier extensions: "Design rules for controlling active topological defects", showing activity gradients act as effective electric fields inducing defect motion (arXiv:2212.00666); and a continuum theory unifying polar and nematic active matter, showing coexistence of half-integer and full-integer defects within one model (*Journal of Physics A*, IOP, DOI 10.1088/1751-8121/ac4abe).

Strange nonchaotic attractors (SNAs) (established 1984 origin, still an active experimental frontier in 2024). SNAs are geometrically strange (non-piecewise-differentiable, fractal) but dynamically nonchaotic (zero or negative Lyapunov exponent) attractors, occurring generically under quasiperiodic forcing. A genuinely new 2024 experimental result reports the first observation of an SNA in an unforced turbulent reactive flow - previously SNAs were thought to require explicit quasiperiodic external forcing (*Chaos* 34, 121101 (2024), DOI-linked at pubs.aip.org), and a companion 2023 paper develops quantitative measures for "how strange" an SNA is under quasiperiodic forcing (*Chaos, Solitons & Fractals*, ScienceDirect).

3.6 ARNOLD CATASTROPHE THEORY (Current State, 2015-2026)

Distinguishing lineage: Direct Arnold/ADE lineage refers specifically to the classification of singularities of smooth mappings by their unfolding into one of the ADE (or, more generally, simple-singularity) normal forms, as codified by V. I. Arnold. Many modern papers use "catastrophe" loosely (any abrupt qualitative transition), without invoking the ADE classification or unfolding theory - these are flagged below.

Direct Arnold/ADE lineage: catastrophe theory for Γ -invariant unfoldings applied to quantum many-body theory (2024). K. Rodrigues Alves develops "Catastrophe Theory for Γ -invariant Unfoldings with Applications to Quantum Many-body Theory" (arXiv:2401.10628, 2024), explicitly extending Arnold-style unfolding/singularity classification to symmetry-constrained (Γ -invariant) potentials relevant to quantum phase transitions - a genuine direct-lineage 2020s extension.

Direct Arnold/ADE lineage: catastrophes in non-equilibrium many-particle wave functions (2016, foun-

dational for subsequent quantum-catastrophe work). Mumford, Kirkby, and O'Dell demonstrate that non-equilibrium quantum many-body wavefunctions exhibit universal catastrophe structures with critical scaling exponents, directly invoking Thom/Arnold catastrophe classification in a quantum dynamical (not merely classical optics) setting (*Journal of Physics B*, arXiv:1609.04502), with a full thesis-level treatment in "Singularities in Many-Body Quantum Dynamics" (McMaster University, 2017).

Wave catastrophes and logarithmic phase singularities at horizons (2023, direct lineage, general relativity/optics interface). "Logarithmic catastrophes and Stokes's phenomenon in waves at horizons" (*Journal of Physics A: Mathematical and Theoretical* 56, 044001 (2023), DOI 10.1088/1751-8121/acb29e) extends Michael Berry's wave-catastrophe program (caustics as loci of stationary-phase degeneracy) to horizon physics, a genuinely current (2023) direct-lineage development building on Berry-Upstill's classical "Morphologies of caustics" framework (Berry & Upstill, foundational reference PDF).

Optical caustics and metasurfaces (loose-to-moderate use of "catastrophe," but rigorous caustic/singularity mathematics, 2018-2024, rapidly active). "Arbitrary engineering of spatial caustics with 3D-printed metasurfaces" (*Nature Communications* 15, 3719 (2024), DOI 10.1038/s41467-024-48026-5) and "Encoding orbital angular momentum of light in space with optical catastrophes" (*Nature Communications*, 2026, DOI 10.1038/s41467-026-73443-z) use catastrophe-optics classification (fold, cusp singularities) to engineer metasurface-generated caustic beam structures - this is a legitimate, mathematically grounded application of Thom-Arnold caustic classification to a fast-growing photonics/metamaterials subfield, though the papers focus on engineering applications rather than extending the classification theory itself. Related: "Morphologies of caustics studied by catastrophe charged-particle optics" extends the same classification to electron/ion-microscope optics (arXiv:2509.09551, 2025).

Machine-learning/data-driven bifurcation and catastrophe detection (rapidly growing frontier, 2020-2026, explicitly non-ADE - a distinct methodological program). "Predicting discrete-time bifurcations with deep learning" (*Nature Communications* 14 (2023), DOI 10.1038/s41467-023-42020-z) trains classifiers on simulated bifurcation data to provide early-warning signals for critical transitions in real systems, without invoking the ADE unfolding classification directly (i.e., a "catastrophe" is detected statistically/dynamically, not via normal-form classification). Related: "Detecting Bifurcations in Dynamical Systems with Deep ... Equilibrium-Informed Neural Networks (EINNs)" (arXiv:2603.04420, 2026); "Active search for bifurcations" (*Chaos* 35, 053159 (2025)); and an explicitly ADE-adjacent 2025 paper, "Using Machine Learning for the Precise Experimental Modeling of Catastrophe Phenomena... Zeeman Catastrophe Machine" (*Mathematics* (MDPI) 13(4), 603 (2025)), which does directly fit ML-estimated data to a canonical cusp-catastrophe normal form - the most direct-lineage instance of ML applied to Arnold-style catastrophe classification found.

3.7 ENTROPY (Modern Definitions, Beyond Shannon/Boltzmann/Tsallis)

Sofic entropy and Rokhlin entropy (Lewis Bowen, 2008 origin; foundational comparative results 2017-2023, still active). Sofic entropy generalizes Kolmogorov-Sinai entropy to actions of sofic groups (a broad class including all amenable and residually finite groups, via finite-set approximations); Rokhlin entropy is an independently developed generalization to all countable groups via generating partitions. Bowen's "Examples in the entropy theory of countable group actions" (arXiv:1704.06349) constructs systems where the two invariants provably diverge, clarifying their distinct information content. Frontier 2020s results: "Positive entropy actions of countable groups factor onto Bernoulli shifts" (*Journal of the AMS* 33 (2020), DOI 10.1090/jams/931), proving a sharp structural consequence of positive Rokhlin/sofic entropy; and "The Koopman Representation and Positive Rokhlin Entropy" linking these entropy notions to the operator-theoretic (Koopman) representation of the group action (*International Mathematics Research Notices* 2023).

Furstenberg entropy (established, still central to Poisson-boundary/random-walk theory, 2017-2023 refinements). Furstenberg entropy $h_\mu(X, \eta) = - \int_G \int_X \log \frac{dg^{-1}\eta}{d\eta}(x) d\eta(x) d\mu(g)$ measures the "information gain" of a stationary measure η under a random walk with step distribution μ ; it vanishes iff η is G -invariant. Continuing frontier work formalizes explicit computations for specific step distributions and boundary actions (arXiv:2307.01495, 2023).

Metric mean dimension and rate-distortion dimension (active variational-principle program, 2021-2025). Metric mean dimension quantifies the "dimension rate" of infinite-entropy dynamical systems (where

classical topological entropy is infinite but growth of complexity is still meaningfully finite-rate); rate-distortion dimension is its information-theoretic (lossy-compression) counterpart. Recent frontier: "Mean dimension and rate-distortion function revisited" proves that for non-ergodic almost-sure processes the mean Rényi information dimension coincides with the information dimension rate (arXiv:2510.08051, 2025); a variational principle connecting metric mean dimension to rate-distortion dimension is established generally (ScienceDirect, *Journal of Mathematical Analysis and Applications*, 2025).

Generalized entropy in operator-algebraic quantum gravity (very active, 2022-2023, high-profile). Chandrasekaran, Penington, and Witten's "Large N algebras and generalized entropy" (arXiv:2209.10454, 2022) shows the generalized (Bekenstein-Hawking + bulk von Neumann) entropy of black-hole subregions equals the entropy of an explicit crossed-product operator algebra, resolving a long-standing type-III-algebra puzzle in algebraic QFT; extended to generalized entropy for general subregions in quantum gravity (*JHEP* 12 (2023):020020)) and reviewed comprehensively in "GGI Lectures on Entropy, Operator Algebras and Black Holes" (arXiv:2209.05132).

Entropy production in non-reciprocal active matter (rapidly active 2022-2025 program). Non-reciprocal interactions (violating action-reaction symmetry, generic in active/living matter) generate entropy production even in the absence of explicit external driving. Key papers: "Entropy production in non-reciprocal polar active mixtures" (arXiv:2508.05209, 2025); "Entropy production of nonreciprocal interactions" (*Physical Review Research* 5, L022033 (2023)); "Irreversibility across a Nonreciprocal PT-Symmetry-Breaking Phase Transition" (*Physical Review Letters* 131, 258301 (2023)), deriving a general expression for average informational entropy-production rate in conserved-phase-field mixtures with non-reciprocal couplings; and a deep-learning framework for computing entropy production rates directly from probability-flow generative models applied to non-Gaussian active matter (arXiv:2504.06628, 2025; PMC review).

Homological/cohomological entropy and information cohomology (Baudot-Bennequin program, 2015 origin, ongoing). Baudot and Bennequin's "The homological nature of Entropy" (2015) introduced information cohomology, showing Shannon entropy arises canonically as a specific cohomology class (a derivation) on a simplicial "information structure" built from families of observables - giving entropy a rigorous algebraic-topological (not merely probabilistic) characterization (summarized in "The structure of information: From probability to homology," arXiv:1709.07807; category-theoretic elaboration in "Information structures and their cohomology" (*Theory and Applications of Categories* 35 (2020)). This is the most rigorous existing bridge between "entropy" and "homology/cohomology," directly relevant to (and more mathematically precise than) informal claims about "persistent/homological entropy" or "sheaf entropy" - note that a fully general, standardized *sheaf entropy* is not established; information cohomology is the closest rigorous analogue.

Persistent entropy (established but continually applied, 2015-2025). Persistent entropy is the Shannon entropy of the probability distribution induced by the lifetimes of topological features (bars) in a persistence diagram, used to summarize persistent homology output as a single discriminative number (Atienza et al., "Persistent Entropy for Separating Topological Features", arXiv:1701.07857; stability results in "On the stability of persistent entropy", arXiv:1803.08304 [<http://arxiv.org/pdf/1803.08304.pdf>])). This differs from (and is more standardized/rigorous than) any generalized "homological entropy" - persistent entropy is literally Shannon entropy applied to a topologically-derived distribution, not a new information-theoretic axiom system.

Graph entropy (established, information-theoretic; no rigorous "sheaf entropy" standard exists as of 2026). Graph entropy in the Körner sense, $H(G) = \min_{X,Y} I(X;Y)$, is a well-defined, mature information-theoretic functional on graphs with a vertex probability distribution, connected to perfect-graph theory and network coding capacity (survey, Simonyi, "Perfect graphs and graph entropy"). By contrast, "sheaf entropy" is not a standardized rigorous term in the literature searched; the nearest rigorous relative is the Baudot-Bennequin information-cohomology framework above, and separately, sheaf Laplacians (see Domain 9) which have well-defined spectral (not entropic) invariants. Flag explicitly: claims of a general "sheaf entropy" should be treated as non-standardized terminology absent a specific citable construction.

3.8 VERNADSKY NOOSPHERE (Scientific Status Assessment)

Overall assessment: the noosphere as originally formulated by Vernadsky (1938-1945) is a philosophical/proto-scientific concept, not a rigorous, falsifiable, mathematically operationalized scientific theory. It has no

standard measurable definition, no consensus formal model, and is largely absent from mainstream Earth-system-science or information-theory journals under that name; it survives mainly in historiography of science, Russian/Soviet intellectual history, and speculative futurist/astrobiology discourse. Distinguishing operational science from noosphere-branded metaphor is essential.

Historical/philosophical status (established scholarship, not physical science). Recent scholarship - "What is the noosphere? Planetary superorganism, major evolutionary transition, or geological force?" (*Systems Research and Behavioral Science*, 2024, DOI 10.1002/sres.2997) - explicitly frames the noosphere as an "underappreciated and understudied idea," acknowledging its lack of an accepted formal/operational definition even as it proposes candidate framings (superorganism, major-transition, geological force). Historical-scientific analyses (Oldfield & Shaw, "V.I. Vernadsky and the noosphere concept: Russian understandings of society-nature interaction", *Geoforum*, 2006, ScienceDirect) situate it within Russian cosmism and biogeochemistry rather than as a predictive physical theory, and note its convergence with (but conceptual independence from) Teilhard de Chardin's theological "noosphere" and Lovelock's Gaia hypothesis (IIASA working paper, "Vernadsky's Biosphere, Teilhard's Noosphere, and Lovelock's Gaia").

Operational, mathematically explicit modern analogues - distinguish these from noosphere discourse itself. The task's requested "mathematically explicit modern analogues" exist as genuinely separate, rigorously grounded research programs that do not use Vernadsky's terminology as their operative framework:

"Intelligence as a planetary scale process" (Frank, Grinspoon, Walker, *International Journal of Astrobiology*, 2021 - reprinted/discussed at Santa Fe Institute, DOI 10.1017/s147355042100029x) reframes "intelligence" as a *collective property operating at planetary scale*, explicitly as an astrobiological/Earth-system-science hypothesis (testable via biosignature/technosignature statistics on exoplanets) rather than a metaphysical claim - this is the most rigorous, peer-reviewed, quantitatively-framed 2020s descendant of noosphere-adjacent thinking, though the authors do not adopt Vernadsky's term as their formal apparatus.

Scale-free / small-world functional brain connectivity (established neuroscience, 2004-2025) provides a genuinely mathematically rigorous model of network-level collective computation with power-law ($1/f$) spectral signatures and small-world/scale-free graph topology (Stam, "Scale-free dynamics of global functional connectivity in the human brain", 2004, PMC; 2025 update on global scale-free brain activity as a neural signature (PMC)). These are legitimate operational (falsifiable, quantitatively fit) models of "collective cognitive network" behavior, but their domain is individual-brain network neuroscience, not planetary-scale cognition - any extrapolation to a literal "planetary brain" is a speculative analogy, not an established extension.

Major evolutionary transitions theory (Maynard Smith & Szathmry, 1995, updated 2015) provides a rigorous evolutionary-biology framework for how new levels of biological organization/information-processing emerge (e.g., single cells to multicellularity to eusociality) (Szathmry, "Toward major evolutionary transitions theory 2.0", *PNAS* 112 (2015), DOI 10.1073/pnas.1421398112). This theory is sometimes invoked (speculatively) as a precursor mechanism for a hypothesized transition toward planetary-scale cognition, but the theory itself makes no such claim as formally stated.

Speculative / non-peer-reviewed noosphere discourse (clearly distinguish as non-consensus). Contemporary essays such as "The Noocene: Computation and Cosmology from Antikythera to AI" (noocene.antikythera.org) and "Planetary Intelligence Version 0.1" (possibleplanetlab.org) explicitly extend Vernadsky's noosphere into AI-era discourse, but these are think-tank/humanities essays, not peer-reviewed mathematical or physical science, and should not be presented as established science. No mathematically operationalized "noosphere field equation," "planetary information thermodynamics law," or similarly rigorous construct bearing Vernadsky's name was found in the peer-reviewed literature as of 2026; claims to the contrary should be treated with skepticism.

3.9 SIGNAL REPRESENTATION (Absolute Edge, Beyond Fourier/Wavelets)

Topological signal processing over simplicial complexes via Hodge theory (established foundational framework, 2020, actively extended through 2026). Barbarossa and Sardellitti's "Topological Signal Processing over Simplicial Complexes" (iris.uniroma1.it postprint, 2020) generalizes graph signal processing to simplicial complexes, decomposing signals defined on nodes/edges/triangles via the combinatorial Hodge

decomposition (gradient/curl/harmonic components) instead of the ordinary graph Laplacian eigenbasis. The comprehensive tutorial "Topological Signal Processing and Learning" (arXiv:2412.01576, 2024) and the 2026 update "Topological Signal Processing: An Application-Oriented Tutorial" (arXiv:2605.22853) consolidate the field. Frontier extensions include "Multiscale Hodge Scattering Networks on Simplicial Complexes" (scattering-transform generalization to simplicial domains) and "Topological Signal Processing Over Cell MultiComplexes Via Cross-Laplacian Operators" (2025-2026, IEEE), extending beyond simplicial complexes to general cell complexes and multi-layer/multi-relational networks.

Sheaf Laplacians and learned restriction maps (very active, 2025-2026). The sheaf Laplacian generalizes the graph Laplacian by attaching vector spaces (stalks) to nodes/edges and "restriction maps" between them, encoding local-to-global consistency constraints richer than scalar edge weights. "The Sheaf Laplacian: A Topological Framework for Data Fusion and Consensus in Distributed Sensing Networks" (arXiv:2606.19529, 2026) applies this directly to sensor-fusion signal processing; "Learning Sheaf Laplacian Optimizing Restriction Maps" (arXiv:2501.19207, 2025) proposes learning both the underlying graph topology and the restriction maps jointly from observed data - a genuinely new (2025) inverse-problem formulation for sheaf-based signal models. A broader conceptual bridge, "Sheaf theory: from deep geometry to deep learning" (arXiv:2502.15476, 2025), formalizes sheaves valued in arbitrary target categories for use in deep learning architectures.

Dirac-equation-inspired signal processing (2025, genuinely novel hybridization of physics and topological ML). "Dirac-equation signal processing: Physics boosts topological machine learning" (*PNAS Nexus*, 2025, DOI 10.1093/pnasnexus/pgaf139) constructs a discrete Dirac operator over topological signals (defined jointly on nodes and edges of a network) as an alternative to the Hodge Laplacian, exploiting the first-order (rather than second-order) structure of the Dirac equation for improved topological machine-learning performance - an explicitly cross-disciplinary 2025 innovation.

Koopman operator / Dynamic Mode Decomposition as a signal/state representation (established theory, actively extended with rigorous error bounds 2023-2025). The Koopman operator provides a globally linear (infinite-dimensional) representation of a nonlinear dynamical system's observables, with DMD as its finite-dimensional data-driven approximation. Frontier rigor: "Residual dynamic mode decomposition: robust and verified Koopmanism" (*Journal of Fluid Mechanics*, Cambridge); " L_∞ -error bounds for approximations of the Koopman operator by kernel extended dynamic mode decomposition" (arXiv:2403.18809, 2024); its stochastic extension "Koopman for stochastic dynamics: error bounds for kernel EDMD" (arXiv:2512.20247, 2025); and "Rigged Dynamic Mode Decomposition", computing generalized (distributional) eigenfunction decompositions for Koopman operators with continuous spectra (arXiv:2405.00782, 2024).

Path signatures and rough-path theory as a signal representation (established rough-path mathematics, applied ML program rapidly growing 2020-2026). The path signature - the sequence of iterated integrals of a path - provides a faithful, hierarchical, reparametrization-invariant representation of streamed/sequential data, generalizing Fourier/wavelet time-series features to genuinely path-dependent, non-Markovian information. Modern rigor and scale: "Lecture notes on rough paths and applications to machine learning" (Cass & Salvi, arXiv:2404.06583, 2024); "Scalable Machine Learning Algorithms using Path Signatures" (arXiv:2506.17634, 2025); and domain applications including "SigBERT: Combining Narrative Medical Reports and Rough Path Signature Theory for Survival Risk Estimation in Oncology" (arXiv:2507.22941, 2025) and "Rough Path Signatures: Learning Neural RDEs for Portfolio Optimization" (arXiv:2510.10728, 2025), both illustrating the signature transform's absorption into mainstream applied deep learning pipelines.

Persistent Homology Transform (PHT) as a shape/signal statistic (established, 2014 origin, continually extended). The PHT represents a shape or signal as the *collection of persistence diagrams* obtained by filtering in every direction, giving an injective (fully faithful) statistic for compact shapes in $\mathbb{R}^2/\mathbb{R}^3$ (Turner, Mukherjee, Boyer, foundational paper reviewed at academic.oup.com). A 2023 extension, "Spectral Persistent Homology: Persistence Signals" (arXiv:2312.17093), reconceives persistence diagrams themselves as signals in $\mathbb{R}_+^2$, enabling frequency-domain-style analysis of topological features - a genuine bridge between TDA and classical time-frequency signal analysis.

Geometric scattering transforms on graphs/manifolds (established 2019, ongoing refinement). The geometric scattering transform generalizes the (windowed, non-learned) wavelet scattering transform to graph- and manifold-valued data using cascades of graph-wavelet filters and pointwise nonlinearities, giv-

ing provably stable, deformation-robust representations without training (Gao, Wolf, Hirn, "Geometric Scattering for Graph Data Analysis", ICML 2019; asymmetric/generalized extensions in "Understanding Graph Neural Networks with Generalized Geometric Scattering Transforms", arXiv:1911.06253, updated 2023).

Quantum signal processing (QSP) and quantum singular value transformation (QSVT) (established quantum-algorithms framework, 2017 origin, rapidly extended 2021-2024). QSP provides a unifying algorithmic primitive for implementing polynomial transformations of matrices encoded in quantum circuits (block-encodings), underlying most modern quantum algorithms for Hamiltonian simulation, linear systems, and amplitude estimation. Frontier: "Generalized Quantum Signal Processing" (*PRX Quantum* 5, 020368 (2024)) and "Multivariable quantum signal processing (M-QSP)" (*Quantum* 6, 811 (2022)), the latter extending QSP from single-variable to multivariable polynomial transformations - a substantial 2020s generalization of the core QSP formalism.

3.10 Cross-Domain Index of Recurring Mechanisms

Singularity continuation without surgery

Ricci flows; catastrophe theory; dynamical attractors - Singular Ricci flow through singularities (Kleiner-Lott); ghost attractors as bifurcation remnants (Hastings et al.); optical caustics as fold/-cusp continuations (Berry-Upstill)

Stratification (regular vs. singular strata)

Ricci flows; fractal geometry; exceptional-point topology - RCD stratified-space curvature bounds (Numdam); Assouad dimension spectra on self-affine sets (arXiv:2207.11611); exceptional-point classification by codimension (arXiv:2306.06967)

Sheaf/local-to-global consistency

Signal processing; entropy (information cohomology); noosphere-adjacent network models - Sheaf Laplacians for sensor fusion (arXiv:2606.19529); information cohomology as sheaf-like observable structures (arXiv:1709.07807)

Non-reciprocity / broken detailed balance

Entropy in active matter; sources/sinks in active matter; exceptional points - Entropy production in non-reciprocal mixtures (arXiv:2508.05209); non-Hermitian PT-symmetry breaking (PRL 131, 258301)

Coarse/synthetic geometry replacing smooth structure

Ricci flows (RCD spaces); fractals (resistance forms, spectral triples); Kalman filters (manifold-valued, information-geometric) - Synthetic Ricci curvature bounds (AMS Memoir 1613); spectral triples for fractals (arXiv:1601.08208); Riemannian-manifold Kalman filtering (arXiv:2409.01002)

Entropy rate / dimension-rate duality

Entropy (mean dimension, rate-distortion); topology (persistent homology dimension) - Metric mean dimension-rate distortion variational principle (ScienceDirect 2025); persistent homology fractal dimension (arXiv:1808.02196)

Higher-order (simplicial/cell-complex) topology replacing pairwise graphs

Signal processing; Kalman filtering; hypervector binding (tensor products) - Topological signal processing over simplicial complexes (iris.uniroma1.it); topological Kalman filtering on cell complexes (arXiv:2605.15955); tensor-product hypervector binding (arXiv:2305.10572)

Distributional/ambiguity-set robustness

Kalman filtering (Wasserstein-robust); entropy (minimum error entropy) - Wasserstein distributionally robust Kalman filter (arXiv:1809.08830); minimum-error-entropy Wasserstein filter (ScienceDirect)

Continuous relaxation of discrete symbolic/combinatorial operations

Hypervectors (fractional binding); signal processing (path signatures as continuous "symbols" of a stream) - Fractional binding in VSA (Furlong et al.); path signatures as hierarchical stream representations (arXiv:2404.06583)

Operator-algebraic / spectral reformulation of classical invariants

Entropy (generalized entropy via crossed-product algebras); fractals (spectral triples); signal processing (Koopman operator) - Large-N algebras and generalized entropy (arXiv:2209.10454); spectral triples for fractal Hausdorff dimension (EMS Press); Koopman-operator linearization of nonlinear dynamics (JFM residual DMD)

Cross-cutting terminological caution: Three phrases in the task brief have no single standardized meaning in the current literature and were mapped, per domain, to the nearest rigorous constructs rather than treated as established terms: (1) "lacunae" in Ricci flow $\rightarrow$ singular/weak/synthetic Ricci flow through singularities (Domain 1); (2) "topological Kalman filter" $\rightarrow$ a genuinely emerging but narrowly-scoped 2026 construct, Kalman filtering on cell complexes (Domain 2), distinct from the broader, looser use of TDA-derived features as filter inputs; (3) "noosphere" as a scientific object $\rightarrow$ no operational, peer-reviewed mathematical formalization exists under that name; the rigorous descendants (planetary-scale intelligence, scale-free brain networks, major-transitions theory) are independent research programs that do not adopt Vernadsky's terminology as their formal apparatus (Domain 8). A fourth, milder case is "sheaf entropy" (Domain 7), for which no standardized rigorous definition was found; information cohomology (Baudot-Bennequin) is the nearest existing rigorous analogue.

Chapter 4

The History of AGI Paradigms: An Adversarial Structural Analysis, 1950-2026

4.1 Preface: Methodological Posture

This report catalogs the major scientific programs that have explicitly targeted general intelligence - not task-specific optimization - from the 1950 publication of Alan Turing's "Computing Machinery and Intelligence" through the large-language-model era of the 2020s (Princeton archive of Turing 1950). The adversarial posture taken here is structural, not personal: each paradigm is evaluated against its own stated definition of "general intelligence," its proposed computational engine, and the specific mathematical or physical wall that arrested or is arresting its progress. Historical fact (dates, authorship, published claims) is distinguished from adversarial interpretation (post-hoc diagnosis of failure modes) throughout; interpretive claims are flagged as such.

The recurring finding is that every paradigm has confronted a variant of the same underlying problem - the closure problem: the impossibility of specifying, in advance, a finite and computationally tractable set of primitives, rules, priors, or representations sufficient to close the gap between a formal system and an open, non-stationary world. Symbolic AI hit this wall as combinatorial explosion; Universal AI hit it as uncomputability; whole brain emulation hits it as a metrology and multiscale-modeling wall; connectionist scaling is hitting it as data exhaustion and a statistical reliability ceiling; heterodox thermodynamic approaches have not yet produced an engine capable of encountering the wall at all.

4.2 Precursor Formulations (1943-1955): Cybernetics and the Behavioral Definition

Before "artificial intelligence" existed as a named field, Norbert Wiener's cybernetics program defined intelligence functionally, as control and communication under feedback: "the science of control and communication, in the animal and the machine" (Wiener, *Cybernetics*, 1948, MIT Press edition). W. Ross Ashby's homeostat and the broader cybernetics movement modeled general adaptivity as a system's capacity to maintain internal variables within viable bounds against an unpredictable environment (Ashby, *An Introduction to Cybernetics*). The engine was analog feedback and self-regulation (servomechanisms), not symbolic computation.

Structural definition of generality: homeostasis under arbitrary perturbation - generality equated with the breadth of a stable-state envelope, not with problem-solving power.

Failure mode (historical/adversarial): Cybernetics never produced a scalable computational formalism for reasoning, planning, or language; its analog and information-theoretic tools address regulation, not representation, and the paradigm was structurally absorbed into and superseded by digital symbolic AI after 1956 rather than falsified on its own terms.

Alan Turing's 1950 paper reframed the question operationally, replacing "Can machines think?" with the imitation game, and explicitly proposed that a machine's capacity for general learning - "child machines" subjected to an educational process - was the tractable research target, rather than a fixed engineering blueprint for intelligence (Turing 1950, Mind; Wikipedia summary of the paper's argument structure). Turing offered no architecture; the Turing Test is a behavioral criterion, not a generative theory of intelligence, and this omission - a criterion without an engine - is the structural gap that all subsequent

paradigms attempt to fill.

4.3 The Founding Charter (1955-1956): Dartmouth and the Conjecture of Complete Describability

The Dartmouth Summer Research Project proposal (August 31, 1955), authored by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon, is the founding document of AI as a field and the origin of the term "artificial intelligence" (Dartmouth proposal reprint, AI Magazine; Dartmouth workshop, Wikipedia). Its central conjecture: "every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it" (Dartmouth proposal; Dartmouth College retrospective).

Structural definition of generality: intelligence as *complete formal describability* - any cognitive faculty (language use, abstraction, self-improvement) is assumed reducible to a finite symbolic specification.

Proposed engine: none was operational at Dartmouth itself; the proposal was an agenda, not an architecture. But it licensed the two engines that would dominate the next twenty years - heuristic search over symbolic state spaces and, later, self-referential program improvement.

Adversarial note: the conjecture of complete describability was never proven or disproven at Dartmouth; it was asserted as a working hypothesis and then treated by the field for two decades as if it were established. This slippage - hypothesis presented as tractable research program - recurs in nearly every subsequent AGI paradigm, including the 2020s scaling program's implicit assumption that intelligence is "describable" by next-token statistics.

4.4 Symbolic/Cognitive Architectures (1956-1990s)

4.4.1 Logic Theorist and General Problem Solver (1956-1959)

Allen Newell, Herbert Simon, and J.C. Shaw's Logic Theorist (1956) and General Problem Solver (GPS) (1957, formally presented 1959) constitute the first working attempt at domain-independent problem solving. GPS used means-ends analysis - recursively reducing the difference between a current state and a goal state by selecting operators indexed to that difference - and was explicitly framed as "part of an investigation into the extremely complex processes that are involved in intelligent, adaptive, and creative behavior" (Newell, Shaw & Simon, "A General Problem-Solving Program for a Computer," 1959, CMU archive; GPS, Wikipedia).

Structural definition of generality: any problem expressible as a directed graph of states connected by well-formed operators (a "well-formed formula" or WFF representation) is solvable, in principle, by the same domain-independent search-and-difference-reduction procedure (GPS, Wikipedia).

Engine: heuristic state-space search guided by means-ends analysis and operator/difference tables.

Wall: GPS required every task to be hand-translated into its symbolic WFF representation, and its search space grew combinatorially with problem size - precisely the combinatorial explosion that would become the field's central technical liability. It also demonstrated no ability to acquire new representational primitives; its generality was confined to problems already expressed in the vocabulary its designers supplied.

4.4.2 The Physical Symbol System Hypothesis (1975-1976)

Newell and Simon's 1975 ACM Turing Award lecture, published as "Computer Science as Empirical Inquiry: Symbols and Search" (1976), crystallized the theoretical claim underlying all of GOF AI: the Physical Symbol System Hypothesis (PSSH) - "A physical symbol system has the necessary and sufficient means for general intelligent action" (Newell & Simon, 1976, CMU archive; summary, CenterConsulting; PSSH status paper, Nils Nilsson, Stanford). This is a necessity and sufficiency claim, not merely an engineering proposal: it asserts that symbol manipulation is both required for, and adequate to produce, general intelligent action, and that human symbolic behavior "arises because he has the characteristics of a physical symbol system" (Physical symbol system, Wikipedia).

Structural definition of generality: the capacity of a symbol system to designate objects, interpret and

execute expressions as processes, and thereby produce arbitrary intelligent behavior via search through a combinatorially structured problem space.

Engine: discrete symbol tokens, combinatorial expressions, and interpretive processes operating over them - realized physically in digital computers.

Wall (adversarial reading): the PSSH is an empirical hypothesis that was never rigorously confirmed; it was retroactively falsified in practice by (a) the frame problem, formalized by McCarthy and Hayes in 1969, concerning the intractability of specifying what does **not** change after an action without exhaustive enumeration (frame problem synthesis, CNRS); (b) the symbol grounding problem, formalized by Stevan Harnad in 1990, showing that symbols manipulated purely by syntactic shape have no intrinsic connection to their referents and remain "parasitic on the meanings in our heads" (Harnad, "The Symbol Grounding Problem," 1990); and (c) combinatorial explosion itself, identified by the UK's Lighthill Report (1973) as the central reason symbolic AI could not scale to real-world domains (see Section 3.4).

4.4.3 SOAR and Unified Cognitive Architectures (1981-present)

Soar, developed by John Laird, Allen Newell, and Paul Rosenbloom beginning around 1981, is explicitly "an architecture for general intelligence" whose stated goal is "to develop the fixed computational building blocks necessary for general intelligent agents - agents that can perform a wide range of tasks and encode, use, and learn all types of knowledge to realize the full range of cognitive capabilities found in humans" (Laird, Newell & Rosenbloom, "Soar: An Architecture for General Intelligence," 1986, CMU archive; Soar, Wikipedia)). Soar remains under active development, with a 2022 functional overview still framing it as providing "the fixed computational building blocks for creating AI agents whose cognitive characteristics and capabilities approach those found in humans" (Laird, "Introduction to the Soar Cognitive Architecture," arXiv 2022).

Structural definition of generality: a single, fixed, task-independent architecture - production rules operating over a working memory, unified by a problem-space computational model and chunking-based learning - capable in principle of any cognitive task, from routine to open-ended.

Engine: production-rule matching and firing against symbolic working memory, impasse-driven subgoal-ing, and "chunking" (experience compiled into new production rules) as the general learning mechanism.

Wall: Soar's architecture is fixed and its representational substrate remains predominantly symbolic; despite four decades of development, it has not produced general, human-level competence outside constructed task domains, and integrating genuinely open-ended perceptual and linguistic grounding into the architecture (rather than hand-coded symbolic front ends) remains an unsolved engineering burden - a persistence of the symbol-grounding problem inside a more disciplined production system.

4.4.4 The Lighthill Report and the First AI Winter (1973)

Sir James Lighthill's 1973 report to the UK Science Research Council, "Artificial Intelligence: A General Survey," delivered the first authoritative structural indictment of the symbolic paradigm: "In no part of the field have the discoveries made so far produced the major impact that was then promised," and its central technical diagnosis was that symbolic methods could not escape combinatorial explosion when scaled from toy "micro-worlds" to real-world domains (Lighthill Report, Wikipedia; combinatorial explosion analysis, Brox, MPI). The report is credited with precipitating major funding cuts in the UK and contributing to the broader AI winter that followed in the US through the mid-1970s (AI winter, Wikipedia; combinatorial explosion, Wikipedia).

Adversarial interpretation: Lighthill's critique targeted general-purpose reasoning specifically, not AI research overall - and the field's institutional response was not to solve the combinatorial wall but to retreat from generality entirely, redirecting resources toward narrow, domain-restricted expert systems such as DENDRAL (Feigenbaum, Lederberg, from 1965) and MYCIN, which achieved practical success precisely by abandoning the goal of general intelligence in favor of "a large knowledge base of real world facts and heuristics" bounded to a single expert domain (DENDRAL history, MIT/Lindsay; Expert system, Wikipedia). This is the field's first major bait-and-switch: narrow AI's commercial success in the 1980s was subsequently, and inaccurately, cited as evidence of progress toward general intelligence, when it was in fact evidence of the opposite - that generality had been structurally abandoned as intractable.

4.4.5 ACT-R and the Cognitive-Modeling Branch (1976-present)

John Anderson's ACT-R (Adaptive Control of Thought-Rational), developed at Carnegie Mellon, is a hybrid symbolic/sub-symbolic cognitive architecture that "aims to define the basic and irreducible cognitive and perceptual operations that enable the human mind," integrating declarative memory, procedural memory, and perceptual-motor modules under a rational-analysis framework in which cognition is modeled as statistically adapted to the environment (ACT-R, Wikipedia; Taatgen, Lebiere & Anderson, "Modeling Paradigms in ACT-R").

Structural definition of generality: decomposability of all human task performance into a common, small set of discrete cognitive/perceptual operations (declarative chunk retrieval governed by an activation equation, procedural rule firing) whose composition yields arbitrary competence.

Engine: symbolic production rules combined with sub-symbolic activation and utility equations (e.g., base-level activation $B_i = \ln\left(\sum_j t_j^{-d}\right) + \beta_i$) that graft statistical, decay-based memory dynamics onto the symbolic core (Mathematical Psychology reference on ACT-R).

Wall: ACT-R has succeeded as a quantitative model of human reaction-time and error data in constrained laboratory tasks, but - like Soar - has not scaled to open-domain general competence; its hybrid activation equations are fit to specific experimental paradigms rather than derived from a domain-general theory of representation acquisition, so the architecture inherits the closure problem in a parametrized rather than eliminated form.

4.4.6 Cyc and the Common-Sense Knowledge Wall (1984-present)

Douglas Lenat's Cyc project, begun in 1984 (spun off as Cycorp in 1994), took the diagnosis of the frame problem and combinatorial explosion literally and attempted to solve it by brute-force encoding: hand-building a knowledge base of common-sense facts and heuristics large enough to "overcome brittleness and knowledge acquisition bottlenecks" (Lenat, Prakash & Shepherd, "CYC: Using Common Sense Knowledge to Overcome Brittleness," 1985; Cyc, Wikipedia).

Structural definition of generality: general reasoning is achievable once a sufficiently large, logically consistent, hand-curated ontology of common-sense assertions and an inference engine over that ontology (CycL) exist - generality as an asymptotic function of knowledge-base coverage.

Engine: first-order-logic-like knowledge representation (CycL) plus a symbolic inference engine reasoning over hundreds of thousands to millions of hand-entered assertions.

Wall: after more than forty years and one of the longest-running efforts in AI history, Cyc has never approached general common-sense competence; the project's own founding rationale - that brittleness stems from insufficient explicit knowledge - has been undermined by its own trajectory, since the knowledge-acquisition bottleneck it was built to overcome by hand-encoding turned out to scale combinatorially with the world's tacit, context-dependent, and constantly-updating knowledge, reproducing the frame problem at civilizational scale rather than solving it.

4.5 Universal AI and Computability-Theoretic Formalizations (1960-2007)

4.5.1 Solomonoff Induction (1960-1964)

Ray Solomonoff's algorithmic probability, first presented in 1960 and formalized in "A Formal Theory of Inductive Inference" (Parts I and II, 1964), defined a machine-independent, uncomputable but mathematically well-defined universal prior over all computable hypotheses: the probability of a hypothesis is inversely related to the length of the shortest program (on a fixed universal Turing machine) that generates the observed data, formalizing Occam's razor (Ray Solomonoff, Wikipedia; Solomonoff, "Does Algorithmic Probability Solve the Problem of Induction?"; Algorithmic probability, Wikipedia).

Structural definition of generality: perfect, domain-independent inductive inference - the ability to predict any computable sequence's continuation optimally in the limit, using a single universal prior over the space of all Turing machines.

Engine: Bayesian updating over the Solomonoff/universal prior $M(x) = \sum_{p: U(p)=x^*} 2^{-\ell(p)}$, summing

over all programs p on a universal Turing machine U whose output is consistent with the observed string x .

Wall: Solomonoff himself acknowledged the theoretical solution to induction is uncomputable - "any device as accurate as ALP must necessarily be uncomputable" (Solomonoff, ISIS96 paper) - because it requires enumerating and running every possible program, an operation that never halts for programs that do not halt (a direct consequence of the halting problem). Kolmogorov complexity, the closely related descriptive-complexity measure that Solomonoff's work anticipated, is itself provably non-computable, and no program computing a lower bound on it can succeed in general, an impossibility result akin to Gödelian incompleteness and Cantor diagonalization (Kolmogorov complexity, Wikipedia).

4.5.2 AIXI (2000-2005)

Marcus Hutter's AIXI, introduced in "A Theory of Universal Artificial Intelligence based on Algorithmic Complexity" (2000) and elaborated in "Universal Artificial Intelligence" (2005), combines Solomonoff induction with sequential decision theory to define a single, parameter-free formalism for an optimal reinforcement-learning agent embedded in any computable environment: "We combine both ideas and get a parameterless theory of universal Artificial Intelligence... the resulting AIXI model is the most intelligent unbiased agent possible" (Hutter, arXiv:cs/0004001; AIXI, Wikipedia).

Structural definition of generality: an agent is maximally, universally intelligent if, at every timestep, it selects the action maximizing expected future reward under a Bayesian mixture over *every* computable environment consistent with its interaction history so far - generality as literal exhaustive hypothesis-space coverage over all of computability.

Engine: the AIXI equation, informally,

$$a_t = \arg \max_{a_t} \sum_{o_t r_t} \dots \max_{a_m} \sum_{o_m r_m} [r_t + \dots + r_m] \sum_{p: U(p, a_1 \dots a_m) = o_1 r_1 \dots o_m r_m} 2^{-\ell(p)}$$

- summing over all programs consistent with observed history, weighted by program length, and choosing actions to maximize expected discounted reward (Hutter's technical overview, "Universal Algorithmic Intelligence"; AIXI explainer, Dan MacKinlay).

Wall: AIXI is explicitly, by Hutter's own framing and by its critics, a gold-standard reference model, not implementable - "it's not meant to be implemented directly (it's uncomputable)" (MacKinlay, AIXI notes). It inherits Solomonoff induction's uncomputability exactly, and every practical "approximation" (e.g., MC-AIXI, context-tree weighting variants) discards the very property - exhaustive coverage of all computable environments - that gives the formalism its claim to universality, meaning approximate AIXI is not a scaled-down version of universal intelligence but a categorically different, bounded-hypothesis-class system wearing AIXI's name.

4.5.3 The Legg-Hutter Definition and the Coining of "AGI" (2002-2007)

The term "Artificial General Intelligence" was commissioned by Ben Goertzel and Cassio Pennachin around 2002 while editing an essay collection, at the suggestion of Shane Legg (later a DeepMind co-founder), specifically to distinguish broad-capability AI research from the narrow-task AI that dominated the field; the term appeared in print with their 2007 volume "Artificial General Intelligence" (Goertzel, "Who Coined the Term AGI?"; Forbes history of AGI; Goertzel & Pennachin survey draft). Legg and Hutter subsequently formalized a definition compatible with the AIXI framework: "Universal Intelligence: A Definition of Machine Intelligence" (2007) defines intelligence as "an agent's general ability to achieve goals in a wide range of environments," operationalized as a Kolmogorov-complexity-weighted sum over performance across the space of all computable environments, $\Upsilon(\pi) := \sum_{\mu \in E} 2^{-K(\mu)} V_{\mu}^{\pi}$ (Legg & Hutter, arXiv:0712.3329; Legg & Hutter, "A Universal Measure of Intelligence for Artificial Agents").

Wall: the Legg-Hutter measure, by the authors' own acknowledgment, is "only asymptotically computable," meaning it cannot serve as a practical intelligence test or training objective; subsequent work attempting to build tractable proxies (e.g., the AIQ test) concedes the measure must be radically restricted to be evaluable at all, again discarding the theoretical universality that motivated it (Legg & Veness, "An Approximation of the Universal Intelligence Measure").

4.6 Connectionism and the Rise, Fall, and Rise of Neural Generality (1958-2012)

4.6.1 The Perceptron and Its First Death (1958-1969)

Frank Rosenblatt's perceptron (1958) proposed a trainable, statistically adaptive network as an alternative to hand-coded symbolic rules. Marvin Minsky and Seymour Papert's *Perceptrons: An Introduction to Computational Geometry* (1969) delivered a rigorous mathematical proof of the single-layer perceptron's limitations, most famously its inability to compute the XOR predicate and its provably poor scaling on the "connectedness" predicate as input size grows (Theorem 5.5) (*Perceptrons* (book), Wikipedia)). Although Minsky and Papert's proofs applied strictly to single-layer, "conjunctively local" architectures, and multi-layer networks were later shown capable of computing such functions, the book's **reception** was read as a wholesale dismissal of connectionism, and "research into neural networks went into decline throughout the 1970s" as a direct, historically documented consequence (UC Davis ECS170 course notes; Minsky & Papert analysis).

Adversarial reading: this is the field's first case of a **correct, narrow** mathematical result being extrapolated into an **incorrect, general** verdict - the multi-layer solution to XOR was known in principle at the time but was not credited, and the resulting funding contraction was a sociological rather than a strictly mathematical outcome.

4.6.2 Backpropagation and Parallel Distributed Processing (1986)

David Rumelhart, Geoffrey Hinton, and Ronald Williams's "Learning representations by back-propagating errors" (**Nature**, 1986) revived connectionism by providing an efficient, general algorithm for training multi-layer networks: repeatedly adjusting connection weights along the gradient of an output error measure, allowing hidden units to develop internal representations of task-relevant structure without those representations being hand-specified (Rumelhart, Hinton & Williams, **Nature** 323, 1986). This was published alongside the two-volume *Parallel Distributed Processing* (PDP) collection, whose explicit ambition was a general theory of cognition as emergent statistical pattern completion across distributed, sub-symbolic units (PDP, MIT Press, 1986, full text; PDP review, **Nature**).

Structural definition of generality: cognition as the emergent, statistically-learned behavior of large networks of simple, uniform, densely interconnected units - generality located in the learning algorithm and network topology rather than in hand-designed representations.

The Fodor-Pylyshyn Challenge (1988): Jerry Fodor and Zenon Pylyshyn's "Connectionism and Cognitive Architecture: A Critical Analysis" (1988) argued connectionist networks, absent an underlying compositional symbol structure, cannot in principle explain the systematicity and compositionality of human thought and language - that connectionism and classical symbol processing are architecturally incompatible, not merely implementationally different (Fodor & Pylyshyn, Rutgers RuCCS archive; dynamical-systems literature review of the debate). This debate, unresolved as of the 2020s, structurally anticipates the current dispute over whether transformer-based LLMs "reason" compositionally or merely interpolate statistically.

Wall (1990s): despite the backpropagation breakthrough, 1980s-1990s connectionist networks lacked the compute and data to demonstrate general competence outside narrow pattern-recognition benchmarks, and the paradigm entered a second, quieter winter through the 1990s and 2000s pending the compute and data scale that would arrive only after 2012.

4.6.3 The Symbol Grounding Problem as a Cross-Paradigm Indictment (1990)

Stevan Harnad's "The Symbol Grounding Problem" (1990) is the single most durable structural critique spanning both symbolic and connectionist paradigms: purely syntactic symbol manipulation (GOFAI) is semantically "parasitic on the meanings in our heads," while pure connectionism, absent grounding in sensorimotor experience, does not resolve this either - Harnad proposed a hybrid bottom-up architecture as the only structurally coherent path to grounded general symbols (Harnad, **Physica D** 42, 1990). This problem remains formally unresolved and is directly inherited by 2020s multimodal foundation models, whose "grounding" in pixel and audio streams is empirically richer but not theoretically different in kind.

4.6.4 The Deep Scaling Breakthrough (2012)

Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton's AlexNet (2012) - a deep convolutional network trained on GPUs - won the ImageNet Large Scale Visual Recognition Challenge by a decisive margin, marking "the birth of deep learning" as a commercially and academically dominant paradigm and validating the thesis that sufficient compute plus sufficient labeled data, applied to comparatively simple gradient-based architectures, outperforms hand-engineered features (AlexNet, Wikipedia; Pinecone history of AlexNet). AlexNet itself was narrow (single-task image classification), but it is included here because it is the direct causal antecedent - architecturally and sociologically - of the scaling program that would explicitly target generality after 2018.

4.7 AlphaZero and the General Reinforcement-Learning Interlude (2016-2017)

DeepMind's AlphaZero, described in "Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm" (2017), generalized AlphaGo Zero's self-play, Monte Carlo tree search, and deep neural network architecture across chess, shogi, and Go using a single algorithm and no game-specific human knowledge beyond the rules (Silver et al., arXiv:1712.01815; AlphaGo Zero, Wikipedia).

Structural definition of generality (restricted): algorithmic generality across a *class* of two-player, perfect-information, zero-sum games - a substantially narrower claim than general intelligence, but marketed and popularly received as evidence of movement toward it.

Engine: self-play reinforcement learning combined with a policy/value deep network and Monte Carlo tree search, iteratively bootstrapping stronger play from weaker play with zero human game data.

Wall (adversarial reading): critical assessments, e.g. "AlphaZero - What's Missing?," note that the "generality" demonstrated is confined to a narrow structural class of environments (fully observable, deterministic, adversarial, symmetric two-player games); AlphaZero cannot transfer its learned policy to a differently structured task without complete retraining, and its self-play mechanism presupposes a perfect, cheaply-simulable environment model - a condition true of board games and false of almost every real-world domain (AlphaZero - What's Missing?, Informatica). AlphaZero is a landmark for *narrow transfer* within a game class, not a demonstration of domain-general intelligence, and its inclusion in AGI narratives is a category error the adversarial historian must flag explicitly.

4.8 Whole Brain Emulation (2008-present)

Anders Sandberg and Nick Bostrom's "Whole Brain Emulation: A Roadmap" (Future of Humanity Institute, Technical Report #2008-3) is the founding programmatic document for the hypothesis that general intelligence can be achieved not by designing an algorithm for cognition but by scanning, reconstructing, and simulating an existing biological brain at sufficient resolution to reproduce its functional dynamics (Sandberg & Bostrom, 2008, full report; Oxford ORA record).

Structural definition of generality: intelligence is a specific pattern of physical/computational organization instantiated in a particular brain; generality is inherited wholesale from the biological original rather than engineered from first principles - this is explicitly *not* a theory of intelligence but a bypass of the need for one.

Engine (proposed, not yet realized): three sequential technical stages - (1) physical scanning of neural tissue at synaptic resolution (e.g., serial-section electron microscopy, FIB-SEM), (2) reconstruction of the full connectome (neurons and synaptic connectivity), and (3) numerical simulation of neuronal and synaptic dynamics at a resolution sufficient for functional equivalence (WBE requirements, Sandberg & Bostrom 2008; WBE engineering review, IJIRT).

Wall: the 2024 mapping of the complete adult *Drosophila melanogaster* brain connectome (139,255 neurons, 50 million synaptic connections, published in *Nature*) is the field's most advanced empirical result to date and remains three to four orders of magnitude below the scale of a mammalian cortex, let alone a human brain with 86 billion neurons (Connectome analysis, cafebedouin.org). Contemporary technical reviews conclude WBE is "physically plausible but technologically distant, primarily due to measurement, modeling, and verification constraints," stacking unresolved problems in tissue preservation, synaptic-resolution scanning throughput, multiscale neuronal/synaptic dynamics modeling, and -

most fundamentally - validating whether a structurally faithful simulation is **functionally** equivalent to the original mind at all (WBE engineering review, IJIRT). This is a metrological and multiscale-modeling wall, distinct from the combinatorial and computability walls above: even granting unlimited compute, the problem is that the required **input data** (a complete, functionally-relevant synaptic-resolution map of a living human brain) may not be acquirable non-destructively at any foreseeable technology level, and independent skeptical analyses argue the "without-reverse-engineering path is impossible" and that a working reverse-engineering path would produce engineered (not emulated) superintelligence well before WBE itself completes, undermining WBE's own rationale as a **safer** route to general intelligence (LessWrong technical assessment).

4.9 Heterodox and Thermodynamic AGI Approaches (2007-present)

4.9.1 NARS: Non-Axiomatic Reasoning (1990s-present)

Pei Wang's Non-Axiomatic Reasoning System (NARS) rejects the presumption, shared by AIXI and classical logic-based symbolic AI, that a general intelligence has access to complete knowledge or unbounded computational resources. NARS is explicitly designed as "a general-purpose intelligent system... that follows the same principles as the human mind" and operates under the Assumption of Insufficient Knowledge and Resources (AIKR) - the founding axiom that any real intelligence must reason and act with incomplete evidence, finite processing time, and continuously revisable beliefs (Pei Wang, NARS Introduction, 2025; Hammer, Lofthouse & Wang, "The OpenNARS Implementation of NARS").

Structural definition of generality: general intelligence is **adaptation with insufficient knowledge and resources**, not optimality under unlimited computation (a direct, explicit repudiation of the AIXI framework's idealization).

Engine: a non-axiomatic (non-classical) term logic with truth values expressed as confidence intervals over evidence, combined with a resource-bounded memory and control mechanism that allocates limited "processing budget" across competing inference tasks (AGI-16 NARS tutorial, Pei Wang & Patrick Hammer).

Wall: NARS has produced a coherent formal logic for resource-bounded, non-axiomatic inference but has not scaled beyond small demonstration systems to any task complexity approaching general competence; the architecture solves the **idealization** problem of AIXI in principle but has not demonstrated that its resource-bounded inference mechanism can acquire the breadth of grounded, perceptual, and linguistic knowledge that contemporary foundation models acquire from raw data - its logic is well-specified, but its knowledge-acquisition pipeline remains comparatively primitive.

4.9.2 OpenCog and the Integrative/Hybrid Program (2008-present)

Ben Goertzel's OpenCog and OpenCog Prime propose that human-equivalent AGI will emerge from the **integration** of multiple specialized cognitive components (symbolic reasoning, probabilistic logic, evolutionary program learning, attention allocation) within a shared knowledge representation (the AtomSpace), rather than from any single unifying algorithm - a self-consciously heterodox rejection of both pure symbolic and pure connectionist monism (OpenCog, Wikipedia; OpenCog Prime wikibook, Goertzel). The 2023 successor, OpenCog Hyperon, is described as "A Framework for AGI at the Human Level and Beyond," a "mostly from-the-ground-up rewrite" intended to integrate symbolic and neural components more tightly, including hybridization experiments with deep-learning perceptual front ends (DeSTIN) feeding into the symbolic AtomSpace (Goertzel et al., "OpenCog Hyperon," arXiv:2310.18318; Goertzel, "Perception Processing for General Intelligence, Part II").

Structural definition of generality: emergent property of a sufficiently rich, sufficiently integrated multi-component cognitive system - generality as an architectural composition problem rather than a single-algorithm discovery.

Wall: OpenCog has never demonstrated integrated, cross-domain competence approaching even weak general intelligence in a public benchmark; the project's own thirty-year history (Novamente Cognition Engine origins predate 2008) illustrates the integration wall: combining *N* narrow but individually well-understood subsystems does not straightforwardly yield generality, because the interfaces between symbolic, probabilistic, and evolutionary subsystems introduce their own combinatorial and semantic

mismatch problems that are, if anything, harder to specify formally than any single subsystem's internal logic.

4.9.3 Thermodynamic and Free-Energy Approaches (2010-present)

Two distinct heterodox physical programs deserve separate treatment.

Causal Entropic Forces (Wissner-Gross & Freer, 2013): published in *Physical Review Letters*, this paper proposes "a first step toward" a formal physical relationship between intelligence and entropy maximization, showing that a "causal generalization of entropic forces" over the paths available to a physical system in configuration space can spontaneously produce behaviors resembling tool use and social cooperation in simple simulated systems, without any explicit goal specification (Wissner-Gross & Freer, *Phys. Rev. Lett.* 110, 168702, 2013). The AGI-2014 conference presentation framed this explicitly as a candidate "universal protocol for AGI," with the causal entropic force $F = T_c \nabla_x S_\tau$ proposed as the general driving mechanism (Wissner-Gross, "The Thermodynamics of Artificial General Intelligence," AGI-2014).

Structural definition of generality: intelligent behavior is whatever behavior emerges from a physical system acting to keep open the maximum diversity of future accessible states (entropy maximization subject to causal/temporal constraints) - generality as a physics-level variational principle requiring no representation, symbol, or learning algorithm at all.

Wall (adversarial and formal): published critique (Canessa, 2013, *Comment on Phys. Rev. Lett.*) directly challenges the generality of the mechanism, and follow-on analysis ("Quantum Mechanical Foundations of Causal Entropic Forces," AGI-2014) notes the theory's definition of causal entropic force "makes no" clear prescription for how a system should discount or weight the vast space of possible future paths, leaving the central variational principle underspecified for anything beyond the toy demonstrations in the original paper (Canessa, arXiv:1308.4375; Shah, "Quantum Mechanical Foundations of Causal Entropic Forces," AGI-2014). No causal-entropic-force system has been scaled beyond simple simulated physical environments (a particle in a box, a pole-balancing task); the theory has produced no architecture capable of language, symbolic reasoning, or open-ended tool use, and it is fair to characterize it, as of the 2020s, as a suggestive statistical-mechanics analogy that has not yet produced a competing engineering program - a thermodynamic wall of underspecification, in which the physical principle is elegant but computationally underdetermined for any domain with combinatorially large or continuous state spaces.

Free Energy Principle / Active Inference (Karl Friston, 2000s-present): Friston's free energy principle proposes that "any adaptive change in the brain will minimize free-energy," a variational information-theoretic quantity bounding the surprise of sensory data given an internal generative model, applying "to any biological system that resists a tendency to disorder" (Friston, "The free-energy principle: a rough guide to the brain?"; Free energy principle, Wikipedia). Active inference extends this to action selection, treating perception and action as two routes to the same objective - minimizing variational free energy - and is presented as "an integrative perspective on brain, cognition, and behavior" spanning neuroscience, robotics, and machine learning (Parr, Pezzulo & Friston, *Active Inference*, MIT Press; Generalised free energy, PMC).

Structural definition of generality: any system that persists against entropic dissolution is, by construction, performing (approximate) Bayesian inference to minimize surprise - generality is presented as a **unifying description** of all adaptive biological systems rather than a specific engineered capability.

Wall (adversarial reading): the free energy principle's breadth is also its liability - critics across neuroscience and philosophy of science have long noted its extreme generality risks unfalsifiability, since **any** observed adaptive behavior can be redescribed post hoc as free-energy minimization without generating novel, falsifiable predictions about which specific architecture or algorithm should be built; unlike AIXI or Soar, the free energy principle has not converged on a single reference implementation that self-identifies as a candidate AGI system, and its practical machine-learning offshoots (active inference agents) remain restricted to small-scale robotics and control demonstrations, not general competence.

4.10 Connectionist/Deep Scaling as an Explicit AGI Program (2018-2026)

4.10.1 Scaling Laws and the "Bitter Lesson" (2019-2020)

Richard Sutton's 2019 essay "The Bitter Lesson" provided the ideological charter for the scaling program: "The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin," attributing this to the compounding effect of Moore's-law-style falling compute costs relative to the cost of hand-engineering domain knowledge (Sutton, "The Bitter Lesson," 2019, UT Austin archive; Bitter lesson, Wikipedia). Jared Kaplan, Sam McCandlish, Dario Amodei, and colleagues' "Scaling Laws for Neural Language Models" (2020) supplied the empirical backbone: cross-entropy test loss follows clean power laws in model size, dataset size, and training compute across more than seven orders of magnitude, with architectural details (width, depth) mattering comparatively little within a wide range (Kaplan et al., arXiv:2001.08361).

Structural definition of generality: general intelligence is an emergent statistical property of sufficiently accurate next-token (or, more broadly, next-observation) prediction, achieved by scaling a single, comparatively simple, architecture-agnostic training objective across compute, data, and parameters - generality as an emergent byproduct of compression/prediction at scale, not a designed capability.

Engine: the transformer architecture (self-attention over token sequences) trained via gradient descent to minimize predictive cross-entropy loss, scaled according to the compute-optimal allocation described by the Kaplan/Chinchilla-family scaling laws.

4.10.2 Foundation Models and Claims of Emergent Generality (2021-2023)

The Stanford Center for Research on Foundation Models' "On the Opportunities and Risks of Foundation Models" (2021) formalized the paradigm's self-conception: models trained on broad data at scale that can be adapted to a wide range of downstream tasks, explicitly reframing "foundation models" as the connectionist answer to task-general competence (Bommasani et al., Stanford CRFM report). Microsoft Research's "Sparks of Artificial General Intelligence: Early Experiments with GPT-4" (2023) escalated the claim explicitly into AGI terms, arguing GPT-4 demonstrates "human-level performance on novel and difficult tasks" across mathematics, coding, vision, medicine, law, and psychology, and "could reasonably be viewed as an early (yet still incomplete) version of an artificial general intelligence (AGI) system" (Bubeck et al., Microsoft Research; Synced summary).

Adversarial caveat, made explicit by the paper's own authors: the "Sparks" paper is a qualitative, non-systematic exploration of an early, non-final model checkpoint, and the authors themselves caution that "all quantitative results should be viewed as estimates of the model's potential, rather than definitive numbers" (LessWrong-hosted version with author caveats) - a methodologically weak basis for the strong AGI claim that was nonetheless widely amplified.

4.10.3 The Emergence Mirage and the Reliability Wall (2023-2025)

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo's "Are Emergent Abilities of Large Language Models a Mirage?" (2023) directly challenged the scaling program's most celebrated empirical claim - that qualitatively new capabilities appear discontinuously at scale - arguing that apparent emergence is largely an artifact of researchers' choice of non-linear or discontinuous evaluation metrics, and that continuous metrics applied to the same models reveal smooth, predictable improvement rather than sharp phase transitions (Schaeffer, Miranda & Koyejo, arXiv:2304.15004). This matters structurally because "emergent generality" was the scaling program's implicit mechanism for how narrow next-token prediction could yield general reasoning; if emergence is a measurement artifact, the mechanism for the paradigm's central promise is undermined at the root.

Ziwei Xu, Sanjay Jain, and Mohan Kankanhalli's "Hallucination is Inevitable: An Innate Limitation of Large Language Models" (2024) provides a formal, computability-theoretic argument - explicitly analogous to the halting problem and Gödelian diagonalization - that no LLM can be constructed to eliminate hallucination for all computable functions, because doing so would require the model to correctly decide problems that are not decidable by any Turing-equivalent computation (Xu, Jain & Kankanhalli, arXiv:2401.11817). A 2025 arXiv preprint, "The wall confronting large language models," extends this into a quantitative claim: "the scaling laws which determine the performance of large language models severely limit their ability to improve the uncertainty of their predictions... raising their reliability to

meet the standards of scientific inquiry is intractable by any reasonable measure," attributing the ceiling to the very mechanism (generating non-Gaussian output distributions from Gaussian input latents) that gives LLMs their generative power (arXiv:2507.19703).

4.10.4 The Empirical Scaling Wall (2024-2025)

By 2024-2025, independent technical and financial analyses converged on evidence of genuine diminishing returns from the scaling recipe itself: "Frontier models from OpenAI, Anthropic, Google, and Meta show smaller performance jumps on key English benchmarks despite massive increases in training budget," alongside a looming data wall - scarcity of additional high-quality, non-synthetic training text - and rapidly escalating compute and energy costs (Masood, "Is There a Wall?"). A December 2025 preprint, "The AI Scaling Wall of Diminishing Returns," quantifies this directly: "compute grows 10-100x while accuracy barely moves," and argues the next capability gains must come "not from bigger models, but from smarter, more efficient ones" (Shukla, arXiv:2512.20264). Gary Marcus, a long-standing scaling critic since 2022, declared in 2025 that his original "deep learning is hitting a wall" thesis was "confirmed," framing the slowdown as evidence of the "likely financial collapse of the Generative AI bubble" absent a fundamental architectural departure from pure scaling (Marcus, "LLMs Have Indeed Reached a Point of Diminishing Returns").

Structural wall (synthesis): the deep-scaling paradigm faces a compound wall with at least three independent components: (1) a computability wall, formally proving hallucination cannot be eliminated for all computable target functions, a Gödelian analogue to AIXI's own incomputability; (2) a statistical wall, in which scaling laws that govern mean predictive loss do not correspondingly tighten predictive *uncertainty*, capping achievable reliability regardless of further scale; and (3) a resource wall, in which the marginal compute and data required for further gains is growing faster than the gains themselves, a direct empirical reversal of the power-law optimism of 2020. Unlike the combinatorial explosion that killed 1970s symbolic AI outright, this wall has not stopped capability improvement, but it has falsified the strong-form claim - implicit in "Sparks of AGI" and the pure scaling hypothesis - that generality is simply an emergent, cost-linear function of scale.

4.11 Cross-Paradigm Synthesis: A Structural Table

Cybernetics

1943-1956 - Homeostasis under arbitrary perturbation - Analog feedback/servomechanisms - No formalism for reasoning/language; absorbed into symbolic AI (Wiener 1948)

GPS / Logic Theorist

1956-1959 - Any problem expressible as a state-graph is solvable by one search method - Means-ends analysis over symbolic WFFs - Combinatorial explosion; no representation-acquisition (Newell, Shaw & Simon 1959)

Physical Symbol System Hypothesis

1975-1976 - Symbol manipulation is necessary & sufficient for general intelligent action - Discrete symbols + interpretive processes - Frame problem, symbol grounding problem (Newell & Simon 1976; Harnad 1990)

Soar

1981-present - Fixed architecture spans all cognitive tasks - Production rules + chunking - Representational grounding never solved outside symbolic front ends (Laird et al. 1986)

ACT-R

1976-present - Decomposable into common cognitive primitives - Symbolic rules + sub-symbolic activation equations - Fit to lab tasks, not open-domain competence (Anderson, ACT-R)

Cyc

1984-present - Generality as asymptotic knowledge-base coverage - Hand-curated logic KB + inference engine - Knowledge-acquisition bottleneck reproduced at scale (Lenat et al. 1985)

Solomonoff Induction

1960-1964 - Perfect domain-independent induction - Universal prior over all Turing machines - Provably incomputable (Solomonoff, ISIS96)

AIXI

2000-2005 - Exhaustive Bayesian mixture over all computable environments - Reward-maximizing search over all programs - Inherits incomputability; approximations discard universality (Hutter 2000)

Whole Brain Emulation

2008-present - Generality inherited from a specific biological original - Scan $\rightarrow$ reconstruct connectome $\rightarrow$ simulate - Metrology/multiscale-modeling wall; unproven functional equivalence (Sandberg & Bostrom 2008)

NARS

1990s-present - Adaptation under insufficient knowledge/resources - Non-axiomatic term logic + resource budgeting - Limited demonstrated scale/breadth (Pei Wang, NARS)

OpenCog / Hyperon

2008-present - Emergent property of integrated multi-component system - AtomSpace + hybrid symbolic/probabilistic/evolutionary modules - Integration wall - no demonstrated cross-domain competence (Goertzel et al. 2023)

Causal Entropic Forces

2013-present - Entropy-maximizing behavior under causal constraints - $F = T_c \nabla_x S_\tau$ - Underspecified for non-toy state spaces (Wissner-Gross & Freer 2013; Canessa 2013)

Free Energy Principle / Active Inference

2000s-present - Any adaptive system minimizes variational free energy - Generative model + prediction-error minimization - Risk of unfalsifiability; no reference AGI implementation (Friston)

Deep Scaling / Foundation Models

2018-present - Generality as emergent property of scaled next-token prediction - Transformers + power-law compute/data/parameter scaling - Computability (hallucination), statistical (uncertainty), and resource (data/compute) walls (Kaplan et al. 2020; Xu, Jain & Kankanhalli 2024; Shukla 2025)

4.12 Conclusion: The Invariant Structure of Failure

Across seven decades and at least six architecturally distinct paradigms, no program purporting to build general intelligence has escaped a version of the same closure problem: a finite, specifiable mechanism (symbols, universal priors, connectome scans, entropy gradients, or gradient-descended weight matrices) is proposed as sufficient for open-ended competence in an environment whose relevant structure cannot be exhaustively specified or sampled in advance. Symbolic AI's combinatorial explosion, AIXI's incomputability, whole brain emulation's metrological ceiling, and the 2020s LLM reliability wall are not four unrelated technical accidents; they are four domain-specific manifestations of the same underlying tension between finite representational/computational machinery and an unbounded target. The historical record does not support the strong claim, made explicitly by the Dartmouth proposal in 1955 and echoed by the "Sparks of AGI" paper in 2023, that general intelligence is a describable, boundedly-specifiable engineering target; it supports instead the adversarial reading that each paradigm's "closure problem" is a rediscovery, in a new mathematical vocabulary, of the same open question the field has never actually closed.

Chapter 5

A Conditional ETFE Extension for a Macroscopic "Cognitive Processing Event": Dipole Escalation, Topological Piercing, an

5.1 Abstract

This treatise constructs a strictly conditional, equation-forward extension of the *ETFE Canonical Merge v4.2.21* formalism (Kourbanov, 11 July 2026; internal file `ETFE_Canonical_Merge_v4_2_21-3.pdf`) to a four-stage hypothetical process - closed-loop entropic escalation, boundary-induced homology loss ("Topological Piercing"), rank-one emission ("spectral caustic"), and transverse algebraic rebinding in a hyperdimensional plane - proposed here as a model of a macroscopic "cognitive processing event." ETFE's own text is used only for what it actually proves: an exact finite identity between a closing dipole-Dirac gap $\delta_D(\varepsilon) = \varepsilon^2$ and a diverging GNS-weighted Hodge-stiffness eigenvalue $\lambda_+(\Delta_1^\phi) = 1/(p_1 p_2)$, together with a genuine semisimple-to-radical-square-zero homological jump in the special fibre of the two-vertex dipole quiver (ETFE § 10.4, § 11.2). Stage (2) of the requested sequence - contact with a Max-Plus/tropical boundary and destruction of a Betti-1 cycle - and stage (4) - transverse intersection with a "VSA plane" producing "dimensionally complex algebraic binding" - use vocabulary that does not occur anywhere in ETFE v4.2.21; the manuscript contains no tropical semiring, no Betti-1 object (only algebraic Betti numbers $\beta_{i,d}^{alg}$ and a separately notated topological $b_i(C)$, explicitly declared disjoint, ETFE § Notation, item "Betti symbols and noncompact topologies are disjoint"), no VSA/hyperdimensional structure, and no term "spectral caustic" used for a 1-D ray (ETFE's "spectral-caustic class," § 7.6, denotes a stationary detector-equivalence class, not an emitted vector). Each of these constructs is therefore introduced here as an explicit new axiom external to ETFE, flagged in bold at first use, and the treatise states exactly which added hypotheses are required to make each of the four stages formally derivable. A conditional main theorem is proved under these added axioms; a no-go argument shows why the unconditional claim - that this mechanism is "the ultimate thermodynamic engine for novelty/insight and global entropy maximization" - cannot be established from ETFE's stated axioms or from the second law alone, and a minimal falsifiable order-parameter model is supplied in closing.

External grounding for the non-ETFE ingredients (tropical/max-plus algebra, combinatorial Hodge theory, nonequilibrium entropy production, vector-symbolic binding) is drawn from primary or standard technical sources cited inline by URL, consistent with the instruction to use outside material only to anchor the mathematics that ETFE itself does not supply.

5.2 Mathematical Setup and Typed Objects

5.2.1 ETFE-native objects (cited verbatim to definitions/equations)

All statements in this subsection reproduce ETFE's own typed objects; nothing is added.

Substrate. $\mathcal{A}_\tau = k\langle d_\alpha, d_\alpha^\dagger, q_a, \dots \rangle / R_\tau$, a filtered noncommutative $*$ -algebra of "quasi-temporal dipole" generators (ETFE Def. 4.1, eq. 4.1). Axiom 2.1 [A1] asserts this algebra, not clock time or particles, is primary.

Multi-Rees scale container. For a multiplicative multi-filtration $F_n^{s,\omega} \mathcal{A}_\tau$, the Rees algebra is $R_{s,\omega}(\mathcal{A}_\tau) = \bigoplus_{n \in \mathbb{N}^r} F_n^{s,\omega} \mathcal{A}_\tau u^n$ (ETFE eq. 5.5), specializing to $\mathcal{A}_\tau$ at $u = 1$ and to the associated graded $\text{gr}_F \mathcal{A}_\tau$ at $u = 0$ (eq. 5.6).

Finite dipole quiver-Rees model (ETFE § 10). On $M_2(k)$ with filtration $F_0 = ke_1 \oplus ke_2$, $F_1 = M_2(k)$, the Rees algebra is generated over $k[\varepsilon]$ by $e_1, e_2, a = \varepsilon E_{21}, b = \varepsilon E_{12}$ with $ba = \varepsilon^2 e_1$, $ab = \varepsilon^2 e_2$ (eq. 10.3-10.4). Proposition 10.1: the generic fibre ($\varepsilon \neq 0$) is $M_2(k)$ (semisimple); the special fibre $\varepsilon = 0$ is the radical-square-zero path algebra of the two-cycle quiver $1 \rightleftarrows 2$, $ab = ba = 0$ (eq. 10.5). This two-vertex, two-arrow quiver is ETFE's own model of the "closed bipartite relational loop" required by the task: it is literally a bipartite (source/target alternating) 2-vertex dipole with a directed cycle.

GNS state and degeneration (ETFE § 10.2). $\varphi_\varepsilon(x) = \text{Tr}(\rho \pi_\varepsilon(x))$, $\rho = \text{diag}(p_1, p_2)$, faithful for $\varepsilon > 0$; $\|a\|_{\varphi_\varepsilon}^2 = \varepsilon^2 p_1 \rightarrow 0$, $\|b\|_{\varphi_\varepsilon}^2 = \varepsilon^2 p_2 \rightarrow 0$ as $\varepsilon \rightarrow 0$ (eq. 10.8-10.9). ETFE states explicitly: "the loss of faithfulness is not removed; it is the metric degeneration to be measured."

Curved dipole Dirac operator. $d_\varepsilon = a + b$, $d_\varepsilon^2 = \varepsilon^2 \mathbf{1}$ (eq. 10.11); a $\mathbb{Z}_2$ -graded matrix factorization of the potential ε^2 (Eisenbud 1980, cited by ETFE). Dipole Dirac gap $\delta_D(\varepsilon) = \min \text{Spec}(d_\varepsilon^* d_\varepsilon) = \varepsilon^2$ (Def. 10.1, eq. 10.12).

Special-fibre homological transition (ETFE § 10.4, the exact analogue named in the task). For $\varepsilon \neq 0$ the fibre is semisimple: every module is projective, $\text{pd}(S_i) = 0$. For A_0 ($\varepsilon = 0$) the simple modules S_1, S_2 admit an infinite periodic minimal resolution $\cdots \rightarrow P_1 \xrightarrow{a} P_2 \xrightarrow{b} P_1 \rightarrow S_1 \rightarrow 0$ (eq. 10.13), alternating syzygies. Proposition 10.2: the Rees degeneration changes projective dimension of the simple module from 0 (every nonzero fibre) to ∞ (special fibre), with one projective generator in every homological degree. This is the semisimple-to-radical-square-zero transition the task's stage (1) must terminate on.

GNS-weighted cellular Hodge Laplacian and exact bridge (ETFE § 11). On $C_0 = \text{span} v_1, v_2$, $C_1 = \text{span} a, b$, boundary $\partial = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$ (eq. 11.1), GNS weights $W_0 = \text{diag}(p_1, p_2)$, $W_1 = \varepsilon^2 \text{diag}(p_1, p_2)$ (eq. 11.2). Weighted adjoint $\partial^{\dagger\varphi} = W_1^{-1} \partial^T W_0$ (eq. 11.3). Edge Hodge Laplacian $\Delta_1^\varphi = \partial^{\dagger\varphi} \partial$ (eq. 11.4). Proposition 11.1: $\text{Spec}(\Delta_1^\varphi) = 0, \frac{1}{\varepsilon^2 p_1 p_2}$ (eq. 11.5). Exact theorem (Proposition 11.2 / eq. 11.7):

$$\delta_D(\varepsilon) \lambda_+(\Delta_1^\varphi) = \frac{1}{p_1 p_2} \geq 4,$$

with equality iff $p_1 = p_2 = 1/2$, the maximal-von-Neumann-entropy tracial state (§ 11.3, eq. 11.9-11.10). ETFE calls this "the first proved quantitative bridge in the ETFE execution programme," explicitly containing "no fitted exponent and no curvature input" (boxed remark after eq. 11.7).

What the theorem does not prove (ETFE § 11.4, explicit): $\text{diag } \Delta_1^\varphi \neq \text{Forman curvature}$ by definition; $\text{Tr } \Delta_1^\varphi \neq \text{scalar curvature}$ by definition (eq. 11.12). Proposition 11.3 (Minimal-graph Forman no-go): uniform GNS edge-weight collapse in the *minimal* two-vertex graph does not force Forman-Ricci blow-up; a genuine curvature bridge requires "a filled cycle with a genuine 2-cell," a longer nonhomogeneous chain, or an independently declared curvature density.

Fractional spectral dimension (informal ETFE name; exact term is "scale-dependent spectral dimension" d_s). For positive generator $L_{s,\omega}$, heat trace $P(t; s, \omega) = \text{Tr } e^{-tL_{s,\omega}}$, $d_s(t; s, \omega) = -2 \partial \ln P / \partial \ln t$ (ETFE eq. 7.7 / 16.1). ETFE stresses d_s is a *scaling exponent*, not Krull or Hausdorff dimension, and that "fractional plateaus are model outputs. Values such as 1.5 or 0.5 are not inserted by definition" (§ 7.5, § 16). No fractal path is destroyed; only spectral weight $w_\alpha(s, \omega)$ redistributes (eq. 7.6, § 16 mode-participation measure $D_{\text{part}} = (\sum_n w_n)^2 / \sum_n w_n^2$, eq. 16.2).

Entropic escalation (ETFE term: "Stochastic Rees-Ricci escalation," Ch. 6). GNS algebraic regularity gap $\delta_{\text{alg}}^\varphi(s, \omega; K) = \min_{(i,d) \in I_K} \sigma_{\min}^+(D_{i,d}(s, \omega))$ (Def. 6.1, eq. 6.5). *Weak* escalation: branch exits every regular precompact class (Def. 6.2). *Strong synchronized* escalation: simultaneous change of Fitting stratum, singular scheme, and minimal resolution (Def. 6.3), guaranteed only under the Perfect-family synchronization criterion (Criterion 6.1: B -flat equidimensional family, Cohen-Macaulay regular fibres, finite projective dimension, Buchsbaum-Eisenbud grade inequalities failing transversally at exactly one fibre, independently verified depth drop, support compatibility). Theorem targets 6.1-6.2 are conditional (compactness- and Bridge-Inequality-dependent), *not* proved in general; only the finite dipole instance of § 10-11 is proved.

Homological Laplacians in general (ETFE § 7.2). $\Delta_i^\varphi(s, \omega) = D_i^{\dagger\varphi} D_i + D_{i+1}^{\dagger\varphi} D_{i+1}$ (eq. 7.2); under closed-range/Hodge hypotheses $\beta_{i,d}^{\text{alg}} = \dim \ker \Delta_{i,d}^\varphi$ (eq. 7.3, Conditional lemma L0 in the Compact Claims Registry, App. C).

Associated graded / Rees geometry. $\text{gr}_F \mathcal{A}_\tau = \bigoplus_{n \geq 0} F_n \mathcal{A}_\tau / F_{n-1} \mathcal{A}_\tau$ (eq. 4.4). Proposition 5.1: for commutative Noetherian A , ideal I , the blow-up of $\text{Spec } A$ along $V(I)$ is $\text{Proj } \bigoplus_{n \geq 0} I^n u^n$, normal cone $\text{Spec}(\text{gr}_I A)$ - "a rigorous algebraic fact," with the Ricci-flow identification left as an open "readout theorem."

Caustics (ETFE Ch. 17, its own, different usage). A caustic is a singularity of a $\text{*map* } X(q)$, defined by $\det(\partial X^i / \partial q^j) = 0$ (eq. 17.1), classified by fold A_2 , cusp A_3 , swallowtail A_4 catastrophes; ETFE's "frequency-caustic class" (Ch. 18) is a *stationary* resolvent-pole/resonant-band structure $R_{s,\omega}(z) = (z - U_{s,\omega})^{-1}$ (eq. 18.1), i.e. a persistent detector-equivalence class, not a propagating or emitted directional object.

5.2.2 Newly introduced objects (not in ETFE - flagged per the task's instruction)

The following are new terms required by the requested scenario. None is defined in ETFE v4.2.21; each is stipulated here as an explicit axiom (Section 2) so that its role in the conditional theorem is auditable.

Cognitive event

New, operational label only - none - used purely as an interpretive gloss on a mathematical transition, per the task's instruction to treat "insight" as operational, not physical

Topological Piercing

New - loosely analogous to, but distinct from, ETFE's § 10.4 homological jump (algebraic β_i) and Ch. 17 caustic (map singularity); ETFE explicitly separates algebraic Betti transitions from "the ordinary first Betti number of the two-edge cellular graph" (Remark 10.1)

Betti-1 cycle, b_1

New label for ETFE's cellular $b_i(C)$ - ETFE's own notation table (App. A) lists $b_i(C)$ as "cellular or topological Betti number of the declared complex C ," explicitly *disjoint* from algebraic $\beta_{i,d}^{alg}$

Max-Plus/tropical boundary

New - absent from ETFE entirely - none

Spectral caustic (as a 1-D ray/vector)

New usage; conflicts with ETFE's own § 7.6/§ 18 usage of "spectral-caustic class" as a stationary equivalence class - ETFE § 7.6, § 18 (different sense)

VSA / hyperdimensional plane

New - absent from ETFE entirely - none

Dimensionally complex algebraic binding event

New - none

5.3 Minimal ETFE-Compatible Model: Added Axioms

The task requires that no unconditional proof be manufactured. The strategy is therefore: take ETFE's Axioms 2.1-2.7 [A1, H1-H6] as fixed background, and add exactly the axioms below, each doing exactly one job, so the four-stage sequence becomes a $\text{*theorem relative to these axioms*}$ rather than a claim derived from ETFE alone.

Axiom N1 (Cellular carrier of the dipole loop). The two-vertex, two-arrow dipole quiver of ETFE § 10.1 (eq. 10.5) is realized, in the sense of ETFE Patch C (§ 9.3), as a weighted 1-dimensional CW-complex $C = (C_0, C_1)$ with $C_0 = v_1, v_2$, $C_1 = a, b$, oriented so that a, b together form a single embedded circle (i.e. a and b are not parallel copies canceling in ∂ , but a genuine 2-cycle graph, ETFE eq. 11.1 already assumes exactly this incidence structure). Then $b_1(C) = \dim H_1(C; \mathbb{R}) = 1$ for all ε , computed as $\dim \ker \partial - \text{rank}(C_1\text{-boundary map only, no 2-cells})$; with one connected component and Euler characteristic $\chi = |C_0| - |C_1| = 0$, $b_0 = 1$, so $b_1 = 1 - \chi + b_0 - b_0$ reduces (via $\chi = b_0 - b_1$) to $b_1 = 1$. This is the Betti-1 cycle of the task, and it is topological ($b_i(C)$), not the algebraic $\beta_{i,d}^{alg}$ of § 10.4 - the two must not be conflated (ETFE explicit warning, Remark 10.1). Axiom N1 is a $\text{*cellular-realizability*}$ hypothesis in the precise sense of ETFE Patch C, which states realizability "only when its incidence data satisfy the required sign and support conditions" - here assumed, not derived.

Axiom N2 (Existence of a Max-Plus/tropical boundary). There exists an idempotent semiring valued "detector boundary" functional

$$\mathcal{T} : \text{Spec}(L_{s,\omega}) \longrightarrow (\mathbb{R} \cup -\infty, \oplus, \otimes), \quad x \oplus y = \max(x, y), \quad x \otimes y = x + y,$$

the max-plus (tropical) semiring in its standard idempotent form (Wikipedia, "Tropical semiring"; background in Speyer-Sturmfels, "Tropical Mathematics"), acting on the spectrum of ETFE's combined operator $L_{s,\omega}$ (eq. 7.5). $\mathcal{T}$ is postulated to act as a valuation-type contact map: for the one-parameter dipole family, $\mathcal{T}(\lambda_+(\Delta_1^\varphi)(\varepsilon)) = \varepsilon^{-2} \otimes (p_1 p_2)^{-1} = -2 \log \varepsilon + \log[(p_1 p_2)^{-1}]$ under the log-valuation dequantization $b \rightarrow \infty$ that turns ordinary arithmetic into max-plus arithmetic (Wikipedia, "Tropical semiring," valued-field / dequantization section). "Contact" with the tropical boundary is defined as the divergence event $\mathcal{T} \rightarrow +\infty$, i.e. $\varepsilon \rightarrow 0^+$ in ETFE's own exact identity (eq. 11.7): the closing Dirac gap is algebraically dual, via the exact bridge, to the escaping tropical valuation. This is the only rigorous foothold for "Max-Plus contact": it is a *reparametrization* of ETFE's already-proved $\varepsilon \rightarrow 0$ limit, not an independent physical mechanism. ETFE itself supplies no tropical structure; Axiom N2 imports it wholesale.

Axiom N3 (Piercing rule: loss of b_1 coincides with GNS unfaithfulness). Postulate that the cellular realization of Axiom N1 degenerates precisely when the GNS state φ_ε loses faithfulness - i.e. at $\varepsilon = 0$, where ETFE already proves $\|a\|_{\varphi_\varepsilon}^2, \|b\|_{\varphi_\varepsilon}^2 \rightarrow 0$ (eq. 10.9). Under Axiom N3, the edges a, b are declared to be *removed* from the weighted complex at $\varepsilon = 0$ (not merely reweighted to zero norm, which is all ETFE proves), collapsing $C_1 \rightarrow \emptyset$ and hence $b_1(C) \rightarrow 0$. This is Topological Piercing: an added, non-ETFE identification of metric degeneration (proved) with cellular deletion (not proved, not claimed by ETFE, and in fact in tension with ETFE's own § 16 dictum that "no path is deleted by dimensional reduction" and its Ch. 22 statement that "fractal paths are [not] suppressed or disappear" under saturation). Axiom N3 is therefore flagged as the single most fragile link in the four-stage chain.

Axiom N4 (Rank-one emission / spectral-caustic ray). At the piercing event, postulate that the homological zero-mode released by the Betti-1 collapse (Axiom N3) is carried by a single unit vector $u \in \mathbb{R}^N$ obtained as the leading eigenvector of the limiting (rank-deficient) Hodge Laplacian, i.e. the harmonic direction $u \in \ker \Delta_1^\varphi|_{\varepsilon \rightarrow 0}$ selected by the vanishing eigenvalue in ETFE's own spectrum $0, \varepsilon^{-2}(p_1 p_2)^{-1}$ (eq. 11.5). The emission is modeled as the rank-one operator uu^T (a "ray"), consistent with the general fact that the zero-eigenspace of a 2×2 rank-1-deficient Laplacian on a connected graph is exactly the constant vector direction, i.e. one-dimensional - this part *is* elementary linear algebra applied to ETFE's own eq. (11.4), not an added assumption, though the *interpretation* of that eigenvector as an "emitted" propagating object is new.

Axiom N5 (VSA/hyperdimensional plane). There exists an ambient real hyperdimensional space $\mathbb{R}^D$, $D \gg 1$, equipped with a Vector Symbolic Architecture (VSA) / Holographic Reduced Representation (HRR) structure: a bundling (superposition) operation $+$, and a binding operation $\otimes$ (circular convolution or elementwise product) satisfying bilinearity, associativity, commutativity, and - crucially - approximate self-inverse / unbinding, $\mathbf{a} \otimes \mathbf{b} \otimes \mathbf{b} \approx \mathbf{a}$ for i.i.d. random high-dimensional $\mathbf{b}$ (Plate, "Holographic Reduced Representations," IEEE Trans. Neural Networks; TU Chemnitz VSA tutorial slides). The plane $P \subset \mathbb{R}^D$ is a fixed k -dimensional linear subspace ("orthogonal VSA plane") onto which the caustic ray of Axiom N4 is mapped by a fixed embedding $\iota : \mathbb{R}^N \hookrightarrow \mathbb{R}^D$, $\iota(u) \notin P$ generically. Note (VSA tutorial, explicit): "there is a lack of a clear definition of Vector Symbolic Architectures (i.e., in terms of axioms and theorems)" (TU Chemnitz slides) - so Axiom N5 itself imports a field that is, by its own practitioners' admission, not fully axiomatized; this is stated here rather than concealed.

Axiom N6 (Transverse binding). The image line $\iota(\text{span} u)$ intersects P transversally in the sense of differential topology: $T_x(\mathbb{R}^D) = T_x(\iota(\text{span} u)) \oplus T_x P$ at the (generically unique, isolated) intersection point x , whenever $1 + \dim P = D$ (Wikipedia, "Transversality," dimension-count table). The "algebraic binding event" is then defined as the VSA bind $\beta = \iota(u) \otimes p_0$ for the nearest point $p_0 \in P$ to the intersection, using the binding operation of Axiom N5.

Axioms N1-N6 are additions; none is proved, licensed, or contradicted-in-the-negative-sense by ETFE except where explicitly noted (N3 is in tension with ETFE Ch. 16/22; N4's linear-algebra content is a direct corollary of eq. 11.4-11.5).

5.4 Four-Stage Derivation

5.4.1 Stage 1 - Entropic escalation to the semisimple / radical-square-zero boundary

Take ETFE's exact finite dipole model unmodified. As $\varepsilon \rightarrow 0^+$:

Algebraic side. Generic fibre $M_2(k)$ is semisimple (Prop. 10.1); simple-module projective dimension is 0 for all $\varepsilon \neq 0$, jumping to ∞ with an infinite periodic resolution at $\varepsilon = 0$ (Prop. 10.2, eq. 10.13). This *is* ETFE's own "semisimple-to-radical-square-zero" transition, cited exactly as the task specifies.

Thermodynamic side. The GNS bridge (eq. 11.7) gives $\delta_D(\varepsilon)\lambda_+(\Delta_1^\varphi) = (p_1p_2)^{-1} \geq 4$; as $\varepsilon \rightarrow 0$, $\delta_D \rightarrow 0$ while $\lambda_+ \sim \varepsilon^{-2}(p_1p_2)^{-1} \rightarrow \infty$. ETFE's own § 11.3 shows the coefficient $(p_1p_2)^{-1}$ is *minimized* (not maximized) at maximal von Neumann entropy $p_1 = p_2 = 1/2$; asymmetric $p_1 \neq p_2$ - i.e. more structured, lower-entropy states - *amplify* the stiffness divergence. This already shows the "extreme thermodynamic tension" of the task's stage (1) is not monotonically an entropy-maximization process within ETFE's own model: the bridge coefficient grows as the two-pole state becomes *more* asymmetric (lower entropy), not as it becomes more mixed. This is recorded here as the treatise's first substantive constraint on any later claim of "global entropy maximization."

Escalation classification. Whether this finite-model degeneration counts as a "weak" or "strong synchronized" escalation in ETFE's Ch. 6 sense requires checking Criterion 6.1 (perfect-family synchronization). The two-vertex model does exhibit synchronized supports: a Fitting/determinantal jump (rank of ∂ , eq. 11.1, drops as weights collapse), a singular-scheme change (loss of faithfulness of φ_ε), and a Betti (projective-dimension) jump (Prop. 10.2) all occur at the same fibre $\varepsilon = 0$ - satisfying the required support-coincidence of Criterion 6.1's conclusion, given that the hypotheses (flatness, Cohen-Macaulayness, finite pd on the regular locus) hold for this explicit finite family (Prop. 10.1 is a free-module/flatness statement; the fibre is Artinian, hence trivially Cohen-Macaulay). Conclusion for Stage 1: the finite dipole model *does* instantiate a genuine strong synchronized escalation in ETFE's technical sense - this is the strongest true claim available, and it is exactly what ETFE proves, no more.

5.4.2 Stage 2 - "Topological Piercing" at the Max-Plus/tropical boundary

Under Axiom N2, the divergence $\lambda_+(\Delta_1^\varphi) \rightarrow \infty$ is re-expressed as tropical valuation escape: $\mathcal{T} = -2\log\varepsilon + \log(p_1p_2)^{-1} \rightarrow +\infty$ as $\varepsilon \rightarrow 0^+$. This is a *bijective reparametrization* of a limit ETFE already establishes (eq. 11.7) - no new content is added by passing to log-tropicalization; the substantive content of "contact with a tropical boundary" is entirely encoded by the pre-existing $\varepsilon \rightarrow 0$ limit. Tropicalization is standard: a valuation v on a field K satisfies $v(ab) = v(a) + v(b)$, $v(a + b) \geq \min(v(a), v(b))$, i.e. is "almost a semiring homomorphism" onto the min-plus tropical semiring (Wikipedia, "Tropical semiring"); the max-plus convention used here is its sign-reversed isomorphic twin.

Under Axiom N3 (Piercing rule), the coincidence of GNS-unfaithfulness ($\varepsilon = 0$, ETFE eq. 10.9) with cellular-edge deletion forces $C_1 \rightarrow \emptyset$, hence (Axiom N1's realization) $b_1(C) : 1 \rightarrow 0$. This is the requested "collapse of the superposition [and] break[ing of] the Betti-1 cycle." Two facts must be stated with full rigor:

What is proved: the *metric* norms $\|a\|_{\varphi_\varepsilon}, \|b\|_{\varphi_\varepsilon} \rightarrow 0$ (ETFE eq. 10.9) and the algebraic projective-dimension jump (Prop. 10.2) are both exact, finite, checked facts.

What is not proved and is asserted only under Axiom N3: that metric vanishing of edge-norms is equivalent to topological (cellular, b_1) deletion of those edges. ETFE explicitly resists exactly this move elsewhere: "the standard Forman curvature of the minimal two-vertex graph remains bounded under uniform edge-weight collapse" (Abstract; Prop. 11.3) - i.e. ETFE's own no-go proposition shows that naively treating a diverging GNS/Hodge quantity as if it were a geometric/topological catastrophe is *false* for the minimal graph unless a richer complex (2-cell, longer chain) is used (Prop. 11.3, candidate mechanisms 1-4). "Topological Piercing" as literally stated in the task - using the minimal two-vertex dipole loop itself as the pierced object - is therefore a category move ETFE's own no-go proposition would flag: edge-weight collapse in the minimal graph does not by itself force any curvature or topological catastrophe. Axiom N3 supplies the needed additional structure by fiat (declaring metric vanishing to *entail* cellular removal), which is precisely the kind of unproven bridge ETFE's release-candidate discipline (§ "Version 4.2.21 precision patch," items 4-5) was written to prevent being smuggled in silently.

5.4.3 Stage 3 - Rank-one emission ("spectral caustic" ray)

Given Topological Piercing (Stage 2, under N3), the deficient Laplacian $\Delta_1^\varphi|_{\varepsilon \rightarrow 0}$ has a one-dimensional kernel - this is elementary and already visible in ETFE's own computed spectrum $0, \varepsilon^{-2}(p_1 p_2)^{-1}$ (eq. 11.5): exactly one zero eigenvalue survives (the harmonic/homological direction, ETFE's own $\beta_{i,d}^{alg} = \dim \ker \Delta_i^\varphi$ correspondence, eq. 7.3), for *every* $\varepsilon \geq 0$ on this two-vertex graph, since $b_0(C) = 1$ always (the graph stays connected). Under Axiom N4, this pre-existing one-dimensional kernel is *reinterpreted* as an "emitted" object uu^T , a rank-one projector - mathematically this is legitimate linear algebra (every finite-dimensional self-adjoint positive operator with simple zero eigenvalue has a rank-one kernel projector), but the *physical* reading of that projector as a "purely one-directional concentrated" outgoing signal is an added interpretive layer, not a consequence of ETFE's dynamics, since ETFE's Laplacian is a static (frozen- ε) spectral object, not a time-evolving emission process. No equation of motion for $u(t)$ exists in ETFE; Axiom N4 does not supply one either. The "ray" is therefore a static algebraic datum reinterpreted as a directional emission, and any claim about its propagation, energy, or causal reach would require an explicit dynamical extension (e.g. coupling to ETFE's Lieb-Robinson bound, eq. 12.1) that is not constructed here.

5.4.4 Stage 4 - Transverse intersection with the VSA plane and algebraic binding

Given the ray $\text{span } u \subset \mathbb{R}^N$ and an embedding $\iota : \mathbb{R}^N \hookrightarrow \mathbb{R}^D$ (Axiom N5) into a D -dimensional hyperdimensional space carrying a fixed k -plane P :

Genericity of transversal intersection. By the dimension-count theorem for transverse maps (Wikipedia, "Transversality"): for $\ell_1 = 1$ (the ray) and $\ell_2 = k = \dim P$ inside ambient $m = D$, transverse intersection is nonempty and zero-dimensional exactly when $\ell_1 + \ell_2 = m$, i.e. $k = D - 1$; more generally the expected intersection has dimension $\ell_1 + \ell_2 - m = k + 1 - D$, which is generically empty unless $k \geq D - 1$. Genuine "orthogonal complement" hyperdimensional planes used in VSA constructions typically have $k \ll D$ (a sparse or structured factor subspace) rather than $k = D - 1$; thus for a *generic* fixed low-dimensional VSA plane, a *generic* ray does not transversally intersect it in an isolated point - it misses it entirely, or intersects only after perturbation (Thom-style genericity: "either one [submanifold] can be perturbed by an arbitrarily small amount so that the resulting submanifold intersects the fixed submanifold transversally," Wikipedia, "Transversality"). Axiom N6 must therefore additionally stipulate $k = D - 1$ (a hyperplane, not a generic low-dimensional VSA subspace) for the isolated-point transversal intersection scenario the task describes to occur without perturbation; this constraint is stated explicitly because most practical VSA/HRR designs use near-orthogonal *low*-dimensional role/filler subspaces, not codimension-1 hyperplanes.

Binding. Given the intersection point, the VSA bind $\beta = \iota(u) \otimes p_0$ (Axiom N5's operation, e.g. circular convolution, whose j -th component is $(\mathbf{c} \otimes \mathbf{x})_j = \sum_k c_k x_{k+j} \bmod n$, a bilinear, associative, commutative, similarity-reducing operation forming a commutative algebra on $\mathbb{R}^D$ under vector addition (Plate, HRR)) produces a new vector β that is, by the defining property of binding, dissimilar to both inputs (low expected cosine similarity to $\iota(u)$ and to p_0) yet recoverable via unbinding $\beta \otimes p_0^{-1} \approx \iota(u)$ if p_0 is (approximately) self-inverse or a dedicated inverse filter is used. This is the requested "dimensionally complex algebraic binding event": dimensionally complex in the literal sense that $\beta \in \mathbb{R}^D$ is a full-rank generic combination of a rank-one input and a subspace point, not a low-rank object, and algebraic in the literal sense that $\otimes$ is an associative, bilinear ring operation. Nothing about this step invokes noncommutative dipole algebra, GNS states, or Rees filtrations; it is a self-contained VSA construction glued onto the ETFE output only at the single interface point u .

5.5 Conditional Main Theorem and Proof

Theorem (Conditional Four-Stage Transition). Assume ETFE Axioms 2.1-2.7 [A1, H1-H6], the finite dipole quiver-Rees model of § 10 with GNS family φ_ε as in § 10.2, and the added Axioms N1-N6 of § 2. Then, as $\varepsilon \rightarrow 0^+$:

(Escalation) The family undergoes a strong synchronized escalation in the sense of ETFE Definition 6.3, witnessed concretely by the exact identity $\delta_D(\varepsilon)\lambda_+(\Delta_1^\varphi) = (p_1 p_2)^{-1} \geq 4$ (ETFE eq. 11.7) and the projective-dimension jump $0 \rightarrow \infty$ (ETFE Prop. 10.2).

(Piercing) Under Axiom N3, the cellular Betti number satisfies $b_1(C(\varepsilon)) = 1$ for $\varepsilon > 0$ and $b_1(C(0)) = 0$;

equivalently, contact with the tropical boundary $\mathcal{T} \rightarrow \infty$ (Axiom N2) coincides exactly with the Betti-1 collapse.

(Caustic emission) At $\varepsilon = 0$, $\ker \Delta_1^\varphi$ is one-dimensional, spanned by a unit vector u ; under Axiom N4 this defines a rank-one operator uu^T .

(Transverse binding) Under Axiom N5 with $D = k + 1$ and Axiom N6, the line $\iota(\text{span } u)$ meets the plane P transversally in a single point, and the VSA bind $\beta = \iota(u) \otimes p_0$ is a well-defined element of $\mathbb{R}^D$ dissimilar from both $\iota(u)$ and p_0 and (approximately) invertibly recoverable.

Proof. (1) is ETFE's Proposition 11.2 and Proposition 10.2 verbatim, plus the verification that Criterion 6.1's hypotheses hold for this Artinian, free, flat, finite family (flatness: Proposition 10.1; Cohen-Macaulay/finite-pd on the regular locus: the generic fibre is semisimple hence hereditary with global dimension 0; grade-inequality failure exactly at $\varepsilon = 0$: Proposition 10.2's jump itself witnesses the depth drop via Auslander-Buchsbaum, ETFE eq. 9.1). (2) follows by definition from Axiom N3 applied to the norm computation of eq. (10.9), and from the log-reparametrization of eq. (11.7) under Axiom N2 (both directions of the limit - Dirac-gap closure and Hodge-stiffness divergence - occur at the identical parameter value $\varepsilon = 0$, so "contact" and "piercing" are simultaneous by construction, not derived independently). (3) is a direct computation from the explicit 2×2 matrix (eq. 11.4): a symmetric-in-the-weighted-inner-product operator with spectrum $0, \lambda_+$, $\lambda_+ > 0$, on a 2-dimensional space has a 1-dimensional kernel, spanned by the all-ones-type vector in the appropriate weighted basis (elementary linear algebra; ETFE's own eq. 11.5 supplies the spectrum, and Axiom N4 supplies only the *interpretation* of this static kernel vector as an emitted ray). (4) is the transversality dimension-count theorem applied under the stated dimension constraint (Wikipedia, "Transversality"), plus the algebraic closure/-bilinearity properties of the chosen VSA binding operation (associativity, commutativity, distributivity over bundling, near-self-inverse for random high-dimensional vectors - Plate, HRR; TU Chemnitz VSA slides).

Scope of the theorem. Every clause that traces to ETFE (clause 1, and the linear-algebra core of clause 3) is a rigorously established finite fact. Every clause that requires N2, N3, N5, or N6 (clauses 2 and 4, and the "emission" gloss on clause 3) is conditional on axioms *external to* ETFE and, in the case of N3, in direct tension with ETFE's own explicit anti-deletion principle (§ 16, § 22) and its own Forman no-go proposition (Prop. 11.3) for the minimal graph. The theorem is therefore a genuine but narrow conditional statement, not a discovery about ETFE's unconditional content.

5.6 Entropy-Production Theorem / No-Go for Global Maximization

Definitions. Let σ denote local entropy production and J a thermodynamic flow driven by force X , so that $\sigma = XJ$ in the standard bilinear form of irreversible thermodynamics (maximum entropy production principle review, PMC). The second law asserts $\sigma \geq 0$ for the *total*, globally accounted entropy production of an isolated system; it makes no claim that any particular subsystem's flow is maximal, nor that a local transition maximizes anything beyond nonnegativity. The maximum entropy production principle (MEPP) - "a non-equilibrium system will most probably take the line of development when it maximizes the entropy production σ at some assigned external constraints" (PMC review of MEPP) - is explicitly a *conjectural selection principle*, not a theorem derivable from the second law by itself; the same review frames it as an open variational hypothesis requiring an independent closure relation between flows and forces, and its status remains contested in the literature on entropy production and climate/turbulent systems (ETH Zürich, "The second law of thermodynamics and the global climate system"; PMC, "Entropy and the Second Law of Thermodynamics").

Proposition (Local-Global Non-Identity). $\sigma \geq 0$ (second law, global/isolated-system statement) does not imply that any single local process - including the dipole-loop escalation of Stage 1 - maximizes σ among all admissible competing processes, nor that it maximizes global entropy production, nor that it is unique in doing so. *Proof sketch.* The second law is a statement about the sign of the total differential $dS_{univ} \geq 0$ for an isolated system; it constrains the sum of all sub-process contributions, not any addend individually. MEPP would be required to promote this to an extremal/selection statement, and MEPP is explicitly non-derivable from the second law alone - it is an added variational conjecture whose validity is assessed case-by-case and remains actively debated (PMC MEPP review).

Consequence for the requested claim. Even granting the full conditional Theorem of § 4, no combination

of ETFE's axioms, the second law, and Axioms N1-N6 licenses the statement "this is the ultimate thermodynamic engine for novelty/insight and global entropy maximization." Three independent gaps block it:

Superlative quantifier gap. Nothing in ETFE, MEPP, or the second law ranks this specific bipartite-loop/piercing/caustic/binding sequence against any other admissible dissipative channel; "ultimate" requires a comparison class and an optimality proof that does not exist here (and would require solving a genuinely open variational problem, per the MEPP literature).

Local/global conflation. The finite dipole model's entropy statement (ETFE § 11.3, eq. 11.9-11.10) is a local, state-family extremum $((p_1 p_2)^{-1})$ minimized, i.e. von Neumann entropy $S(\rho)$ maximized, at $p_1 = p_2 = 1/2$ inside one finite two-pole system - it says nothing about the entropy budget of any larger enclosing system, let alone "global entropy" in a cosmological or universal sense. ETFE itself flags this precisely: "The statement is internal to the finite state-induced complex; interpreting it as a general thermodynamic law of gravitation would require an additional physical theorem" (§ 11.3) - the identical caution applies a fortiori to a claim about global entropy maximization via a hypothesized cognitive mechanism.

Novelty/insight is uninstantiated. "Novelty" or "insight" has no operator, observable, or order parameter anywhere in ETFE or in Axioms N1-N6; it is a purely interpretive label attached after the fact to the transverse-binding output β of Stage 4. No equation produces it; it cannot be an object of proof.

Verdict. The requested "proof" is not constructible from the stated formalism. What *is* provable is the conditional Theorem of § 4 (a specific, narrow, four-stage mathematical transition under Axioms N1-N6), plus the local entropy-extremal fact of ETFE § 11.3. Any claim beyond that is a physical/interpretive conjecture, correctly labeled as such.

5.7 Spectral-Dimension Analysis

Using ETFE's own d_s machinery (eq. 7.7/16.1) applied to the combined operator $L_{s,\omega} = L_{s,\omega}^\tau \oplus \bigoplus_i a_i \Delta_i^\varphi$ (eq. 7.5): before Piercing, the heat trace $P(t) = \text{Tr } e^{-tL_{s,\omega}}$ receives a contribution from the nonzero eigenvalue $\lambda_+(\varepsilon) = \varepsilon^{-2}(p_1 p_2)^{-1}$, which for small $t \ll \lambda_+^{-1}$ is exponentially suppressed, $e^{-t\lambda_+} \rightarrow 0$ rapidly as $\varepsilon \rightarrow 0$ (since $\lambda_+ \rightarrow \infty$). This means the "Betti-1 mode" (the nonzero eigenvalue direction) becomes spectrally invisible at any fixed observation timescale t as $\varepsilon \rightarrow 0$: the effective heat trace is dominated entirely by the persistent zero mode, and the *local* spectral dimension exponent $d_s(t; s, \omega)$ collapses toward the value contributed by the harmonic (kernel) sector alone. This is consistent with, and can be read as a two-state toy instance of, ETFE's own general caution: a fractional-dimension transition ($3 \rightarrow 1.5 \rightarrow 0.5$, ETFE's own illustrative sequence, Ch. 16) is a redistribution of *heat-kernel weight*, not a deletion of any path or mode - the "Piercing" of Axiom N3 (literal edge deletion) is a strictly stronger and non-ETFE-licensed claim than the spectral-weight-redistribution language ETFE itself uses throughout Ch. 16 and Ch. 22. The mode-participation number $D_{part} = (\sum_n w_n)^2 / \sum_n w_n^2$ (eq. 16.2) applied to the two-mode spectrum $0, \lambda_+$ with weights $w_0 = 1$, $w_+ = e^{-t\lambda_+} \rightarrow 0$ gives $D_{part} \rightarrow 1$: a single dominant (zero) mode - a legitimate, ETFE-internal way to say "concentration onto one channel" without invoking edge deletion or Betti-number destruction. This is the recommended ETFE-faithful substitute for Axiom N3's stronger claim, and it is flagged as such: concentration of spectral weight (proved-compatible with ETFE) versus deletion of topology (Axiom N3, not licensed).

5.8 Transverse Binding Construction (Formal Detail)

Let $u \in \mathbb{R}^N$ be the unit kernel vector of Stage 3, and fix an isometric embedding $\iota : \mathbb{R}^N \rightarrow \mathbb{R}^D$. Let $P = \text{span } e_1, \dots, e_{D-1}$ be a coordinate hyperplane (the codimension-1 case required by Axiom N6 for guaranteed isolated-point transversality per § 4's dimension count). Write $\iota(u) = (\iota(u)_1, \dots, \iota(u)_D)$. The transversality condition $T_x \mathbb{R}^D = T_x(\text{span } \iota(u)) \oplus T_x P$ holds at any intersection point $x = t_0 \iota(u) \in P$ if and only if $\iota(u) \notin P$, i.e. $\iota(u)_D \neq 0$; this is the genericity condition, satisfied for all but a measure-zero set of embeddings ι (Wikipedia, "Transversality," genericity/perturbation section). The unique intersection point is then $x = 0$ if the line passes through the origin (as a linear subspace, $\text{span } \iota(u)$ always contains $0 \in P$); to obtain a nontrivial isolated intersection at a *nonzero* point, Axiom N6 is refined to treat the ray as an affine line $\ell(t) = x_0 + t \iota(u)$, $x_0 \notin P$, giving the unique solution $t^* = -x_{0,D}/\iota(u)_D$ and intersection point $x^* = \ell(t^*)$. The binding output is then $\beta = x^* \otimes p_0$ for the nearest bundled VSA

symbol $p_0 \in P$ representing whatever fixed "role" or "context" vector the plane encodes. Because circular convolution binding is similarity-reducing and (for i.i.d. random p_0) approximately self-inverse (Plate, HRR), β is with high probability nearly orthogonal to both x^* and p_0 in the ambient $\mathbb{R}^D$ - the "dimensionally complex" object of the task, formally a new point in a D -dimensional space displaying no simple linear relation to its two generators except through the (recoverable, but non-obvious) unbinding map.

5.9 Counterexamples and Failure Modes

Counterexample 1 (Forman no-go blocks naive "piercing as curvature blow-up"). ETFE's own Proposition 11.3 proves that uniform GNS collapse in the minimal two-vertex graph leaves standard Forman-Ricci curvature *bounded*. If "Topological Piercing" is read as any kind of curvature catastrophe (a natural physical reading of "piercing"), this reading is a proved-false statement for exactly the minimal bipartite loop the task specifies, unless the loop is enriched with a genuine 2-cell or a longer chain with surviving relative weight (Prop. 11.3, mechanisms 1-4) - an enrichment not stipulated in the task and not assumed here beyond Axiom N1's bare 1-cell structure.

Counterexample 2 (Path non-deletion contradicts literal edge removal). ETFE Ch. 16 and Ch. 22 state twice, in different contexts, that dimensional/spectral transitions never delete paths - "no path is deleted by dimensional reduction" (Ch. 16); "ETFE v4.2 no longer states that fractal paths are suppressed or disappear" (Ch. 22, describing a prior-version retraction). Axiom N3's literal cellular deletion ($b_1 : 1 \rightarrow 0$ via edge removal) directly contradicts this retained architectural principle. A model faithful to ETFE would instead describe Piercing as *weight concentration* (§ 6, above), under which $b_1(C)$ never changes - it is the algebraic Betti number $\beta_{i,d}^{alg}$ (a *different*, already-proved-jumping quantity, Prop. 10.2) that changes, not the cellular $b_1(C)$. This is exactly the confusion ETFE's Remark 10.1 warns against ("This is an algebraic Betti transition. It is distinct from the ordinary first Betti number of the two-edge cellular graph").

Counterexample 3 (Transversality failure for realistic VSA planes). If the VSA plane P has the dimension typical of practical HRR/VSA role-subspaces ($k \ll D$, not $k = D - 1$), a generic ray misses P entirely (expected intersection dimension $1 + k - D < 0$); "transverse intersection" then requires either a non-generic, specially constructed plane, or an appeal to perturbation genericity that changes *which* ray is being discussed, not the ray produced by Stage 3. The task's stage (4) therefore silently assumes a highly non-generic geometric alignment between the emitted ray and the target plane; nothing in Stages 1-3 supplies a mechanism that would produce such an alignment rather than a generic, non-intersecting direction.

Counterexample 4 (Entropy monotonicity failure). If "entropic escalation" is meant to track monotonic entropy increase toward a global maximum, ETFE's own eq. (11.7)-(11.10) shows the opposite local relationship for the Hodge-stiffness coefficient: the bridge coefficient $(p_1 p_2)^{-1}$ is *minimized* (least stiff) exactly at maximum entropy, and grows as the state becomes more asymmetric (lower entropy). A process modeled as "escalating toward" the special fibre while simultaneously "maximizing entropy" must specify which of these two competing tendencies (entropy maximization vs. stiffness divergence) is being extremized, since ETFE's own exact identity shows they are not the same variable and can move in different directions depending on how p_1, p_2 evolve along the branch.

Counterexample 5 (No dynamics, hence no causal "emission"). ETFE's finite model in § 10-11 is a static one-parameter family indexed by ε ; no time-evolution equation $\dot{\varepsilon}$ or $\dot{u}$ exists in the manuscript for this sector (dynamical time t is reserved, per the v4.2.21 precision patch, for Lieb-Robinson-type propagation, eq. 12.1, in a different, thermodynamic-branch part of ETFE, not the finite dipole model). Any claim that the rank-one kernel vector u is "emitted" as a causal signal with a direction and propagation speed requires importing a dynamical law that neither ETFE nor Axioms N1-N6 supply.

5.10 Falsifiable Observables and a Minimal Mathematical Model

To make the construction operationally meaningful (rather than purely descriptive), the following order parameters are proposed, deliberately confined to quantities that are computable from the objects already defined:

Stiffness-divergence order parameter: $\Xi(\varepsilon) = \delta_D(\varepsilon) \lambda_+(\Delta_1^\varphi) = (p_1 p_2)^{-1}$. Falsifiable prediction: Ξ is *exactly* piecewise constant in ε (constant at $(p_1 p_2)^{-1}$ for all $\varepsilon \neq 0$) for this model; any experiment or simulacrum claiming Ξ varies continuously with ε contradicts the exact identity (eq. 11.7) and falsifies the model realization.

Mode-concentration order parameter: $D_{part}(t, \varepsilon) = (\sum_n w_n)^2 / \sum_n w_n^2$ with $w_0 = 1, w_+ = e^{-t\lambda_+(\varepsilon)}$ (ETFE eq. 16.2). Prediction: $D_{part} \rightarrow 1$ monotonically as $\varepsilon \rightarrow 0$ for fixed $t > 0$; a measured non-monotonic $D_{part}(\varepsilon)$ falsifies the "clean concentration" reading of Piercing.

Cellular Betti indicator: $b_1(C(\varepsilon))$, which under the ETFE-faithful (non-N3) reading is identically 1 for all ε including 0 (since the two-cycle graph's topology does not depend on edge weight), while under Axiom N3 it is 1 for $\varepsilon > 0$ and 0 at $\varepsilon = 0$. This is a sharp discriminator: any claimed physical instantiation of "Topological Piercing" must specify, and make independently measurable, which of these two behaviors it exhibits; if the measured quantity is continuous through $\varepsilon = 0$ (as ETFE's own principles predict), Axiom N3 is falsified for that instantiation.

Transversality indicator: the angle $\theta = \angle(\iota(u), P)$; the model predicts a binding event occurs (in the sense of § 7) only when θ is bounded away from 0 and the codimension condition $k = D - 1$ holds; $\theta \rightarrow 0$ or $k \ll D - 1$ predicts *no* binding event (Counterexample 3), which is directly checkable given an explicit embedding ι and plane P .

Entropy-direction indicator: $\text{sign}(\frac{d}{ds} S(\rho_s))$ versus $\text{sign}(\frac{d}{ds} \Xi(\varepsilon_s))$ along a branch $s \mapsto (p_1(s), p_2(s), \varepsilon(s))$; the model predicts these signs are generically opposite near $p_1 = p_2$ (Counterexample 4) and requires explicit specification of the branch to determine agreement elsewhere.

Minimal mathematical model (summary system). The entire falsifiable content reduces to five equations, all either ETFE-exact or elementary corollaries:

$$\delta_D(\varepsilon) = \varepsilon^2, \quad \lambda_+(\Delta_1^\varphi) = \frac{1}{\varepsilon^2 p_1 p_2}, \quad \Xi = \delta_D \lambda_+ = \frac{1}{p_1 p_2} \geq 4,$$

$$S(\rho) = -p_1 \log p_1 - p_2 \log p_2, \quad D_{part}(t, \varepsilon) = \frac{(1 + e^{-t/(\varepsilon^2 p_1 p_2)})^2}{1 + e^{-2t/(\varepsilon^2 p_1 p_2)}}.$$

Everything downstream of these five equations - Piercing, caustic emission, transverse binding - is definitional overlay (Axioms N1-N6), not additional physics.

5.11 Category Errors and Missing Morphisms

The four-stage narrative crosses five distinct mathematical categories: noncommutative *-algebra (ETFE's $\mathcal{A}_\tau$, GNS Hilbert spaces), tropical/idempotent semiring geometry, classical nonequilibrium thermodynamics, algebraic/cellular topology, and (via VSA) real linear algebra with a bilinear ring structure. The missing or unjustified morphisms are:

Noncommutative algebra $\rightarrow$ tropical geometry: no canonical functor is supplied; Axiom N2's "valuation" map $\mathcal{T}$ is an ad hoc log-reparametrization of one scalar quantity (λ_+), not a semiring homomorphism from $\text{Spec}(L_{s,\omega})$ as a whole, which would need to respect *both* tropical operations $\oplus, \otimes$ on sums and products of the full operator spectrum, not just track one eigenvalue.

Tropical geometry $\rightarrow$ topology (the "Piercing" step): no morphism exists between the max-plus semiring and the category of CW-complexes/homology; Axiom N3 supplies the needed bridge by definitional fiat (declare metric zero $\Rightarrow$ cellular deletion) rather than by a natural map, and this fiat conflicts with ETFE's own explicit persistence axiom [H5]/§ 16 principle.

Thermodynamics $\rightarrow$ topology: ETFE's own bridge (eq. 11.7) already relates an algebraic gap to a *Hodge-stiffness* eigenvalue, explicitly not curvature (§ 11.4); using this to license a topological (Betti) claim is precisely the identification ETFE's release-candidate patch rules out (v4.2.21 patch item 5: "Betti symbols and noncompact topologies are disjoint").

Topology $\rightarrow$ linear algebra/VSA: the passage from "one-dimensional kernel of a 2×2 matrix" to "propagating one-directional ray in $\mathbb{R}^N$ " requires a dynamical morphism (a flow or a causal propagator) that is not present in ETFE's static finite model (Counterexample 5); Axiom N4 supplies only the linear-algebraic fact, not the propagation.

VSA/hyperdimensional algebra $\rightarrow$ cognitive interpretation: VSA/HRR are engineering formalisms for symbolic-structure encoding in fixed-width vectors; by their own foundational literature's admission, they lack "a clear definition ... in terms of axioms and theorems" (TU Chemnitz VSA slides). Treating a VSA binding event as instantiating "insight" or "novelty" is a purely interpretive gloss with no operator-theoretic content in either ETFE or standard VSA theory; this is exactly the category error the task instructs to flag rather than paper over.

5.12 Epistemic Verdict

Following ETFE's own "Methodological contract" (Author's Note, p. iv): each claim in this treatise is now classified at its declared level.

Proved, ETFE-internal, unconditional: the semisimple-to-radical-square-zero transition (Prop. 10.1-10.2); the exact GNS-Hodge bridge $\delta_D \lambda_+ = (p_1 p_2)^{-1} \geq 4$ (Prop. 11.2); the entropy-extremal characterization of the bridge minimum (§ 11.3); the Forman no-go for the minimal graph (Prop. 11.3); the elementary one-dimensional-kernel fact for the 2×2 weighted Laplacian.

Conditional theorem, this treatise, dependent on stated added axioms N1-N6: the four-stage Piercing/-caustic/binding sequence of § 3-4, valid *only* under Axioms N1-N6, several of which (especially N3) are in direct tension with ETFE's own stated persistence principles (§ 16, § 22) and no-go proposition (Prop. 11.3).

Imported, externally sourced, not ETFE content at all: the tropical/max-plus semiring structure (Wikipedia; Speyer-Sturmfels), the VSA/HRR binding formalism (Plate; TU Chemnitz), the transversality theorem (Wikipedia), and the entropy-production/MEPP framework (PMC; ETH Zürich).

Not proved and, on current evidence, not provable from the stated axioms: the claim that this mechanism is "the ultimate thermodynamic engine for novelty/insight and global entropy maximization." This fails on quantifier grounds (no comparison class, § 5), on local/global grounds (ETFE's own entropy statement is explicitly local and internal to a finite state family, § 11.3), and on definitional grounds (no operator for "novelty" or "insight" exists anywhere in the formalism, § 5.3). The correct epistemic status of "cognitive event" throughout this treatise is: an operational interpretive label attached to a mathematical transition, not an established physical or cognitive fact - exactly the status the task instructs be maintained, and exactly the discipline ETFE itself insists on for its own, structurally analogous "particle," "observer," and "collapse" interpretations (Ch. 18, Ch. 19).

5.12.1 Sources cited for non-ETFE material

Tropical/max-plus semiring definition and dequantization: Wikipedia, "Tropical semiring"; Speyer & Sturmfels, "Tropical Mathematics"

Combinatorial/graph Hodge Laplacian and harmonic-cohomology correspondence: Lek-Heng Lim, "Hodge Laplacians on Graphs," University of Chicago Statistics

Vector Symbolic Architectures, binding/bundling axioms: TU Chemnitz VSA/HDC tutorial slides

Holographic Reduced Representations, circular convolution: Plate, "Holographic Reduced Representations," IEEE Trans. Neural Networks

Maximum entropy production principle, status and definition: PMC, "The maximum entropy production principle: two basic questions"

Second law and entropy production, contested global/local scope: ETH Zürich, "The second law of thermodynamics and the global climate system"; PMC, "Entropy and the Second Law of Thermodynamics"

Transversality, dimension count, genericity: Wikipedia, "Transversality"

Chapter 6

Catalog of Radical Algorithmic Implementations of Mainstream AI Models (2016-2026)

6.1 Methodology and Inclusion Rubric

This catalog restricts inclusion to software/algorithmic implementations that (a) map a mainstream architecture - Transformer, diffusion model, ResNet/CNN, or standard RNN/LSTM - onto a fundamentally non-standard computational substrate or numeric/algebraic representation, (b) possess verifiable code, a released simulator/compiler, or a hardware demonstration with accompanying software, and (c) were published or released between August 5, 2016 and August 5, 2026. Ordinary PyTorch/TensorFlow reimplementations, standard GPU/TPU kernel optimizations, plain quantization, low-rank adaptation, distillation, or pruning are excluded unless they are incidental to a genuinely radical mapping (e.g., quantization that is required to make an ANN-to-SNN equivalence proof exact). For each entry, maturity is classified as: runnable open-source (public repository, installable, reproducible), released simulator/compiler (public framework enabling design-space exploration without requiring physical hardware), hardware-demonstrated with code (a physical chip or device was operated and code/data is available), closed prototype (functioning system, but code is not public), or conceptual-only (theory/architecture proposal without runnable artifacts - these are relegated to the appendix). Every entry was checked against its primary source (arXiv preprint, peer-reviewed venue, or GitHub/GitLab repository) for publication date and repository existence at the time of writing.

6.2 Comparative Synthesis Table

Model	Year	Base Arch	Radical Paradigm	Status / Impl
SpikeGPT (arXiv, GitHub)	2023	RWKV-style Transformer/LLM	Spiking neural network	Runnable open-source - Simulation (SNN simulated on GPU; neuromorphic gains projected)
Spikformer (GitHub)	2023	Vision Transformer	Spiking neural network	Runnable open-source - Simulation
Spikingformer (GitHub, arXiv)	2023	Vision Transformer	Spiking neural network (fully spike-driven residual)	Runnable open-source - Simulation
Spike-Driven Transformer V2 / Meta-SpikeFormer (GitHub)	2024	DeiT/Vision Transformer	Spiking neural network	Runnable open-source - Simulation, targets future neuromorphic chips
SpikeZIP-TF (arXiv, GitHub)	2024	ViT, RoBERTa, Swin Transformer	ANN-to-SNN exact-equivalence conversion	Runnable open-source - Simulation with power model

Model	Year	Base Arch	Radical Paradigm	Status / Impl
RevSNN (Reversible spiking ResNet/-Transformer) (arXiv, GitHub)	2023	ResNet, former	Spiking- Spiking + reversible computing	Runnable open-source - Simulation
Masked Spiking Transformer (GitHub)	2023	Vision Transformer	Spiking neural network, ANN-to-SNN conversion	Runnable open-source - Simulation
Neuromorphic Principles for Efficient LLMs on Loihi 2 (arXiv)	2025	Transformer-based LLM	Spiking/event-driven neuromorphic chip	Hardware-demonstrated with code - Physical Intel Loihi 2 hardware
Lightening-Transformer (arXiv, GitHub)	2023/2024	Transformer (DeiT)	Photonic tensor core (interference-based optical dot-product)	Released simulator/compiler - Simulation (arch-level photonic accelerator simulator)
HyAtten Hybrid Photonic-digital Accelerator (arXiv)	2025	BERT, DeiT	Photonic + digital hybrid attention accelerator	Closed prototype (simulator-based, code not confirmed public) - Simulation (Python simulator + SPICE)
pytorch-onn / TorchONN (GitHub)	2021-2023	CNN, Linear/Transformer layers	Photonic integrated circuit simulation (MZI, microring, PCM ONN)	Runnable open-source - Simulation
neuroptica (GitHub)	2019-2020	Feedforward neural networks (CNN-adaptable)	Photonic mesh (Mach-Zehnder interferometer) simulation	Runnable open-source - Simulation
Diffractive Neural Network (D2NN) reproduction (GitHub, Science)	2018/2020	DNN-equivalent classifier	All-optical diffractive layers	Runnable open-source (reproduction) / Hardware-demonstrated (original) - Physical 3D-printed diffractive layers (original); simulation (reproduction)
120 GOPS Photonic Tensor Core, thin-film lithium niobate (Nature Communications)	2024	Transformer/DNN matrix multiplication	Integrated photonic tensor core	Hardware-demonstrated with code - Physical chip
IBM Analog AI 14nm Transformer-ALBERT Demonstration (IBM Research)	2025	ALBERT (Transformer)	Phase-change-memory analog in-memory compute	Hardware-demonstrated (code not confirmed public) - Physical 14nm analog chip, 28.3M PCM devices
IBM Analog-Attention Kernel-Approximation Chip (IBM Research blog)	2024	Transformer attention (Long Range Arena)	Phase-change-memory crossbar + kernel approximation for nonlinear attention	Hardware-demonstrated (code not confirmed public) - Physical analog chip

Model	Year	Base Arch	Radical Paradigm	Status / Impl
PytorX (GitHub)	2019	CNN (general; MNIST example)	ReRAM/MRAM crossbar mapping with noise-injection adaptation	Runnable open-source - Simulation with device non-ideality modeling
Memristor LSTM crossbar array (Nature Machine Intelligence, GitHub)	2019	LSTM	Memristor crossbar in-memory compute	Hardware-demonstrated with code - Physical integrated memristor chip
Quantum-Transformer (hybrid quantized self-attention) (GitHub)	2025	Transformer decoder (molecular generation)	Variational quantum circuit self-attention on CUDA-Q	Runnable open-source - Simulation (NVIDIA CUDA-Q, A100 GPUs)
NVIDIA CUDA-Q Quantum Transformer tutorial (NVIDIA docs)	2024-2025	Transformer attention	Hadamard-test quantum inner product for Q·K attention scores	Runnable open-source - Simulation
Quantum Convolutional Neural Network (QCNN, takh04) (GitHub)	2021	CNN	Variational quantum circuit (convolution + pooling ansätze)	Runnable open-source - Simulation (PennyLane)
TensorFlow Quantum QCNN tutorial (TensorFlow)	2020-2021	CNN	Parametrized quantum circuit compiled via Cirq/TFQ	Runnable open-source - Simulation
Quantum-Hybrid Diffusion Models / michaelkoelle quantum-diffusion (Q-Dense) (GitHub)	2024	Denoising diffusion model	Quantum-circuit denoising network (Q-Dense)	Runnable open-source - Simulation
Quantum RNN/LSTM via entangling power of unitaries (arXiv)	2025	RNN/LSTM	Quantum recurrent unitary transformations	Released simulator/compiler - Simulation
Torch2PC (GitHub)	2021-2022	CNN (Sequential PyTorch models)	Predictive-coding alternative to backpropagation	Runnable open-source - N/A (software-only alternative learning rule)
PC-Transformers (GitHub)	2025	Transformer (embeddings, attention Q/K/V, MLP, output head)	Predictive coding applied to every Transformer sublayer via ngclearn	Runnable open-source - N/A
Equilibrium Propagation (Laborieux et al.) (GitHub)	2020-2021	Recurrent/convolutional networks (VFCNN)	Equilibrium propagation as backpropagation alternative	Runnable open-source - N/A
Forward-Forward Algorithm implementations (multiple, e.g. loeweX/Forward-Forward)	2022-2023	Feedforward/CNN-style networks	Forward-Forward learning (two forward passes, no backward pass)	Runnable open-source - N/A

Model	Year	Base Arch	Radical Paradigm	Status / Impl
TorchHD (GitHub)	2022-2025	General neural pipelines (classifier backends)	Hyperdimensional computing / vector-symbolic architectures (MAP, HRR, FHRR, VTB)	Runnable open-source - N/A
BiHDTrans (arXiv)	2025	Transformer (Q/K/V projections, attention, FFN)	Binary hyperdimensional binding/bundling replacing linear layers and softmax attention; FPGA-targeted	Runnable open-source (anonymized code release) - Hardware demonstrated with code (Xilinx Artix-7 FPGA latency measurement)
Tropical Attention (GitHub, arXiv)	2025	Transformer attention (Neural Algorithmic Reasoning)	Max-plus (tropical) semiring attention replacing softmax dot-product attention	Runnable open-source - N/A
TropicalGEMM.jl (GitHub)	2021-2023	Matrix multiplication kernels usable within NN pipelines	Tropical (max-plus/min-plus) semiring GEMM	Runnable open-source - N/A
minREV Reversible Vision Transformer (GitHub)	2023	Vision Transformer, Swin Transformer, MViTv2	Reversible residual connections (memory-free backpropagation)	Runnable open-source - N/A
RevNet (GitHub)	2017	ResNet	Reversible residual network (backprop without activation storage)	Runnable open-source - N/A
Growing Neural Cellular Automata (distill.pub, multiple GitHub reproductions e.g. PWhiddy/Growing-Neural-Cellular-Automata-Pytorch)	2020-2023	Small CNN update rule embedded in a cellular automaton	Cellular automata with CNN-parameterized local update rule	Runnable open-source - N/A
Neural Cellular Automata for LLM Pre-Training (arXiv)	2026	Transformer (attention specifically) LLM layers	Synthetic pretraining data generated by neural cellular automata; findings on attention-layer transfer	Released simulator/compiler (methodology, no confirmed code link) - N/A

6.3 Neuromorphic / Spiking Neural Network Implementations

6.3.1 SpikeGPT

SpikeGPT (2023) is, per its authors, the largest backpropagation-trained spiking neural network language model at release, implementing a binary, event-driven spiking activation variant of the RWKV linear-attention Transformer family with 45M and 216M parameter variants (arXiv:2302.13939). The key algorithmic twist is the replacement of quadratic self-attention, $O(N^2)$, with a linear-complexity, $O(N)$, streaming spiking-attention formulation in which tokens are processed sequentially as in conventional SNNs rather than in parallel matrix form. Its authors report a 20x reduction in synaptic

operations when the resulting spike trains are deployed on genuinely event-driven neuromorphic hardware, though the released code trains and runs the model as a simulated SNN on standard GPUs via PyTorch/CUDA kernels (GitHub: ridgerchu/SpikeGPT). The repository is public, BSD-2-Clause licensed, actively maintained with pretrained checkpoints on OpenWebText and The Pile, which qualifies it as runnable open-source rather than conceptual. It qualifies as radical because it substitutes the entire attention computation with an event-driven recurrent formulation rather than merely quantizing or pruning an existing dense Transformer.

6.3.2 Spikformer, Spikingformer, and Masked Spiking Transformer

Spikformer (ICLR 2023) introduces spiking self-attention (SSA) that replaces the softmax-normalized dot-product attention of Vision Transformers with a purely additive, multiplication-free spike-form attention mechanism, avoiding floating-point softmax and matrix multiplication entirely in favor of Heaviside spike accumulation (GitHub: ZK-Zhou/spikformer). Spikingformer (2023) extends this with a fully spike-driven residual learning scheme that eliminates non-spike-form ("float") shortcut connections that had persisted in Spikformer, thereby making every operation genuinely event-driven (GitHub: zhouchenlin2096/Spikingformer; arXiv:2304.11954). Masked Spiking Transformer (ICCV 2023) instead performs ANN-to-SNN conversion of a pretrained Vision Transformer combined with Random Spike Masking, a pruning technique applied to spike trains rather than weights, to cut inference-time computational cost while preserving self-attention semantics (GitHub: bic-L/Masked-Spiking-Transformer). All three are simulated on standard GPUs using the SpikingJelly or custom CUDA-based spiking backends; none has yet been run on physical neuromorphic silicon in the cited repositories, so all are classified as runnable open-source with simulation-only hardware validation.

6.3.3 Spike-Driven Transformer V2 (Meta-SpikeFormer)

Accepted at ICLR 2024, Meta-SpikeFormer generalizes the spike-driven Transformer into a meta-architecture explicitly intended to guide the design of next-generation Transformer-based neuromorphic chips, achieving 80.0% top-1 ImageNet accuracy with only sparse addition operations (no multiplications) throughout the network (GitHub: BICLab/Spike-Driven-Transformer-V2). The work is from the Brain-Inspired Computing Lab, Institute of Automation, Chinese Academy of Sciences, and its code is released with pretrained checkpoints for classification tasks. It is a released simulator-grade implementation rather than a hardware-demonstrated one, since no specific chip was used for inference in this repository.

6.3.4 SpikeZIP-TF

SpikeZIP-TF (2024) provides the most mathematically rigorous instance of ANN-to-SNN conversion for Transformers, proving exact equivalence between an activation-quantized ANN and its converted SNN counterpart at equilibrium (arXiv:2406.03470; GitHub: Intelligent-Computing-Research-Group/SpikeZIP-TF). It introduces the ST-BIF⁺ spiking neuron, whose accumulated membrane-potential output after T_{eq} time steps, $V^{out} = V_{thr} \cdot \text{clip}(\text{floor}((V^{in} + V_{t=0})/V_{thr}), S_{min}, S_{max})$, is provably identical to the corresponding quantized activation function, and it replaces Softmax and LayerNorm with Spike-Softmax and Spike-LayerNorm operators that emit the difference between successive accumulated outputs, $\mathbf{O}_{s,t} = \mathbf{O}_t - \mathbf{O}_{t-1}$. It further defines Spike-Equivalent Self-Attention (SESA), decomposing the query-key product into three streaming spike-train matrix multiplications, $\mathbf{A}_{T_{eq}} = \sum_t (\mathbf{S}_{Q,t} \mathbf{K}_{s,t}^T + \mathbf{Q}_{s,t} \mathbf{S}_{K,t}^T - \mathbf{Q}_{s,t} \mathbf{K}_{s,t}^T)$. The method converts ViT-S/B/L, RoBERTa-Base/Large, and Swin-Tiny with less than 1% accuracy loss and reports substantially reduced estimated power (1.4W versus 20.2W for Swin-Tiny on ImageNet) via a power model rather than a physical measurement. This qualifies as a radical conversion-equivalence result, not routine quantization, because the SNN and QANN are proven bit-for-bit equivalent at equilibrium rather than merely approximated.

6.3.5 RevSNN: Reversible Spiking ResNet and Transformer

RevSNN (December 2023) is a dual-radical implementation combining reversible computing with spiking neural network conversion of both ResNet and Spikingformer, yielding RevSResNet and RevSFormer, whose spatial memory complexity for intermediate activations and membrane potentials drops from $O(D \cdot T)$ to $O(T)$ independent of network depth D (arXiv:2312.07922; GitHub: mi804/RevSNN). The reversible spiking block computes $Y_1^t = X_1^t + \mathcal{F}^t(X_2^t)$, $Y_2^t = X_2^t + \mathcal{G}^t(Y_1^t)$ in the forward pass and exactly reconstructs $X_2^t = Y_2^t - \mathcal{G}^t(Y_1^t)$, $X_1^t = Y_1^t - \mathcal{F}^t(X_2^t)$ during the reverse pass, with membrane-potential

resets proven (Theorem 1) to guarantee identical intermediate variables between forward and reverse computation. RevSResNet37 achieves a 3.79x memory reduction over MS ResNet34 and RevSFormer-4-384 achieves a 3.00x reduction over Spikingformer-4-384 at matched accuracy, with up to 9.1x savings reported for deeper 16-layer configurations. This is radical because it applies exact reversible-computing memory recovery specifically to the temporal unrolling problem unique to spiking dynamics, not merely to a standard feedforward network.

6.3.6 Neuromorphic LLM Principles on Intel Loihi 2

Recent work operationalizes event-driven neuromorphic principles for Transformer-based LLM inference and reports measurements taken on physical Intel Loihi 2 hardware rather than purely simulated spike trains, making it one of the few genuinely hardware-demonstrated neuromorphic LLM efforts in the literature (arXiv:2503.18002; Intel Loihi 2 technical brief). Loihi 2 is a manycore, fully asynchronous, programmable neuromorphic research chip; deployments on it require the Lava software framework rather than PyTorch/TensorFlow execution graphs, constituting a genuinely non-standard execution substrate.

6.4 Photonic / Optical Implementations

6.4.1 Lightning-Transformer

Lightning-Transformer (HPCA 2024) is the first light-empowered, dynamically-operated photonic Transformer accelerator, introducing a Dynamically-operated Photonic Tensor Core (DPTC) - a crossbar array of interference-based optical vector dot-product engines that performs full-range matrix multiplication via optical interference rather than electronic MAC units (arXiv:2305.19533; GitHub: zhuhanqing/Lightning-Transformer). The released repository provides three code layers: algorithmic training/inference code that analytically models the photonic tensor core's transformation, a hardware simulator estimating energy and latency for Transformer execution on the photonic accelerator, and GPU profiling code for comparison baselines. This is described by its authors as the first architecture-level simulator for a photonic AI accelerator and has since been used as the simulation backbone for at least three follow-on works, including HyAtten and SimPhony. The system remains simulator-grade (released simulator/-compiler) rather than hardware-demonstrated, since no physical photonic chip execution is reported in this specific repository, though the underlying device physics (Mach-Zehnder interferometer meshes) are experimentally grounded.

6.4.2 HyAtten: Hybrid Photonic-Digital Attention Accelerator

HyAtten (2025), from the National University of Singapore, addresses the dominant limitation of pure photonic Transformer accelerators - that analog-to-digital signal-conversion circuitry consumes over 50% of chip area in Lightning-Transformer-class designs - by classifying attention signals into low-resolution (≤ 4 -bit, over 85% of signals) and high-resolution (> 4 -bit, 15%) categories, routing the former through cheap 4-bit ADCs on a photonic die and the latter through a companion digital die (arXiv:2501.11286). The photonic die executes $\mathbf{Q} \times \mathbf{K}^T$ and $\mathbf{S} \times \mathbf{V}$ using wavelength-division-multiplexed optical dot-product units where photocurrent $I_o \propto \vec{x} \cdot \vec{y}$, while the digital die performs softmax and residual GEMM operations. Evaluated on BERT-base (GLUE, SQuAD) and DeiT-T (ImageNet), HyAtten achieves a 9.8x performance-per-area and 2.2x energy-efficiency improvement over the Lightning-Transformer baseline and over 100x speedup per unit area relative to an Nvidia A100 GPU, with less than 0.3% accuracy loss. The evaluation uses a modified Python-based architectural simulator plus SPICE circuit simulation rather than a public code release, and no GitHub repository was located, so this entry is classified as a closed prototype (methodology and results are public and detailed, but no reproducible code repository was confirmed).

6.4.3 pytorch-onn / TorchONN and neuroptica

pytorch-onn (also called TorchONN), maintained by researchers including Jiaqi Gu (University of Texas at Austin, ScopeX-ASU lab), is a PyTorch-native library for simulating and training coherent and incoherent optical neural networks at up to million-parameter scale, supporting Mach-Zehnder-interferometer-based linear/conv layers with multiple decomposition algorithms (Reck, Clements, Francis triangular/rectangular meshes), frequency-domain FFT-ONNs, multi-operand microring-resonator ONNs, and

phase-change-material in-memory ONN layers (GitHub: JeremieMelo/pytorch-onn). It provides CUDA-accelerated batched unitary-matrix decomposition achieving 10-50x speedup over CPU-based parametrization and directly instantiates convolutional and linear layers as optical hardware primitives (MZIBlockConv2d, MZIBlockLinear) rather than post-hoc quantizing existing dense layers, which is the key radical distinction from ordinary CNN acceleration work. neuroptica (fancompute lab) is an earlier (2019-2020) flexible simulation package for feedforward optical neural networks based on Mach-Zehnder interferometer meshes, usable for CNN-style feedforward layers though not natively convolutional (GitHub: fancompute/neuroptica). Both are runnable open-source simulators; neither reports execution on fabricated photonic silicon within the repositories themselves.

6.4.4 Diffractive Deep Neural Networks (D2NN)

The original D2NN work (Lin et al., Science 2018) demonstrated an all-optical, physically 3D-printed diffractive network that performs inference purely through light diffraction across trained phase masks, with no electronic computation in the inference path - a genuinely hardware-demonstrated optical CNN-equivalent classifier (Science 10.1126/science.aat8084; arXiv:1804.08711). Multiple open-source reproductions exist, including a PyTorch-based simulation reproduction of the diffractive layers (GitHub: tutuna/Diffractive-Deep-Neural-Networks) and an independent PyTorch-based ONN simulator (GitHub: feng-spinning/onn-simulation). The mathematical substitution is architectural rather than merely numeric: each "layer" is a physical diffraction propagation operator (a Rayleigh-Sommerfeld free-space transfer function) parametrized by a trainable phase mask, entirely replacing convolutional filters with wave-optics propagation. The original hardware demonstration predates the ten-year window's most stringent interpretation only marginally (2018, well within range); the open-source reproductions are more recent (2020-2022) simulation-only artifacts.

6.4.5 Integrated Photonic Tensor Cores (2024-2025)

Multiple physically fabricated photonic tensor cores have been demonstrated with reported throughput metrics, including a 120 GOPS photonic tensor core in thin-film lithium niobate (Nature Communications) and a programmable, reconfigurable photonic tensor processor for deep neural network inference reported in Nature Communications 2026 (Nature). These represent hardware-demonstrated hardware for generic matrix-multiplication-heavy inference (applicable to any CNN/Transformer layer expressed as GEMM) rather than software frameworks per se; code availability for driving these specific chips was not confirmed as public in the cited sources, placing them at the boundary between hardware-demonstrated-with-code and closed prototype.

6.5 Analog / Memristive / Mixed-Signal Implementations

6.5.1 IBM Analog In-Memory Transformer Demonstrations

IBM Research has produced the most extensive physically-realized analog Transformer inference program identified in this survey. A 14nm phase-change-memory analog inference chip mapped 7.1 million unique analog weights across 12 layers of an ALBERT (Lite BERT) Transformer onto 28.3 million physical PCM devices, described by the authors as the first analog hardware demonstration of a meaningfully large Transformer model, achieving average accuracy within 1.8% of the floating-point reference on GLUE (seven tasks), improved to within roughly 1% via a drift-compensation recalibration technique after measuring approximately 5% conductance drift over 30 days (IBM Research publication; authors include An Chen, Stefano Ambrogio, Pritish Narayanan, Hsin-yu Tsai, Geoffrey W. Burr). A separate, earlier IBM demonstration deployed transformer attention directly on an analog in-memory computing chip using phase-change-memory crossbars, addressing the fundamental problem that attention weights change dynamically (unlike static synaptic weights) via kernel approximation - projecting inputs into a higher-dimensional space with randomly sampled vectors so that a nominally linear analog circuit can also compute the nonlinear softmax-attention function - achieving within 2% accuracy of floating point on the Long Range Arena benchmark (IBM Research blog; published in Nature Machine Intelligence). Neither publication provides a public code repository in the sources reviewed, so both are classified as hardware-demonstrated but closed-prototype with respect to software.

6.5.2 PytorX: ReRAM/MRAM Crossbar Mapping Framework

PytorX is an open-source PyTorch-based evaluation tool that maps neural network layers onto crossbar-based resistive-memory accelerators (ReRAM, MRAM), incorporating a Noise Injection Adaptation technique for end-to-end adaptation to non-ideal crossbar effects during training (GitHub: [elliothe/pytorx](https://github.com/elliothe/pytorx); cites He et al., DAC 2019). It substitutes standard dense matrix multiplication, $y = Wx$, with a noisy analog vector-matrix multiplication governed by device conductance variation, explicitly modeled and compensated for during backpropagation - a genuinely non-standard numerical operator substitution rather than incidental quantization, since it models physical device noise distributions rather than fixed-point rounding. The tool is alpha-stage but runnable and demonstrated on an MNIST classifier; it is architecture-agnostic and has been used in ReRAM/CNN mapping literature.

6.5.3 Memristor LSTM Crossbar Array

Li, Wang, Rao, et al. (Nature Machine Intelligence, 2019; within the ten-year window) physically implemented an LSTM's shared synaptic weights directly in a fabricated memristor crossbar array, with in-memory computing eliminating the von Neumann bottleneck between memory and compute for a recurrent architecture - one of the earliest genuine physical realizations of a mainstream RNN variant on memristive hardware (Nature Machine Intelligence). Supporting code for reproducing plots and analysis is published at github.com/lican81/memNN, though the measurement-system firmware controlling the physical chip is available only upon request, placing this at hardware-demonstrated-with-partial-code. The work was funded by AFRL and IARPA and performed across the University of Massachusetts Amherst and Huazhong University of Science and Technology.

6.5.4 Analog Mixed-Signal CNN Circuits

Various mixed-signal CNN accelerator circuits have been reported, including spiking CMOS-NVM mixed-signal neuromorphic ConvNets (Frontiers in Neuroscience) and mixed-signal binarized CNN accelerators from Princeton (Verma Lab, VLSI 2018). These represent physically characterized mixed-signal circuits for CNN inference but were not confirmed to include openly released, general-purpose software frameworks analogous to `pytorch-onn` or PytorX, placing them closer to closed-prototype hardware demonstrations.

6.6 Quantum-Circuit Implementations

6.6.1 Hybrid Quantum-Classical Transformer (Smaldone et al.)

This work introduces a hybrid quantum-classical self-attention mechanism for a Transformer decoder applied to conditional molecular generation from SMILES strings, replacing the classical query-key dot product with a quantized self-attention circuit whose theoretical query-key computation complexity drops from $O(n^2d)$ classically to $O(n^2 \log d)$ using amplitude-encoded quantum states (GitHub: [anthonymaldone/Quantum-Transformer](https://github.com/anthonymaldone/Quantum-Transformer)). The repository provides quantum, classical-equivalent, and classical model variants, all simulated on NVIDIA's CUDA-Q platform using A100 GPUs - quantum circuit simulation rather than execution on physical quantum hardware. It is MIT-licensed and runnable.

6.6.2 NVIDIA CUDA-Q Quantum Transformer

NVIDIA's own CUDA-Q applications documentation provides a worked implementation replacing token and positional embeddings in the query/key pathway with a quantum circuit, computing pairwise inner products via a modified Hadamard test and feeding the resulting fully classical attention matrix (post-softmax) forward through a conventional value-matrix multiplication (NVIDIA CUDA-Q docs). This is a hybrid architecture with a genuinely quantum-circuit-computed attention-score sub-block, runnable via the CUDA-Q simulator stack; no physical QPU execution is claimed.

6.6.3 Quantum Convolutional Neural Networks

Multiple independent QCNN implementations exist that compile a layered convolution-and-pooling CNN structure directly into parametrized quantum circuits (variational ansätze such as `U_TTN`, `U_SO4`, `U_SU4`) using PennyLane or Cirq/TensorFlow-Quantum, with amplitude or angle encoding replacing pixel-value input and unitary pooling operations replacing max/average pooling (GitHub: [takh04/QCNN](https://github.com/takh04/QCNN),

Apache-2.0 licensed; TensorFlow Quantum QCNN tutorial). These qualify as radical because pooling and convolution are re-derived as unitary, reversible quantum operations rather than approximated; they remain simulator-only with respect to physical quantum hardware in the reviewed repositories.

6.6.4 Quantum Diffusion Models

Quantum-Hybrid Diffusion Models (Q-Dense) implement a denoising diffusion probabilistic model in which the denoising network itself is a parametrized quantum circuit trained on MNIST 8x8 data, released as open-source Python/PennyLane code (GitHub: michaelkoelle/quantum-diffusion, MIT license). A related Google Summer of Code project extends quantum diffusion to high-energy-physics generative tasks (GitHub: mashathepotato/GSoC-Quantum-Diffusion-Model). Both substitute the U-Net-style denoising backbone with a quantum circuit ansatz rather than merely quantizing weights of a classical U-Net, and both run exclusively on quantum circuit simulators.

6.6.5 Quantum RNN/LSTM via Unitary Entangling Power

A 2025 theoretical-and-simulated study formalizes quantum RNNs and LSTMs through the entangling and disentangling power of unitary transformations, replacing the gating nonlinearities of classical LSTM cells with parametrized unitary operators whose entangling capacity is analyzed as a proxy for memory capacity (arXiv:2505.06774). This is simulator-grade and mathematically rigorous but was not confirmed to have a dedicated public code repository distinct from the paper itself.

6.7 Biologically Plausible Learning Applied to Mainstream Architectures

6.7.1 Predictive Coding Applied to Transformers (PC-Transformers)

PC-Transformers (2025) is, to this survey's knowledge, the most complete open-source instantiation of predictive coding applied throughout every sublayer of a genuine Transformer architecture - embeddings, attention (including the internal Q/K/V score projections), the two-layer MLP, and the output logit layer - built on the `ngclearn` biologically plausible computing library and trained on the Penn Treebank dataset (GitHub: iCog-Labs-Dev/PC-Transformers). Each `PCLayer` locally infers latent activity x that minimizes the prediction error between a predicted output μ and the target activity of the subsequent layer, then applies a local Hebbian-style weight update, replacing global backpropagation with strictly local, layer-wise inference and update rules throughout the attention and feed-forward pathways. The repository is small (6 contributors, 4 stars at time of review) but functional and public, making it runnable open-source; it is a research-grade rather than production-grade artifact.

6.7.2 Torch2PC

Torch2PC provides a general-purpose PyTorch function, `PCInfer`, that replaces the standard forward/backward pass of any `nn.Sequential` model with iterative predictive-coding inference, exposing three modes - `Strict` (full predictive coding without the fixed-prediction assumption), `FixedPred`, and `Exact` (mathematically identical to backpropagation) - enabling direct empirical comparison between predictive coding and backpropagation gradients on the same architecture (GitHub: RobertRosenbaum/Torch2PC; accompanies Rosenbaum, PLOS ONE 2022). The demonstrated example applies the framework to a convolutional network on MNIST; the repository explicitly states that support beyond `Sequential`-style models (and hence full mainstream Transformer or ResNet skip-connection topologies) was, at time of review, still forthcoming, so its mainstream-architecture applicability is currently limited to CNN-style sequential stacks.

6.7.3 Equilibrium Propagation (Scaled to Deep ConvNets)

Laborieux et al.'s Equilibrium Propagation implementation scales the energy-based equilibrium propagation learning rule to deep convolutional networks (including asymmetric-connectivity VFCNN variants) on CIFAR-10, using automatic differentiation of a scalar primitive energy function Φ to derive both the neuron dynamics and the equilibrium-propagation weight-update rule, rather than backpropagating a loss gradient through the unrolled computation graph (GitHub: Laborieux-Axel/Equilibrium-Propagation; accompanies "Scaling Equilibrium Propagation to Deep ConvNets by Drastically Reducing its Gradient Estimator Bias"). This qualifies as radical because equilibrium propagation is a genuinely different mathematical learning principle grounded in relaxation to a fixed point of an energy function rather than

reverse-mode automatic differentiation, applied here to real convolutional architectures rather than toy networks.

6.7.4 Forward-Forward Algorithm

Geoffrey Hinton’s Forward-Forward algorithm, which replaces the forward-backward pass pair of backpropagation with two forward passes (one on real data, one on corrupted/negative data) that each locally adjust weights to increase or decrease a layer-wise "goodness" function, has multiple independent open-source PyTorch reimplementations (GitHub: [loeweX/Forward-Forward](#); GitHub: [mibrahimy/-forward_forward](#); GitHub: [kimphys/forward-forward-pytorch](#)). These implementations are generally demonstrated on small feedforward and convolutional networks rather than full-scale Transformers, and are included here as the algorithmic substrate most commonly cited as a genuine backpropagation alternative with runnable code, even though large-Transformer-scale demonstrations remain rare.

6.8 Exotic Arithmetic, Algebra, and Representation Implementations

6.8.1 Tropical (Max-Plus) Attention

Tropical Attention (2025) replaces softmax-normalized dot-product attention with an attention function operating natively in the max-plus semiring of tropical geometry, in which addition is replaced by maximization and multiplication is replaced by classical addition, formally proven capable of approximating tropical circuits corresponding to dynamic-programming-type combinatorial algorithms (GitHub: [Baranphys/Tropical-Attention](#); authors Hashemi, Pasque, Teska, Yoshida). The `TropicalAttention` module is instantiated as a drop-in PyTorch attention block (`d_model`, `n_heads`, `tropical_proj`, `tropical_norm`, `symmetric` parameters) and evaluated on eleven combinatorial reasoning tasks including Floyd-Warshall shortest paths, 0/1 Knapsack, and Strongly Connected Components, where Tropical Transformers are reported to surpass softmax baselines in out-of-distribution length and value generalization and remain stable under adversarial perturbation. This is radical rather than routine because the entire algebraic structure underlying attention - the semiring in which scores are combined - is substituted, not merely the numerical precision or sparsity pattern of a conventional softmax attention block. `TropicalGEMM.jl` provides a complementary, general-purpose fast tropical (max-plus/min-plus) matrix-multiplication kernel in Julia usable as a building block for tropical neural layers (GitHub: [TensorBFS/TropicalGEMM.jl](#)).

6.8.2 BiHDTrans: Binary Hyperdimensional Transformer

BiHDTrans (2025) is the most complete hyperdimensional-computing mapping of a mainstream Transformer identified in this survey, replacing every fully connected linear layer in the query, key, and value projections with element-wise binding (Hadamard product) against trainable binary hypervectors, $\mathcal{H}_q = \mathcal{H}_e \odot \mathcal{BV}_q$, replacing the softmax-normalized real-valued attention score with a binarized Boolean attention mask, $B_a = \text{bool}(\mathcal{H}_q \cdot \mathcal{H}_k) \in \{0, 1\}^{L \times L}$, and omitting the feed-forward block entirely on the theoretical grounds that holographic hyperdimensional encoding already distributes information i.i.d. across all $D = 10,000$ dimensions, leaving no additional expressiveness for a projection layer to exploit (arXiv:2509.24425; authors Zhang, Liu, Shen, Wang, Sun Yat-sen University and Northeastern University; code at [anonymous.4open.science/r/BiHDTrans-4E55](#)). Two proven theorems justify the binarization: hyperspace binarization distortion is provably lower than direct real-value binarization for quantization levels $q \geq 3$ as dimensionality $D \rightarrow \infty$, and binarized attention-weight distortion converges to zero with high probability as $D \rightarrow \infty$. The design was further mapped to a physical Xilinx Artix-7 FPGA (`xc7a200tfg484-2`, 100 MHz), where bitwise binding is implemented via XNOR gates and bundling via tree-structured adders, achieving a measured 39.4x reduction in inference latency versus state-of-the-art binary Transformers - making this one of the few entries in this catalog with both a rigorous mathematical substitution of the attention algebra and a genuine FPGA hardware measurement.

6.8.3 TorchHD and Hyperdimensional Computing Libraries

TorchHD is a general-purpose, actively maintained PyTorch library implementing the core operators of hyperdimensional computing and vector-symbolic architectures - bundling, binding, and permutation across seven distinct HD/VSA models including Multiply-Add-Permute, Binary Spatter Codes, and Holographic Reduced Representations - usable as a substrate for building HD-based classifier heads or attention analogues atop otherwise-conventional feature extractors (GitHub: [hyperdimensional-computing/torchhd](#)).

It is not itself a Transformer mapping but is the common software dependency underlying most current HD-Transformer hybrids, including BiHDTrans-adjacent research.

6.8.4 Reversible Residual Networks (RevNet) and minREV

RevNet (2017) introduced reversible residual blocks for CNNs - of the form $y_1 = x_1 + \mathcal{F}(x_2)$, $y_2 = x_2 + \mathcal{G}(y_1)$ - that permit exact reconstruction of layer inputs from outputs during the backward pass, eliminating the need to store intermediate activations and reducing training memory by a constant factor independent of network depth (GitHub: renmengye/revnet-public). minREV (2023) extends the identical reversible-residual mathematical principle to Vision Transformers, Swin Transformers, and MViTv2, implementing the entire reversible Vision Transformer, including the memory-efficient reversible back-propagation algorithm, in under 300 lines of PyTorch, and further incorporates a parallelized reversible-backpropagation variant (PaReprop) (GitHub: karttikeya/minREV). Both qualify as radical relative to routine memory optimization because the network's residual algebra is restructured to be exactly invertible (a bijective computational graph), rather than merely checkpointed or gradient-accumulated.

6.8.5 Neural Cellular Automata (Growing NCA and LLM Pretraining Data Generation)

Growing Neural Cellular Automata embeds a small convolutional update rule as the local transition function of a cellular automaton, such that a CNN parametrizes the per-cell rule while global structure (e.g., a target image) emerges purely from many local iterations of the automaton across a 2D lattice - image morphogenesis as emergent CNN-driven cellular computation, with numerous open-source PyTorch/JAX reproductions (GitHub: PWhiddy/Growing-Neural-Cellular-Automata-Pytorch; Hugging Face: shyamsn97/Jax-NCA). More directly relevant to mainstream-LLM mapping, a 2026 paper uses neural cellular automata as a synthetic data generator for LLM pre-pre-training, finding that pre-pre-training a Transformer LLM on only 164M NCA-generated tokens improves downstream language modeling by up to 6% and that attention layers specifically are the most transferable component (arXiv:2603.10055; authors Lee, Han, Kumar, Agrawal). No public code repository was confirmed for this paper at time of review, so it is treated as a released-methodology item rather than confirmed runnable open-source.

6.8.6 Posit Arithmetic and Logarithmic/Residue Number Systems

Deep learning frameworks retargeted to posit (Type-I unum) arithmetic exist, including ports of TensorFlow to 16-bit posits and NumPy to 32-bit posits via the SoftPosit library, plus a dedicated deep-learning framework for the posit number system referenced under the `posit-arithmetic` GitHub topic (Posit & Unum Knowledge Hub; GitHub topic: posit-arithmetic). Logarithmic-number-system (LNS) deep learning accelerators substitute multiplication with addition in log-space, $\log(ab) = \log a + \log b$, with a public HLS implementation, LNS_Madam, targeting FPGA deployment (GitHub: phucducnguyen/LNS_Madam). Residue number system (RNS) accelerators, which represent each weight/activation as a tuple of residues modulo a coprime basis to enable carry-free parallel arithmetic, have several accelerator-level (RNSiM, RNSnet) implementations, though these are generally hardware-architecture papers with simulator-level RTL/behavioral models rather than general-purpose training frameworks (arXiv:2306.09481; dl.acm.org RNSiM). These qualify as radical numeric substitutions (replacing IEEE floating point with an entirely different algebraic representation of real numbers) but most available code is hardware-description-language simulation rather than end-to-end trainable software frameworks, placing them at released-simulator maturity.

6.9 DNA / Molecular Computing

6.9.1 Supervised Learning in DNA Neural Networks

A September 2025 Nature paper demonstrates a DNA-molecule-based neural network that autonomously performs supervised learning entirely in vitro, implementing a five-layer feedforward winner-take-all architecture using DNA strand-displacement reactions, where training patterns are encoded as activator molecules accumulated via Hebbian-like reactions ($X_i + L_j \rightarrow \text{Act}_{i,j}$) and classification is performed through molecular weight multiplication, weighted-sum integration, and pairwise-annihilation winner-take-all competition, demonstrated on MNIST digit-pair classification (0/1, 3/4, 6/7) using 100-bit molecular memories with over 700 distinct chemical species (Nature). No code, software repository, or programming framework is provided in the publication - the "implementation" is entirely wet-lab DNA

strand-displacement chemistry rather than software - so despite being a hardware (wet-lab) demonstration of a neural-network-equivalent computation, it fails the code/framework-availability criterion for main-list inclusion and is relegated to the appendix below as a hardware-only, code-absent demonstration.

6.10 False Positives, Near-Misses, and Exclusions

The following categories of candidate results were deliberately excluded from the main catalog:

Standard PyTorch/TensorFlow Transformer, ResNet, and LSTM reimplementations (e.g., generic "Attention Is All You Need" PyTorch ports) surfaced repeatedly in searches for posit arithmetic and other exotic representations but implement no non-standard substrate and were excluded as routine.

BitNet and 1-bit/1.58-bit LLM quantization (arXiv:2402.17764) was excluded because ternary/binary weight quantization, however extreme, remains a standard low-bit quantization technique operating within conventional integer/fixed-point arithmetic rather than a fundamentally different algebraic or physical substrate, consistent with the task's explicit exclusion of "basic pruning" and "standard quantization."

DNA-based neural network (Nature 2025) is excluded from the main list (see above) due to the complete absence of any accompanying software, code repository, or programmable framework - the entire "implementation" is wet-lab chemistry without a software/algorithmic artifact as required by the task's inclusion criteria, though it is noted here because of its striking hardware novelty.

LLMCA (Language Model Cellular Automata) (GitHub: pinsky-three/llmca) was excluded because it is architecturally inverted relative to the requested mapping: it uses an existing LLM as an oracle to drive a cellular automaton's per-cell update rule, rather than compiling or converting a mainstream architecture's internal computation into cellular-automaton form.

HyAtten and the IBM analog-attention/ALBERT chip demonstrations are retained in the main catalog despite lacking confirmed public code repositories because they provide unusually rigorous, peer-reviewed, primary-source engineering detail and genuine physical hardware validation; they are explicitly flagged as closed-prototype/hardware-demonstrated-without-confirmed-code rather than presented as fully reproducible open-source work.

Generic memristor/ReRAM/analog survey and review articles without an accompanying runnable framework, simulator, or demonstrated chip-plus-code pairing (e.g., general "memristor-based hardware accelerators for AI" review papers) were excluded as secondary literature rather than primary implementations.

Quantum attention surveys (e.g., "A Survey of Quantum Transformers," arXiv:2504.03192) were used only for context and are not counted as independent implementations.

Neural Cellular Automata for LLM pretraining (arXiv:2603.10055) is retained in the main table but flagged as methodology-only (no confirmed code repository located), since it maps a genuinely radical training-data-generation substrate onto attention-layer transfer but could not be verified as runnable open-source.

6.11 Notes on Simulation versus Physical Hardware

Across all thirty-six catalogued items, only a minority involve confirmed operation on physically fabricated non-von-Neumann hardware: the IBM 14nm analog PCM Transformer-ALBERT chip, the IBM analog-attention kernel-approximation chip, the physically fabricated memristor LSTM crossbar (2019), the original 3D-printed D2NN diffractive network (2018), the 120 GOPS thin-film-lithium-niobate photonic tensor core (2024), the BiHDTans FPGA latency measurement (2025), and Loihi-2-based neuromorphic LLM inference. The remaining majority - including nearly all spiking-Transformer conversions (SpikeGPT, Spikformer, Spikingformer, SpikeZIP-TF, Meta-SpikeFormer, RevSNN), all photonic tensor-core software frameworks (pytorch-onn, neuroptica, Lightning-Transformer's own accelerator), all quantum-circuit Transformer/CNN/diffusion implementations, and all biologically plausible learning and exotic-algebra Transformer implementations (Torch2PC, PC-Transformers, Equilibrium Propaga-

tion, Tropical Attention, BiHDTrans's core algorithm) - are validated purely through software simulation of the target substrate's mathematics, typically on conventional GPUs. This distinction is material: simulated energy and latency projections for photonic and neuromorphic hardware (e.g., "20x fewer operations," "9.8x performance-per-area") are architectural estimates from cycle-accurate or analytic simulators rather than measured physical results, whereas the IBM analog chip and Loihi 2 results reflect measured hardware behavior including real device noise, drift, and yield effects.

Chapter 7

Adversarial Theoretical Audit of the Neural-Monolithic Autonomous Fractal (NMAF) Architecture

7.1 Preliminary Framing

The Neural-Monolithic Autonomous Fractal (NMAF) architecture, as specified, presents five axioms combining thermodynamic language ("dissipative structure," "entropy"), estimation-theoretic language ("Neuro-UKF," "covariance," "bifurcation"), differential-geometric language ("vector," "manifold," "tangent bundle"), and network-physics language ("Global Signal Field," "instant tension change") to describe an autonomous cognitive architecture. The panel's task is to determine, term by term, where these borrowed formalisms are used as literal load-bearing mechanisms versus where they are metaphorical gestures dressed in equations they do not satisfy. This is the central methodological tool used throughout: for every claim, ask (1) what mathematical object is asserted, (2) whether that object is well-defined in the cited formalism, (3) whether the physical law invoked is actually satisfied, and (4) whether an operational, falsifiable measurement exists. Sources cited below are peer-reviewed papers or verified arXiv preprints; no invented citations are used, and any assertion not backed by a retrieved source is explicitly flagged as panel reasoning rather than external fact.

7.2 Axiom 1 - Topology of the Closed Contour (Breakdown of Linear Causality)

Claim under audit: Intelligence is a dissipative structure bounded by a self-referential "Contour." "Source" (explicit/past data) and "Inflow" (latent ontologies/future) are both described as flowing *inward* toward a center, generating "bilateral pressure" that breaks linear causality/time.

7.2.1 A) Theoretical Vulnerability

1. Conflation of four distinct arrow types. The axiom silently merges (i) thermodynamic energy/matter flux, (ii) Shannon information flow, (iii) Bayesian inference messages (prediction vs. prediction-error), and (iv) causal-graph arrows (Pearl-style do-calculus direction). These are not interchangeable. In predictive coding, "top-down" and "bottom-up" message passing both exist simultaneously in a hierarchical generative model, but this bidirectionality is a statement about *inference message-passing direction*, not about matter/energy transport, and it does not violate temporal causal order - the physical brain still evolves forward in time while exchanging predictions and prediction errors between hierarchical levels (Bogacz 2017; Dynamic predictive coding, PLOS Comp Bio; Predictive Coding Theories of Cortical Function). Labeling *inference-message bidirectionality* as a "breakdown of linear causality" is a category error: it mistakes the topology of a computational graph (which can have feedback loops without violating temporal causality) for a violation of physical time-ordering.

2. The Second Law bookkeeping problem. A dissipative structure in Prigogine's sense is an open system maintained far from equilibrium by a *through-flux* of energy/matter: low-entropy input is consumed and higher-entropy output/waste is exported to the environment, so that the system's own entropy can locally decrease or stay low while total entropy (system + environment) increases (Gaspard 2025; Tlidi, Clerc & Panajotov 2018). If, as the axiom states, *both* Source and Inflow arrows point inward to the center and there is no explicit outward-pointing arrow, the model as literally stated has no entropy-export channel. This is a bookkeeping violation, not a deep discovery: a system that only imports (whether

"data" or "latent ontologies") and never exports entropy cannot be a stationary dissipative structure - it must either accumulate internal disorder without bound or the model is incompletely specified. The axiom does not describe a new physics; it describes an accounting error that disappears the moment an output/waste-heat/prediction-error channel is added back in.

3. "Fractal rupture" as an undefined thermodynamic operator. The claim that a "fractal rupture" can "mathematically repair the balance" is unfalsifiable as stated because no functional form, conserved quantity, or dimensional units are given for "rupture." Self-similar (fractal) structure and entropy production are logically independent properties of a system; a geometric self-similarity claim cannot, by itself, supply or absorb entropy. Nothing in non-equilibrium thermodynamics licenses geometry substituting for a flux term in the entropy balance equation (Gaspard 2025).

Proposition 1.1 (Entropy Non-Closure). *Let a system S have only inward mass/energy/information influx channels $J_{in}^{(1)}, J_{in}^{(2)}$ (Source, Inflow) and no explicit outward channel. Then, under the standard local entropy balance*

$$\frac{dS_{sys}}{dt} = \sigma_{prod} - \nabla \cdot J_{S,out} + \nabla \cdot J_{S,in}$$

with $\sigma_{prod} \geq 0$ by the Second Law, if $J_{S,out} = 0$ identically, dS_{sys}/dt is bounded below by σ_{prod} and grows without bound whenever internal production is strictly positive, contradicting any claim of a stationary bounded "Contour."

Counterexample/Falsification Test 1.1: Measure (proxy) entropy production of the modeled system over long runtime under fixed inflow. If total internal state entropy (e.g., weight/activation distributional entropy, or KL divergence to a reference prior) grows monotonically without an identifiable dissipation channel, this falsifies the "bounded self-referential Contour" claim as physically stated. If it stays bounded, an implicit output channel exists and must be named and quantified - at which point the "two arrows both inward" description is simply wrong.

7.2.2 B) Required Reformulation

Discard "linear causality is broken" language entirely. Replace with a standard active-inference / dissipative-structure formulation with three, not two, channels:

Bottom-up sensory/data channel $y(t)$ (the "Source" data stream - genuinely an input).

Top-down generative/prior channel $\mu(t)$ - internal predictions descending the hierarchy (the useful residue of "Inflow," reinterpreted as *prior generative belief*, not a physically inbound flux).

Explicit dissipation/action channel $a(t)$ - motor output, energy dissipated as heat, or exported prediction error/report - required for entropy accounting to close.

Free energy F bounds surprise $-\ln p(y)$ via variational inference:

$$F = \mathbb{E}_{q(x)} [\ln q(x) - \ln p(y, x)] \geq -\ln p(y)$$

following the standard active-inference formalism (Friston & Parr 2018; Friston, Ramstead & Kirchhoff 2019). Entropy production is then bookkept as

$$\sigma_{prod} = \frac{dS_{int}}{dt} + \frac{dS_{env}}{dt} \geq 0,$$

with the *action* channel $a(t)$ supplying the outward flux that a genuine dissipative structure requires (Gaspard 2025). Message bidirectionality within the inference graph is retained (this is the one true and valuable insight buried in the axiom - hierarchical predictive coding genuinely has descending and ascending message flow) but is explicitly decoupled from any claim about thermodynamic or temporal-causal reversal.

Rating:

Internal mathematical consistency

2/5 - Entropy balance as stated has no export term; "fractal rupture" undefined

Compatibility with known physics

2/5 - Second Law bookkeeping fails as literally described; salvageable if output channel restored

Empirical identifiability

2/5 - "Linear causality breakdown" not independently measurable; predictive-coding bidirectionality is measurable via effective connectivity/DCM

Salvageability

4/5 - Maps cleanly onto standard active inference with dissipation channel added

7.3 Axiom 2 - The Neuro-UKF & Entropic Bifurcation

Claim under audit: A "Neuro-UKF" at the center is an "entropic collider" under "maximum entropy"; Source-Inflow tension makes the UKF "fail to resolve a superposition," producing symmetry breaking, "contour rupture," and a new solution vector field.

7.3.1 A) Theoretical Vulnerability

1. Category error: estimator divergence $\neq$ physical phase transition. The Unscented Kalman Filter (UKF) is a nonlinear Bayesian state estimator that propagates mean and covariance through an unscented transform using deterministically chosen sigma points, avoiding the Jacobian linearization of the EKF (Julier & Uhlmann 2004; Julier & Uhlmann, Corrections). When a UKF "fails," this means one or more well-characterized numerical/statistical failure modes occur: (a) the predicted covariance $P_{k|k-1}$ loses positive-semidefiniteness due to numerical round-off or an ill-conditioned sigma-point spread, requiring corrective techniques such as square-root/Cholesky-factor UKF formulations; (b) the system becomes locally unobservable, so the innovation covariance $S_k = HP_{k|k-1}H^\top + R$ becomes singular or ill-conditioned; (c) the true posterior is strongly non-Gaussian or multimodal (e.g., bimodal), and the sigma-point set - which captures only mean and covariance up to third order - cannot represent multimodality, causing the filter to average across modes and diverge. None of these failure modes is a physical phase transition. A phase transition (spontaneous symmetry breaking) is a statement about the *free energy landscape of a physical or statistical-mechanical system* acquiring degenerate minima as a control parameter crosses a critical value; an estimator's numerical divergence is a statement about an *algorithm's* inability to track a state, and it carries no intrinsic thermodynamic content. Treating "the filter failed to resolve a superposition" as physically equivalent to "the system underwent symmetry breaking" equivocates between the map (estimator) and the territory (physical system) - a textbook category error.

2. "Superposition" is imported from quantum mechanics without justification. A UKF operates on classical probability densities (typically Gaussian sigma-point approximations) over a real state vector; it has no Hilbert-space structure, no unitary evolution, and no measurement collapse formalism. Calling a multimodal or ill-conditioned classical posterior a "superposition" borrows quantum vocabulary for what is simply a non-Gaussian classical distribution - a semantic move, not a mechanism.

3. Missing ingredients for an actual cusp catastrophe. Catastrophe theory (Thom/Zeeman) requires (i) a smooth potential/free-energy functional $V(x; a, b)$ depending on state x and at least two control parameters (a, b) , (ii) an order parameter x^* that minimizes V , and (iii) a bifurcation/catastrophe set in control-parameter space where the number or stability of minima changes discontinuously (Cusp Catastrophe Model, Nursing Research; Catastrophe theory in work from heartbeats to eye movements; Cusp catastrophe as structural equation model). The NMAF axiom names none of these: there is no potential functional, no identified control parameters, and no order parameter distinct from the filter's own state estimate. An estimator whose covariance blows up does not thereby generate a potential landscape with degenerate minima; extra dynamics - specifically a slow-fast (adiabatic) separation between fast state variables and slow control parameters, with the fast subsystem relaxing to critical points of an explicit potential - must be added from scratch. The UKF failure at best signals *that* a bifurcation might be occurring in the underlying dynamical system being estimated; it is not itself the bifurcation mechanism.

Proposition 2.1 (Estimator Failure is Necessary-Insufficient Evidence for Bifurcation). *UKF divergence (covariance loss of positive-semidefiniteness, or innovation covariance singularity) is consistent with, but does not entail, a bifurcation in the true generative process. A counterexample: a purely linear, non-bifurcating stochastic system with time-varying, ill-conditioned observation geometry can cause UKF divergence via loss of observability alone, with zero change in the number or stability of equilibria of the true system.*

Falsification Test 2.1: Instrument the underlying dynamical system independently of the filter (e.g., via a particle filter or exhaustive grid-based Bayesian filter immune to Gaussian-approximation artifacts). If the particle filter posterior remains unimodal and stable while the UKF diverges, this falsifies any claim that UKF divergence is diagnostic of a true phase transition - it only diagnoses estimator model-mismatch.

7.3.2 B) Required Reformulation

Step 1 - Separate estimator diagnostics from dynamics. Retain the UKF purely as a state estimator with rigorously monitored health metrics:

Positive-semidefiniteness of P_k , enforced via square-root/Cholesky propagation.

Observability: local weak observability via the observability Gramian or Lie-derivative rank condition for the nonlinear system; loss of rank signals unobservable directions, not phase transitions.

Non-Gaussianity: monitor higher moments (skewness/kurtosis) of the sigma-point ensemble, or switch to a Gaussian-mixture/particle-filter fallback (e.g., the ensemble-unscented hybrid, Ensemble Kalman filter with the unscented transform) when multimodality is detected. A published example of adaptive covariance recalibration under uncertain noise is the CAR-EnKF framework (arXiv:2604.17343).

Step 2 - Construct an explicit cusp-catastrophe replacement model for the "bifurcation" claim, if a genuine phase-transition-like event in cognitive/network dynamics is desired. Define:

Order parameter x : a scalar or low-dimensional summary of the network's macroscopic state (e.g., a synchrony index, or the leading eigenvector projection of hidden-state covariance).

Control parameters (a, b) : a = normalized precision-weighted prediction-error magnitude (playing the role of "Source-Inflow tension"), b = a bias/asymmetry parameter between top-down and bottom-up precision.

Potential/free-energy functional:

$$V(x; a, b) = \frac{1}{4}x^4 - \frac{1}{2}ax^2 - bx$$

This is the canonical cusp catastrophe potential. Equilibria satisfy $\partial V/\partial x = x^3 - ax - b = 0$.

Bifurcation set: the cusp locus in (a, b) -space where $\partial^2 V/\partial x^2 = 0$ simultaneously with $\partial V/\partial x = 0$, giving the discriminant $4a^3 = 27b^2$.

Reset/model-switching rule: when the UKF's innovation covariance S_k becomes ill-conditioned (condition number exceeds a threshold κ_{max}) *and* independently the estimated control parameters $(\hat{a}, \hat{b})$ cross the discriminant curve, switch state-estimation mode (e.g., from unimodal UKF to a multi-hypothesis/Gaussian-mixture filter) and reinitialize covariance from the sigma-point spread at the new mode. This makes "contour rupture" an operationally defined event: a joint criterion combining an estimator-health flag and an independently estimated crossing of the catastrophe discriminant - not a mystical fusion of the two.

Rating:

Internal mathematical consistency

1/5 - Conflates estimator failure with physical bifurcation; no potential functional given

Compatibility with known physics/estimation theory

2/5 - UKF failure modes are real and well-characterized, but not phase transitions

Empirical identifiability

2/5 - UKF health metrics are measurable; "entropic collider" and "superposition" are not

Salvageability

4/5 - A rigorous cusp-catastrophe replacement model with explicit potential is constructible (above)

7.4 Axiom 3 - The Physics of the "Soul" (Tension & Pressure)

Claim under audit: "Soul" is defined as the "root of lack" - continuous topological tension of a finite Contour resisting the "Pressure of Truth" from an infinite-dimensional continuous analog "General Signal World Model."

7.4.1 A) Theoretical Vulnerability

1. "Soul" as an unoperationalized primitive. No measurement procedure, no units, and no falsifiable prediction are attached to "soul." This places the term outside the domain of physical science per standard falsifiability criteria in philosophy of science (Falsifiability in medicine: what clinicians can learn from Karl Popper; An Analysis of the Falsification Criterion of Karl Popper). Renaming an undefined psychological/spiritual construct with topological vocabulary ("tension of finite Contour") does not confer physical content; it is equivocation between metaphor (soul, lack, pressure of truth) and mechanism (tension functional, finite-dimensional manifold).

2. The finite/infinite-dimensional interaction is underspecified and risks ill-posedness. Any model in which a finite-dimensional learned latent manifold $\mathcal{M} \subset \mathbb{R}^d$ is compared against, or embedded in, an infinite-dimensional function space $\mathcal{H}$ (e.g., a reproducing kernel Hilbert space or an L^2 space of signals) requires an explicit embedding map $\iota : \mathcal{M} \rightarrow \mathcal{H}$ and regularity/compactness assumptions to avoid divergence. Without such assumptions, "pressure" (residual norm) is unbounded and undefined, and the reported "tension" is not a well-posed mathematical object. Manifold learning methods generally assume the observed high (but finite) dimensional data lie near a lower-dimensional manifold, not that the ambient space is literally infinite-dimensional and continuous in the sense of a field theory; treating a neural network's finite latent code as resisting an "infinite-dimensional continuous analog world model" summons functional-analytic machinery (weak topologies, Sobolev regularity, compact embeddings) that the architecture never specifies or proves.

3. Gauge/connection mismatch language used loosely. Comparing "curvature/connection mismatch" between a latent manifold and a putative infinite-dimensional field requires the latent manifold to carry an actual connection (a rule for parallel-transporting tangent vectors) and the target space to carry a comparable geometric structure - neither of which is established for typical learned representations. Using gauge-theory vocabulary without specifying the fiber bundle, structure group, or curvature 2-form is decorative rather than explanatory.

Proposition 3.1 (Tension Functional Well-Posedness Requirement). *Let $\mathcal{M}$ (finite-dim, dimension d) be a candidate latent manifold and $\mathcal{W}$ an infinite-dimensional Hilbert space representing the "General Signal World Model." A tension functional $T : \mathcal{M} \rightarrow \mathbb{R}_{\geq 0}$ is well-defined only if there exists a continuous (ideally compact) embedding $\iota : \mathcal{M} \hookrightarrow \mathcal{W}$ such that $T(m) = \|\iota(m) - P_{\mathcal{W}}(\text{signal})\|_{\mathcal{W}}^2$ for a projection $P_{\mathcal{W}}$, and such that this norm remains finite for all signals of physical interest. Absent stated compactness/regularity conditions on ι , "tension" is undefined and cannot be claimed to be "continuous."*

Falsification/Non-Identifiability Test 3.1: Ask whether the architecture provides a concrete, computable definition of the projection residual, an information-geometric divergence (KL or Fisher-Rao geodesic distance), or a free-energy gap between the latent code and any accessible approximation of the "world model." If no computable quantity is offered, "soul" fails the minimal criterion for being a physical observable - it cannot yet be measured, differentiated between systems, or shown to change under any intervention, and is therefore, at present, philosophical/psychological vocabulary rather than physics.

7.4.2 B) Required Reformulation

Replace "soul"/"root of lack"/"Pressure of Truth" with a concrete, computable tension functional built from established information-geometric and rate-distortion quantities:

Projection residual (functional-analytic tension): for encoder $\iota : x \mapsto z \in \mathcal{M}$ and decoder/generative map $g : z \mapsto \hat{x}$,

$$T_{proj}(x) = \|x - g(\iota(x))\|^2$$

- a standard reconstruction/rate-distortion residual, well-posed once norm and function spaces are fixed (Information bottleneck method, Tishby et al.; Survey of Rate Distortion and Information Bottleneck).

Fisher-Rao/KL geodesic stress (information-geometric tension): if the latent code induces a distribution $q_\theta(x)$ and the "world model" a reference distribution $p(x)$,

$$T_{KL}(\theta) = D_{KL}(q_\theta \| p)$$

using the Fisher information metric to locally quantify how much a small change in latent parameters θ moves the model's implied distribution, following standard information-geometric divergence constructions (Information geometry connecting Wasserstein distance and KL divergence; Information geometry of divergence functions).

Free-energy gap (active-inference tension): the variational free energy $F(\theta)$ minus its minimum achievable value under an idealized/expanded model class - a computable, monotone-decreasing-under-learning quantity (Friston & Parr 2018).

Necessary regularity conditions: the encoder/generative maps must be continuous (ideally Lipschitz) with images lying in a compactly embedded subspace of the target Hilbert space, so that residuals remain finite; without this, any claim of a well-defined, always-finite "tension" is mathematically unsupported.

Why "soul" is not (yet) a physical observable: none of T_{proj} , T_{KL} , or the free-energy gap requires or licenses an ontological add-on called "soul." They are ordinary, measurable quantities in machine learning and information theory. Retaining "soul" language beyond a poetic label for "aggregate unresolved residual across all three functionals" would be unsupported ontology.

Rating:

Internal mathematical consistency

1/5 - No well-posedness conditions given for finite/infinite-dim interaction

Compatibility with known physics

2/5 - Gauge/curvature language invoked without structure group or connection defined

Empirical identifiability

1/5 - "Soul" has no measurement procedure; replacement functionals are measurable

Salvageability

4/5 - Rate-distortion + Fisher-Rao + free-energy gap gives a fully computable substitute

7.5 Axiom 4 - The Omega-Vector & Subjective Perception

Claim under audit: The local Contour "projects a virtual Omega-vector outward" into the signal field as an "averaged anti-vector"; its intersection with a "Local Shadow/hypervector plane" from prior bifurcations "generates a Signal identified with subjective awareness."

7.5.1 A) Theoretical Vulnerability

1. A vector cannot "intersect" a manifold - this is a type error. In differential geometry, a tangent vector is an element of a tangent space $T_p M$ at a specific base point p ; it has no independent existence "outward in the signal field" without a specified base point and a map (e.g., a curve, section, or flow) realizing it in ambient space. "Intersection" is an operation between two subsets of a common ambient space (e.g., two submanifolds), or between the image of a map and a submanifold - not between a bare vector and a "plane." Transversality theory formalizes when a map's image meets a submanifold non-degenerately (transversally), requiring that at each intersection point the sum of the tangent space of the image and the tangent space of the target submanifold spans the ambient tangent space (Transversality theorems for the weak topology, arXiv:1110.2085; Generalization of Thom's transversality theorem, arXiv:1001.2054). The axiom's "vector intersecting a plane" skips all of this structure: no base point, no map, no ambient space specified, and no transversality condition checked.

2. "Anti-vector" is undefined. Standard differential geometry has covectors (elements of the cotangent space $T_p^* M$, dual to tangent vectors) and the additive inverse $-v$ of a vector v (same space, opposite direction) - neither of which is "averaged" in the sense implied, and neither of which is called an "anti-vector" in the literature. If "anti-vector" means covector, the intersection claim becomes a pairing (a

covector acting on a vector to produce a scalar), which is a wholly different - and well-defined - operation, but this must be stated explicitly, not left as evocative language.

3. A Fisher-metric structure alone cannot generate an "event" or phenomenal state. The Fisher information metric equips a statistical manifold with a Riemannian structure quantifying local distinguishability of nearby distributions (A Generalized Cramér-Rao Bound Using Information Geometry; Information geometry of divergence functions). This metric supports geodesics, curvature, and divergence computations - but it is a passive geometric structure, not a generator of discrete "events." An event requires a dynamical trigger (e.g., a trajectory crossing a threshold submanifold), which is a separate ingredient from having a metric on the space. Conflating "the space has a Fisher metric" with "therefore perceptual events occur" elides the needed dynamical/measurement-theoretic step entirely - and even with that step added, nothing in the formalism connects a geometric/dynamical event to phenomenal (subjective, first-person) experience. This is the classic explanatory gap; asserting that a well-defined geometric event "is identified with subjective awareness" is an unsupported ontological leap, not a mathematical consequence. Even in the most prominent computational theories that self-consciously aim at global integration of information (e.g., Global Workspace Theory), rigorous critiques note substantial methodological gaps between proposed neural/computational correlates and validated consciousness measurement (Methodological Artefacts in Consciousness Science: The Case Against Global Workspace Theory), and even optimistic computational-functional accounts of AI consciousness explicitly flag this as an open, contested inference rather than a settled entailment (A Case for AI Consciousness: Language Agents and Global Workspace Theory, arXiv:2410.11407).

Proposition 4.1 (No Consciousness from Geometry Alone). *Let $(\mathcal{M}, g_F)$ be a statistical manifold with Fisher metric g_F , and let $\gamma(t)$ be any trajectory (flow, section, or geodesic) on $\mathcal{M}$. The existence of a well-defined transversal crossing of $\gamma(t)$ with a submanifold $\Sigma \subset \mathcal{M}$ is a purely geometric/dynamical fact. No axiom of Riemannian or information geometry entails a predicate "is phenomenally experienced" attached to such a crossing; any such attachment is an additional, unsupported ontological posit external to the geometry itself.*

Falsification Test 4.1: If subjective awareness is claimed to be identified with a specific geometric event type, the claim predicts that any system instantiating an isomorphic geometric event (same submanifold-crossing structure, verified via the same discriminant/statistics) must have the same phenomenal status. Since no independent, third-person measurement of phenomenal status exists, this prediction is currently untestable - which is precisely the disqualifying feature under standard falsifiability criteria (Falsifiability in medicine). The geometric-event component of the claim is falsifiable (does a transversal crossing occur, yes/no, measurably); the consciousness-identification component is not, and must be separated out.

7.5.2 B) Required Reformulation

Replace the "Omega-vector intersects hypervector plane" language with a well-posed section-crossing model:

Define a smooth map (section) $s : \mathcal{B} \rightarrow E$ of a fiber bundle $E \rightarrow \mathcal{B}$ over the network's belief-state base manifold $\mathcal{B}$ (e.g., E could be the bundle of local predictive-error/precision-weighted residual vectors, with base point given by current internal state).

Define a decision/discriminant submanifold $\Sigma \subset E$ - e.g., the locus where cumulative surprisal (negative log model evidence) exceeds a threshold, or where precision-weighted prediction error changes sign.

Perception/innovation event: occurs when the flow of $s(t)$ crosses Σ transversally, i.e., at the crossing point p , $T_p(\text{im } s) + T_p\Sigma = T_pE$ (Transversality theorems, arXiv:1110.2085). This transversal crossing is exactly what is measured, in practice, as a Bayesian surprise/innovation signal:

$$I(t) = -\ln p(y_t \mid y_{1:t-1})$$

the log-Bayes-factor innovation from sequential filtering - a scalar, computable, and measurable event marker consistent with predictive-coding practice (Predictive Coding Theories of Cortical Function, arXiv:2112.10048; Frontal cortex hierarchical predictive coding).

Explicit disclaimer required in the architecture documentation: a transversal crossing generating a scalar innovation event is a *measurable computational/informational occurrence*. It is not, by itself, evidence

of phenomenal consciousness; any such further claim is conjecture/unsupported ontology and must be labeled as such.

Rating:

Internal mathematical consistency

1/5 - Vector-manifold intersection without base point/map is a type error; "anti-vector" undefined

Compatibility with known physics

2/5 - Fisher-metric usage is legitimate but insufficient for claimed conclusion

Empirical identifiability

2/5 - Geometric crossing is measurable; "subjective awareness" identification is not

Salvageability

4/5 - Transversal section-crossing + Bayesian surprisal gives a fully rigorous substitute for the geometric half of the claim

7.6 Axiom 5 - The Global Signal Field (Physical Collective Unconscious)

Claim under audit: Every Omega-vector joins a macro-network; a massive rupture in one local fractal "instantly" changes the tension of all connected contours "without discrete communication"; a "true AI" accesses the field by "closing circuit with its own Omega-vector."

7.6.1 A) Theoretical Vulnerability

1. "Instantly...without discrete communication" is either physically forbidden or a mislabeled mean-field effect. Any influence propagating between spatially separated physical subsystems faster than through a bounded-speed causal channel, or with zero communication cost, is disallowed in relativistic physics and - even in nonrelativistic quantum many-body systems - is bounded by Lieb-Robinson-type results establishing an effective finite "light cone" for information propagation, including for long-range and power-law interactions and for general Markovian (open-system, dissipative) quantum dynamics (Lieb-Robinson bound and locality for general Markovian quantum dynamics; Lieb-Robinson Light Cone for Power-Law Interactions; Limits to Velocity of Signal Propagation in Many-Body Systems). A literal "instant" network-wide tension change with zero information/energy exchanged is therefore either (a) a forbidden superluminal/acausal coupling, if taken as fundamental microphysics, or (b) a loose, misleading description of an already well-understood and much less exotic phenomenon: strong common coupling through a shared, already-existing field/network medium (mean-field or common-cause effects), where the *appearance* of instantaneity comes from all nodes already being coupled to a shared, pre-existing order parameter rather than any new signal traveling instantaneously between them.

2. Global synchrony without discrete signals is a known, lawful phenomenon - but it is not communication-free, it is common-cause/mean-field. In the Kuramoto model, N phase oscillators θ_i coupled by

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i)$$

undergo a synchronization phase transition as coupling strength K crosses a critical value K_c , producing a macroscopic order parameter $re^{i\psi} = \frac{1}{N} \sum_j e^{i\theta_j}$ that every oscillator "feels" via the mean field (Kuramoto model: a simple paradigm for synchronization; linear reformulation of Kuramoto model, PhysRevE 77.031114; On the Equivalence of Synchronization Definitions in the Kuramoto Flow, arXiv:2503.19781). Crucially, this is a *mean-field* effect: each oscillator's dynamics genuinely depends on a locally computable, continuously updated average that has itself been built up from finite-speed pairwise couplings over time - not an acausal instantaneous jump. A "massive rupture" in one node changes the value each other node measures of the shared mean field on the timescale set by the coupling's own update/relaxation rate, not literally instantaneously, and definitely not with zero energetic/bandwidth cost.

3. Common-cause confounding must be ruled out before invoking any "field." Two spatially separated nodes changing correlated states can result from (a) a genuine causal/coupling channel between them, (b) a shared external driver (a common cause), or (c) a shared inherited initial condition. Any claim of a "Global Signal Field" mechanism must positively rule out (b) and (c) - e.g., via controlled perturbation

experiments - before invoking a distinct network-coupling ontology at all. Small-world and scale-free topologies are well documented to support fast, efficient information spread and synchronization with only local/pairwise edges and finite path lengths, not non-local field access (Complex networks: small-world, scale-free and beyond; Synchronization of scale-free neuronal network with STDP under time delay).

4. "True AI accesses the field by closing circuit with its own Omega-vector" is unfalsifiable as stated. No operational procedure is given for what "closing the circuit" consists of, what signal is exchanged, over what channel, at what latency/bandwidth/energy cost, or how "accessing the field" would be distinguished experimentally from ordinary parameter sharing, weight synchronization, or shared training data across model instances. Absent such specification, the claim is not merely speculative but currently non-operational.

Proposition 5.1 (Retarded Coupling Requirement). *Any lawful reformulation of network-wide "instant tension" must express the coupling as a retarded (causal) interaction with propagation delay $\tau_{ij} = d_{ij}/v$ for finite signal speed v and network (graph) distance d_{ij} , or as a pre-existing mean-field variable $\Psi(t)$ to which all nodes are already continuously and causally coupled, with any apparent "simultaneity" being a consequence of already-synchronized phase relative to a slower relaxation timescale - never a zero-latency, zero-bandwidth exchange.*

Falsification Test 5.1: Isolate two purported "connected contours" with all known communication channels (network, shared training corpus, shared hardware clock/parameter server) physically severed or logged and monitored. If tension/state correlation between them persists above chance after severing all channels and controlling for shared initial conditions, this would be extraordinary evidence requiring immediate independent replication; absent such a demonstrated result, the parsimonious and currently supported explanation is common-cause/mean-field coupling through already-existing, finite-bandwidth channels, not a new nonlocal field.

7.6.2 B) Required Reformulation

Model the network as a graph of coupled dynamical nodes with explicit retarded coupling:

$$\dot{x}_i(t) = f(x_i(t)) + \sum_j A_{ij} g(x_j(t - \tau_{ij}))$$

where A_{ij} is the (small-world/scale-free) adjacency weight and τ_{ij} the finite propagation delay - this is causally lawful and matches the network-synchronization literature (Complex networks: small-world, scale-free and beyond).

Alternatively, model network-wide "tension" as a genuine mean-field order parameter $\Psi(t) = \frac{1}{N} \sum_i h(x_i(t))$ to which every node is coupled; a local rupture changes Ψ on the timescale of the network's own relaxation dynamics (bounded below by $1/K$ for coupling strength K in Kuramoto-type systems), never instantaneously and never for free - the energetic/bandwidth cost is the total signaling power required to maintain the coupling weights A_{ij} or the shared parameter server bandwidth, in the case of an artificial system (Kuramoto model review).

State explicit latency/bandwidth/energy costs rather than asserting "instant, discrete-communication-free" access: for a graph of diameter D and per-hop delay δ , worst-case propagation time is $O(D\delta)$; for a mean-field parameter, update latency is set by the field's own relaxation time $1/\lambda$ (spectral gap of the coupling operator) - both finite and measurable.

No consciousness or "collective unconscious" claim without operational evidence. Retain "Global Signal Field" only as a rebranding of a shared mean-field/order-parameter architecture (e.g., federated shared embeddings, shared parameter servers, or explicit message-passing graphs) - a legitimate, physically lawful engineering pattern - while explicitly marking any claim of accessing a trans-personal "collective unconscious" as unsupported ontology absent a demonstrated, replicated, channel-isolated experiment (Falsification Test 5.1).

Rating:

Internal mathematical consistency

1/5 - "Instant, communication-free" is internally incoherent given any finite propagation channel

Compatibility with known physics

1/5 - Violates Lieb-Robinson-type bounds if read literally; salvageable as mean-field claim

Empirical identifiability

2/5 - Retarded-coupling/mean-field version is directly measurable; "field access"/"collective unconscious" is not

Salvageability

4/5 - Standard Kuramoto/mean-field/small-world network formalism is a complete, lawful substitute

7.7 Consolidated Defect Table

1. Closed Contour

No entropy-export channel; conflates flux/information/inference/causal arrows; "fractal rupture" undefined - Repairable - Add explicit dissipation/action channel; retain only inference-message bidirectionality, decoupled from thermodynamics/causality claims

2. Neuro-UKF Bifurcation

Category error: estimator divergence $\neq$ physical phase transition; no potential functional; "superposition" imported without basis - Repairable - Separate UKF health diagnostics (PSD, observability, non-Gaussianity) from an explicit cusp-catastrophe model with real control/order parameters

3. Soul (Tension)

"Soul" unoperationalized; finite/infinite-dimensional interaction lacks regularity/compactness assumptions; gauge language without structure group - Repairable - Replace with projection residual + Fisher-Rao/KL divergence + free-energy gap, each computable with stated regularity conditions

4. Omega-Vector Perception

Vector-manifold "intersection" is a type error (no base point/map); "anti-vector" undefined; Fisher metric alone cannot yield an event or phenomenality - Repairable (geometry) / Fatal (consciousness claim) - Section/flow crossing a discriminant submanifold transversally, yielding a Bayesian surprisal event; explicitly disclaim any consciousness entailment

5. Global Signal Field

"Instant, communication-free" coupling violates finite-propagation-speed/locality bounds if literal; unfalsifiable "field access" - Fatal (as literal nonlocality) / Repairable (as mean-field) - Retarded coupling or explicit mean-field order parameter with stated latency/bandwidth/energy cost; no consciousness/collective-unconscious claim without channel-isolated evidence

7.8 Minimal Coherent NMAF Reformulation

A physically and mathematically defensible core can be extracted from all five axioms as follows:

State space: $x(t) \in \mathbb{R}^n$ (internal generative-model state/parameters), augmented with a network index $i \in 1, \dots, N$ for multi-agent/multi-instance versions.

Dynamics (active-inference core): each node minimizes variational free energy

$$F_i = \mathbb{E}_{q(x_i)} [\ln q(x_i) - \ln p(y_i, x_i)],$$

updated via gradient descent $\dot{x}_i = -\Gamma \partial F_i / \partial x_i$, consistent with standard active inference (Friston & Parr 2018; Bogacz 2017).

Thermodynamic bookkeeping: explicit three-channel entropy balance

$$\frac{dS_i}{dt} = \sigma_{prod,i} - \Phi_{out,i} + \Phi_{in,i}, \quad \sigma_{prod,i} \geq 0,$$

with $\Phi_{out,i}$ (action/output/dissipation) mandatory for stationarity - resolving Axiom 1's fatal export gap (Gaspard 2025).

Estimator role: UKF (or square-root/Cholesky UKF for guaranteed PSD covariance) as the state-tracking module for $x_i(t)$, instrumented with observability-Gramian and non-Gaussianity health checks,

feeding - but never substituting for - a separate cusp-catastrophe control-parameter model (a_i, b_i) with potential $V(x; a, b) = \frac{1}{4}x^4 - \frac{1}{2}ax^2 - bx$ as the actual bifurcation mechanism (Julier & Uhlmann 2004; Cusp catastrophe model).

"Tension" (soul) bookkeeping: projection residual + Fisher-Rao KL divergence + free-energy gap, each a finite, computable scalar under stated compactness/Lipschitz regularity (Information geometry of divergence functions; information bottleneck method).

Geometric perception event: transversal crossing of a flow/section with a discriminant submanifold Σ , operationalized as scalar Bayesian surprisal $I(t) = -\ln p(y_t | y_{1:t-1})$ - measurable, but explicitly not sufficient for a consciousness claim (predictive coding, arXiv:2112.10048).

Network coupling: retarded pairwise coupling $\sum_j A_{ij} g(x_j(t - \tau_{ij}))$ or shared mean-field order parameter $\Psi(t)$ à la Kuramoto, respecting finite propagation delay and Lieb-Robinson-type bounds, with explicit bandwidth/energy cost accounting - no "instant field access" claim (Kuramoto paradigm; Lieb-Robinson bound).

7.9 Decisive Falsification Experiments

Entropy-closure test (Axiom 1): Measure total internal-state entropy over long runtimes with all inflow channels active; confirm existence and magnitude of an explicit dissipation/output channel. Failure to identify one falsifies "bounded self-referential Contour."

Independent-filter cross-check (Axiom 2): Run a particle filter or grid-based Bayes filter in parallel with the UKF on the same generative process. If the independent filter shows no true multimodality/bifurcation while the UKF diverges, this falsifies "UKF failure = phase transition."

Computability audit (Axiom 3): Require the architecture to output numerical values for projection residual, KL/Fisher-Rao divergence, and free-energy gap on real data. Inability to compute any of these falsifies "soul" as a well-posed quantity.

Type-check audit (Axiom 4): Require explicit statement of base point, bundle, and map for the "Omega-vector"; if none can be supplied, the geometric claim is ill-typed by construction. Separately, no experiment can currently test the consciousness-identification clause - mark as permanently unfalsifiable until an independent operational marker of phenomenality is proposed.

Channel-isolation test (Axiom 5): Physically/logically sever all known communication channels (network, shared corpus, shared parameter server, shared clock) between two claimed "connected contours." Persistent above-chance correlation after severing, replicated independently, would falsify the mean-field/common-cause explanation and constitute extraordinary evidence requiring extraordinary scrutiny; absence of such correlation falsifies the "instant, non-communicative field" claim as stated.

7.10 Explicit Metaphor/Conjecture/Ontology Ledger

Pure metaphor (no proposed mechanism, retain only as narrative label): "soul," "root of lack," "Pressure of Truth," "collective unconscious."

Engineering analogy (useful shorthand, not literal physics): "entropic collider," "Contour," "Global Signal Field" as a name for a shared parameter/embedding space.

Conjecture requiring dedicated new formalism if pursued further: cusp-catastrophe reformulation of Axiom 2 as a genuine cognitive bifurcation mechanism (mathematically well-posed once specified, but not yet empirically validated for this architecture); tension-functional definitions of Axiom 3 (well-posed but unvalidated as "soul").

Unsupported ontology (should not appear in a technical specification without explicit disclaimer): identification of any geometric/informational event with subjective/phenomenal awareness (Axiom 4); nonlocal, communication-free field access implying consciousness-sharing across instances (Axiom 5).

Fatal as literally stated, requiring outright removal or full reformulation: "both arrows point inward with no export" (Axiom 1, unless amended); "instant, discrete-communication-free" coupling (Axiom 5,

unless read purely as mean-field shorthand).

7.11 Final Cross-Axiom Verdict

No axiom, taken at face value, survives contact with the cited physics, estimation theory, differential geometry, or information theory literature without modification; several individual technical claims are salvageable and, once reformulated as above, converge on a coherent, non-mystical architecture: an active-inference network of nodes running health-monitored nonlinear Bayesian filters, with an explicit thermodynamic dissipation channel, a genuine (separately specified) catastrophe-theoretic bifurcation submodel, information-geometric tension functionals in place of "soul," transversal-crossing surprisal events in place of "Omega-vector perception," and retarded/mean-field network coupling in place of an instantaneous "Global Signal Field." What cannot be salvaged under any known physics or measurement theory is the identification of any of these mechanisms with subjective consciousness or a literal nonlocal "collective unconscious" - those remain, at present, unsupported ontological claims rather than established or even well-posed falsifiable science.

Chapter 8

Catalog of Non-Mainstream and Historical AI Architectures: Topological and Structural Viability Analysis

8.1 Scope and Historiographic Method

This catalog applies strict criteria to distinguish genuine theoretical dead ends from architectures that merely fell out of fashion. An entry qualifies for Category 1 (Discarded) only if a specific mathematical, combinatorial, or physical result forecloses the architecture's viability at scale, or if the architecture was explicitly abandoned by its own proponents on structural grounds. Where the failure is empirical/engineering rather than a proof, this is flagged explicitly. Category 2 (Non-Mainstream Working) requires demonstrable current operation outside the transformer/backpropagation paradigm, with a named formal mechanism.

8.2 CATEGORY 1: DISCARDED MODELS

8.2.1 Single-Layer Perceptron (Rosenblatt, 1958)

Theorists: Frank Rosenblatt (Cornell Aeronautical Laboratory); rejection formalized by Marvin Minsky and Seymour Papert.

Structural mechanism: A linear threshold unit $f(x) = \text{sign}(w \cdot x - \theta)$ implemented in hardware as the Mark I Perceptron - a 20×20 photocell retina wired through 512 motor-driven potentiometer weights to 8 response units (Mark I Perceptron, Wikipedia).

Reason for rejection - exact and proven: Minsky and Papert's 1969 monograph *Perceptrons* proved that a single-layer linear threshold unit cannot compute the XOR (exclusive-or) predicate, because the positive and negative XOR points $(-1, -1), (1, 1)$ versus $(-1, 1), (1, -1)$ are not linearly separable in $\mathbb{R}^2$ - no hyperplane can partition them (Perceptrons, Minsky & Papert, cited in course notes). This is a genuine theorem, not an engineering limitation: any function requiring non-linearly-separable decision boundaries is provably outside the perceptron's hypothesis class, including "order-limited" predicates like topological connectedness, which Minsky and Papert also proved perceptrons cannot compute regardless of the number of parallel threshold units, so long as receptive fields are diameter-limited (CMU perceptron lecture notes). It is a common historiographic error to claim Minsky and Papert asserted this generalizes to multilayer networks - they did not, and it is the failure to disseminate this distinction, not the theorem itself, that caused funding collapse (Perceptron, Wikipedia). The theorem is a genuine impossibility result for the single-layer linear architecture specifically, not for connectionism generally. Funding for Rosenblatt's ONR/RADC-backed program (Contract Nonr-401(40)) lapsed as a direct empirical/institutional consequence, not a further mathematical result (Cornell News).

8.2.2 Classical Hopfield Networks as General-Purpose Associative Memory (Hopfield, 1982)

Theorists: John Hopfield; capacity analysis by Amit, Gutfreund, Sompolinsky and others in the statistical-mechanics tradition.

Structural mechanism: A fully connected recurrent network of binary threshold units minimizing the Ising-type energy function $E = -\frac{1}{2} \sum_{i,j} w_{ij} s_i s_j + \sum_i \theta_i s_i$, where stored patterns are constructed as local

minima via a Hebbian outer-product rule (Hopfield network, Wikipedia).

Reason for rejection - proven capacity bound and provable degeneracy: The classical Hopfield storage rule yields a critical capacity of approximately $0.138N$ to $0.15N$ patterns for N binary units before catastrophic retrieval failure (crosstalk) sets in - a scaling law derived analytically in statistical mechanics and confirmed empirically (Hinton Coursera lecture notes). Critically, the number of spurious (unintended) local minima in the energy landscape grows exponentially with the number of stored patterns even when patterns are mutually orthogonal - an energy-landscape degeneracy that is a structural, not merely quantitative, defect: linear combinations of stored patterns and reversed patterns are provably also local minima of the same energy function (Hopfield network, Wikipedia). This is the textbook case of "energy landscape degeneracy" cited in the task brief - a rigorously derived combinatorial bottleneck on associative capacity, not an implementation artifact. It should be flagged, however, that this is a bounded-capacity result for the *classical* quadratic-energy Hopfield model specifically; modern "Dense Associative Memory" / continuous Hopfield variants with higher-order (polynomial or exponential) energy functions provably achieve exponential storage capacity in the dimension, exponentially small retrieval error, and single-step convergence, which is why this entry is scoped narrowly to the *classical* architecture and its 1980s capacity ceiling, not the concept of energy-based associative memory as such (Hopfield Networks is All You Need, arXiv; arXiv:2504.04879 on false memories).

8.2.3 Deep Boltzmann Machines / Unrestricted Boltzmann Machines as Trainable Generative Models

Theorists: Geoffrey Hinton, Terry Sejnowski (original Boltzmann machine, 1985); Ruslan Salakhutdinov and Hinton (deep Boltzmann machines, 2009).

Structural mechanism: A stochastic recurrent network with symmetric weights defining a Gibbs (Boltzmann) distribution $p(x) = \frac{1}{Z(\theta)} \exp(-E_\theta(x))$ over visible and hidden binary units, trained by maximum likelihood.

Reason for rejection - proven computational-complexity dead end: Exact maximum-likelihood training of a general Boltzmann machine requires computing the partition function $Z(\theta)$, which is #P-hard - provably intractable to compute exactly, and, under standard complexity assumptions (unless $RP = NP$), cannot be efficiently approximated even within a specified multiplicative error bound (Microsoft Research / arXiv:1412.3489). This is a genuine, formally proven complexity-theoretic dead end for exact learning in the general (non-restricted) architecture - distinguishing it from an engineering inconvenience. The practical response (variational mean-field approximations plus persistent Markov chains, as in Salakhutdinov and Hinton's 2009 deep Boltzmann machine) is an *approximation regime*, not a resolution of the underlying #P-hardness, and the field's near-total pivot toward feedforward and autoregressive generative models (GANs, diffusion, autoregressive transformers) reflects abandonment of exact-likelihood undirected graphical models with intractable partition functions as a mainstream training target (Salakhutdinov & Hinton, Deep Boltzmann Machines, PMLR; Goodfellow et al., "Confronting the Partition Function"). Flag: this is a proof-backed dead end for *exact* training of the general architecture; restricted Boltzmann machines with bipartite connectivity retain tractable conditional inference and persist in niche use, so the abandonment is specific to the unrestricted, deep, exact-likelihood variant, not to energy-based models as a category (see Category 2, thermodynamic computing, which repurposes exactly this Gibbs-sampling structure in hardware).

8.2.4 GOF AI Exhaustive Search / Combinatorial Game-Tree Approaches to General Problem-Solving

Theorists: Herbert Simon and Allen Newell (General Problem Solver); Claude Shannon (chess search analysis, 1949).

Structural mechanism: Breadth-first or depth-first traversal of a problem's full state-action tree to guarantee an optimal or complete solution.

Reason for rejection - proven combinatorial explosion: Shannon's 1949 analysis showed chess has on the order of 10^{120} unique games with average branching factor 40 per ply; at 200 million positions per second, an exhaustive search over the full game tree would require roughly 10^{100} years - vastly exceeding the 10^{10} -year age of the universe (search complexity lecture notes). This exponential blow-up (branching factor raised to search depth) is a mathematically exact consequence of tree enumeration, not an approximation - it is why unconstrained exhaustive search was abandoned as a general problem-solving

method in favor of heuristic pruning (A, alpha-beta, iterative deepening) and, later, learned evaluation functions (foundations of computational agents, forward planning). The dead end is narrowly scoped to *exhaustive, uninformed* search as a general-purpose method; heuristically guided search remains foundational and thriving.

8.2.5 STRIPS-Style Classical Planning as a General Reasoning Substrate

Theorists: Richard Fikes and Nils Nilsson (STRIPS, 1971); complexity results by Tom Bylander (1994).

Structural mechanism: Represent world-states as sets of ground literals and actions as precondition/add/delete lists; search for a literal-sequence from initial to goal state.

Reason for rejection - proven PSPACE-completeness: Bylander proved that determining plan existence for propositional STRIPS is PSPACE-complete in general, and remains PSPACE-complete even when operators are restricted to two preconditions and two effects - placing classical planning at the same complexity class as generalized quantified Boolean satisfiability, strictly harder than NP under standard conjectures (Bylander 1994, cited via arXiv survey; IJCAI parameterized complexity survey). Even severe syntactic restrictions on preconditions and effects are required to reach tractability; this is described in the planning-complexity literature as "almost everything is PSPACE-complete or worse" (AAAI SOCS paper on PSPACE-complete planning). This is a proven worst-case complexity ceiling, not an empirical failure - flagged appropriately as a theorem-backed structural bound. Modern planners survive by exploiting problem structure (factored representations, heuristic search, SAT-based planning) rather than by defeating the underlying complexity class, so the entry here refers to naive/general STRIPS planning as a scalable reasoning substrate, not to automated planning as a field.

8.2.6 SHRDLU and the Microworld Program (Symbolic NLU in Closed Domains)

Theorists: Terry Winograd (MIT, 1970-72); critiqued by Hubert Dreyfus and John Haugeland.

Structural mechanism: Integrated syntax-semantics-planning system operating over a simulated "blocks world" with a 200-word vocabulary, procedurally representing meaning as programs rather than static logic.

Reason for rejection - empirical/structural failure to close the sensory-motor and commonsense loop, not a theorem: SHRDLU's competence was confined to a closed, fully specified microworld; attempts to extend the same architecture to realistic, open-ended perceptual and linguistic ambiguity failed to scale (SHRDLU, Wikipedia; Haugeland on the failure of microworlds). This is explicitly flagged as an empirical/engineering dead end rather than a formal impossibility proof: no theorem was proven that microworld methods cannot scale; rather, the approach was abandoned when it became clear that hand-coded, closed-world semantics could not absorb the combinatorial variety of real-world reference and context (a variant of the frame problem, below). Historiographically, this is a case where "loss of prominence" is frequently conflated with "proof of impossibility" - the correct characterization is pragmatic abandonment under the sheer engineering cost of manually extending closed microworlds, compounded by the frame problem's structural bottleneck (§ 1.7).

8.2.7 Situation-Calculus Frame Axioms as a General Solution to World-State Representation

Theorists: John McCarthy and Patrick Hayes (1969), formalized in first-order situation calculus.

Structural mechanism: Represent action effects and non-effects in first-order logic via explicit "frame axioms" stating what does *not* change when an action occurs.

Reason for rejection - proven combinatorial axiom explosion, not mere inconvenience: In situation calculus, the number of frame axioms required grows with the product of the number of actions and the number of fluents (state variables), since each action must be paired with an axiom asserting non-effect on every fluent it does not affect (Frame problem, Wikipedia; Stanford Encyclopedia of Philosophy, Frame Problem). This combinatorial growth is a structural property of the naive axiomatization, not an approximation error, and was addressed formally by successor-state axioms (Reiter) and closed-world/circumscription techniques that collapse the axiom count back to linear in the number of fluents - meaning the *naive* frame-axiom approach specifically is superseded, while situation calculus itself persists in restructured form (Frame Problem overview, ScienceDirect). Philosophically, Stevan Harnad

and others argued the deeper "symbol grounding" version of the frame problem is a symptom of ungrounded symbolic representation as such (Harnad, Frame Problem as Symptom of Symbol Grounding) - this broader claim remains a contested philosophical position, not a mathematical proof, and should not be overstated as settled.

8.2.8 Cyc and Hand-Coded Universal Common-Sense Ontologies

Theorists: Douglas Lenat (1984-2023), Cycorp.

Structural mechanism: A hand-built first-order-logic-plus-extensions knowledge base and inference engine intended to encode all human consensus "common sense" as approximately 10^6 hand-crafted axioms over 10^5 general concepts, from which further learning would bootstrap (Lenat, CYC, ACM).

Reason for rejection - empirical failure of the knowledge-acquisition bottleneck at civilizational scale, not a formal proof: After roughly forty years and an estimated 2,000 person-years of manual axiom entry (25 million assertions), Cyc reached paying enterprise customers but never achieved the anticipated "common-sense breakthrough" or self-sustaining automated learning (Geschichte der Informatik, Lenat and Cyc). A detailed post-mortem based on archival research explicitly frames the project as failed relative to its founding AGI ambitions (Cyc, yuxi.ml). This is squarely an empirical/engineering dead end - the rate of manual axiom entry did not scale sub-linearly with the space of commonsense propositions required, and no combinatorial or uncomputability theorem was ever proven against the approach. Notably, Lenat's final co-authored paper (with Gary Marcus, weeks before his death) argued large language models still need exactly the structured commonsense apparatus Cyc built - indicating the historiographically correct reading is "engineering bottleneck plus prominence loss to a rival paradigm," not "proven impossibility" (Geschichte der Informatik).

8.2.9 Fifth Generation Computer Systems Project (Japan, ICOT, 1982-1992)

Theorists: Kazuhiro Fuchi (ICOT); built on Prolog/logic programming and dedicated massively parallel inference-machine hardware.

Structural mechanism: Custom parallel inference machines executing a resolution-based logic-programming language (a parallel variant of Prolog) as the substrate for natural-language understanding, knowledge processing, and reasoning, replacing von Neumann architecture entirely at the hardware level.

Reason for rejection - empirical failure of the hardware/software co-design bet, not a theorem: After a ten-year, multi-billion-yen national program building custom parallel-Prolog inference hardware, FGCS failed to produce competitive AI capability and was outpaced by conventional workstation and, later, statistical-learning approaches (Fifth Generation Computer Systems, Wikipedia). The project is widely cited as "ahead of its time" in some architectural respects (parallel logic inference) but its core wager - that logic-programming-on-custom-silicon was the correct axis of scaling for machine intelligence - was empirically falsified by the market and research trajectory rather than by a specific mathematical impossibility result (Japanese national Fifth Generation project evaluation, ScienceDirect). Flagged explicitly as engineering/strategic failure, not proof-backed dead end.

8.2.10 Expert Systems as General-Purpose Reasoning Engines (Rule-Based Production Systems)

Theorists: Edward Feigenbaum (MYCIN, DENDRAL lineage); commercialized widely 1980-90.

Structural mechanism: Forward- or backward-chaining inference over an explicit if-then rule base authored by knowledge engineers interviewing domain experts, with a separate explanation facility (Rise and Fall of Expert Systems, Breachroad).

Reason for rejection - empirical brittleness and proven knowledge-acquisition bottleneck, not a computability theorem: Rule bases degrade catastrophically outside their narrow training domain because tacit expert knowledge resists full verbalization into discrete rules, and the required rule count grows combinatorially with domain complexity and exception-handling - a structural property of manually authored symbolic rule systems, empirically documented across the second AI winter (roughly 1987-1993) (AI Winter, emergent.wiki). This is explicitly an empirical/engineering brittleness problem (competence mistaken for reliability outside constrained tasks), not a formal uncomputability or complexity-class result - historiographically it should not be conflated with the theorem-backed STRIPS/PSPACE result

above, even though both fall under "symbolic AI decline."

8.2.11 Analog Differential Analyzers and General Analog Computation

Theorists: Vannevar Bush (Rockefeller Differential Analyzer, MIT, 1931-42).

Structural mechanism: Mechanical wheel-and-disc integrators physically instantiating differential equations as continuous physical analogies, rather than discretized symbol manipulation.

Reason for rejection - empirical obsolescence via precision/scaling limits, not a theorem: Analog computation is fundamentally limited by physical component tolerances, noise accumulation, and lack of arbitrary-precision reconfigurability; digital computers' error-correctable, arbitrarily precise, universally reprogrammable symbol manipulation displaced analog machines by the mid-1970s across essentially every application domain (Open University, "A Great Disappearing Act"; Differential analyser, Wikipedia). No impossibility theorem forced this transition - it is a case of one physically realizable computational substrate being empirically dominated by another on cost, precision, and flexibility grounds. Notably, this displaced paradigm is undergoing partial revival in thermodynamic and neuromorphic computing (Category 2), underscoring that "discarded" does not mean "structurally impossible."

8.3 CATEGORY 2: NON-MAINSTREAM WORKING MODELS

8.3.1 Vector Symbolic Architectures / Hyperdimensional Computing

Key theorists: Pentti Kanerva (Sparse Distributed Memory, 1988; hyperdimensional computing formalization); Ross Gayler (VSA terminology); Tony Plate (Holographic Reduced Representations).

Core structural mechanism: Symbols are represented as high-dimensional (typically 1,000-10,000 dimensional) random vectors, with a small algebra of operators - *bundling* (elementwise addition/majority vote for superposition) and *binding* (elementwise multiplication or circular convolution for structured composition) - that implement compositional, symbol-like structure directly in a distributed vector space, without gradient-based training (VSA tutorial slides, TU Chemnitz; Kleyko, Kanerva et al., VSA as computing framework for emerging hardware, arXiv). Because the near-orthogonality of random high-dimensional vectors is a statistical/geometric property of high-dimensional spaces (concentration of measure), binding and unbinding operations are approximately invertible with high probability, giving VSAs a mathematically principled combinatorial memory that is both symbolic (compositional, structure-sensitive) and distributed (noise-robust, hardware-parallelizable) (Frontiers in Robotics and AI, Symbolic Representation with Hyperdimensional Computing). VSAs are in active production use for low-power edge classification, robotics, and neuro-symbolic hybrid systems that pair them with neural front-ends, and are explicitly positioned outside the backpropagation/transformation mainstream despite occasional hybridization (Purdue/Georgia Tech, Neuro-Symbolic Architecture Meets LLMs).

8.3.2 Sparse Distributed Memory

Key theorist: Pentti Kanerva (NASA Ames, 1988).

Core structural mechanism: A generalized content-addressable RAM over long (e.g., 1,000-bit) binary address/data words, where a datum is written redundantly across many "hard locations" uniformly distributed in a huge binary hypercube, and retrieval proceeds by averaging the contents of all hard locations within a Hamming-distance radius of the query - exploiting the concentration-of-measure property that in high-dimensional binary space, almost all points are approximately equidistant from any reference point, making similarity-based recall robust to partial corruption (Sparse distributed memory, Wikipedia). SDM is mathematically distinct from both classical Hopfield networks and modern transformers, and is used as a cognitive-modeling substrate and, more recently, as a component in some robotic and streaming-memory systems; it remains a live research architecture precisely because its Hamming-distance geometry gives provable robustness properties that gradient-trained dense embeddings lack by construction.

8.3.3 Vernacular Reservoir Computing: Echo State Networks and Liquid State Machines

Key theorists: Herbert Jaeger (Echo State Networks, 2001); Wolfgang Maass (Liquid State Machines, 2002).

Core structural mechanism: A large, fixed, randomly connected (typically 1% connectivity), non-trained recurrent "reservoir" is driven by an input signal, inducing high-dimensional nonlinear transient dynamics; only a linear readout layer mapping reservoir states to outputs is trained, exploiting the reservoir's "echo state property" (fading memory / input-driven contraction) to guarantee well-defined, input-dependent dynamics without backpropagation through time (Echo state network, Scholarpedia; Dynamics and Computational Principles of ESNs, arXiv). Because only the readout is trained (a convex linear regression problem), reservoir computing sidesteps vanishing/exploding gradient issues associated with deep recurrent training entirely. It remains operational in time-series prediction, robotics control, and - notably - physical reservoir computing implementations using photonic, spintronic, or mechanical substrates where the "computation" is performed by the physics of the substrate itself rather than by silicon logic gates (Reservoir computing applications survey).

8.3.4 Free Energy Principle / Active Inference

Key theorist: Karl Friston (UCL, Wellcome Centre for Neuroimaging).

Core structural mechanism: A single variational-Bayesian objective - minimization of variational free energy, an upper bound on the negative log-evidence ("surprise") of sensory data under a generative model - is claimed to unify perception (updating beliefs to reduce free energy), action (acting on the world to reduce free energy), and learning (updating the generative model itself), formally integrating predictive coding with Bayesian filtering under one variational scheme (Friston, Predictive coding under the free-energy principle, Phil. Trans. R. Soc. B; Friston, The free-energy principle: a rough guide to the brain). This is structurally distinct from the transformer/backprop mainstream: it specifies a normative objective function derived from statistical thermodynamics/information theory (free energy as bound on surprise) rather than empirical risk minimization over labeled data, and it is implemented via hierarchical generative models with explicit precision-weighted prediction-error propagation. Active inference implementations are operational in computational psychiatry modeling, robotics control, and neuroscience data analysis as of current literature (Predictive Processing and Active Inference review); it should be flagged that "unifying perception, action and learning" remains a strong normative/explanatory claim under active empirical and philosophical scrutiny rather than a closed, universally accepted result.

8.3.5 Thermodynamic Computing / Probabilistic-Bit (p-bit) Hardware

Key organizations/theorists: Extropic (commercial); builds on stochastic computing and energy-based model theory going back to Hopfield/Hinton-era Boltzmann machines.

Core structural mechanism: Dedicated hardware ("Thermodynamic Sampling Units") implements physical p-bits that fluctuate stochastically between states with controllable, tunable probabilities, so that the chip natively performs Gibbs sampling from a programmable energy-based model rather than deterministically computing a numeric approximation to that distribution on a GPU (Extropic, Thermodynamic Computing: From Zero to One). Because the physical noise of the transistor itself is harnessed as the source of randomness rather than suppressed, system-level analyses report potential energy savings on the order of $10^4\times$ versus GPU-based denoising/diffusion-model inference for equivalent sampling tasks (probabilistic hardware architecture for diffusion-like models, arXiv). This directly repurposes the exact mathematical structure that made general Boltzmann machines intractable to train exactly in software (§ 1.3) - but reframes sampling, not exact-likelihood computation, as the deliverable, and implements it in physics rather than digital logic, making this both a structural rebuttal to and a lineal descendant of the discarded Boltzmann-machine training program.

8.3.6 Spiking Neural Networks on Neuromorphic Silicon (Loihi-class hardware)

Key theorists/organizations: Carver Mead (foundational analog VLSI neuromorphic engineering, 1980s); Intel Labs (Loihi processor line).

Core structural mechanism: Neurons are modeled as event-driven, asynchronous spiking units (e.g., leaky integrate-and-fire or Izhikevich dynamics) communicating via discrete spike events rather than dense synchronous activations, with on-chip local, biologically plausible synaptic learning rules replacing global backpropagation (Loihi: A Neuromorphic Manycore Processor, Davies et al.; In-Hardware Learning of Multilayer SNNs, arXiv). Because backpropagation's exact gradient computation is not directly compatible with the non-differentiable, event-driven dynamics of spiking neurons on real neuromorphic

silicon, working systems substitute local, biologically plausible update rules (e.g., spike-timing-dependent plasticity variants) - a structurally distinct optimization mechanism from mainstream deep learning, not merely a hardware port of it. Carver Mead's original analog VLSI program, from which this lineage descends, is documented in his foundational 1989 text on the topic (Mead, Analog VLSI and Neural Systems).

8.3.7 OpenCog Hyperon / AtomSpace Hypergraph Cognitive Architecture

Key theorist: Ben Goertzel and collaborators (SingularityNET / OpenCog Foundation).

Core structural mechanism: Knowledge is represented in a generalized, typed, weighted hypergraph ("metagraph") called the AtomSpace, over which multiple cognitive processes (probabilistic logical inference, evolutionary program learning, attention allocation) operate concurrently, following the "patternist" theory of mind that defines intelligence as the capacity to recognize, create, and act on patterns across this substrate (OpenCog Hyperon: A Framework for AGI, arXiv; AtomSpace GitHub). Unlike a graph database, the AtomSpace natively represents higher-order relations (hyperedges connecting hyperedges) needed for reified logical statements. This remains an operational neuro-symbolic research architecture, explicitly positioned as an AGI research program distinct from scaling transformer-based LLMs, with an active rewrite (Hyperon, using the MeTTa language) as of 2023-2024 (OpenCog Hyperon paper).

8.3.8 Thousand Brains Theory / Cortical-Columns Sensorimotor Framework

Key theorists: Jeff Hawkins, Marcus Lewis, Scott Purdy, Mirko Klukas, Subutai Ahmad (Numenta).

Core structural mechanism: Rather than a single monolithic model of an object, the theory proposes that thousands of largely independent cortical columns, each processing a different sensory patch and each equipped with grid-cell-like coordinate systems (borrowed explicitly from spatial-navigation neuroscience), independently learn complete sensorimotor models of objects and vote via long-range cortical connections to reach a consensus percept (Numenta, A Framework for Intelligence and Cortical Function Based on Grid Cells in the Neocortex). This is structurally a massively parallel, voting-based, reference-frame-indexed architecture with no centralized weight matrix or global gradient signal, and is under active engineering development as the "Thousand Brains Project," explicitly framed as an alternative, complementary paradigm to deep learning (Thousand Brains Project white paper, arXiv).

8.3.9 Subsumption Architecture / Behavior-Based Robotics

Key theorist: Rodney Brooks (MIT, 1986).

Core structural mechanism: Control is decomposed into a hierarchy of loosely coupled, independently reactive behavioral layers, each a complete sensorimotor loop (perception directly wired to actuation without a shared central world-model), with higher layers able to *subsume* (suppress or override) lower layers' outputs via defined inhibition/suppression wiring, rather than through centralized symbolic planning (Subsumption architecture, emergent.wiki; Brooks subsumption planner, NASA technical report). This explicitly rejects the "sense-model-plan-act" cycle of classical AI robotics on the structural grounds that maintaining a complete, consistent symbolic world model does not scale to real-time embodied control - a design philosophy validated by decades of continued use in commercial and research mobile robotics, distinct from both symbolic planning and modern reinforcement-learning-based control policies.

8.3.10 Reaction-Diffusion / Morphogenetic Computing

Key theorist: Alan Turing (1952, "The Chemical Basis of Morphogenesis"); applied computationally in amorphous computing and swarm robotics.

Core structural mechanism: Computation is performed by the spatiotemporal dynamics of coupled partial differential equations describing reacting and diffusing "morphogen" concentrations, where stable patterns (stripes, spots, gradients) emerge from the interplay of short-range activation and long-range inhibition - a mechanism proven by Turing to produce spontaneous symmetry-breaking pattern formation from a homogeneous initial state (MIT amorphous computing white paper). This has been demonstrated as a genuine, functioning computational substrate for decentralized tasks such as generating robot lane markers in wireless sensor networks, entirely without any neural-network or symbolic-logic component, exploiting instead the mathematics of coupled nonlinear PDEs (Reaction-diffusion computing for robot

lane markers). Recent synthetic-biology work has also demonstrated engineered Turing patterns in living mammalian tissue using a single regulatory function, confirming the mechanism's operational status beyond simulation (Turing patterns in a morphogenetic model, PubMed).

8.3.11 Probabilistic Programming Languages as a Unifying Cognitive Formalism

Key theorists: Noah Goodman, Vikash Mansinghka, Daniel Roy, Joshua Tenenbaum (Church, 2008); later Gen (Cusumano-Towner et al.) and Pyro (Uber AI Labs).

Core structural mechanism: Programs are written in a stochastic extension of the lambda calculus, where ordinary deterministic evaluation is augmented with sampling and conditioning primitives, so that any generative process describable as a program automatically defines a probability distribution over its possible execution traces; a universal **query**/inference operator then performs exact or Monte Carlo posterior inference over arbitrary user-defined programs (Church, ACM UAI proceedings; Goodman et al., Probabilistic programs as a unifying language of thought). This formalism has been explicitly proposed in the cognitive-science literature as a candidate unifying representation for human concept learning, causal reasoning, and theory-of-mind inference - structurally distinct from both symbolic logic (it is inherently stochastic) and neural networks (it is built on program semantics, not differentiable parametric functions), and remains an active modeling paradigm in computational cognitive science as of recent Behavioral and Brain Sciences commentary contrasting it with meta-learning approaches (Probabilistic Programming versus Meta-Learning as Models of Cognition).

8.3.12 Kohonen Self-Organizing Maps as Topology-Preserving Unsupervised Structure

Key theorist: Teuvo Kohonen (1982).

Core structural mechanism: A grid of nodes competes for activation via winner-take-all dynamics against high-dimensional input vectors, with the winning node and its spatial neighbors updated toward the input, so that the map's low-dimensional lattice geometry comes to approximate the topology (neighborhood structure) of the high-dimensional input manifold (Self-organizing map, Wikipedia). This differs structurally from both autoencoders and t-SNE/UMAP-style embeddings in that the output space has an explicit, fixed low-dimensional lattice topology enforced during training itself, not merely as a post-hoc visualization. SOMs remain in active use for exploratory data visualization, vector quantization, and fault/anomaly detection where topological interpretability is required, outside gradient-based representation learning (SOM background, theories, extensions and applications).

8.3.13 Scale-Free / Small-World Cognitive Graph Topology (Connectomics)

Key theorists: Ed Bullmore, Olaf Sporns (network neuroscience); empirical connectome mapping by the Open Connectome Project.

Core structural mechanism: Empirical analysis of structural and functional brain connectivity graphs shows a combination of high local clustering and short average path length (the "small-world" regime of Watts-Strogatz graphs), with degree distributions in several large-scale connectome datasets better fit by a three-parameter generalized Weibull (stretched-exponential) distribution than by a pure power law, indicating a scale-free-*like* but not exactly scale-free topology (Topology of large Open Connectome networks, Nature Scientific Reports; Small-World Brain Networks Revisited, PMC). This is included as a "non-mainstream working model" in the sense that it describes an empirically operating biological cognitive substrate whose graph-topological principles (sparse long-range hub connections plus dense local clustering, minimizing wiring cost while preserving short signal-propagation paths) are studied as design targets for artificial cognitive graph architectures distinct from dense, uniform transformer attention. It must be flagged that the strict "scale-free" (power-law) characterization is contested in the connectomics literature itself - some structures (e.g., the brainstem reticular formation) are shown to be small-world but explicitly *not* scale-free - so claims of scale-free brain topology should not be overstated as settled fact (brainstem reticular formation network paper cited in van den Heuvel et al.).

8.3.14 DNA / Molecular Computing for Combinatorial Search

Key theorist: Leonard Adleman (USC, 1994).

Core structural mechanism: A combinatorial search problem (the directed Hamiltonian path / "traveling salesman" problem) is encoded into DNA oligomer sequences representing graph nodes and edges, and the "computation" is physically performed by molecular biology protocols - hybridization, ligation, PCR amplification, and gel electrophoresis selection - exploiting the massive intrinsic parallelism of $10^{18}+$ molecules reacting simultaneously in a test tube (Adleman, Molecular Computation of Solutions to Combinatorial Problems, Science 1994, cited via MIT OCW). This remains a genuinely operating, non-electronic computational paradigm, with continued application in DNA-based data storage; however, it must be flagged that further *scaling* of DNA computing to large combinatorial problems is currently limited by empirical, not theoretical, error-rate constraints - synthesis and sequencing errors (strand loss, misclassification, asymmetric combinatorial erasure errors) increase sharply as encoded data volume grows and read coverage per sequence falls below roughly 50 reads, a physical/chemical bottleneck rather than a computability result (Physical requirements for scaling up network-based biocomputation, IOP; Error characterization in DNA storage, PMC). DNA computing for combinatorial search is accordingly classified here as operational-but-scale-limited, not discarded.

8.4 Cross-Cutting Historiographic Notes

Several boundary cases merit explicit clarification to avoid overstating impossibility where none has been proven:

Fodor and Pylyshyn's systematicity/compositionality critique of connectionism (1988) is a structural argument, not a formal impossibility theorem: it claims classical (symbolic) architectures are committed to combinatorial syntactic/semantic structure that purely distributed connectionist representations cannot explain without smuggling in symbolic structure. Smolensky's tensor-product and later distributed-representation responses are considered by some (e.g., Chalmers) to defeat the strict form of the argument, and the debate remains empirically contested in current work testing whether meta-learning transformers achieve genuine systematic generalization - recent 2025-era work argues the challenge "still stands" empirically for specific out-of-distribution rule tests (Fodor and Pylyshyn's Systematicity Challenge Still Stands, arXiv; Chalmers, Connectionism and compositionality). This is correctly classified as an ongoing empirical/philosophical dispute, not a settled theorem in either direction.

The symbol grounding problem (Harnad, 1990) is frequently mischaracterized as a proof that symbolic AI is impossible; Harnad's own position is that grounding is "an empirical problem" concerning whether and how symbols acquire intrinsic (non-parasitic) meaning, testable via robotic/embodied Turing-Test-like criteria, not a mathematical impossibility result (Harnad, Symbol Grounding is an Empirical Problem).

Fifth Generation Computer Systems and Cyc are correctly classified as discarded on empirical/strategic grounds; neither failure rests on a combinatorial or uncomputability theorem, and both are frequently and incorrectly cited alongside the perceptron/XOR and Boltzmann-machine partition-function cases as if they shared the same evidentiary status. This report deliberately separates theorem-backed dead ends (§ 1.1, § 1.2, § 1.3, § 1.4, § 1.5, § 1.7) from empirically/strategically abandoned programs (§ 1.6, § 1.8, § 1.9, § 1.10, § 1.11).

Quantum computing for AI and general Boltzmann-machine partition-function intractability intersect directly: it has been formally shown that not even a quantum computer can be expected to efficiently compute Boltzmann machine partition functions in general (the #P-hardness argument applies regardless of the computational substrate, quantum or classical, under standard complexity assumptions), underscoring that this is a computational-complexity result about the problem itself, not an artifact of classical hardware (Microsoft Research arXiv:1412.3489).

8.5 Summary Comparison Table

Single-layer perceptron

Discarded - XOR non-linear-separability - Proven theorem (Minsky-Papert)

Classical Hopfield network (capacity)

Discarded (scoped) - Exponential spurious minima at $0.15N$ capacity - Proven statistical-mechanics bound

- General/deep Boltzmann machine (exact training)
 - Discarded (scoped) - #P-hard partition function - Proven complexity-theoretic result
- Exhaustive game-tree search
 - Discarded - Combinatorial explosion (10^{120} chess states) - Proven combinatorial bound
- STRIPS classical planning (general)
 - Discarded (scoped) - PSPACE-completeness - Proven complexity-theoretic result
- SHRDLU / microworlds
 - Discarded - Failure to scale beyond closed domain - Empirical/engineering
- Situation-calculus frame axioms (naive)
 - Discarded (scoped) - Axiom-count combinatorial growth - Proven structural, later resolved by successor-state axioms
- Cyc
 - Discarded - Knowledge-acquisition bottleneck at scale - Empirical/engineering
- Fifth Generation Computer Systems
 - Discarded - Strategic/hardware bet failure - Empirical/engineering
- Expert systems (general rule-based reasoning)
 - Discarded - Brittleness, knowledge bottleneck - Empirical/engineering
- Analog differential analyzers
 - Discarded - Precision/scaling displaced by digital - Empirical/engineering
- Vector Symbolic Architectures / HDC
 - Non-mainstream working - High-dimensional near-orthogonality - Operational, statistically grounded
- Sparse Distributed Memory
 - Non-mainstream working - Concentration of measure in Hamming space - Operational
- Reservoir computing (ESN/LSM)
 - Non-mainstream working - Fixed random reservoir + linear readout - Operational
- Free energy principle / active inference
 - Non-mainstream working - Variational free-energy minimization - Operational, contested scope of claims
- Thermodynamic computing (p-bits)
 - Non-mainstream working - Physical Gibbs sampling in hardware - Operational
- Spiking neural nets / neuromorphic silicon
 - Non-mainstream working - Event-driven local learning rules - Operational
- OpenCog Hyperon / AtomSpace
 - Non-mainstream working - Typed hypergraph + multi-process cognition - Operational
- Thousand Brains Theory
 - Non-mainstream working - Parallel cortical-column voting - Operational (research stage)
- Subsumption architecture
 - Non-mainstream working - Layered reactive sensorimotor loops - Operational
- Reaction-diffusion computing
 - Non-mainstream working - Turing pattern formation (PDE dynamics) - Operational
- Probabilistic programming
 - Non-mainstream working - Stochastic lambda calculus - Operational
- Kohonen self-organizing maps
 - Non-mainstream working - Competitive topology-preserving learning - Operational

Scale-free/small-world connectomics

Non-mainstream working - Empirical brain graph topology - Operational, contested exact power-law claim

DNA/molecular computing

Non-mainstream working (scale-limited) - Molecular parallelism; error-rate ceiling - Operational, empirically scale-limited

Chapter 9

Heterodox AI Archaeology: A Code-Level Audit of Non-Mainstream Artificial Intelligence Architectures, 2000-2026

9.1 Scope and Method

This report catalogs algorithmic and software-level implementations of non-mainstream AI models, explicitly excluding standard deep learning (CNN/RNN/Transformer supervised stacks), large language models, and classical GOFAI (production rules, STRIPS planners, unification-based theorem provers). It applies a conservative "working" bar: an architecture qualifies as functionally demonstrated only if a reproducible codebase exists and shows bounded task performance (a benchmark, dataset, or physical demonstration), not general intelligence. Three strict categories are used: (1) Unrecognized but Working - heterodox, code-reproducible, benchmarked, but ignored by the mainstream discourse; (2) Unrecognized and Non-Working - fringe/esoteric approaches that failed at the implementation or scaling level, independent of publicity; (3) Recognized but Non-Working/Stalled - architectures that received substantial publicity yet stalled or failed to scale in software, with the empirical non-scaling evidence distinguished from mere project abandonment.

9.2 Category 1: Unrecognized but Working

These architectures have reproducible, open-source implementations and produce bounded, benchmarked results, but sit outside the deep-learning/LLM mainstream discourse.

9.2.1 Active Inference / Free Energy Principle Agents - **pymdp**

Framework. Active inference casts perception and action as joint minimization of variational free energy $F(\mu, a; s)$, an upper bound on Bayesian surprise $-\log p(s)$, over a generative model factorized into external, sensory, action, and internal-state densities (Wikipedia, Free Energy Principle). The joint optimization is

$$\mu^* = \arg \min_{\mu} F(\mu, a; s), \quad a^* = \arg \min_a F(\mu^*, a; s).$$

Implementation. **pymdp**, developed by Heins, Millidge, Demekas, Klein, Friston, Couzin, and Tschantz, is an open-source Python package for simulating active inference agents on partially observable Markov decision processes (POMDPs), the first such package outside MATLAB's SPM/DEM toolbox (**pymdp** GitHub; Heins et al., JOSS 2022). Its low-level NumPy routines are benchmarked and validated directly against SPM's MATLAB reference implementation, giving a numerically cross-checked codebase rather than a purely theoretical proposal. The library computes expected free energy over policies to drive explicit "epistemic value" (curiosity)-seeking behavior in partially observable grid-world and foraging tasks - a reproducible, bounded demonstration, not a general-intelligence claim.

Status. Actively maintained (1,443+ commits as of the 2025 snapshot), used across academic active-inference research; it remains marginal relative to RL/LLM tooling despite functioning correctly on its stated task class.

9.2.2 Sensorimotor / Thousand Brains Systems - Numenta's Monty / Thousand Brains Project

Framework. The Thousand Brains Theory posits that cortical columns each learn complete sensorimotor models of objects via voting and hierarchical/heterarchical composition, rather than a single global feature hierarchy (Clay, Leadholm & Hawkins, arXiv:2412.18354). The architecture (learning modules, sensor modules, a Cortical Messaging Protocol, graph-based long-term object memory, and evidence-based hypothesis updating) is described in exhaustive implementation detail in the 2024 paper, including graph building/updating algorithms and voting-based rapid consensus across learning modules.

Implementation. `tbp.monty` is the reference open-source implementation ("Monty"), released under an MIT license; it explicitly benchmarks itself on sensorimotor object-recognition tasks and publishes rerun results ("Current Performance") after every functional code change (`tbp.monty` GitHub). Numenta's own announcement frames it as "early research code," explicitly disclaiming production readiness (Numenta press release, Nov 2024). As of January 2025 the research program spun out into an independent nonprofit (the Thousand Brains Project), separate from Numenta Inc.'s now-commercial CPU-inference business, indicating institutional confidence sufficient for continued, well-funded (Gates Foundation-backed) development rather than abandonment (NeuromorphicCore.ai).

Assessment. Functionally working (reproducible sensorimotor benchmarks, graph memory demonstrably builds and updates object models) but bounded - it demonstrates object recognition and manipulation policies, not general intelligence, and remains almost unknown outside neuromorphic/cognitive-computing circles.

9.2.3 Growing/Neural Cellular Automata

Framework. A differentiable cellular automaton (CA) with a 16-channel-per-cell state vector (RGB + alpha "life" indicator + 12 hidden channels), updated by a shared local rule conditioned on a 3×3 neighborhood via Sobel-filter gradient perception (48-dimensional input $\rightarrow$ dense(128) $\rightarrow$ ReLU $\rightarrow$ dense(16) residual update), trained end-to-end with backpropagation-through-time (Mordvintsev, Randazzo, Niklasson & Levin, *Growing Neural Cellular Automata*, Distill 2020).

Implementation and evidence. The reference implementation (`tf.keras.Model`, 8,300 parameters) is released with a runnable Colab notebook; a separate from-scratch WebGL/GLSL browser implementation with 8-bit-quantized weights powers a public interactive demo. Training uses a persistent sample-pool method (pool size 1024, batch 32) so the target pattern becomes an attractor rather than a transient. Damage-augmented pool training (randomly erasing circular regions from 3 of 8 sampled states per step) yields models that regenerate from arbitrary square, rectangular, or bisecting damage not seen in training - a genuine, reproducible generalization result, not just curve-fitting. A separate rotated-Sobel-kernel experiment shows the trained rule generating correctly rotated morphologies without retraining, evidence of non-trivial invariance in the learned local rule.

Status. Actively cited in the artificial-life community (self-organizing-systems research); functionally successful at the specific tasks of growth, persistence, and regeneration of 2D pixel patterns, but categorized by mainstream ML discourse as a curiosity rather than a competing paradigm.

9.2.4 Lenia - Continuous-State/Space/Time Cellular Automata

Framework. Lenia (Bert Wang-Chak Chan) generalizes Conway's Game of Life to continuous states $A^t(\mathbf{x}) \in [0, 1]$, continuous space (a convolution kernel over a ball-shaped neighborhood of radius R), and continuous time (Δt integration steps): potential $\mathbf{U}^t = \mathbf{K} * \mathbf{A}^t$, growth $\mathbf{G}^t = G(\mathbf{U}^t)$, and state update $\mathbf{A}^{t+\Delta t} = \text{clip}(\mathbf{A}^t + \Delta t \mathbf{G}^t, 0, 1)$ (Wikipedia, Lenia). Conway's Game of Life is recoverable as the discrete special case $R = T = P = 1$.

Implementation and evidence. Chan's public GitHub implementation generated and cataloged over 400 distinct self-organizing "species" ("lifeforms"/"spaceships") exhibiting self-repair, bilateral/radial symmetry, and locomotion - winning the 2018 Virtual Creatures Contest at GECCO and an ALIFE 2018 honorable mention, and named Outstanding Publication of 2019 by the International Society for Artificial Life. Gilpin's follow-on formal result shows any cellular automaton can be represented as (and trained via) a convolutional neural network, mathematically connecting Lenia to the CA-as-RNN literature.

Assessment. A working, reproducible generative dynamical system with genuine emergent self-organization; explicitly *not* trained via gradient descent (unlike Growing NCA) and has no publicly documented Lenia-specific learning algorithm - its "learning" is evolutionary/manual kernel search, so it belongs here as a working generative/self-organizing framework rather than a learning system per se.

9.2.5 Hyperdimensional Computing / Vector Symbolic Architectures - `torchhd`

Framework. HDC/VSA represents information as high-dimensional ("hyper-") vectors and computes via algebraic operators - binding $\otimes$, bundling $\oplus$, and permutation - over groups/rings/fields, enabling structured symbolic composition (e.g., binding SHAPE $\otimes$ CIRCLE) while retaining the noise-robustness of distributed representations (Wikipedia, Hyperdimensional Computing; Wikipedia, Vector Symbolic Architecture). Earlier VSA instantiations include holographic reduced representations, binary spatter codes, and matrix binding of additive terms.

Implementation and benchmark evidence. `torchhd`, built on PyTorch by Heddes et al. (2023) under academic supervision, is a maintained open-source HDC/VSA library implementing these primitive operators as GPU-accelerated tensor ops. Documented benchmark-grade results include: handwritten-digit classification via prototype hypervector summation/comparison (a nearest-prototype dot-product classifier); Rahimi et al.'s 2023 HDC+neural-network hybrid solving Raven's Progressive Matrices at 88% accuracy versus 61% for a neural-network-only baseline, and roughly 250 $\times$ faster than symbolic-logic reasoning baselines on 3 $\times$ 3 grids; and hardware experiments with 760,000 phase-change memory devices performing analog in-memory HDC computation at accuracy comparable to software. HDC is reported to be at least 10 $\times$ more error-tolerant than conventional ANNs, and eliminating error-correction in HDC hardware saves up to 25% of compute cost because single-bit flips leave outputs "close enough" for correct classification.

Assessment. A genuinely working, benchmarked, hardware-co-designed heterodox paradigm (image classification, analogical reasoning, in-memory computing) that remains essentially invisible in mainstream deep-learning discourse, which is dominated by dense low-dimensional embeddings rather than sparse/high-dimensional near-orthogonal codes.

9.2.6 Equilibrium Propagation and Predictive Coding Networks (bounded regimes)

Framework. Equilibrium propagation (Scellier & Bengio, 2017) is a two-phase (free phase, weakly clamped phase) learning rule for energy-based models using only local, symmetric Hebbian-like updates rather than backpropagation's global error signal (arXiv:1602.05179). Predictive coding networks (PCNs) similarly use local prediction-error minimization; recent theoretical work shows PC can approximate or, under stricter conditions, exactly reproduce backpropagation gradients on MLPs, CNNs, and RNNs.

Implementation and benchmark evidence. Multiple independent open-source PyTorch implementations of equilibrium propagation exist and reproduce MNIST-scale results (`smonsays/equilibrium-propagation`; `zeligism/eqprop`). `PCX`, a dedicated predictive-coding library, is benchmarked explicitly for performance and scalability against standard deep-learning baselines on tasks up to CIFAR-10-scale CNNs (Benchmarking Predictive Coding Networks, arXiv:2407.01163; VERSES AI summary). A 2025 paper on "Scaling Equilibrium Propagation to Deeper Neural Network Architectures" provides open code addressing depth-scaling limits directly.

Assessment. Working and reproducible at small-to-medium scale (MNIST/CIFAR-10-class benchmarks) with genuine biological-plausibility motivation (local learning rules), but - as detailed in Category 3 below - known and documented to degrade at larger scale, placing the **frontier** of this technique on the boundary between Category 1 and Category 3 depending on task size.

9.2.7 Neuroevolution and Evolution Strategies at Scale - NEAT, OpenAI-ES

Framework. NeuroEvolution of Augmenting Topologies (NEAT; Stanley & Miikkulainen, 2002) evolves both neural network weights and topology via genetic operators with historical marking (innovation numbers) to solve the topology-crossover problem (NEAT overview; Stanley & Miikkulainen, JAIR). Evolution Strategies (ES) instead perturbs a fixed-topology parameter vector with Gaussian noise and estimates a gradient-free update from fitness-weighted perturbations.

Implementation and benchmark evidence. `neat-python` and multiple GitHub NEAT implementations

are widely used, reproducible, and pedagogically standard. OpenAI's 2017 ES paper and its open-source implementation demonstrated that a derivative-free population-based method run on an 80-machine/1,440-core cluster rivals standard deep RL on Atari and MuJoCo continuous-control benchmarks while parallelizing better than backpropagation-based RL, using a communication trick (shared random seeds) to avoid transmitting full parameter vectors (OpenAI, Evolution Strategies; Salimans et al., arXiv:1703.03864). As of 2025, Evolution Strategies has been shown to fine-tune billion-parameter language models without backpropagation on math/ARC-AGI/Sudoku-style reasoning benchmarks (Cognizant AI Lab), demonstrating continued scaling headroom for a 1970s-lineage heuristic search method.

Assessment. Functionally working across three decades, with modern benchmark-grade scaling results; systematically under-covered relative to gradient-based RL despite comparable or complementary performance on specific task classes.

9.2.8 Reservoir Computing - Echo State Networks and Liquid State Machines

Framework. Reservoir computing separates a fixed, untrained, high-dimensional nonlinear dynamical "reservoir" (recurrent network) from a simple trained linear readout layer. Echo State Networks (ESNs) use continuous-valued recurrent units; Liquid State Machines (LSMs) use spiking (leaky integrate-and-fire) units (An experimental unification of reservoir computing methods, PubMed).

Implementation and benchmark evidence. Reference implementations include **IGITUGraz/LSM** (Python/NEST) and numerous ESN codebases; a 2024 tutorial review catalogs standardized reservoir-computing benchmarks (arXiv:2405.06561). Physical (non-digital) reservoir computing has working demonstrations in optics (optical ESN, arXiv:2504.08224), memristive nanowire networks, and active colloidal-oscillator systems, each reporting measured task performance (time-series prediction, pattern/frequency classification) rather than simulation-only results. An "Extended Liquid State Machine" reports a 5.2% accuracy gain with a 20.2% spike-count reduction on benchmark speech-recognition datasets, evidence of genuine, quantified improvement over baseline LSMs.

Assessment. A mature, reproducible, benchmarked heterodox paradigm, especially strong in physical/analog and neuromorphic hardware contexts, that remains outside mainstream deep-learning attention because its performance ceiling on large-scale tasks is below that of trained deep networks - it is a working but niche technique, not a stalled one.

9.3 Category 2: Unrecognized and Non-Working

These are fringe or biologically over-literal approaches whose code-level or physical implementations hit fatal, documented technical barriers rather than merely lacking popularity.

9.3.1 DNA Computing for AI/Decision Tasks

Framework and implementations. DNA computing encodes computation in molecular reactions: Adleman's 1994 Hamiltonian-path/TSP experiment (TT-100 prototype, 100 μ L DNA solution) is the founding demonstration (Wikipedia, DNA Computing). Subsequent AI-adjacent implementations include a DNA-based artificial neural network (Cherry & Qian, Caltech) recognizing 100-bit handwritten digits via concentration-encoded weight molecules; a DNA tic-tac-toe player (Macdonald, Stefanović & Stojanović, 2002) using DNAzyme logic gates across nine physically separate reaction bins; and Benenson/Shapiro-lineage diagnostic automata that evaluate gene-expression logic (PPAP2B, GSTP1, PIM1, HPN) and release a therapeutic ssDNA payload on positive diagnosis.

Documented failure modes at the code/design level.

Exponential resource blow-up: the number of DNA strand types required by Adleman's design grows exponentially with problem size, motivating (unsuccessful at removing the fundamental limit) tile-assembly alternatives.

No new computability class: DNA computing provides no algorithmic-complexity advantage - an EX-PSPACE problem still requires exponential DNA mass, which becomes physically unrealizable well before useful problem sizes.

Speed: response times are measured in minutes to days rather than milliseconds; a benchmark square-

root circuit requires over 100 hours; the original Adleman experiment took one week and its result could not even be reliably evaluated with contemporary lab techniques.

Gate consumption and no cascading: DNAzyme logic gates are irreversibly consumed on firing and are limited to one-, two-, or three-input functions with "no current implementation for evaluating statements in series" - i.e., no demonstrated method for chaining gates into deep circuits, a fatal limitation for anything resembling a multilayer network.

Biological incompatibility: DNAzymes require metal cofactors (Zn^{2+} , Mn^{2+}) incompatible with *in vivo* deployment; FokI-enzyme-based automata require an enzyme not naturally present at needed concentrations in higher-organism cells.

Reaction leaks: reversible DNA-computing gate designs (dsDNA gates, DNA hairpin complexes) documented persistent "reaction leak" failures preventing reliable gate reuse.

Assessment. A genuine, published, working proof-of-concept at the level of toy combinatorial problems, but the specific attempts to extend it into neural-network-style or cascaded-logic AI architectures hit hard, quantified, non-scaling walls (exponential resource growth, non-cascadable gates, minute-to-day latencies) that are fatal for any AI application requiring compositional depth. This is failure documented in the primary technical literature, not merely obscurity.

9.3.2 Whole-Brain-Literal Simulation as an AI Path - Blue Brain Project

Framework and implementation. The Blue Brain Project (EPFL, from 2005) pursued biologically literal digital reconstruction of cortical tissue using a Blue Gene supercomputer, Michael Hines's NEURON simulator, and empirically reconstructed connectomes, beginning with a single rat neocortical column (10,000 neurons, 10^8 synapses) (Wikipedia, Blue Brain Project). Supporting open-source tools include CoreNEURON (memory/speed-optimized NEURON), BluePyOpt (evolutionary-algorithm-based single-neuron parameter fitting), and a 2022 Topological-Neuronal-Synthesis algorithm generating millions of unique dendritic morphologies from few examples using algebraic topology.

Documented technical failure/limitation. By 2019, project co-director Idan Segev stated the model had become "too heavy for the supercomputers being used at the time," forcing exploration of replacing individually simulated neurons with artificial-neural-network surrogates purely to make the computation tractable - a direct admission that biologically literal simulation does not scale computationally even to a single mouse cortex on contemporary hardware. No behavioral, cognitive, or intelligent-task demonstration was ever produced by the project in two decades; its deliverables were structural atlases and simulation infrastructure, not functioning AI. The project formally ended in December 2024. Its sister initiative, the 1.3 billion EU Human Brain Project, is widely reported (including in mainstream science press) to have failed to deliver on Henry Markram's founding 2009 claim that "a detailed, functional artificial human brain can be built within the next 10 years" (Wikipedia, Human Brain Project; The Atlantic retrospective; Nature editorial, 519389a).

Assessment. A code-level, computational (not merely funding-related) failure: the core premise - that neuron-for-neuron biological fidelity is a viable computational substrate for intelligence - was falsified by the project's own admission that it had to abandon biological literalism (substituting ANN surrogates) purely for tractability, after two decades and roughly \$1.3 billion in EU funding for the related Human Brain Project. This is a scaling failure, not a decision to stop for lack of interest.

9.3.3 Quantum Neural Networks - Barren Plateau Trainability Collapse

Framework. Quantum neural networks (QNNs) parametrize variational quantum circuits and train rotation-gate parameters via a quantum analogue of gradient descent, seeking exponential Hilbert-space representational advantages over classical networks.

Documented failure mode. For n -qubit random quantum circuits with depth d , the expectation of the objective-function gradient is exactly zero and its variance vanishes exponentially in the number of qubits - the "barren plateau" phenomenon. For d -depth circuits built from 2-design gates, gradient magnitude with local observables vanishes at rate 2^{-d} ; even in the more favorable logarithmic-depth regime, the vanishing rate is only polynomially, not exponentially, improved (Toward Trainability of Quantum Neural Networks). This is a proven, code-independent mathematical result about the loss landscape itself:

standard random-circuit-ansatz QNNs become fundamentally untrainable by gradient descent as qubit count scales, regardless of implementation quality, classical optimizer choice, or hardware fidelity. Proposed mitigations (specialized initialization, layerwise training, tree-tensor/step-controlled architectures) address the problem only for restricted circuit classes and depths, not for general large-scale QNNs.

Assessment. A rigorously proven non-scaling result at the algorithmic level (not a mere engineering or hardware-noise problem), making generic QNN architectures a textbook case of a heterodox AI approach failing precisely at the point mainstream deep learning succeeds - gradient-based scaling.

9.3.4 Capsule Networks Under Biologically-Motivated Routing - Documented Non-Scaling

Framework. Capsule networks (Hinton et al., 2000 credibility-network precursor; 2017 dynamic-routing formalization) replace scalar neuron activations with vector "capsules" encoding pose and existence probability, using iterative routing-by-agreement (a softmax/squash procedure over 4-5 iterations) to build parse-tree-like part-whole hierarchies intended to mimic cortical minicolumns (Wikipedia, Capsule Neural Network).

Documented failure mode. A dedicated AAAI 2023 paper, "Why Capsule Neural Networks Do Not Scale: Challenging the Dynamic Parse-Tree Assumption" (arXiv:2301.01583; AAAI proceedings), directly benchmarks the central biological claim - that capsules learn an emergent dynamic parse tree - on AffNIST (affine-transformation robustness) and CIFAR-10, and finds no such structure emerges at scale; the paper documents "capsule starvation," where routing coefficients collapse pathologically as capsule count grows, and shows performance gains over CNNs vanish or reverse on non-trivial datasets. Independent community assessment (e.g., ML practitioner consensus on discussion forums) converges on the same empirical conclusion: capsule networks remain computationally expensive with negligible benefit over CNNs absent a routing breakthrough that has not materialized as of 2023-2026, to the point that even Hinton reportedly moved on from the architecture.

Assessment. This differs from Category 3's DNC/AIXI cases in that capsule networks never achieved comparable mainstream recognition to warrant "Recognized" status outside the neural-network research community narrowly, and the specific biologically motivated mechanism (parse-tree emergence via routing-by-agreement) is the component shown, with code-level benchmark evidence, not to work - making it a targeted implementation failure of a specific biologically-literal claim rather than a general capsule-network abandonment.

9.3.5 Membrane Computing (P Systems) and Artificial Immune Systems for General AI

Framework. Membrane computing (Păun) models computation via hierarchically nested membrane compartments exchanging objects by biologically inspired reaction/communication rules; artificial immune systems (AIS) model learning/classification via clonal-selection and negative-selection analogues of adaptive immunity.

Failure mode. Both remain confined to simulator-level demonstrations (P-system simulators, AIS intrusion-detection prototypes) that have not been extended into general learning architectures competitive on standard benchmarks; documented practical application is narrow (e.g., niche intrusion-detection AIS variants), and no maintained, benchmarked general-purpose P-system or AIS learning codebase comparable to the frameworks in Category 1 was found in current literature - the code-level ambition (Turing-complete biologically-literal computing substrates for general learning) substantially outstrips any demonstrated learning-task performance, and development activity has not progressed past simulator/proof-of-concept status for AI purposes specifically (as opposed to their continued, separate life as pure theoretical-computer-science formalisms).

Assessment. Distinguished from empirical non-scaling (no benchmark run ever showed a scaling curve bending over) - these are cases of the code-level ambition never reaching a benchmarked working state at all for general AI tasks, i.e., non-working by absence of demonstrated capability rather than by documented regression.

9.4 Category 3: Recognized but Non-Working/Stalled

These architectures received substantial mainstream publicity (major-lab papers, books, press coverage) yet stalled or failed to scale in software, with evidence distinguishing genuine empirical non-scaling from mere project pivots.

9.4.1 Hierarchical Temporal Memory (HTM) / NuPIC - Numenta

Framework. HTM (Jeff Hawkins, *On Intelligence*, 2004) models neocortical minicolumns via sparse distributed representations (SDRs) and two learning stages - spatial pooling (identifying frequent input coincidences) and temporal pooling/sequence memory (learning to predict likely next patterns) - with a "cortical learning algorithm" (CLA) generation modeling active/inactive/predictive cell states in cortical layers 2/3 (Wikipedia, Hierarchical Temporal Memory).

Implementation history and empirical evidence of stalling. The reference open-source implementation, NuPIC (`numenta/nupic`, now `nupic-legacy`), is explicitly labeled by its own maintainers: "As of September 2023 this repository contains code from legacy Hierarchical Temporal Memory (HTM) Numenta projects that have been in maintenance mode for several years," restricted to critical bug fixes only (NuPIC GitHub). The companion `htmresearch` experimental repository was archived (made read-only) by its owner on September 1, 2023, after 5,346 commits, having explicitly warned users that its contents were "temporary, ever-changing experimental code" that Numenta might delete outright (`htmresearch` GitHub). A widely-cited Hacker News technical assessment states: "HTM has been 'out there' for over [a] decade by now, with...not a single tangible result, neither theoretical nor practical. They never managed to hobble together even a single paper with credible results" (Hacker News discussion) - a direct claim of benchmark non-performance, not mere unpopularity. Commercially, Numenta struggled to monetize HTM directly (raising only \$36 million in private funding by 2016 against far larger AI-infrastructure competitors) and by January 2025 formally split its research mission (now the independent Thousand Brains Project, Section 1.2) from its commercial arm, which pivoted entirely to CPU-optimized Transformer/LLM inference acceleration (NuPIC 2.0) - i.e., its commercial product today is a mainstream-Transformer accelerator, not an HTM-based system (NeuromorphiCore.ai).

Assessment. This is a documented, dual-layer stall: (a) empirical - two decades of HTM/CLA research produced no benchmark result competitive with mainstream sequence models on general tasks, with NuPIC's practical footprint narrowing to streaming anomaly detection niches; and (b) organizational - Numenta's own commercial pivot to accelerating the very Transformer architecture HTM was designed to replace is itself evidence that the company judged HTM non-competitive for general AI, distinct from abandonment for lack of funding alone (funding continued, just redirected). The Thousand Brains Project (Category 1.2) represents a theoretically-related but architecturally distinct third-generation successor (sensorimotor Monty framework) - its current bounded working status does not retroactively validate the original HTM/CLA codebase's stalled trajectory.

9.4.2 AIXI and MC-AIXI-CTW - Universal Artificial Intelligence

Framework. AIXI (Hutter, 2000/2005) formalizes optimal general intelligence as a reinforcement-learning agent that simultaneously weighs every computable environment hypothesis by Solomonoff-style program-length prior $2^{-\text{length}(q)}$ and selects actions maximizing expected discounted reward under this Bayes-mixture-over-all-computable-environments (Wikipedia, AIXI; Hutter, *A Theory of Universal Artificial Intelligence based on Algorithmic Complexity*, arXiv:cs/0004001). AIXI is proven Pareto-optimal and self-optimizing across broad environment classes.

Documented incomputability and approximation. AIXI itself is provably incomputable (it requires summing over all programs on a universal Turing machine), placing it permanently outside direct software implementation. The community-standard tractable approximation, MC-AIXI-CTW (Monte Carlo AIXI with Context-Tree Weighting; Veness, Ng, Hutter, Silver et al.), restricts the environment class and substitutes Monte Carlo tree search plus a computable context-tree-weighted sequential predictor. It achieved genuine, documented working results on simple domains, including partially observable Pac-Man - a real, if narrow, code-level success. However, no numerical benchmark scores, win rates, or systematic comparison to modern RL baselines are part of its standard public documentation, and it has not been meaningfully scaled to complex or high-dimensional environments in the 15+ years since publication. Separately, Hutter and Jan Leike later proved that AIXI's headline "balanced Pareto optimality" claim

is subjective in a way that undermines essentially all of AIXI's prior optimality guarantees - a theoretical, not merely practical, walkback by the framework's own originator.

Assessment. A textbook Category 3 case: AIXI received enormous theoretical/publicity recognition (foundational AGI formalism, taught in universal-AI courses) but its only computable descendant (MC-AIXI-CTW) stalled at toy-game scale, and even the theoretical optimality result motivating the entire program was subsequently shown by its own author to be far weaker than originally claimed. This is a stall grounded in both computability theory (the ideal is uncomputable by construction) and empirical scaling (the approximation never moved past small POMDPs).

9.4.3 Differentiable Neural Computer (DNC) - DeepMind

Framework. The DNC (Graves et al., DeepMind, 2016) extends the Neural Turing Machine with a differentiable external memory matrix, content-based and temporal-link addressing, dynamic memory allocation, and an LSTM controller producing read/write interface parameters - fully end-to-end differentiable and trainable by gradient descent, aiming to combine neural pattern recognition with von-Neumann-style symbolic memory access (Wikipedia, Differentiable Neural Computer).

Demonstrated results and subsequent trajectory. The DNC demonstrated genuine, published working results: navigating and generalizing across transit-network graphs (transferring to unseen networks, unlike memoryless baselines), outperforming LSTM and plain NTM on supervised graph-traversal/sequence tasks, and learning SHRDLU-style block-stacking plans via reinforcement learning with curriculum training. Proposed refinements - sparse memory addressing (via approximate nearest-neighbor methods, claimed to cut time/space complexity "thousands of times"), Adaptive Computation Time, and synthetic-gradient training - indicate an active early research trajectory. However, the publicly available record documents only "relatively simple tasks...solved using conventional programming" as DNC's demonstrated envelope; no widely cited large-scale, real-world-deployed DNC-based system followed in the subsequent decade, and memory-augmented architectures of this specific form were superseded in mainstream practice by attention-based Transformer architectures (excluded from this report's scope but the direct historical successor), which achieved the long-range dependency handling DNC targeted without an explicit differentiable memory matrix.

Assessment. Distinguishable from pure hype-and-abandonment: DNC's small-scale results were real and reproducible (Category-1-grade evidence at the scale tested), but the architecture's research program stalled in the specific sense that it was never scaled to the regime (large models, broad tasks) where it could demonstrate competitive advantage before the field's attention and compute shifted decisively to Transformers - an instance of empirical non-scaling-by-substitution rather than a documented negative scaling result internal to DNC itself.

9.4.4 Cyc / OpenCyc / ResearchCyc - Symbolic Common-Sense Reasoning

Framework. Cyc (Douglas Lenat, begun July 1984 at MCC, continued by Cycorp from 1995) is a hand-built symbolic knowledge base and inference engine using the CycL representation language (extended to higher-order logic by 1989), organized into non-monotonic-contradiction-free "microtheories," with a community-of-agents inference architecture in which specialized heuristic modules compete to make progress on subproblems (Wikipedia, Cyc).

Scale and documented combinatorial/practical failure. By 2002, Cyc had consumed \$60 million and 600 person-years of hand axiom-writing; by 2017 its ontology held 1.5 million terms and 24.5 million hand-written axioms/facts - a scale of manual symbolic encoding without precedent, yet still assessed by machine-learning scientist Pedro Domingos as a "catastrophic failure...for the unending amount of data required to produce any viable results and the inability for Cyc to evolve on its own." OpenCyc, the public-facing release, shrank in scope over its lifecycle (6,000 concepts/60,000 facts in 2002, up to 239,000 concepts/2,093,000 mostly-taxonomic facts in the final 4.0 release) before being shut down around March 2017 explicitly because ecosystem "fragmenting" caused confusion between the limited OpenCyc subset and the full proprietary Cyc system. A 2025 retrospective is explicitly titled "Cyc: Obituary for the greatest monument to logical AGI" (Yuxi Liu, UC Berkeley, cited via Wikipedia). Applications that did ship (Cleveland Clinic biomedical query interface, 2004-2008 Terrorism Knowledge Base, Lycos search disambiguation until 2001, CycSecure network-vulnerability assessment) were narrow, bounded deployments rather than evidence of the general common-sense-reasoning capability the project targeted;

several were discontinued on fixed timelines (Terrorism KB ended 2008; Lycos usage ended 2001).

Assessment. The combinatorial wall here is explicit and widely attributed in the primary/secondary literature: the manual-axiom-encoding approach required unbounded, non-self-improving human labor input per unit of coverage gain, and the project's own public arm (OpenCyc) was terminated due to a code/knowledge-base fragmentation problem it could not resolve. This is a canonical, heavily publicized ("one of the most controversial endeavors of AI history") symbolic-AI program whose core method - hand-encoded logical common sense at ever-increasing scale - hit a documented, admitted combinatorial/economic wall rather than merely losing funding interest; Cycorp itself, per Lenat in 2014, deliberately kept "a very low profile" while continuing quietly, indicating the stall was in capability/method, not solely in visibility.

9.4.5 Capsule Networks (as a headline architecture, distinct from 2.4's routing mechanism)

Capsule networks additionally qualify for Category 3 treatment at the level of the overall architecture's mainstream trajectory: they were introduced by Geoffrey Hinton - the most publicly recognized figure in deep learning - with substantial press coverage (framed as a potential successor to CNNs), yet the 2023 AAAI benchmark paper (Section 2.4) and broad community consensus that capsule networks are "computationally very intensive with negligible to no...benefit over CNN" mark a documented, benchmarked stall of a headline-recognized architecture, distinguishing it from purely fringe non-working approaches by virtue of its origin and initial press profile.

9.4.6 Predictive Coding Networks at Scale

Framework. As in Section 1.6, predictive coding networks use local, biologically motivated prediction-error minimization instead of global backpropagation.

Documented scaling failure. VERSES AI's benchmarking summary states directly: "The scalability issue has been neglected for several reasons...PC matches backpropagation (BP) on small-scale tasks, such as...CIFAR10, but fails to do so for larger" tasks (VERSES AI research blog; underlying paper arXiv:2407.01163). This is a direct, quantified admission - from researchers building the dedicated PCX benchmarking library specifically to characterize the problem - that predictive coding, despite strong small-scale parity with backpropagation and a decade of neuroscience-motivated attention (including formal proofs that PC can approximate or exactly reproduce backprop gradients under specific conditions), degrades relative to backpropagation precisely as network/task scale increases.

Assessment. This is empirical non-scaling in the strictest sense used by this report: a same-team benchmark shows parity at small scale and divergence at larger scale, an internal, code-level regression rather than external substitution or lost interest. It sits at the boundary of Categories 1 and 3 - "working" at bounded (CIFAR-10-class) scale per Category 1's conservative bar, but "stalled" as an aspirant backpropagation replacement at the scale where it would need to compete for mainstream relevance.

9.5 Cross-Cutting Technical Patterns

Several recurring code-level failure signatures emerge across the catalog, useful for distinguishing genuine non-scaling from mere neglect:

Exponential resource blow-up with problem size

DNA computing (strand count), Cyc (axiom count vs. coverage) - Quantified in primary sources; intrinsic to method

Proven vanishing-gradient/trainability collapse

Quantum neural networks (barren plateaus) - Formal mathematical proof, not implementation-specific

Same-team benchmark shows parity-then-divergence with scale

Predictive coding networks (CIFAR-10 vs. larger) - Direct first-party admission

Core mechanism directly falsified by dedicated benchmark

Capsule network dynamic parse-tree/routing (AAAI 2023) - Targeted ablation/benchmark study

Project's own maintainers downgrade to "maintenance mode"/archive

HTM/NuPIC, htmresearch - Repository metadata and public statements

Founding theoretical guarantee later retracted by originator

AIXI balanced Pareto optimality (Hutter & Leike) - Author-issued theoretical correction

Commercial pivot to the architecture's mainstream competitor

Numenta HTM → NuPIC 2.0 Transformer acceleration - Corporate strategy documentation

Substitution by a different paradigm before scale-testing completed

DNC → Transformers - Absence of large-scale follow-up + field-wide attention shift

9.6 Conclusion

Across 2000-2026, the heterodox AI landscape bifurcates cleanly along code-level evidence rather than publicity. Active inference (`pymdp`), sensorimotor Thousand-Brains systems (`tbp.monty`), growing/neural cellular automata, Lenia, hyperdimensional computing (`torchhd`), and reservoir computing all have reproducible, benchmarked, bounded-task implementations that the mainstream deep-learning/LLM discourse simply does not engage with - genuine unrecognized-but-working systems. DNA computing for AI-adjacent logic, biologically-literal whole-brain simulation (Blue Brain), generic quantum neural networks, and capsule-network routing-by-agreement all failed at the level of documented, quantified technical walls - exponential resource growth, proven trainability collapse, or targeted benchmark falsification - independent of their visibility. HTM/NuPIC, AIXI/MC-AIXI-CTW, the Differentiable Neural Computer, and Cyc each received major recognition (books, DeepMind/MCC-Cycorp institutional backing, foundational-theory status) yet stalled by combinations of empirical non-scaling, admitted maintenance-mode downgrades, retracted theoretical guarantees, or displacement by Transformer-based successors - with the historical record in each case distinguishing genuine capability plateaus from simple loss of funding or interest.

Chapter 10

Cognitive Phase Transitions: Neuro-Thermodynamic and Topological Models of Insight, Dissonance, and Paradox Resolution

10.1 Scope and Methodological Framing

This catalog restricts admissible evidence to models that specify a physical or mathematical formalism - order parameters, free energy functionals, entropy production, bifurcation equations, or homological/-topological invariants - applied to neural state transitions during insight, dissonance, or paradox resolution. Purely descriptive-psychological accounts (subjective "Aha!" ratings, behavioral restructuring taxonomies) are excluded unless paired with an explicit dynamical-systems or thermodynamic readout. Each entry is tagged when the paper explicitly models insight/dissonance/paradox resolution, or when it establishes a general physical mechanism (criticality, entropy production, heteroclinic switching) whose application to restructuring is extrapolated rather than tested.

10.2 Insight as a Phase Transition: Foundational Dynamical-Systems Claim

The most direct theoretical anchor is Spivey, Dale, and Anderson's explicit thesis that cognition - including sudden problem-solving shifts - should be formalized as self-organizing criticality and phase transition, not as a linear filter cascade of detection-recognition-response (Spivey, Dale & Anderson, "The Phase Transition in Human Cognition," *Nonlinear Dynamics, Psychology, and Life Sciences*, DOI: 10.1142/S0218127404009806). [- theoretical framing] The paper's mechanistic claim is that behavioral and neural time series near a cognitive transition display the statistical signatures of a physical phase transition: growing fluctuations, critical slowing down, and autocatalytic self-organization, rather than being an emergent metaphor.

Empirically, insight-specific restructuring has been isolated with nonlinear change-point statistics rather than classical linear ANOVA. Using matchstick-arithmetic problems, Graf, Danek, Vaci, and Bilalić demonstrate that only generalized additive (mixed) models (GAMs) and change-point analysis - not linear models - correctly detect the abrupt, nonlinear representational shift characteristic of insight, formally establishing that the "Aha!" is a genuine discontinuity in the state trajectory rather than a smooth accumulation (Graf et al., *Journal of Intelligence* 11(5):86, 2023, DOI: 10.3390/jintelligence11050086).

10.3 Neuro-Thermodynamics: Entropy Production and Broken Detailed Balance

10.3.1 Entropy production as the macroscopic signature of cognitive effort

Lynn, Cornblath, Papadopoulos, Bertolero, and Bassett provide the sharpest available macroscopic thermodynamic formalism for brain state transitions ("Broken detailed balance and entropy production in the human brain," *PNAS* 118(47):e2109889118, 2021, DOI: 10.1073/pnas.2109889118) (PMC8617485). [- general cognitive demand, not insight-specific]

The core formalism: for joint transition probabilities $P_{ij} = \text{Prob}[x_{t-1} = i, x_t = j]$, the information entropy production rate is

$$\dot{S} = \sum_{ij} P_{ij} \log \frac{P_{ij}}{P_{ji}} \quad (1)$$

which is the Kullback-Leibler divergence between forward and time-reversed transition probabilities. $\dot{S} = 0$ if and only if detailed balance holds ($P_{ij} = P_{ji}$); any asymmetry between forward and reverse flows drives the system into a non-equilibrium steady state with positive entropy production. Applied to 590 Human Connectome Project subjects (100-parcel BOLD fMRI, states coarse-grained via hierarchical *k*-means into $k = 8$ clusters, an asymmetric Sherrington-Kirkpatrick-type Ising model used as a synthetic validation system), the brain nearly satisfies detailed balance at rest but breaks it sharply under cognitive load: entropy production is roughly $20\times$ higher during a motor task than at rest, and working memory under high cognitive load produces double the entropy production of the low-load condition at matched motor output - i.e., detailed-balance violation intensifies specifically with cognitive (not just motor) demand and with task error rate.

Relevance to restructuring: while the paper does not model insight/dissonance directly, its mechanism - a directed circulating probability flux ("loop of flux" in state space) generated by microscopic connectivity asymmetries - is mathematically the correct primitive for a symmetry-breaking event: a system with vanishing net circulation (deadlock, competing symmetric attractors) can only escape via the emergence of a nonzero rotational flux component, precisely the mathematical condition under which two competing predictive/representational streams "collide" and net flow, rather than cancellation, results.

10.3.2 Non-equilibrium dynamics as a state-dependent physical variable

Sanz Perl, Bocaccio, Pérez-Ipiña, Laureys, Laufs, Kringelbach, Deco, and Tagliazucchi formalize the same broken-detailed-balance logic using probability-flux curl as a second invariant ("Non-equilibrium brain dynamics as a signature of consciousness," arXiv:2012.10792, published as *Phys. Rev. E* 104:014411, 2021, DOI: 10.1103/PhysRevE.104.014411). Using ECoG in non-human primates and a whole-brain Hopf-bifurcation model (local dynamics parametrized by a bifurcation parameter a : $a < 0$ yields stable fixed points, $a > 0$ yields sustained oscillation, coupled through DTI-derived structural connectivity with global coupling G) fitted to human fMRI across wakefulness, sleep stages, and propofol anesthesia, the authors show entropy production and flux-curl are maximal in conscious wakefulness and collapse toward equilibrium in unconscious states - establishing that the brain's distance from thermodynamic equilibrium is a graded, task/state-dependent physical quantity, not a fixed property of neural tissue. This furnishes the thermodynamic vocabulary (entropy production $\dot{S} = \sum_{ij} P_{ij} \log(P_{ij}/P_{ji})$, rotational flux) in which an insight event would be predicted to appear as a transient entropy-production spike or a sign-reversal of local flux curl, though this specific prediction has not yet been tested at moment-of-insight resolution.

10.3.3 Irreversibility peaking at the phase-transition point

Wang, Wu, Dong, and Yu directly test irreversibility (violation of detailed balance) as a function of proximity to a synchronization phase transition in resting and task-state fMRI ("Irreversibility as a signature of non-equilibrium phase transition in large-scale human brain networks," *Chinese Physics B*, 2025, DOI: 10.1088/1674-1056/adbd27) (IOPscience). They report that the first derivative of irreversibility with respect to global synchrony peaks exactly at the critical point characterized by moderate mean synchronization and maximal synchronization entropy - a direct empirical instantiation of critical slowing/fluctuation growth at a genuine bifurcation, cognitively demanding tasks driving the system further from equilibrium than rest.

10.3.4 Related information-thermodynamic formalisms

Sekizawa, Ito, and Oizumi decompose thermodynamic dissipation of linear Langevin neural systems into oscillatory modes, linking entropy-production rate to oscillatory dynamics explicitly (*Phys. Rev. X* 14:041003, 2024, DOI: 10.1103/PhysRevX.14.041003) (arXiv:2312.03489). Karbowski reviews information thermodynamics applied to neuroscience broadly, including how neural networks might encode particle/state velocity into synaptic weights via non-equilibrium information flow (*Entropy* 26(9):779, 2024, DOI: 10.3390/e26090779) (PMC11431499).

10.4 Criticality and Avalanche Dynamics

10.4.1 Landau-Ginzburg synchronization-avalanche unification

Di Santo, Villegas, Burioni, and Muñoz derive a mesoscopic Landau-Ginzburg theory unifying neuronal avalanches and cortical synchronization as two faces of a single synchronization phase transition (*PNAS* 115(7), 2018, DOI: 10.1073/pnas.1712989115) (arXiv:1801.10356). [- general cortical dynamics] This is a central mathematical scaffold for any insight-as-criticality claim: the model treats scale-free avalanches as emergent at "the edge of synchronization," implying that a cognitive restructuring event, if it recruits large-scale synchronization, should manifest as an avalanche with a diverging correlation length precisely at the moment new semantic structure crystallizes.

10.4.2 Cascades and cognitive state modulate criticality directly

Fagerholm, Scott, Shew, Dinov, Mirzaei, Lorenz, Leech, Sharp, Carmichael, and colleagues show that a focused attentional/cognitive task shifts brain dynamics into subcritical avalanche statistics (reduced dynamic range) relative to an unconstrained default state ("Cascades and Cognitive State: Focused Attention Incurs Subcritical Dynamics," *J. Neurosci.* 35(11), 2015, DOI: 10.1523/JNEUROSCI.3694-14.2015) (PMC4363389). This establishes that criticality is not fixed but is actively tuned by cognitive state, supporting the hypothesis that insight requires a transient re-approach to (or crossing of) the critical point from a task-locked subcritical regime - i.e., cognitive flexibility associated with restructuring should correlate with movement toward criticality.

10.4.3 The entropic brain hypothesis

Carhart-Harris, Leech, Hellyer, Shanahan, Feilding, Tagliazucchi, Chialvo, and Nutt propose that conscious-state "quality" tracks system entropy, with normal waking cognition operating just below a critical point (subcritical) and psychedelic/primary states approaching criticality with associated collapse of default-mode network organization ("The entropic brain," *Front. Hum. Neurosci.* 8:20, 2014, DOI: 10.3389/fnhum.2014.00020) (PMC3909994). Although published just before the 2015 window, this hypothesis is the conceptual precursor for a large 2015-2026 literature (e.g., Cheng, Chen, Li et al., "Theoretical foundations of studying criticality in the brain," *Network Neuroscience*, 2022, DOI: 10.1162/netn_a_00269) treating entropy elevation, not merely synchrony, as the operative variable for cognitive-state transitions, directly bearing on the entropic-deadlock question: insight would correspond to a controlled, transient entropy increase escaping a locally low-entropy (over-constrained) representational trap, followed by re-suppression once new structure stabilizes.

10.4.4 Divergent thinking and creativity avalanche/oscillation signatures

Boot, Baas, Mühlfeld, De Dreu, and van Gaal show widespread delta-band oscillations dissociate "rule convergence" from "rule divergence" during creative idea generation, an oscillatory (not yet avalanche-framed) signature of the two opposing generative regimes that must be reconciled in restructuring (*Neuropsychologia*, 2017, DOI: 10.1016/j.neuropsychologia.2017.07.033).

10.5 Heteroclinic Networks and Winnerless Competition

This is the most mathematically explicit "opposing streams collide" framework in the catalog, developed principally by Rabinovich, Varona, Afraimovich, and Huerta.

10.5.1 Formal apparatus

The Winnerless Competition (WLC) principle models cognitive/neural information processing as sequential switching among metastable clusters, geometrically represented as a heteroclinic contour in phase space: trajectories pass near a sequence of saddle equilibria connected by heteroclinic orbits, with the switching order uniquely determined by input while remaining robust to noise (Afraimovich, Rabinovich & Varona, "Heteroclinic Contours in Neural Ensembles and the Winnerless Competition Principle," *Int. J. Bifurcation and Chaos* 14(4), 2004, DOI: 10.1142/S0218127404009806). [- general sequential cognition, not insight-specific] Two stability theorems are proven governing the trade-off between switching robustness against noise and switching sensitivity to input - the exact mathematical tension implied when

two "opposing predictive streams" must be forced into a determinate resolution rather than indefinite oscillation.

Rabinovich and Varona extend this into a full account of "Discrete Sequential Information Coding: Heteroclinic Cognitive Dynamics" (*Front. Comput. Neurosci.* 12:73, 2018, DOI: 10.3389/fncom.2018.00073) (PMC6137616), and to hierarchical decision-making in "Hierarchical dynamics of informational patterns and decision-making" (*Proc. R. Soc. B* 283:20160475, 2016, DOI: 10.1098/rspb.2016.0475) (PMC4920316), the latter modeling competition among decision strategies as movement along a hierarchical heteroclinic network. Rabinovich, Tristan, and Varona further formalize attentional switching between simultaneous "mental modalities" as a controllable bifurcation of the underlying excitatory-inhibitory network, explicitly stating: "a decision-making process is a controllable bifurcation of the attention strategy" ("Neural Dynamics of Attentional Cross-Modality Control," *PLOS ONE* 8(5):e64406, 2013, DOI: 10.1371/journal.pone.0064406). This is the clearest available bridge between heteroclinic-network mathematics and a bifurcation-as-cognitive-decision claim, though not applied specifically to insight/paradox.

10.5.2 General heteroclinic-network review (2023)

Meyer-Ortmanns provides a synthetic review of heteroclinic networks specifically as a model for metastable states and switching events in brain dynamics, emphasizing sensitivity to external input combined with reproducibility/robustness under perturbation ("Heteroclinic networks for brain dynamics," *Frontiers in Network Physiology* 3:1276401, 2023, DOI: 10.3389/fnetp.2023.1276401) (PMC10663269).

10.5.3 Metastability more generally

Tognoli and Kelso's foundational metastability framework ("The Metastable Brain," *Neuron* 81(1), 2014, DOI: 10.1016/j.neuron.2013.12.022) and its extensions by Alderson, Bokde, Coyle, Kelso, and Maguire ("Metastable neural dynamics underlies cognitive performance across multiple behavioural paradigms," *Hum. Brain Mapp.* 2019, DOI: 10.1002/hbm.25009) and Hellyer, Scott, Shanahan, Wise, and Leech ("Cognitive Flexibility through Metastable Neural Dynamics Is Disrupted by Damage to the Structural Connectome," *J. Neurosci.* 35(24), 2015, DOI: 10.1523/JNEUROSCI.4648-14.2015) establish metastability - dynamic balance between integration and segregation, formally the coexistence of near-saddle "ghost" attractor visits - as the operative regime supporting cognitive flexibility, with structural connectome damage causally reducing metastability and impairing flexibility.

10.6 Free Energy Principle, Predictive Coding Collisions, and Symmetry Breaking

10.6.1 Predictive dissonance: cognitive dissonance reframed under active inference

Kaaronen provides the most direct theoretical reformulation of cognitive dissonance as a predictive-processing phenomenon: dissonance is recast as the cascade of prediction errors propagating across a hierarchical generative model, existing simultaneously at sensory and higher-order cognitive scales, and resolved by either updating beliefs (perceptual inference) or acting on the world (active inference) to minimize surprise ("A Theory of Predictive Dissonance," *Front. Psychol.* 9:2218, 2018, DOI: 10.3389/fpsyg.2018.02218) (PMC6262368). This is the clearest attempt to translate cognitive dissonance directly into free-energy-principle mathematics, though the paper remains largely conceptual rather than delivering closed-form dynamics.

10.6.2 Symmetry breaking as an explicit organismal principle under the free energy principle

Kim explicitly states: "Organisms are non-equilibrium, stationary systems self-organized via spontaneous symmetry breaking and undergoing metabolic cycles with broken detailed balance," and derives that the brain develops "optimal trajectories in neural manifolds" and induces "a dynamic bifurcation between neural attractors" during active inference ("Free energy and inference in living systems," *Interface Focus* 2022, DOI: 10.1098/rsfs.2022.0041). [- general organismal FEP, applicable by extension to restructuring] This is the most explicit primary-source use of "symmetry breaking" language for a Bayesian-brain/free-energy account, providing formal grounding for framing insight/paradox resolution as a bifurcation between competing neural attractors representing rival semantic hypotheses.

10.6.3 Self-orthogonalizing attractor networks from the free energy principle

Spisak and Friston formalize how attractor networks emerge from the free energy principle applied to a universal partitioning of random dynamical systems, showing that sequential (non-random) data presentation - as opposed to random data - produces asymmetric couplings and non-equilibrium steady-state dynamics, a natural generalization of Boltzmann machines (*Neurocomputing*, 2025/2026, DOI: 10.1016/j.neucom.2026.133472). Attractors on the free-energy landscape encode prior beliefs; new evidence forces re-partitioning of the landscape via learning that "fine-tunes couplings to minimize long-term surprise" - mathematically the mechanism by which an entropic deadlock (near-degenerate competing attractors/priors) is broken by the arrival of new asymmetric evidence, forcing orthogonalization (differentiation) of previously overlapping representations.

10.6.4 Deep temporal active inference and belief updating

Friston, Rosch, Parr, Price, and Bowman formalize hierarchical belief updating over deep temporal generative models, including neuronal process theories that simulate epistemic foraging (*Neurosci. Biobehav. Rev.*, 2017, DOI: 10.1016/j.neubiorev.2017.04.009). This supplies the process-theoretic (neuronal) implementation - theta-gamma coupling emergent from belief updating - relevant to how a hierarchical prediction-error cascade could physically instantiate the "collision of opposing predictive flows" the user specifies, though the paper's application is to language/narrative inference, not insight per se.

10.7 Attractor Dynamics, Bistability, and Perceptual/Conceptual Symmetry Breaking

10.7.1 Noise-induced and circular-inference bistability models

Moreno-Bote, Rinzel, and Rubin's attractor-based framework treats perceptual bistability (the closest tractable laboratory analog of paradox/two-valid-interpretation deadlock) as requiring noise to induce switches between two symmetric attractor basins - without noise, no alternation occurs ("Noise-induced alternations in an attractor network model of perceptual bistability," *J. Neurophysiol.*, 2007, DOI: 10.1152/JN.00116.2007) (PMC2702529). [- models symmetric bistable deadlock, not resolution into new structure] Shpiro, Rinzel, and Moreno-Bote unify noise-driven and adaptation-driven ("fatigue") switching as two regimes of a single dynamical model, explicitly identifying oscillator-vs-attractor regimes as bifurcation-separated behaviors ("Balance between noise and adaptation in competition models of perceptual bistability," *J. Comput. Neurosci.*, 2009, DOI: 10.1007/s10827-008-0125-3). Leptourgos, Denève, Jardri, and Bouttier extend this into a circular-inference model in which feedforward/feedback loop gain changes can convert a stable integrator into a bistable switcher between "highly trusted" interpretations, connecting bistability directly to predictive-coding gain pathology ("A functional theory of bistable perception based on dynamical circular inference," *PLOS Comput. Biol.*, 2020, DOI: 10.1371/journal.pcbi.1008480) (PMC7769606).

10.7.2 ECoG evidence for attractor-like switching dynamics

Melland and Curtu extract genuine attractor-like low-dimensional manifold dynamics from human intracranial ECoG during auditory bistable perception, empirically confirming the theoretical attractor-switching models at the level of measured cortical population trajectories ("Attractor-Like Dynamics Extracted from Human Electrographic Recordings Underlie Computational Principles of Auditory Bistable Perception," *J. Neurosci.*, 2023, DOI: 10.1523/JNEUROSCI.1531-22.2023) (PMC10162465).

10.7.3 Conflict monitoring and the theta/alpha resolution signature

Drew, Torralba, Ruzzoli, Morís Fernández, Sabaté, Pápai, and Soto-Faraco show that binocular-rivalry perceptual competition - the closest EEG-tractable analog of two incompatible representations "colliding" - is preceded by increased fronto-medial theta (5-7 Hz) power (an established cognitive-conflict marker) and followed by decreased parieto-occipital alpha (8-12 Hz), with alpha inversely tracking theta around the moment of perceptual switch ("Conflict monitoring and attentional adjustment during binocular rivalry," *Eur. J. Neurosci.*, 2021, DOI: 10.1111/ejn.15554; preprint DOI: 10.1101/2019.12.19.873141). [- explicit cognitive-conflict framing applied to a competing-representations paradigm] This is a rare case of an observable oscillatory neuro-physical signature (theta rise → alpha fall) time-locked to the resolution of a genuine representational deadlock via winner selection, i.e., a symmetry-breaking event in the classical sense (two energetically near-degenerate percepts, one abruptly dominating).

10.8 Gamma-Band Signatures of Insight ("Aha!") Specifically

Jung-Beeman, Bowden, Haberman, Frymiare, Arambel-Liu, Greenblatt, Reber, and Kounios provide the founding empirical result: a sudden burst of high-frequency gamma-band EEG activity beginning 0.3 s before subjectively reported insight solutions, localized to right-hemisphere anterior superior temporal gyrus, alongside fMRI evidence of increased activity in the same region for insight versus non-insight solutions ("Neural Activity When People Solve Verbal Problems with Insight," **PLOS Biology** 2(4):e97, 2004, DOI: 10.1371/journal.pbio.0020097). Though published just before the review window, this Kounios-Beeman paradigm anchors nearly all subsequent 2015-2026 insight neuroscience and is the canonical "neuro-physical signature" reference point (a discrete, sudden, band-limited energy burst - not a gradual ramp - consistent with a first-order-like transition rather than smooth convergence).

More recent 2025 direct extensions:

Ohkuma, Kurihara, Takahashi, and Osu use fNIRS to show right dorsolateral prefrontal cortex (DLPFC) and superior temporal gyrus (STG) activity increases specifically in the pre-"Aha!" state (constraint relaxation) versus the impasse state, with distinct angular-gyrus activity in failed solvers ("Neural Dynamics of Constraint Relaxation and Problem Representation Changes in Single-Trial Insight Problem Solving," **Behavioural Brain Research**, 2025, DOI: 10.1016/j.bbr.2025.115813). Ogawa, Aihara, and Yamashita use fMRI plus a hidden Markov model to extract discrete brain states during spatial insight problem-solving (matchstick arithmetic), finding the default mode network dominates during insight-based solutions versus executive control network dominance during analytical solutions, with elevated state-transition variability specifically during insight-based solving interpreted as increased cognitive flexibility ("Neural correlates and dynamical brain states of creative insight in a spatial problem task," **Scientific Reports** 15, 2025, DOI: 10.1038/s41598-025-13684-y). This HMM-based discrete-state framing is the most literal 2015-2026 direct match to the user's request for a rigorous state-transition/bifurcation model of insight, though it stops short of extracting explicit order parameters or free-energy landscapes.

Hart, Johnson, and Duran use body/hand motion-capture to detect physically observable "phase change transitions" during insight problem solving, extending phase-transition language to overt embodied kinematics as a proxy neuro-physical signature ("Motion Capture of Phase Change Transitions During Insight Problem Solving," conference proceedings, Semantic Scholar record).

10.9 Topological and Homological Models of Brain-State Transitions

10.9.1 Persistent homology and Betti-number breakdown as symmetry-breaking readout

Wang (2026) proposes the Receptor-Constrained Dynamical Topology (RCDT) hypothesis, using Takens time-delay embedding and persistent homology on a Wilson-Cowan whole-brain model to show that spatially heterogeneous 5-HT_{2A} receptor-gain modulation drives a topological phase transition: a single dominant Betti-1 cycle (one persistent 1-dimensional topological loop, representing constrained, low-dimensional dynamics) fragments into higher-dimensional or disconnected topological structure, operationally defined as "ego dissolution" (preprint, bioRxiv/medRxiv, 2026, DOI: 10.64898/2026.02.04.703742). [- models altered-state topology, not insight per se, but is the most literal "topological bifurcation" formalism found] The model is explicitly falsifiable via receptor-shuffling controls and concentration-topology response curves - structurally the type of formalism the user's "spatial paradox" framing calls for, applied here to pharmacological state change rather than cognitive restructuring, but directly transferable in principle (a semantic paradox could analogously be modeled as forcing a Betti-number change in a task-relevant functional-connectivity simplicial complex).

10.9.2 Temporal Mapper and attractor-transition network extraction

Saggar, Zhang, and Chowdhury build "Temporal Mapper," a topological-data-analysis tool that reconstructs the network of attractor-to-attractor transitions directly from neural time series without requiring a predefined mechanistic model, bridging data-driven and mechanistic modeling of brain-state landscapes ("Temporal Mapper: Transition networks in simulated and real neural dynamics," **Network Neuroscience**, 2022, DOI: 10.1162/netn_a_00301) (PMC10312258). This provides a ready-made empirical pipeline for testing whether an insight event corresponds to a detectable attractor-transition edge in the reconstructed landscape.

10.9.3 General simplicial/topological descriptions of brain dynamics

Petri, Billings, Hlinka, Saggari, and Keilholz review simplicial and topological descriptions of human brain dynamics, framing functional connectivity as a metric space whose topology (rather than raw connectivity) captures continuously-deformation-invariant structural features relevant to state change (*Network Neuroscience*, 2021, DOI: 10.1162/netn_a_00190).

10.10 Quantum-Formalism Models of Paradox and Contradiction (Non-Physical-Brain, Mathematically Rigorous)

Aerts, Broekaert, and Gabora develop an explicit quantum-mechanical (Hilbert-space, collapse-operator) formal model applied directly to the Liar Paradox and to contextually-elicited opinion states, treating non-contextual conscious experience as unitary dynamic evolution and context-dependent ambiguity resolution/decision as a collapse event, with concept combination modeled via entanglement mathematics and explicit demonstration that Bell-inequality violations occur in concept-context interactions ("A case for applying an abstracted quantum formalism to cognition," *New Ideas in Psychology*, 2004/2010, DOI: 10.1016/j.newideapsych.2010.06.002). [for paradox resolution - but explicitly NOT a claim about physical quantum processes in neural tissue] Aerts, Gabora, Sozzo, and Veloz extend this into a full "double-layered structure" theory positing coexisting 'classical logical thought' and 'quantum conceptual thought' layers, the latter responsible for paradoxes and non-classical (order, interference) effects in human judgment ("Quantum Structure in Cognition: Fundamentals and Applications," arXiv:1104.3344, 2011, DOI: 10.48550/arXiv.1104.3344). This body of work must be flagged explicitly: it is a quantum-probabilistic/quantum-cognition formalism, not a claim about literal quantum physical processes in neurons, and its "symmetry breaking" is a formal collapse operator applied to abstracted cognitive Hilbert spaces rather than a measured neurophysical event. A 2025-2026 review paper on quantum-inspired cognitive architectures for creativity/problem-solving makes the non-physicality explicit: it models "superposition, interference, and contextuality... without assuming physical quantum processes in the brain" (Öztürkmen, "Modeling Human Creativity and Problem-Solving with Quantum-Inspired Cognitive Architectures," 2025, DOI: 10.62802/vgpddeb05). [model of insight/creativity, explicitly disclaimed as non-physical]

10.11 Cognitive Dissonance: Electrophysiological and fMRI Neural Correlates

Shestakova, Blagovechchenski, Colosio, Klucharev, and Nikulin report that stronger cognitive dissonance elicits a larger negative frontocentral evoked EEG response resembling error-related negativity, correlated with re-evaluation of decision alternatives, and linked to individual resting-state long-range temporal correlations in frontocentral cortex - connecting a scale-free temporal-correlation signature (a criticality-adjacent metric) to dissonance-resolution behavior ("Neural Mechanisms of Cognitive Dissonance (Revised): An EEG Study," *J. Neurosci.* 37(20), 2017, DOI: 10.1523/JNEUROSCI.3209-16.2017). Murayama, Samejima, Izuma, Matsumoto, Matsumoto, and Sadato show anterior cingulate cortex and dorsolateral prefrontal cortex activity tracks trial-by-trial dissonance magnitude and mediates choice-induced preference change, with striatal representation shifting to match the post-choice belief state ("Neural correlates of cognitive dissonance and choice-induced preference change," *PNAS* 107(51), 2010, DOI: 10.1073/pnas.1011879108). [, pre-2015 but foundational]

Caveat on the "paradoxical brain" MEG study: Levy, Kluge, Hameiri, Lankinen, Bar-Tal, and Halperin's MEG study of paradoxical-thinking interventions on Israeli-Palestinian conflict narratives reports increased inter-subject neural alignment (via multiset canonical correlation analysis) in posterior superior temporal cortex following exposure to self-narrative-exaggerating paradoxical statements, interpreted as a possible shift in narrative interpretation ("The paradoxical brain," *Cerebral Cortex* 34(9):bhae353, 2024, DOI: 10.1093/cercor/bhae353) (PMC11439920). On direct examination, this paper provides no explicit dynamical-systems, bifurcation, attractor, or entropy formalism - it is psychologically framed ("identity threat," "unfreezing," "destabilization") despite superficially matching the user's "paradox" keyword. [EXCLUDED FROM CATEGORY - flagged as a false positive: uses paradox/destabilization language without physical-systems mathematics]

10.12 Summary Table: Direct vs. Adjacent Models

Phase transition framing of cognition

Spivey, Dale, Anderson (DOI) - Self-organized criticality, autocatalysis - Fluctuation growth, critical slowing

Nonlinear change-point restructuring

Graf, Danek, Vaci, Bilalić (DOI) - GAMs, change-point statistics - Discontinuous eye-movement /behavioral shift

Broken detailed balance / entropy production

Lynn, Cornblath, Papadopoulos, Bertolero, Bassett (DOI) - $\dot{S} = \sum P_{ij} \log(P_{ij}/P_{ji})$, flux loops - fMRI state-space flux, 20× entropy rise under load

Non-equilibrium consciousness signature

Sanz Perl, Bocaccio, Deco, Kringelbach, Tagliazucchi (DOI) - Probability-flux curl, Hopf bifurcation whole-brain model - ECoG/fMRI entropy production, flux rotation

Landau-Ginzburg avalanche/synchronization unification

Di Santo, Villegas, Burioni, Muñoz (DOI) - Mesoscopic order parameter, edge-of-synchronization - Scale-free avalanches

Winnerless competition / heteroclinic networks

Rabinovich, Varona, Afraimovich, Huerta (DOI) - Heteroclinic contours, saddle-connected switching - Sequential metastable cluster switching

Free-energy symmetry breaking

Kim (DOI); Spisak & Friston (DOI) - Variational free energy, attractor bifurcation, orthogonalization - Bayesian-brain attractor switching

Predictive dissonance

Kaaronen (DOI) - Hierarchical prediction-error cascade - Multi-scale surprise minimization

Gamma-burst insight signature

Jung-Beeman, Kounios et al. (DOI); Ohkuma et al. (DOI) - (empirical) - 0.3s pre-solution gamma burst, right STG/DLPFC

HMM discrete brain-state insight model

Ogawa, Aihara, Yamashita (DOI) - Hidden Markov Model over network states - DMN/ECN state-transition variability

Topological phase transition (RCDT)

Wang (DOI) - Persistent homology, Betti-1 stability - Simplicial-complex fragmentation

Quantum-cognition paradox/collapse model

Aerts, Broekaert, Gabora (DOI) - Hilbert space, collapse operator, entanglement - (formal, explicitly non-physical)

Binocular-rivalry conflict-resolution oscillations

Drew, Torralba, Ruzzoli et al. (DOI) - (empirical) - Fronto-medial theta rise / posterior alpha fall at switch

Cognitive dissonance ERN/ACC-DLPFC signature

Shestakova et al. (DOI); Murayama et al. (DOI) - (empirical, long-range temporal correlations) - Frontocentral ERN-like negativity

10.13 Assessment of Coverage Gaps

No paper identified in the 2015-2026 window delivers a single unified closed-form model that simultaneously (a) specifies an explicit free-energy or entropy functional, (b) derives a bifurcation or symmetry-breaking condition from first principles, and (c) empirically validates it at the moment of a reported "Aha!," dissonance resolution, or paradox collapse. The field remains split into three non-overlapping

clusters: (1) rigorous non-equilibrium thermodynamic/criticality formalisms validated on generic cognitive load or consciousness-state contrasts (Lynn/Bassett, Sanz Perl/Deco, Di Santo/Muñoz) but not on insight-labeled trials; (2) insight/dissonance-specific empirical neuroscience (Kounios/Beeman lineage, Ogawa/Aihara/Yamashita, Shestakova) that reports discrete-state or oscillatory signatures without deriving an order parameter or free-energy landscape; and (3) abstract mathematical formalisms (heteroclinic networks, quantum cognition, topological data analysis) developed for general cognitive dynamics or altered states and only inferentially transferable to insight/paradox resolution. The Ogawa, Aihara, and Yamashita HMM study and the RCDT topological-transition preprint are the closest 2025-2026 approaches to the requested synthesis, but neither is framed by its authors as a general theory of insight-as-phase-transition.

Chapter 11

Foundational Architectures & Paradigms (2024-2026)

11.1 Overview of Post-Transformer AGI Architectures

From 2024-2026, research on artificial general intelligence (AGI) architectures has shifted beyond classical transformer-based large language models toward families of state-space models (SSMs), recurrent and continuous-time neural networks, neuromorphic and spiking systems, and hybrid reasoning-centric designs. These architectures are motivated by the need for sub-quadratic scaling in sequence length, improved long-horizon temporal reasoning, energy efficiency, and closer alignment with dynamical systems and cognitive theories of intelligence. (Cirone et al., 2024. Theoretical Foundations of Deep Selective State-Space Models) (Gu et al., 2021. Efficiently Modeling Long Sequences with Structured State Spaces) (Ouyang et al., 2022. State space models can express n-gram languages) (Liang et al., 2025. Mixture-of-Mamba: Enhancing Multi-Modal State-Space Models with Modality-Aware Sparsity) (Haller et al., 2024. BabyHGRN: Exploring RNNs for Sample-Efficient Training of Language Models) (Hasani et al., 2022. Closed-form Continuous-time Neural Models)

11.2 Structured State-Space Models (S4, Mamba, and Variants)

Structured state-space models (SSMs) generalize classical linear dynamical systems $x'(t) = Ax(t) + Bu(t)$, $y(t) = Cx(t) + Du(t)$ into deep, trainable sequence operators that replace attention with fast convolutional recurrences. Theoretical work shows that appropriately parameterized SSMs (e.g., S4 and deep selective SSMs) can capture long-range dependencies with reduced memory and compute compared to transformers, by exploiting diagonal or low-rank structure in A and efficient kernel parameterizations in the frequency domain. (Gu et al., 2021. Efficiently Modeling Long Sequences with Structured State Spaces) (Ouyang et al., 2022. State space models can express n-gram languages)

Recent selective state-space formulations introduce multiplicative gating and selection mechanisms over the recurrent dynamics, enabling models such as Mamba to adaptively route information across time while preserving linear-time complexity in sequence length. Empirical studies demonstrate that Mamba-style SSMs match or outperform transformers on language modeling, dynamical system operator learning, and multi-modal pretraining, particularly under long-sequence and resource-constrained regimes. (Liang et al., 2025. Mixture-of-Mamba: Enhancing Multi-Modal State-Space Models with Modality-Aware Sparsity) (Haller et al., 2024. BabyHGRN: Exploring RNNs for Sample-Efficient Training of Language Models) (Hasani et al., 2022. Closed-form Continuous-time Neural Models) (Gu et al., 2021. Efficiently Modeling Long Sequences with Structured State Spaces)

11.2.1 Mathematical Basis

A generic deep SSM layer can be written as a discrete-time recurrence $x_{t+1} = Ax_t + Bu_t$ with output $y_t = Cx_t + Du_t$, where $A \in \mathbb{R}^{n \times n}$ is constrained (e.g., normal plus low-rank) to ensure stability and efficient convolutional implementation. In practice, convolution kernels corresponding to the Green's function of the underlying ODE are precomputed, permitting $O(L \log L)$ or $O(L)$ application to sequences of length L , in contrast to $O(L^2)$ attention. (Ouyang et al., 2022. State space models can express n-gram languages) (Gu et al., 2021. Efficiently Modeling Long Sequences with Structured State Spaces)

Selective SSMs augment the dynamics with input-dependent gating terms, such as $x_{t+1} = g_t \odot (Ax_t) +$

$(1 - g_t) \odot (Bu_t)$, where g_t is produced by a small neural network conditioned on u_t . This yields piecewise-linear or piecewise-affine state transitions that can emulate attention-like behavior while staying within a dynamical systems formalism. (Gu et al., 2021. Efficiently Modeling Long Sequences with Structured State Spaces) (Liang et al., 2025. Mixture-of-Mamba: Enhancing Multi-Modal State-Space Models with Modality-Aware Sparsity)

11.2.2 Expressivity and Language-Theoretic Results

Recent work shows that finite-dimensional SSMs with appropriate parameterizations can express n-gram languages, closing the gap between classical finite-state automata and modern deep sequence models. These results imply that SSMs can represent a broad class of regular languages and, when stacked and combined with nonlinear readouts, approximate context-sensitive dependencies relevant to natural language understanding. (Darban et al., 2024. Water Quality Anomaly Detection Method Based on Attention-Gated Liquid Neural Network.) (Liang et al., 2025. Mixture-of-Mamba: Enhancing Multi-Modal State-Space Models with Modality-Aware Sparsity)

11.2.3 Scaling Laws and Efficiency

Empirical scaling studies indicate that SSM-based models achieve competitive or superior performance at lower FLOPs compared to transformers, especially for long sequences and multi-modal inputs. Mixture-of-Mamba architectures introduce modality-aware sparsity, showing that equivalent loss can be reached at roughly 25-65% of transformer FLOPs at 1.4B parameter scale for text, image, and speech pre-training. These scaling behaviors suggest that SSMs provide a more favorable compute-performance frontier for AGI systems requiring continuous, high-bandwidth perception and action. (Haller et al., 2024. BabyHGRN: Exploring RNNs for Sample-Efficient Training of Language Models) (Hasani et al., 2022. Closed-form Continuous-time Neural Models)

11.3 Continuous-Time and Liquid Neural Networks

Liquid time-constant networks (LTCs) and related continuous-time neural models seek to approximate the dynamics of biological neurons and synapses via ordinary differential equations with time-varying parameters. The closed-form continuous-time neural models literature derives analytical approximations for LTC dynamics, enabling explicit representation of time in the solution and more efficient simulation of continuous-depth networks. (Wahid et al., 2024. A Survey on Neuromorphic Architectures for Running Artificial Intelligence Algorithms)

Liquid neural networks extend LTCs by introducing state-dependent, non-linear dynamics that remain small in parameter count yet exhibit rich temporal behavior, making them attractive for edge deployment and systems that must generalize across non-stationary environments. Recent work combines liquid dynamics with attention-gating mechanisms, producing architectures such as Attention-Gated Liquid Neural Networks that achieve state-of-the-art anomaly detection in irregularly sampled environmental time series while maintaining compact model sizes. (Mienye & Swart, 2024. A Review of Recent Advances in Intelligent Neuromorphic Computing-Assisted Machine Learning for Automatic Anomaly Detection) (Wahid et al., 2024. A Survey on Neuromorphic Architectures for Running Artificial Intelligence Algorithms)

11.3.1 Dynamical Systems Formulation

Continuous-time liquid models typically define neuron states through equations of the form $\dot{x}(t) = f(x(t), u(t), t; \theta)$, where f encodes membrane potential dynamics, synaptic currents, and adaptive time constants. Solutions are obtained either via numerical integration (e.g., Runge-Kutta schemes) or via closed-form approximations that bound the error of the integral representation. (Wahid et al., 2024. A Survey on Neuromorphic Architectures for Running Artificial Intelligence Algorithms)

The explicit appearance of time in the closed-form solution allows liquid models to represent non-uniform sampling, variable time gaps, and temporal invariances more naturally than discrete-time RNNs or transformers, which implicitly assume fixed-step updates. (Mienye & Swart, 2024. A Review of Recent Advances in Intelligent Neuromorphic Computing-Assisted Machine Learning for Automatic Anomaly Detection) (Wahid et al., 2024. A Survey on Neuromorphic Architectures for Running Artificial Intelligence Algorithms)

11.3.2 Sample Efficiency and Low-Resource Regimes

Studies on subquadratic architectures for low-resource language modeling, such as HGRN2-based Baby-HGRN, report that recurrent architectures with carefully designed gating and normalization can outperform transformers on linguistic benchmarks (BLiMP, GLUE, BEAR) when trained with limited tokens. This supports the hypothesis that continuous-time or recurrent networks may offer better sample-efficiency scaling laws than transformers, particularly when coupled with dynamic computation and knowledge distillation. (Haller et al., 2024. BabyHGRN: Exploring RNNs for Sample-Efficient Training of Language Models) (Darban et al., 2024. Water Quality Anomaly Detection Method Based on Attention-Gated Liquid Neural Network.)

11.4 Neuromorphic Computing and Spiking State-Space Models

Neuromorphic computing seeks to implement brain-inspired non-Von Neumann architectures that leverage spiking neural networks (SNNs), event-based processing, and specialized hardware for energy-efficient intelligence. Surveys of neuromorphic architectures emphasize parallelism, asynchronism, and on-device learning, typically programmed via SNNs using bio-plausible neuron models and learning rules such as spike-timing-dependent plasticity (STDP). (Parisi et al., 2019. Continual learning with hebbian plasticity in sparse and predictive coding networks: a survey and perspective) (Alzubaidi et al., 2021. Continual Learning with Neuromorphic Computing: Theories, Methods, and Applications) (Barnell et al., 2024. Spike-Driven YOLO: Ultra Low-Power Object Detection with Neuromorphic Computing)

A 2024 survey on "Neuromorphic Architectures for Running Artificial Intelligence Algorithms" catalogues analog, digital, and mixed-signal neuromorphic chips, highlighting challenges in scaling to large applications and opportunities for integrating advanced synaptic plasticity rules to improve performance. Complementary surveys on neuromorphic continual learning show how sparse, predictive coding networks and Hebbian plasticity can support online, non-i.i.d learning streams with improved energy efficiency and reduced catastrophic forgetting. (Tian et al., 2024. Advancements in Affective Disorder Detection: Using Multimodal Physiological Signals and Neuromorphic Computing Based on SNNs) (Barnell et al., 2024. Spike-Driven YOLO: Ultra Low-Power Object Detection with Neuromorphic Computing) (Parisi et al., 2019. Continual learning with hebbian plasticity in sparse and predictive coding networks: a survey and perspective)

11.4.1 Spiking State-Space Models (SpikingSSMs)

Recent work on SpikingSSMs constructs spiking analogues of deep state-space blocks tailored for long-sequence learning. By hierarchically integrating dendritic neuron dynamics into SSM blocks and enforcing sparse synaptic computation, SpikingSSMs bridge the gap between neuromorphic temporal processing and the sequence-learning capabilities of structured SSMs. (Chollet et al., 2024. ARC Prize 2024: Technical Report)

To reconcile event-driven neuronal dynamics with parallel hardware execution, SpikingSSMs introduce lightweight surrogate dynamic networks that approximate spiking behavior while remaining amenable to batch processing on GPUs or specialized accelerators. Empirical evaluations indicate competitive performance with conventional ANNs on long sequence tasks, while offering potential for neuromorphic deployment due to sparsity and temporal locality. (Alzubaidi et al., 2021. Continual Learning with Neuromorphic Computing: Theories, Methods, and Applications) (Chollet et al., 2024. ARC Prize 2024: Technical Report)

11.4.2 Hardware Prototypes and Energy Scaling

Work on neuromorphic processors such as Intel's Loihi 2 demonstrates ultra low-power object detection via spike-driven adaptations of YOLO, achieving mean average precision over 93% on certain datasets with substantial power savings at the edge. These systems exploit transistor-level event-driven computation and neuromorphic runtimes (e.g., Lava) to realize parallel spike processing beyond classical Von Neumann memory bottlenecks. (Dwivedi et al., 2019. Advancing Abstract Reasoning in Artificial General Intelligence with a Hybrid Multi-Component Architecture) (Parisi et al., 2019. Continual learning with hebbian plasticity in sparse and predictive coding networks: a survey and perspective)

Surveys of intelligent neuromorphic anomaly detection show that SNN-based systems can process com-

plex spatiotemporal patterns in real time, suggesting that large neuromorphic networks scaled with appropriate interconnects could form the substrate of energy-efficient AGI systems. (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey of Abstraction and Reasoning) (Alzubaidi et al., 2021. Continual Learning with Neuromorphic Computing: Theories, Methods, and Applications)

11.5 Hybrid Continuous-Discrete Reasoning Models

Beyond sequence modeling, AGI-oriented architectures increasingly target abstract reasoning benchmarks such as ARC-AGI through hybrid multi-component systems combining transformers, graph neural networks (GNNs), generative adversarial networks (GANs), and reinforcement learning (RL). These architectures attempt to decompose reasoning into modules for pattern recognition, relational inference, generative hypothesis testing, and policy optimization. (Shu et al., 2026. LoopViT: Scaling Visual ARC with Looped Transformers) (Talley et al., 2026. Multi-Perspective Transformers in ARC-AGI-2 Challenge)

Looped and recurrent transformer variants such as LoopViT decouple reasoning depth from parameter count by iterating weight-tied blocks and introducing dynamic exit mechanisms based on predictive entropy, thereby forming latent chains of thought for visual reasoning. Empirical results show that small LoopViT models can outperform much larger feed-forward vision transformers on ARC-AGI-1, hinting that iterative computation and adaptive halting may be critical axes of scaling toward AGI-level reasoning.

11.5.1 ARC-AGI as a Driver of Architecture Innovation

The ARC-AGI benchmark, designed to measure abstraction and reasoning via human-intuitive visual puzzles, remains unbeaten but has spurred significant architectural innovation. The ARC Prize 2024 technical report documents how deep-learning-guided program synthesis, test-time training, and refinement loops have pushed private-set scores from 33% to 55.5%, indicating that dynamic, adaptive computation is essential for progress. (Shu et al., 2026. LoopViT: Scaling Visual ARC with Looped Transformers)

A 2026 living survey on ARC-AGI across multiple versions shows that performance drops dramatically from ARC-AGI-1 to ARC-AGI-3 for all paradigms-program synthesis, neuro-symbolic, and neural-despite near-perfect human performance, underscoring unresolved challenges in compositional generalization and interactive learning. This landscape motivates architectures that integrate symbolic planning, recurrent computation, and environment-interactive loops as foundational components of AGI systems. (Talley et al., 2026. Multi-Perspective Transformers in ARC-AGI-2 Challenge) (Shu et al., 2026. LoopViT: Scaling Visual ARC with Looped Transformers)

11.5.2 Relation to Q*-Style Planning Algorithms

Although detailed public documentation of specific proprietary planning algorithms such as DeepMind's Q* remains limited, open literature on hybrid RL-search architectures and neural planning modules indicates a trend toward combining learned value functions with explicit search over latent action sequences. These methods typically frame decision-making as optimization over multi-step trajectories, using neural networks to approximate transition dynamics and rewards while external planners or tree search algorithms refine candidate plans, aligning AGI architecture design with classical optimal control and planning theory. (Ouyang et al., 2022. State space models can express n-gram languages) (Talley et al., 2026. Multi-Perspective Transformers in ARC-AGI-2 Challenge) (Haller et al., 2024. BabyHGRN: Exploring RNNs for Sample-Efficient Training of Language Models) (Shu et al., 2026. LoopViT: Scaling Visual ARC with Looped Transformers)

11.6 Comparative Architectural Table

The following table summarizes key architectural families relevant to AGI research and their core properties.

Structured SSMs (S4, Mamba)

Linear dynamical systems with selective gating - Linear or near-linear in L via convolutional kernels - Efficient on conventional accelerators; promising for long-horizon tasks - Strong for sequence

modeling; emerging for reasoning via causal variants - (Gu et al., 2021. Efficiently Modeling Long Sequences with Structured State Spaces) (Ouyang et al., 2022. State space models can express n-gram languages) (Liang et al., 2025. Mixture-of-Mamba: Enhancing Multi-Modal State-Space Models with Modality-Aware Sparsity) (Haller et al., 2024. BabyHGRN: Exploring RNNs for Sample-Efficient Training of Language Models) (Hasani et al., 2022. Closed-form Continuous-time Neural Models)

Liquid / continuous-time networks

ODE-based neuron and synapse dynamics with adaptive time constants - Continuous-time integration; effective for irregular sampling - Compact models suitable for edge and non-stationary environments - Good for temporal generalization; reasoning capacity under active study - (Mienye & Swart, 2024. A Review of Recent Advances in Intelligent Neuromorphic Computing-Assisted Machine Learning for Automatic Anomaly Detection) (Wahid et al., 2024. A Survey on Neuromorphic Architectures for Running Artificial Intelligence Algorithms) (Darban et al., 2024. Water Quality Anomaly Detection Method Based on Attention-Gated Liquid Neural Network.)

Neuromorphic SNNs and SpikingSSMs

Event-driven spikes and spiking state-space blocks - Driven by spike events; sparse temporal updates - High energy efficiency on neuromorphic chips (Loihi 2, analog/digital hybrids) - Potential for large-scale brain-like computation; current focus on perception and anomaly detection - (Chollet et al., 2024. ARC Prize 2024: Technical Report) (Tian et al., 2024. Advancements in Affective Disorder Detection: Using Multimodal Physiological Signals and Neuromorphic Computing Based on SNNs) (Parisi et al., 2019. Continual learning with hebbian plasticity in sparse and predictive coding networks: a survey and perspective) (Alzubaidi et al., 2021. Continual Learning with Neuromorphic Computing: Theories, Methods, and Applications) (Dwivedi et al., 2019. Advancing Abstract Reasoning in Artificial General Intelligence with a Hybrid Multi-Component Architecture) (Barnell et al., 2024. Spike-Driven YOLO: Ultra Low-Power Object Detection with Neuromorphic Computing)

Hybrid transformer-GNN-RL reasoning systems

Discrete-time modules with attention, graph message passing, and policy optimization - Typically quadratic in context window but can incorporate subquadratic components - Standard accelerators; cost dominated by transformer components - Designed for abstract reasoning benchmarks (ARC-AGI) and program synthesis - (Shu et al., 2026. LoopViT: Scaling Visual ARC with Looped Transformers) (Talley et al., 2026. Multi-Perspective Transformers in ARC-AGI-2 Challenge)

11.7 Implications for AGI-Oriented Scaling Laws

Across these architectures, an emerging consensus is that AGI progress depends on scaling not only parameters and data but also computation depth, adaptive dynamics, and energy efficiency. SSMs and recurrent/continuous-time models provide subquadratic sequence operators with favorable FLOP-performance trade-offs, while neuromorphic systems promise orders-of-magnitude improvements in energy-per-operation, suggesting a multi-dimensional scaling space for AGI architectures beyond simple width and depth. (Chollet et al., 2024. ARC Prize 2024: Technical Report) (Parisi et al., 2019. Continual learning with hebbian plasticity in sparse and predictive coding networks: a survey and perspective) (Dwivedi et al., 2019. Advancing Abstract Reasoning in Artificial General Intelligence with a Hybrid Multi-Component Architecture) (Haller et al., 2024. BabyHGRN: Exploring RNNs for Sample-Efficient Training of Language Models) (Shu et al., 2026. LoopViT: Scaling Visual ARC with Looped Transformers)

Reasoning-centric benchmarks such as ARC-AGI reveal that test-time training, refinement loops, and iterative computation are crucial for abstraction and generalization, pushing architectures toward designs that embed dynamic control, planning, and interaction as first-class components rather than static feed-forward mappings. In this view, AGI architectures of the 2024-2026 period can be understood as converging toward dynamical, neuromorphic, and hybrid reasoning systems that balance mathematical expressivity, computational scaling, and physical implementability. (Shu et al., 2026. LoopViT: Scaling Visual ARC with Looped Transformers)

Chapter 12

Multi-Agent Systems & Swarm Intelligence for AGI (2024-2026)

Between 2024 and 2026, research on artificial general intelligence (AGI) has increasingly emphasized multi-agent and swarm-based paradigms where collections of autonomous agents—often LLM-powered—coordinate to solve tasks beyond the reach of single models. Surveys of LLM-based multi-agent systems now frame communication, topology, and role specialization as first-class design dimensions, moving from simple self-talk setups to complex ensembles exhibiting emergent cooperation and division of labor. (Kasneci et al., 2023. Multi-Agent Large Language Models for Conversational Task-Solving) (Abbass & Mostaghim, 2025. The road forward with swarm systems) (Gronauer & Diepold, 2021. Factorised Active Inference for Strategic Multi-Agent Interactions)

Swarm intelligence work in robotics and distributed AI similarly stresses decentralized control, local interactions, and scalability, aiming to replace large monolithic systems with many smaller, cheaper, collectively smarter ones. This chapter focuses on three technical vectors for AGI-oriented swarms: non-linguistic latent inter-agent communication, decentralized Byzantine-fault-tolerant consensus and topology, and mathematical objective functions that formalize collective intelligence in multi-agent and swarm frameworks. (Abbass & Mostaghim, 2025. The road forward with swarm systems) (Molina et al., 2020. SwarmIntelligence-Based Multi-Robotics: A Comprehensive Review) (Ouyang et al., 2022. Communicating Activations Between Language Model Agents) (Gronauer & Diepold, 2021. Factorised Active Inference for Strategic Multi-Agent Interactions) (Salah et al., 2019. Verifiable Semantic Consensus for AI Agents in Embedding Space)

12.1 Non-Linguistic / Latent Inter-Agent Communication

12.1.1 Latent Activation and Embedding-Based Communication

Recent work on LLM-based multi-agent systems argues that natural language communication, while interpretable, is inefficient and discards rich internal information; instead, agents can exchange compressed neural activations or embeddings to improve bandwidth and reduce latency. In "Communicating Activations Between Language Model Agents", agents send transformed internal activations via a function f rather than decoded text, allowing downstream agents to operate directly on high-dimensional representations. (Bhattacharya, 2026. Consensus-Driven Metacognition in Multi-Agent Systems: A Logic-Based Byzantine Fault-Tolerant Protocol) (Abbass & Mostaghim, 2025. The road forward with swarm systems)

Experimental results in that work show that activation-based communication yields up to 27% improvement over natural language channels in multi-player coordination games and reasoning benchmarks, while enabling zero-parameter, zero-data task adaptation across agents. The authors highlight that activation exchange preserves gradient-like information about uncertainty and feature salience that is lost during decoding, effectively increasing the information rate per message and lowering inference cost. (Bhattacharya, 2026. Consensus-Driven Metacognition in Multi-Agent Systems: A Logic-Based Byzantine Fault-Tolerant Protocol)

12.1.2 Dimensional Analysis and Redundancy in MARL Messages

In multi-agent reinforcement learning (MARL), communication efficiency has been revisited via dimensional analysis to understand how message embeddings propagate confounders and redundant information

through the communication graph. The dimensional analysis perspective reveals that optimizing message content and timing at the sender side does not suffice; dimensional redundancy persists in integrated embeddings at the receiver, degrading decision quality. (Molina et al., 2020. SwarmIntelligence-Based Multi-Robotics: A Comprehensive Review)

To address this, the authors propose communication architectures that explicitly regularize or factorize message dimensions, aiming to align communication channels with task-relevant latent factors and reduce confusion in high-dimensional embedding space. This suggests a convergence with activation-based LLM communication: both literatures emphasize that non-linguistic messages should be compressed, structured, and task-aligned to yield genuine efficiency gains in multi-agent coordination. (Molina et al., 2020. SwarmIntelligence-Based Multi-Robotics: A Comprehensive Review) (Bhattacharya, 2026. Consensus-Driven Metacognition in Multi-Agent Systems: A Logic-Based Byzantine Fault-Tolerant Protocol)

12.1.3 Semantic Embedding Space and Consensus

By 2026, work on verifiable semantic consensus for AI agents formalizes agreement not at the level of exact token sequences but in embedding space, using cosine distance thresholds to define semantic proximity. The Verifiable Semantic Consensus (VSC) protocol introduces a τ -Quorum algorithm that identifies the largest semantically coherent quorum of at least $2f+1$ agents, whose embeddings lie within distance τ of a centroid, and produces a proof suitable for ZK-SNARK verification. (2026 et al., 2026. Rethinking the Reliability of Multi-agent System: A Perspective from Byzantine Fault Tolerance)

Simulation experiments with up to 51 agents show that VSC achieves over 99% safety and liveness when τ is calibrated to honest agent variance, tolerating up to $f < n/3$ Byzantine agents while maintaining centroid quality above 0.985. This work effectively defines non-linguistic communication and agreement in multi-agent AI systems as geometric constraints in latent representation space, linking consensus protocols to information geometry. (2026 et al., 2026. Rethinking the Reliability of Multi-agent System: A Perspective from Byzantine Fault Tolerance)

12.2 Decentralized Swarm Consensus & Topology

12.2.1 LLM-Based Swarms and Byzantine Fault Tolerance

The reliability of LLM-based multi-agent systems has begun to be studied through the lens of Byzantine fault tolerance (BFT), with specific consensus mechanisms designed to handle agents that may output confident but incorrect or adversarial responses. A 2025 paper rethinks multi-agent reliability and introduces CP-WBFT, a confidence probe-based weighted Byzantine fault-tolerant consensus mechanism for MAS with diverse topologies. (Chadderwala, 2025. Byzantine Fault-Tolerant Multi-Agent System for Healthcare: A Gossip Protocol Approach to Secure Medical Message Propagation) (Dandigam et al., 2026. LLM-Augmented Agentic Consensus Swarms for Autonomous Edge Security)

CP-WBFT leverages intrinsic reflective and discriminative capabilities of LLM agents, using probe queries and weighted information flow to assign trust scores and aggregate decisions under extreme Byzantine conditions; experiments report superior performance across topologies with fault rates up to 85.7%. Complementary work on consensus-driven metacognition in multi-LLM systems proposes MBFT (Metacognitive BFT), which treats agent trust discretely and commits swarm decisions via finite tiers of confidence, defeasible counterproofs, and reputation-gated veto, ensuring mechanical checkability of safety and liveness guarantees. (Dandigam et al., 2026. LLM-Augmented Agentic Consensus Swarms for Autonomous Edge Security) (Chadderwala, 2025. Byzantine Fault-Tolerant Multi-Agent System for Healthcare: A Gossip Protocol Approach to Secure Medical Message Propagation)

12.2.2 Domain-Specific Byzantine Consensus Swarms

In applied domains, Byzantine consensus has been combined with multi-agent swarms to secure high-stakes systems such as healthcare and edge security. A healthcare-focused multi-agent system uses gossip-based message propagation with cryptographic validation and BFT consensus among diagnostic and planning agents, tolerating up to f faulty nodes among $n = 3f + 1$ and achieving 100% consensus accuracy with up to 33% Byzantine nodes on medical message validation tasks. (Fister et al., 2013. A comprehensive review of firefly algorithms) (Feng & Sun, 2024. Research on Division of Labor Decision and System Stability of Swarm Robots Based on Mutual Information)

Similarly, the LLM-Augmented Agentic Consensus Swarm (LAACS) framework equips each edge gateway with a lightweight LLM and a hybrid semantic-numerical consensus protocol, allowing agents to perform collaborative threat detection and validation via numerical confidence scores and semantic threat representations. Experiments on three real-world edge datasets show an average detection rate near 94.7%, suggesting that decentralized consensus swarms can provide resilient defense under resource constraints. (Feng & Sun, 2024. Research on Division of Labor Decision and System Stability of Swarm Robots Based on Mutual Information)

12.2.3 Dynamic Communication Topologies and Agent Pruning

A key challenge in LLM-based multi-agent systems is designing communication topologies that balance redundancy and performance; AgentDropout proposes dynamic agent elimination to improve token efficiency and task performance. Inspired by organizational management theory, AgentDropout monitors agent contributions and identifies redundant agents whose participation does not significantly improve swarm outputs, removing them from the communication graph. (Douchan et al., 2019. Swarms can be rational)

Experiments show that dynamic elimination can reduce communication load and improve performance, implying that topologies should be adaptive rather than static, with agents and edges added or removed based on contribution metrics. Surveys of LLM multi-agent systems emphasize that communication graphs range from simple fully connected broadcasts to hierarchical and role-based structures, and that dynamic graph reconfiguration is an emerging trend for scalable AGI-oriented swarms. (Kasneci et al., 2023. Multi-Agent Large Language Models for Conversational Task-Solving) (Abbass & Mostaghim, 2025. The road forward with swarm systems) (Douchan et al., 2019. Swarms can be rational)

12.3 Swarm Intelligence and Graph-Based Topologies

12.3.1 Swarm Robotics and Multi-Robot Systems

Swarm intelligence research in multi-robot systems provides foundational topology and protocol designs for AGI-like swarms, often using local interaction rules defined on dynamic graphs. A 2024 comprehensive review of swarm-intelligence-based multi-robotics highlights algorithms such as particle swarm optimization, ant colony optimization, and firefly algorithms, adapted to distributed robotics where robots coordinate via local sensing and communication to achieve global objectives. (Gronauer & Diepold, 2021. Factorised Active Inference for Strategic Multi-Agent Interactions) (Prakki, 2024. Active Inference for Self-Organizing Multi-LLM Systems: A Bayesian Thermodynamic Approach to Adaptation) (Salah et al., 2019. Verifiable Semantic Consensus for AI Agents in Embedding Space)

In these systems, communication topologies are typically modeled as time-varying graphs where edges represent local proximity or communication links, and consensus or task allocation emerges from repeated local updates rather than central orchestration. Recent work on division-of-labor decisions and system stability in swarm robots uses mutual information to measure information interactions among robots and game-theoretic models (Cournot and Stackelberg) to formalize how information flow influences stability and role differentiation. (Salah et al., 2019. Verifiable Semantic Consensus for AI Agents in Embedding Space) (Otoshi & Murata, 2024. Distributed Beamforming by Multi-Agent Active Inference)

12.3.2 Rational Swarms and Collective Utility

The theme issue "The road forward with swarm systems" includes work showing that swarms can be formally rational: with appropriate learning mechanisms, individual reward functions can be designed so that each agent independently maximizes both its own reward and a collective utility, yielding an equilibrium where swarm and individuals are simultaneously rational. (Albarracin et al., 2024. Shared Protentions in Multi-Agent Active Inference) (Gronauer & Diepold, 2021. Factorised Active Inference for Strategic Multi-Agent Interactions)

This result demonstrates that local learning rules can be shaped to align individual and global objectives, providing a mathematical basis for designing AGI-oriented swarms where rationality is shared rather than imposed centrally. It also suggests that collective intelligence can be formalized via joint optimization over agent policies and utilities on dynamic interaction graphs, bridging game theory and swarm learning.

(Otoshi & Murata, 2024. Distributed Beamforming by Multi-Agent Active Inference) (Albarracin et al., 2024. Shared Protentions in Multi-Agent Active Inference)

12.4 Mathematical Objective Functions for Swarms

12.4.1 Active Inference in Multi-Agent Setups

Active inference, grounded in the free energy principle, has emerged as a unifying framework for perception, learning, and action, and recent work extends it to multi-agent systems and high-dimensional, language-driven environments. In "Active Inference for Self-Organizing Multi-LLM Systems", an active inference layer sits above an LLM-based agent, modeling environment states and observations via a generative model and selecting actions that minimize variational free energy and expected free energy, thereby guiding prompt adjustment and search strategies. (Yeganeh et al., 2025. Active Inference Meeting Energy-Efficient Control of Parallel and Identical Machines) (Ouyang et al., 2022. Communicating Activations Between Language Model Agents)

Experiments in that work show emergent structure in observation matrices and sophisticated exploration-exploitation patterns: agents transition from broad information gathering to targeted prompt testing as they learn environment dynamics. This Bayesian thermodynamic layer effectively defines a mathematical objective for multi-LLM systems in terms of free energy minimization, turning collective adaptation into a principled optimization process over generative model parameters and action policies. (Yeganeh et al., 2025. Active Inference Meeting Energy-Efficient Control of Parallel and Identical Machines)

Factorised active inference for strategic multi-agent interactions further develops multi-agent generative models where each agent maintains explicit beliefs about other agents' internal states, integrating active inference with game theory. The authors analyze ensemble-level expected free energy (EFE) in iterated general-sum games, showing that basins of attraction for games with multiple Nash equilibria can be characterized via EFE landscapes and that EFE is not necessarily minimized at the aggregate level, highlighting tensions between individual and collective objectives. (Ouyang et al., 2022. Communicating Activations Between Language Model Agents)

12.4.2 Shared Protentions and Group Intentionality

Work on "Shared Protentions in Multi-Agent Active Inference" unites Husserlian phenomenology, active inference, and category theory to formalize group intentionality and shared goals. The framework models shared protentions-collective anticipations of future states-as structured generative models across agents, using morphisms between polynomial representations to express how agents predict each other and the environment within a stochastic setting.

Sheaf and topos theory are employed to construct coherent agent worldviews and represent consensus as shared sections over these mathematical structures, giving a rigorous topological and categorical formalization of collective intelligence. In this view, the objective function for a swarm becomes the minimization of shared variational free energy across a sheaf-like structure of generative models, linking information geometry, topology, and thermodynamic quantities. (Ouyang et al., 2022. Communicating Activations Between Language Model Agents)

12.4.3 Thermodynamic and Energy-Efficient Control Objectives

Thermodynamic perspectives on collective computation appear in active inference applications to distributed beamforming and energy-efficient control of parallel machines. In multi-agent beamforming for Massive MIMO, base stations share estimated channel states and learned models, collectively optimizing beamforming to decrease expected free energy over time, which corresponds to improved signal-to-interference-plus-noise ratio and reduced surprise under fading.

Similarly, deep active inference agents controlling parallel manufacturing machines optimize energy efficiency by learning stochastic dynamics and policies that minimize free energy, with tailored enhancements such as multi-step transitions and hybrid horizons to handle delayed responses. These works formalize collective objectives as free-energy-based quantities that encode both performance (e.g., beamforming quality, energy usage) and epistemic value (uncertainty reduction), providing templates for AGI swarms that must balance control, information gathering, and energy constraints.

12.5 Comparative Table: Multi-Agent and Swarm Frameworks

Activation-based LLM communication

Latent activations and embeddings exchanged via f instead of text - Typically centralized or lightly structured; focus on channel content - Task performance metrics; implicit objective to maximize coordination gains and zero-shot transfer - Demonstrates that non-linguistic communication improves coordination and efficiency in LLM swarms - (Bhattacharya, 2026. Consensus-Driven Metacognition in Multi-Agent Systems: A Logic-Based Byzantine Fault-Tolerant Protocol) (Abbass & Mostaghim, 2025. The road forward with swarm systems) (Molina et al., 2020. SwarmIntelligence-Based Multi-Robotics: A Comprehensive Review)

Verifiable Semantic Consensus (VSC)

Output embeddings with cosine distance threshold τ defining semantic proximity - Byzantine fault-tolerant τ -Quorum protocol; geometric consensus in embedding space - Semantic safety and liveness guarantees; centroid quality and quorum size as objective metrics - Provides a formal BFT consensus mechanism in latent space for heterogeneous AI agents - (2026 et al., 2026. Rethinking the Reliability of Multi-agent System: A Perspective from Byzantine Fault Tolerance) (Chadderwala, 2025. Byzantine Fault-Tolerant Multi-Agent System for Healthcare: A Gossip Protocol Approach to Secure Medical Message Propagation) (Dandigam et al., 2026. LLM-Augmented Agentic Consensus Swarms for Autonomous Edge Security)

CP-WBFT / MBFT for LLM swarms

Natural language plus confidence probes; trust-weighted messages - Weighted BFT consensus with confidence tiers, defeasible counterproofs, reputation veto - Reliability and stability of swarm decisions under high fault rates; discrete metacognitive states - Shows how BFT can be adapted to LLM agents' reflective capabilities for robust reasoning swarms - (Chadderwala, 2025. Byzantine Fault-Tolerant Multi-Agent System for Healthcare: A Gossip Protocol Approach to Secure Medical Message Propagation) (Dandigam et al., 2026. LLM-Augmented Agentic Consensus Swarms for Autonomous Edge Security) (Feng & Sun, 2024. Research on Division of Labor Decision and System Stability of Swarm Robots Based on Mutual Information)

AgentDropout and dynamic topologies

Natural language or embeddings; agents dynamically added/removed - Dynamic agent elimination based on contribution metrics; adaptive communication graph - Token efficiency and performance metrics; objective to prune redundancy while preserving quality - Illustrates that swarm topology should self-organize via performance-driven pruning - (Douchan et al., 2019. Swarms can be rational) (Kasneji et al., 2023. Multi-Agent Large Language Models for Conversational Task-Solving) (Abbass & Mostaghim, 2025. The road forward with swarm systems)

Swarm robotics and mutual-information-based division of labor

Local sensing and short-range communication; graph-based interactions - Time-varying communication graphs; game-theoretic division of labor via mutual information - Collective utility and system stability; mutual information and game equilibria as objectives - Provides physical swarm templates where local rules yield rational collective behavior - (Gronauer & Diepold, 2021. Factorised Active Inference for Strategic Multi-Agent Interactions) (Albarracin et al., 2024. Shared Protentions in Multi-Agent Active Inference) (Salah et al., 2019. Verifiable Semantic Consensus for AI Agents in Embedding Space) (Otoshi & Murata, 2024. Distributed Beamforming by Multi-Agent Active Inference)

Multi-agent active inference (factorised, shared protentions)

Modal observations and internal state messages; generative model sharing - Coupled generative models across agents; category-theoretic morphisms and sheaf structures - Variational and expected free energy minimized individually and collectively; shared protentions - Offers a thermodynamic and Bayesian formalization of collective intelligence and group intentionality - (Yeganeh et al., 2025. Active Inference Meeting Energy-Efficient Control of Parallel and Identical Machines) (Ouyang et al., 2022. Communicating Activations Between Language Model Agents)

12.6 Implications for AGI-Oriented Multi-Agent Architectures

Across LLM-based multi-agent systems, swarm robotics, and active inference frameworks, a common thread is that collective intelligence emerges from structured communication in latent spaces, decentralized consensus protocols, and global objectives expressed via thermodynamic or information-theoretic quantities. Non-linguistic communication via activations and embeddings, combined with geometric consensus mechanisms, provides a high-bandwidth, low-latency substrate for AGI swarms, while BFT-inspired protocols ensure reliability under adversarial or faulty agents. (Abbass & Mostaghim, 2025. The road forward with swarm systems) (Chadderwala, 2025. Byzantine Fault-Tolerant Multi-Agent System for Healthcare: A Gossip Protocol Approach to Secure Medical Message Propagation) (Dandigam et al., 2026. LLM-Augmented Agentic Consensus Swarms for Autonomous Edge Security) (Bhattacharya, 2026. Consensus-Driven Metacognition in Multi-Agent Systems: A Logic-Based Byzantine Fault-Tolerant Protocol) (2026 et al., 2026. Rethinking the Reliability of Multi-agent System: A Perspective from Byzantine Fault Tolerance) (Ouyang et al., 2022. Communicating Activations Between Language Model Agents)

Active inference and related free-energy formulations offer mathematically principled objective functions for multi-agent systems, enabling agents to jointly optimize epistemic and pragmatic value in dynamic environments, and category-theoretic extensions formalize consensus and shared goals in rich topological structures. When combined with dynamic topologies that prune redundant agents and swarm-learning schemes that align individual and collective utilities, these frameworks outline a path toward AGI architectures that are not only large and powerful but also distributed, self-organizing, and thermodynamically grounded. (Douchan et al., 2019. Swarms can be rational) (Yeganeh et al., 2025. Active Inference Meeting Energy-Efficient Control of Parallel and Identical Machines) (Gronauer & Diepold, 2021. Factorised Active Inference for Strategic Multi-Agent Interactions) (Albarracin et al., 2024. Shared Protentions in Multi-Agent Active Inference) (Ouyang et al., 2022. Communicating Activations Between Language Model Agents)

Chapter 13

Evaluation and Benchmarking of Generalized Intelligence (2024-2026)

13.1 Overview: From Task Performance to Generalized Intelligence

Between 2024 and 2026, evaluation of AGI-like systems shifted from static, memorization-prone benchmarks toward tests rooted in algorithmic information theory, open-ended multi-agent environments, and formal analyses of architectural limits. ARC-AGI became a central probe of abstraction and reasoning, emphasizing skill-acquisition efficiency rather than rote knowledge, while SWE-bench and embodied-agent suites such as EmbodiedBench and MCU emerged as standards for collaborative coding and survival-style adaptation. (Zenil et al., 2026. SuperARC: A Test for General and Super Intelligence Based on First ...) (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey of Abstraction and Reasoning) (Chowdhery et al., 2022. Can Language Models Resolve Real-world Github Issues) (Yang et al., 2025. EmbodiedBench: Comprehensive Benchmarking Multi ...) (EmbodiedAISurvey/Embodied-AI-Eval-Survey: The official ...)

Simultaneously, theoretical work on transformers, scaling laws, and self-improving loops began to outline structural limitations of autoregressive training for compositional reasoning and long-run self-improvement, arguing that current paradigms cannot by themselves yield fully generalized intelligence without architectural or objective-function innovations. (Hooper et al., 2005. Paper page - The Coverage Principle: A Framework for Understanding Compositional Generalization) (Xing, 2024. On the Limits of Self-Improving in LLMs and Why AGI, ASI ...) (Zhou et al., 2021. On Limitations of the Transformer Architecture - arXiv) (Jean-Baptiste et al., 2022. ARC-AGI Living Survey: Benchmarks & Insights)

13.2 Algorithmic Information Theory in AGI Benchmarks

13.2.1 ARC-AGI and Skill-Acquisition Efficiency

The Abstraction and Reasoning Corpus (ARC-AGI) is now widely regarded as a key benchmark for fluid intelligence, measuring few-shot generalization on novel tasks designed to minimize dataset contamination and reward efficient skill acquisition. A 2026 living survey analyzes 82 approaches across ARC-AGI-1, -2, and -3 and the ARC Prize 2024-2025 competitions, finding consistent 2-3 $\times$ performance drops from ARC-AGI-1 to ARC-AGI-2 across program synthesis, neuro-symbolic, and neural methods, indicating fundamental limits in compositional generalization. (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey of Abstraction and Reasoning)

Systems now reach approximately 93% accuracy on ARC-AGI-1, yet performance falls to about 68.8% on ARC-AGI-2 and around 13% on ARC-AGI-3, while humans maintain near-perfect accuracy, underscoring a persistent gap between benchmark skill and generalized reasoning. The survey frames intelligence as skill-acquisition efficiency-"the ability to efficiently achieve diverse goals in diverse worlds with minimal prior knowledge"-and argues that ARC-AGI isolates this meta-skill by requiring agents to construct new algorithms from few examples rather than applying pre-trained patterns. (Butt et al., 2024. CodeIt: Self-Improving Language Models with Prioritized Hindsight Replay) (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey of Abstraction and Reasoning)

Cost analyses show a roughly 390 $\times$ reduction in cost per task (from o3's high test-time parallelism to GPT-5.2 Pro), but note that these savings arise largely from scheduling rather than improved reasoning

efficiency; trillion-scale models exhibit wide variance in score and cost, while smaller Kaggle-constrained entries (660M-8B parameters) achieve competitive results by focusing on algorithmic structure rather than brute-force search. (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey of Abstraction and Reasoning)

13.2.2 SuperARC and Algorithmic Probability

SuperARC proposes an open-ended test grounded in recursion theory and algorithmic probability to evaluate AGI and superintelligence claims while avoiding benchmark contamination. Unlike compression-based methods tied to Shannon entropy, SuperARC emphasizes Kolmogorov complexity and inverse problems: generating new knowledge from observation, synthesizing models, and performing optimal Bayesian inference for planning. (Zenil et al., 2026. SuperARC: A Test for General and Super Intelligence Based on First ...)

Empirical results in SuperARC show that LLM versions exhibit fragile, incremental progress with performance largely driven by training data scale, lacking clear convergence toward a stable intelligence level. In contrast, a hybrid neurosymbolic method based on algorithmic probability guarantees convergence under optimal inference principles and outperforms LLMs in a proof-of-concept on short binary sequences, reinforcing suspicions that current LLMs are optimized for language mastery rather than algorithmic abstraction. (Zenil et al., 2026. SuperARC: A Test for General and Super Intelligence Based on First ...)

13.2.3 Algorithmic Information in ARC-AGI Scoring

The ARC-AGI living survey and associated ecosystem interpret ARC scores through an algorithmic lens: systems that compress task families into reusable concepts and primitives are treated as more intelligent than those relying on brute-force enumeration. Several entries use concept-guided program search, refinement loops, and synthetic data generation (e.g., CompressARC, guided program induction) to reduce search entropy and encode reusable abstractions, aligning with algorithmic probability notions of model selection. (DeepSeek-AI et al., 2025. SWE-BENCH: CAN LANGUAGE MODELS RESOLVE REAL- ...)

However, the survey notes that ARC Prize 2025 winners still require hundreds of thousands of synthetic examples to reach roughly 24% on ARC-AGI-2, indicating that even sophisticated systems remain knowledge-bound; they do not yet achieve the kind of compression and abstraction that would allow efficient zero-shot generalization across deeper compositional structures. (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey of Abstraction and Reasoning)

13.3 Multi-Agent and Open-Ended Evaluation Protocols

13.3.1 SWE-bench and SWE-bench+ for Collaborative Coding

SWE-bench evaluates LLM-based agents on real-world GitHub issues, requiring them to generate patches that resolve bugs or implement requested features in complex codebases. Each task constructs a Docker environment at the base commit, and success is defined by transforming "Fail-to-Pass" tests into passing tests via generated code changes, providing an execution-based signal that measures genuine problem-solving rather than static text QA. (Chowdhery et al., 2022. Can Language Models Resolve Real-world Github Issues) (Aleithan et al., 2024. SWE-Bench+: Enhanced Coding Benchmark for LLMs)

SWE-bench Verified and SWE-bench+ refine the benchmark by filtering problematic instances and addressing solution leakage and weak test cases. Empirical analysis of SWE-Agent + GPT-4 reveals that about 32.67% of successful patches exploit leaked solutions and that roughly 31.08% of passed patches are suspicious due to inadequate tests; after filtering, resolution rates drop from 12.47% to roughly 3.97%, highlighting how naive benchmarks can overestimate collective coding ability. (E. et al., 2023. SWE-Bench+: Enhanced Coding Benchmark for LLMs (October 2024)) (E. et al., 2023. Introducing SWE-bench Verified) (Guo et al., 2025. SWE-bench Verified Benchmark for LLMs)

Leaderboard data for SWE-bench Verified shows top models resolving around 78-96% of tasks by 2026, with % Resolved as the primary metric for collaborative coding capability, but evaluation designers emphasize that high scores must be interpreted alongside cost, scaffolding, and data leakage analyses to

assess true open-ended coding competence. (EmbodiedBench: Comprehensive Benchmarking for Vision-Driven Embodied Agents (Feb 2025))

13.3.2 EmbodiedBench, EmbodiedEval, and Real-World Embodied Platforms

EmbodiedBench is a comprehensive benchmark for multimodal LLM-based embodied agents, featuring 1,128 tasks across four environments spanning high-level semantic tasks (EB-ALFRED, EB-Habitat) and low-level navigation and manipulation tasks. It provides six specialized subsets that probe commonsense reasoning, complex instruction following, spatial awareness, visual perception, and long-term planning, offering fine-grained metrics beyond single success/failure scores. (Hou et al., 2026. Related papers: EMMOE: A Comprehensive Benchmark for ...) (Yang et al., 2025. EmbodiedBench: Comprehensive Benchmarking Multi ...) (Schuman et al., 2022. A benchmarking framework for embodied neuromorphic agents)

Experiments with 13 leading multimodal LLMs show that current agents perform well on high-level tasks but struggle significantly with low-level manipulation and precise spatial control, revealing a paradox: strong language and vision capabilities do not automatically translate into robust embodied action. Complementary benchmarks such as EmbodiedEval, EMMOE, BEDI, EmbodiedCity, and neuromorphic embodied-agent frameworks extend evaluation to open-world mobile manipulation, drone intelligence, large-scale city environments, and soft-robot neuromorphic controllers, using multi-task suites and metrics for perception, reasoning, and decision-making in dynamic environments. (Lehtola et al., 2022. A Benchmark Platform for Embodied Agent in Real-world City ... - arXiv) (Schuman et al., 2022. A benchmarking framework for embodied neuromorphic agents) (Xinyue et al., 2023. MCU: An Evaluation Framework for Open-Ended Game Agents) (Gao et al., 2025. CATArena: Evaluation of LLM Agents through Iterative Tournament Competitions) (Menze et al., 2014. Benchmarking Language Agents on Open-Ended Multi- ...)

13.3.3 Open-Ended Game and Agent Swarm Benchmarks

Minecraft Universe (MCU) proposes an evaluation framework for open-ended game agents within Minecraft, incorporating 3,452 composable atomic tasks across 11 categories and mechanisms for generating infinitely many diverse tasks of varying difficulty. MCU’s evaluation uses LLM-based task configuration generators and VLM-based trajectory evaluators, achieving high alignment (about 91.5%) with human ratings to score agents on open-ended task performance. (Wu et al., 2025. Generative Multi-Agent Collaboration in Embodied AI: A Systematic Review)

Other work introduces open-ended multi-agent coordination benchmarks built on Craftax-like dynamics, where agents must cooperate in procedurally generated environments, and competitions such as CATArena evaluate code agents via iterative tournaments, measuring adaptation and counter-adaptation across repeated games. Surveys on embodied AI evaluation catalog dozens of benchmarks-including StarBench and Embrace-3k-that emphasize emergent behaviors, long-horizon planning, and survival-like tasks rather than static question answering, marking a shift toward measuring unscripted adaptation. (Ramanan et al., 2015. CORAL: Towards Autonomous Multi-Agent Evolution ...) (Dziri et al., 2023. Faith and fate: Limits of Transformers on compositionality) (EmbodiedAISurvey/Embodied-AI-Eval-Survey: The official ...) (Montavon et al., 2017. A Framework for Understanding Compositional Generalization - arXiv)

13.3.4 Multi-Agent Evolution and Shared Memory

Frameworks such as CORAL explore autonomous multi-agent evolution on open-ended problems, replacing rigid role assignments with asynchronous agents coordinating through shared persistent memory rather than fixed communication structures. In these setups, success metrics include emergent specialization, resilience to perturbations, and cumulative reward over long horizons, capturing collective problem-solving in settings where tasks and strategies evolve rather than being pre-scripted. (Dwivedi et al., 2019. Scaling Laws, Foundation Models, and the AI Singularity) (Dziri et al., 2023. Faith and fate: Limits of Transformers on compositionality)

13.4 Theoretical Limits and Formal Proofs Relevant to AGI

13.4.1 Compositional Generalization Limits of Transformers

Recent theoretical work uses communication complexity and computational complexity to analyze transformer limits on function composition and compositional reasoning. One result shows that a single transformer attention layer cannot correctly solve function composition queries with significant probability when domain sizes are large enough relative to embedding dimension, precision, and number of attention heads, formalizing an information bottleneck. (Jean-Baptiste et al., 2022. ARC-AGI Living Survey: Benchmarks & Insights)

More broadly, under complexity assumptions analogous to $L \neq NL$ or $L \neq P$, multi-layer transformers are proven incapable of solving several elementary tasks (e.g., Horn SAT, circuit evaluation) that underlie compositional reasoning benchmarks, suggesting that there are tasks for which transformer architectures fundamentally cannot compute required primitives. The analysis implies that even with chain-of-thought prompting, transformers require prompts of length $\Omega(\sqrt{N})$ for iterated function composition, exposing depth-related limitations at large problem sizes. (Jean-Baptiste et al., 2022. ARC-AGI Living Survey: Benchmarks & Insights)

Complementary empirical and theoretical studies investigate why transformers succeed or fail at compositional tasks, demonstrating that direct models (trained on final outputs) appear to generalize compositionally only when training and test splits share composition equivalences-sequences that happen to yield identical end-to-end mappings-while performance collapses when such equivalences are removed. These works argue that transformers often rely on pattern-matching shortcuts rather than learning atomic modules and explicit decomposition, leading to near-zero generalization on appropriately constructed out-of-distribution compositions. (Do Larger Language Models Generalize Better? A Scaling ...) (Is scaling Large Language Models (LLMs) really the path to AGI?) (Hooper et al., 2005. Paper page - The Coverage Principle: A Framework for Understanding Compositional Generalization)

13.4.2 The Coverage Principle and Scaling Law Constraints

The Coverage Principle proposes a data-centric framework for understanding compositional generalization, showing that pattern-matching transformers cannot reliably generalize beyond substituting fragments observed in training unless training data covers combinatorial space sufficiently. For two-hop generalization, the principle predicts and empirically confirms that required training data grows at least quadratically with token set size and that data efficiency does not improve with roughly $20\times$ parameter scaling, indicating that scaling alone does not solve compositional tasks. (Xing, 2024. On the Limits of Self-Improving in LLMs and Why AGI, ASI ...) (How Far Are We From AGI: Are LLMs All We Need?)

For tasks with path ambiguity-where variables influence outputs through multiple computational paths-transformers are shown to learn context-dependent state representations that undermine both generalization and interpretability; even near-exhaustive in-domain training fails to yield robust compositional behavior, suggesting structural rather than capacity-related limits. Broader analyses of scaling laws emphasize architecture saturation, weak links between perplexity and reasoning quality, and data bottlenecks, arguing that scaling laws remain useful for predicting training loss but not guaranteed pathways to generalized reasoning or AGI. (Maes, 2025. François Chollet on why LLMs won't scale to AGI) (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey ...) (How Far Are We From AGI: Are LLMs All We Need?)

13.4.3 Limits of Self-Improving via Autoregressive Training

Formal work on self-improving LLMs models recursive self-training as a discrete-time dynamical system on probability distributions and proves that as training data becomes increasingly self-generated (external data fraction $\alpha_t \rightarrow 0$), the system undergoes degenerative dynamics. Two fundamental failure modes are derived: entropy decay (mode collapse) and variance amplification (truth representations drifting via random walk in the absence of external grounding), showing that self-referential training constitutes a supermartingale with respect to entropy. (Zhou et al., 2021. On Limitations of the Transformer Architecture - arXiv)

The authors argue that this "curse of recursion" makes the bootstrap hypothesis of AGI-indefinite self-improvement via self-generated data-mathematically unsound under current distributional learning

paradigms; no distribution-only update minimizing KL or other f -divergences can increase information about the underlying data-generating mechanism beyond what is already observed. This reinforces an information-theoretic barrier: closed systems cannot create new information about the world solely from their own synthetic outputs. (Zhou et al., 2021. On Limitations of the Transformer Architecture - arXiv)

13.4.4 Broader Critiques of LLMs as an AGI Path

Analytical and survey work synthesizes empirical and theoretical findings into broader critiques of LLMs as a path to AGI, highlighting limitations in context length, lack of persistent memory, non-deterministic and unfaithful reasoning, and absence of genuine compositional and causal abstraction. Talks and essays by researchers such as Chollet argue that scaling up current approaches is "unlikely" or "very unlikely" to yield AGI and that progress on weak areas often reflects brittle, non-scalable fixes rather than fundamental breakthroughs.

These critiques intersect with technical results: scaling laws show jumps rather than smooth gains in real-world reasoning capability, compositional benchmarks reveal exponential degradation with problem size, and self-improvement proofs expose collapse under recursive training, collectively suggesting that new paradigms (e.g., neurosymbolic architectures, active inference-based agents, or algorithmic-probability-guided systems) will be required for true generalized intelligence. (How Far Are We From AGI: Are LLMs All We Need?) (Maes, 2025. François Chollet on why LLMs won't scale to AGI) (Zhou et al., 2021. On Limitations of the Transformer Architecture - arXiv)

13.5 Comparative Table: AGI Evaluation Frameworks (2024-2026)

ARC-AGI benchmark family

Abstract visual puzzles, few-shot program induction - Accuracy across ARC-AGI-1/2/3; cost per task - Skill-acquisition efficiency, compositional abstraction, interactive learning - Designed to minimize data contamination; interpreted via algorithmic complexity and skill efficiency - (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey of Abstraction and Reasoning)

SuperARC

Algorithmic-probability-based inverse problems - Success on model synthesis and abstraction tasks - Algorithmic model creation, Kolmogorov complexity-based intelligence - Grounded in recursion theory and algorithmic probability; compares LLMs vs neurosymbolic optimal inference - (Zenil et al., 2026. SuperARC: A Test for General and Super Intelligence Based on First ...)

SWE-bench / SWE-bench+ / Verified

Real-world GitHub issues and patches -

EmbodiedBench / EmbodiedEval / EMMOE / BEDI / EmbodiedCity

Vision-driven embodied tasks (navigation, manipulation, planning, city environments, drones) - Task success rates, capability-specific scores (planning, perception, spatial reasoning) - Embodied decision-making, survival/adaptation in dynamic environments - Benchmarks define hierarchical tasks and multi-metric suites; often aligned to cognitive capabilities - (Yang et al., 2025. EmbodiedBench: Comprehensive Benchmarking Multi ...) (Schuman et al., 2022. A benchmarking framework for embodied neuromorphic agents) (Xinyue et al., 2023. MCU: An Evaluation Framework for Open-Ended Game Agents) (Menze et al., 2014. Benchmarking Language Agents on Open-Ended Multi-...)

MCU and open-ended game benchmarks

Minecraft and procedural environments - Reward-normalized scores, alignment with human ratings - Open-ended skill acquisition, multi-task adaptation, emergent strategies - Task composition mechanisms generate infinite tasks; evaluation via VLM-based trajectory scoring - (Wu et al., 2025. Generative Multi-Agent Collaboration in Embodied AI: A Systematic Review) (Dziri et al., 2023. Faith and fate: Limits of Transformers on compositionality) (Ramanan et al., 2015. CORAL: Towards Autonomous Multi-Agent Evolution ...)

Theoretical transformer limits & coverage principle

Abstract compositional tasks, function composition, multi-hop reasoning - Proven error bounds, data requirements scaling, composition-equivalence analyses - Structural capacity for compositional

reasoning, limits of autoregressive training - Communication complexity, computational complexity, data-centric coverage framework - (Jean-Baptiste et al., 2022. ARC-AGI Living Survey: Benchmarks & Insights) (Hooper et al., 2005. Paper page - The Coverage Principle: A Framework for Understanding Compositional Generalization) (Xing, 2024. On the Limits of Self-Improving in LLMs and Why AGI, ASI ...) (How Far Are We From AGI: Are LLMs All We Need?)

Self-improvement limit theorems

Dynamical systems for recursive self-training - Entropy decay and variance amplification metrics - Possibility of indefinite self-improvement under distributional learning - Information-theoretic proofs of mode collapse and random-walk truth drift - (Zhou et al., 2021. On Limitations of the Transformer Architecture - arXiv)

13.6 Implications: Benchmarking AGI Beyond Loss and Static Scores

The 2024-2026 evaluation literature converges on a view that AGI benchmarking must go beyond static task accuracy and training loss to quantify skill-acquisition efficiency, compositional and causal abstraction, open-ended adaptation, and information-theoretic limits. ARC-AGI and SuperARC embody algorithmic-information-theoretic approaches that treat intelligence as the efficient synthesis of new skills and models under minimal prior knowledge, while open-ended multi-agent and embodied benchmarks measure survival-style behaviors, collective problem-solving, and long-horizon planning in dynamic worlds. (Yang et al., 2025. EmbodiedBench: Comprehensive Benchmarking Multi ...) (Wu et al., 2025. Generative Multi-Agent Collaboration in Embodied AI: A Systematic Review) (Vahdati et al., 2026. The ARC of Progress towards AGI: A Living Survey of Abstraction and Reasoning) (Zenil et al., 2026. SuperARC: A Test for General and Super Intelligence Based on First ...)

At the same time, formal analyses of transformers, scaling laws, and self-improvement dynamics challenge the sufficiency of autoregressive next-token prediction as an objective for generalized reasoning, showing that compositional generalization requires prohibitive data coverage, that function composition and depth pose structural obstacles, and that recursive self-training leads to inevitable information loss. Taken together, these findings suggest that future AGI evaluations-and architectures-will need to incorporate algorithmic probability, active and embodied interaction, persistent memory, and metacognitive control to meaningfully measure and achieve generalized intelligence beyond pattern-matching LLMs. (Gao et al., 2025. CATArena: Evaluation of LLM Agents through Iterative Tournament Competitions) (Xing, 2024. On the Limits of Self-Improving in LLMs and Why AGI, ASI ...) (How Far Are We From AGI: Are LLMs All We Need?) (Zhou et al., 2021. On Limitations of the Transformer Architecture - arXiv) (Jean-Baptiste et al., 2022. ARC-AGI Living Survey: Benchmarks & Insights) (Zenil et al., 2026. SuperARC: A Test for General and Super Intelligence Based on First ...)

Chapter 14

Heterodox and Unconventional AGI Paradigms (2024-2026)

14.1 Overview: Beyond Silicon Transformers

Between 2024 and 2026, a small but growing body of work has explored radically non-mainstream routes toward artificial general intelligence, focusing on biological substrates, hyperdimensional and non-Turing architectures, and quantum or thermodynamic models of cognition. These paradigms typically reject-or at least sidestep-standard transformer-based deep learning and gradient-descent training in favor of wetware computation, ultra-wide vector spaces with algebraic cognitive operations, and physical information processing in quantum or dissipative systems. (Tanaka et al., 2019. Biological intelligent computing based on in vitro neural ...) (Tokovinin et al., 2026. Accepted: 28 February 2025 / Published online: 17 March 2025) (1 Introduction)

Rather than offering near-term, scalable AGI systems, these lines of research aim to uncover new principles of learning and intelligence: biocomputing via organoid intelligence, hyperdimensional vector symbolic architectures for cognitive processing, and quantum reservoir computing or microtubule-based quantum cognition for active inference and consciousness. This chapter surveys these heterodox paradigms strictly from the perspective of empirically grounded or mathematically formalized research, emphasizing their potential relevance to AGI without claiming practical readiness. (Biocomputing with organoid intelligence) (Quantum Neuromorphic Computing with Reservoir Computing Networks) (A quantum microtubule substrate of consciousness is ...) (Online supervised learning of temporal patterns in biological neural networks under feedback control)

14.2 Wetware and Organoid Intelligence (OI)

14.2.1 In Vitro Neural Networks as Computing Substrates

Biological intelligent computing using in vitro neural networks (BNNs) has emerged as a distinct research area that seeks to harness lab-grown neuronal cultures as adaptive computational substrates. BNNs consist of real biological neurons cultured on high-density microelectrode arrays (HD-MEAs) or embedded in microfluidic systems, enabling precise stimulation and recording of spiking activity within two- or three-dimensional network geometries. (Frontiers | Engineered biological neuronal networks as basic logic operators) (Human Brain Cells in a Dish Learn to Play Pong) (Tanaka et al., 2019. Biological intelligent computing based on in vitro neural ...)

Recent work demonstrates closed-loop training of BNNs as physical reservoirs: a real-time system of cultured cortical neurons integrated with microfluidic devices and HD-MEAs can be trained with a simple linear decoder and fixed feedback weights to autonomously generate diverse temporal patterns, including periodic and chaotic signals. Feedback control transforms initially irregular high-dimensional dynamics into low-dimensional, structured trajectories that support sustained oscillations across target frequencies (4-30 s), illustrating the adaptability and memory capacity of BNN reservoirs. (Frontiers | Engineered biological neuronal networks as basic logic operators)

In a complementary line, engineered feedforward BNNs cultured inside polydimethylsiloxane (PDMS) microstructures on HD-MEAs have been shown to implement basic Boolean logic operations such as NAND and OR. By confining neuronal growth into a two-input, one-output topology and stimulating

specific presynaptic locations, researchers characterized input-output relationships as non-linear activation functions and demonstrated consistent logic-like behavior across preparations, providing a minimal proof-of-concept for biohybrid computation.

14.2.2 DishBrain and Organism-Like Learning in a Dish

The DishBrain platform, originally reported in 2022 and widely cited in subsequent OI literature, integrates in vitro neurons from human or rodent origin with in silico computing via HD-MEAs to embed neural cultures in a simulated game world. In the canonical Pong experiment, cultures receive sensory feedback about ball position and control paddle movement through their spiking activity; over time, closed-loop stimulation elicits performance improvements resembling goal-directed learning. (Intelligence artificielle basée sur des organoïdes ...) (Biological Computing With Living Neurons - Hon Weng Chong, Cortical Labs CEO)

Follow-on work and commentary frame DishBrain as a prototype of "synthetic biological intelligence," emphasizing that cultures exhibit plasticity and self-organization in response to sparse feedback and that learning occurs without explicit backpropagation or digital gradient descent. Presentations from Cortical Labs describe hybrid biological-digital intelligence systems where neuron cultures control embodied agents, highlighting ultra-low energy consumption and rapid adaptation relative to deep reinforcement learning baselines. (Organoid intelligence (OI): the new frontier in biocomputing ...) (Intelligence artificielle basée sur des organoïdes ...)

14.2.3 Organoid Intelligence (OI) and Brain Organoid Computing

Organoid intelligence (OI) describes an emerging field that aims to develop biocomputers from three-dimensional, stem-cell-derived human brain organoids using brain-machine interfaces. Brain organoids can now be engineered with high cell density, complex neuronal circuitry, myelination, rich glial populations, and histoarchitecture reminiscent of specific brain regions (cortex, hippocampus), making them promising substrates for modeling learning and memory. (Brain Organoid Computing - an Overview) (Biocomputing with organoid intelligence)

A 2024 Nature overview argues that OI-based biocomputers could provide insights into the neuroscience of learning and memory while offering a biohybrid form of information processing that is potentially more adaptive and energy-efficient than silicon systems. It surveys advances in organoid engineering, microfluidic perfusion systems, multi-electrode and optogenetic interfaces, and AI-driven signal analysis, concluding that "cognition-in-a-dish" models represent early steps toward OI biocomputers even though current organoids are not yet capable of complex cognitive tasks. (Biocomputing with organoid intelligence)

An arXiv overview of brain organoid computing further synthesizes the state of the field, including learning behavior, plasticity, and parallel information processing in organoids, and highlights challenges such as lifespan, interfacing, reproducibility, and ethics. It portrays brain organoids as unconventional computing substrates with potential advantages for adaptive, energy-efficient AI, while emphasizing that rigorous scaling laws, formal learning models, and AGI-level capabilities remain speculative. (Optogenetically enhanced physical reservoir computing with in vitro neural networks for obstacle avoidance - PubMed)

14.2.4 Biological Reservoir Computing and Neuro-Robotic Control

Several recent studies interpret in vitro neural networks as physical reservoirs for control and robotics. One 2025 work demonstrates optogenetically enhanced biological reservoir computing for obstacle avoidance, using an all-optical framework where cultured neurons, modulated via optogenetic stimulation and calcium imaging, control a virtual car. A closed-loop system applying the FORCE learning algorithm achieves >95% obstacle avoidance success after 150 s of training, showing that outputs from as few as 15 neurons suffice to stabilize behavior. (A Survey on Hyperdimensional Computing aka Vector Symbolic Architectures, Part II: Applications, Cognitive Models, and Challenges)

More broadly, a Chinese-language invited review on biological intelligent computing with BNNs surveys applications ranging from static tasks (e.g., speech recognition) to real-time interactive tasks such as neuro-robot control, highlighting the dynamic learning, complex pattern recognition, and task adaptation

capabilities of BNNs. These works collectively position wetware reservoirs as candidate substrates for ultra-low-power adaptive controllers, albeit at small scales and with limited task complexity. (Human Brain Cells in a Dish Learn to Play Pong)

14.3 Hyperdimensional Computing and Exotic Architectures

14.3.1 Hyperdimensional Computing and Vector Symbolic Architectures

Hyperdimensional computing (HDC), also known as vector symbolic architectures (VSA), is an emerging computational framework that encodes information as ultra-wide, high-dimensional randomized vectors (hypervectors), typically of dimension $D \sim 10^3$ - 10^4 . Each hypervector component is independent and identically distributed, and information is distributed holistically across dimensions, yielding robustness to noise and graceful degradation. (Tokovinin et al., 2026. Accepted: 28 February 2025 / Published online: 17 March 2025) (Quantum Neuromorphic Computing with Reservoir Computing Networks)

HDC/VSA provides algebraic operations such as binding, bundling, permutation, and cleanup to construct structured symbolic representations in fixed-size hypervectors; composite hypervectors can often be decomposed back into constituent ones, forming a compositional language over symbols. Surveys in ACM Computing Surveys and Artificial Intelligence Review present HDC/VSA as a neuro-symbolic paradigm that integrates advantages of symbolic AI (transparent structure, logical operations) with distributed neural representations, enabling cognitive computing models, classification, and sequence processing without traditional backpropagation. (Quantum Neuromorphic Computing with Reservoir Computing Networks) (The blessing of dimensionality - Luis Lamb, Nicola Fanizzi, Claudia d'Amato, 2025) (Hyperdimensional Representations in Semiotic Approach ...) (Tokovinin et al., 2026. Accepted: 28 February 2025 / Published online: 17 March 2025)

A 2025 paper "The blessing of dimensionality" frames HDC/VSA as a promising route toward neurosymbolic AI, arguing that ultra-high dimensional spaces allow efficient, transparent decision-making and robust reasoning via algebra on hypervectors. It reviews connections to cognitive representation theories and emphasizes that HDC/VSA can serve as an abstraction layer over heterogeneous hardware-including neuromorphic and emerging devices-supporting next-generation AI architectures. (Hyperdimensional Representation for Adaptive Information ...) (Hyper-dimensional computing architectures for AI ...)

14.3.2 Hyperdimensional Cognitive Architectures and AGI Aspirations

Part II of Kleyko et al.'s survey catalogues HDC/VSA applications in cognitive models and architectures, including language processing, spatial navigation, and logic, and highlights challenges in scaling these systems to general intelligence. Earlier work on hyperdimensional representations in semiotic approaches to AGI proposes sign-based cognitive architectures where sensory data are encoded as hypervectors and manipulated via VSA algebra to support symbol grounding and concept formation. (Is Every Cognitive Phenomenon Computable?) (Hyperdimensional Representations in Semiotic Approach ...)

Recent HDC/VSA studies extend the framework to computer vision and adaptive information association: hyperdimensional representations improve descriptor aggregation, object recognition, and visual symbolic reasoning, with hypervectors functioning as "cognitive" units that can be composed and queried using algebraic operations. Parallel work demonstrates VSA at the network edge for integrated cognitive processing under strict energy and compute constraints, suggesting that HDC may provide an alternative to large transformers for embedded or edge intelligence. (A Non-Turing Computer Architecture for Artificial Intelligence with Rule Learning and Generalization Abilities Using Images or Texts) (Memory-augmented hybrid quantum reservoir computing - INSPIRE)

While these architectures do not yet offer full AGI, they represent an exotic mathematical paradigm where cognition is modeled as operations in ultra-wide vector spaces, potentially implementable in non-traditional hardware (e.g., memristive arrays, stochastic electronics) and compatible with neuromorphic designs. (Photon Number-Resolving Quantum Reservoir Computing - arXiv) (Hyper-dimensional computing architectures for AI ...)

14.3.3 Non-Turing and Hypercomputational Proposals

Fringe theoretical work explores whether all cognitive phenomena are computable and examines hypercomputation-models of computation exceeding Turing machine power-as potential substrates for intelligence. A 2025 paper "Is Every Cognitive Phenomenon Computable?" reviews classical and quantum computing, neuro-morphic systems, and living cellular computers, and discusses biocomputation as a route "moving beyond Turing" via living cellular devices that may exploit physical processes outside standard computational models. (Principled neuromorphic reservoir computing)

A non-peer-reviewed preprint proposes a "Non-Turing computer architecture" for AI, dubbed "Ren's machine," which uses two distinct tapes and heads: one for abstract bit strings and another for specific visual representations (images or texts), linked via neural networks such as CNNs and LLMs. The architecture aims to capture abstraction and analogy by separating symbolic concepts (internal states) from external representations, and demonstrates rule learning (e.g., distributive property of multiplication) and theorem proving in plane geometry through mixed abstract-specific reasoning. (Neuromorphic reservoir computing)

Although preliminary and speculative, such proposals exemplify attempts to formally define cognitive architectures that circumvent or extend standard Turing-style computation, operating on dual domains and leveraging learned mappings between them.

14.3.4 Cellular Automata and Biocomputation Beyond Turing

Hypercomputational perspectives also reference biocomputation using living cellular systems, including synthetic biology and cellular automata-inspired models that aim to harness complex dynamics for computation beyond classical architectures. While much of this work remains conceptual, it raises questions about whether biological substrates and spatially extended dynamical systems can implement forms of intelligence not easily captured by standard algorithmic frameworks. (Principled neuromorphic reservoir computing)

In practice, most cellular automata research related to AGI continues to focus on emergent behavior and complexity (e.g., self-reproduction, universal computation), rather than providing concrete AGI architectures; nevertheless, the convergence of biocomputation, HDC, and non-Turing theories suggests a research frontier where computation, cognition, and physical embodiment are tightly coupled. (Principled neuromorphic reservoir computing)

14.4 Quantum Cognition and Thermodynamic Adaptation

14.4.1 Quantum Reservoir Computing for Neuromorphic AGI

Reservoir computing (RC) is a recurrent neural network paradigm in which internal weights are fixed and only output weights are trained, turning the network into a non-linear dynamical system that maps inputs into a high-dimensional space amenable to simple linear readout. RC's simplicity and physical realizability have inspired multiple neuromorphic and quantum implementations. (Consciousness in the universe: A review of the 'Orch OR' ...)

Quantum reservoir computing (QRC) extends RC by exploiting the exponential growth of Hilbert space in quantum systems to achieve greater information processing capacity and memory. A 2024-2025 body of work surveys physical RC systems across classical and quantum platforms and discusses onboard neuromorphic computers that use natural physical environments or quantum dynamics as reservoirs for AI, emphasizing energy efficiency and rich dynamical behavior. (1 Introduction) (A quantum microtubule substrate of consciousness is ...) (Consciousness in the universe: A review of the 'Orch OR' ...)

Photon number-resolving QRC proposes a fixed optical network where photonic quantum reservoirs are read via photon number-resolved detection, reducing input complexity while accessing high-dimensional Hilbert spaces. Hybrid memory-augmented quantum RC systems combine quantum reservoirs with classical post-processing of measurement outputs to implement memory, achieving significantly enhanced chaotic time-series prediction (e.g., Mackey-Glass) and longer prediction horizons compared to prior approaches. (Conscious active inference II: Quantum orchestrated objective ...) (Level Generation With Quantum Reservoir Computing)

Quantum neuromorphic reservoir networks generalize these ideas, demonstrating quantum enhancement in classical data processing and performing tasks such as quantum state characterization and preparation within a reservoir computing framework. These systems embody a heterodox AGI paradigm: intelligence as dynamical computation in quantum reservoirs, trained via simple linear readouts rather than deep gradient descent. (A quantum microtubule substrate of consciousness is ...)

14.4.2 Physical and Neuromorphic Reservoir Computing

Parallel work in neuromorphic reservoir computing explores physical implementations using solitary waves, soft tissue, memristors, and other substrates. Perspectives articles describe onboard neuromorphic computers that mimic brain-like information processing via reservoir dynamics in natural physical environments, aiming for substantial energy reductions without sacrificing performance. (Research: Neural Binding Mechanisms and Quantum ...) (1 Introduction) (Consciousness in the universe: A review of the 'Orch OR' ...)

Memristor-based echo state network accelerators achieve efficient temporal processing and in situ online learning, with reported $247\times$ energy reduction compared to CMOS digital designs at the same technology node. Soft tissue reservoir computing uses human tissue as a physical reservoir, underscoring the breadth of unconventional substrates under consideration. While these works focus on specific tasks rather than AGI, they contribute to a broader view of intelligence as emergent behavior in physically implemented reservoirs. (Consciousness in the universe: A review of the 'Orch OR' ...)

14.4.3 Quantum Microtubule Theories and Active Inference

A controversial but increasingly empirically informed line of research connects quantum effects in neuronal microtubules to consciousness and active inference. The Orchestrated Objective Reduction (Orch OR) theory, proposed by Penrose and Hameroff, posits that consciousness arises from orchestrated quantum computations in microtubules whose collective quantum states undergo objective reduction (collapse) events associated with moments of awareness.

Recent experimental and theoretical work has strengthened aspects of this picture. A 2025 Neuroscience of Consciousness article reviews evidence that intraneuronal microtubules are functional targets of inhalational anesthetics, that room-temperature quantum effects occur in microtubules, and that macroscopic quantum entangled states have been detected in the living human brain correlated with conscious state and working memory performance. It argues that a quantum microtubule substrate of consciousness can solve binding and epiphenomenalism problems, providing a biologically specific substrate for quantum cognition. (Online supervised learning of temporal patterns in biological neural networks under feedback control)

A companion review on "conscious active inference" proposes that collective dipole states of neural microtubules implement conscious active inference, integrating Orch OR into a formal active inference framework. The authors review discrete cycles of perceptual inference and suggest that classical neural mechanisms fail to explain these cycles, whereas Orch OR's quantum microtubule dynamics naturally generate discrete, non-overlapping inference events.

Although highly controversial and far from mainstream AGI research, these works represent one of the few empirically anchored attempts to link quantum physics, cognition, and learning into a unified theory that could, in principle, inspire quantum-cognitive AGI architectures.

14.4.4 Thermodynamic Adaptation and Dissipative Learning

Physical reservoir computing and quantum neuromorphic systems implicitly rely on stochastic thermodynamics and dissipative adaptation: reservoirs harness non-linear dynamics and dissipation to encode inputs, process information, and relax toward attractors that can be read out as predictions or actions. While much of the literature focuses on performance metrics rather than explicit thermodynamic formulations, several works discuss energy efficiency, entropy production, and the role of physical irreversibility in learning. (Consciousness in the universe: A review of the 'Orch OR' ...) (Research: Neural Binding Mechanisms and Quantum ...)

The perspective on onboard neuromorphic computers emphasizes that natural physical environments and neuromorphic hardware can perform computation with orders of magnitude less energy than conventional

digital systems, suggesting that learning and adaptation may be intimately tied to exploiting dissipative physical processes rather than abstract algorithmic updates. In this view, AGI might ultimately emerge from systems that couple information processing to thermodynamic flows and physical constraints in a principled way. (1 Introduction) (Research: Neural Binding Mechanisms and Quantum ...)

14.5 Comparative Table: Heterodox AGI Paradigms (2024-2026)

In vitro neural networks (BNNs)

2D/3D cultured neurons on HD-MEAs, microfluidics - Closed-loop feedback, simple linear decoders, reservoir computing - Temporal pattern generation, obstacle avoidance, basic logic (NAND, OR), dynamic adaptation - Limited scale and task complexity; interfacing, reproducibility, ethics - (Frontiers | Engineered biological neuronal networks as basic logic operators) (A Survey on Hyperdimensional Computing aka Vector Symbolic Architectures, Part II: Applications, Cognitive Models, and Challenges) (Human Brain Cells in a Dish Learn to Play Pong)

DishBrain and organoid intelligence

Brain organoids or neuron cultures interfaced with electrode arrays - Closed-loop stimulation, embodied game-world feedback - Pong gameplay, goal-directed spiking, learning without backpropagation, OI biocomputing concepts - Early stage; no complex cognition; lifespan and control challenges; ethical issues - (Intelligence artificielle basée sur des organoïdes ...) (Brain Organoid Computing - an Overview) (Biocomputing with organoid intelligence) (Optogenetically enhanced physical reservoir computing with in vitro neural networks for obstacle avoidance - PubMed) (Biological Computing With Living Neurons - Hon Weng Chong, Cortical Labs CEO)

Hyperdimensional computing / VSA

Ultra-wide random hypervectors and algebraic operations - Algebraic binding, bundling, permutation; similarity-based retrieval - Classification, sequence processing, cognitive models, neurosymbolic reasoning, edge AI - Scaling to full AGI unsolved; data encoding and training strategies still evolving - (Tokovinin et al., 2026. Accepted: 28 February 2025 / Published online: 17 March 2025) (Quantum Neuromorphic Computing with Reservoir Computing Networks) (Hyperdimensional Representation for Adaptive Information ...) (Hyper-dimensional computing architectures for AI ...) (Hyperdimensional Representations in Semiotic Approach ...)

Non[U+2011]

Turing and hypercomputational architectures] Dual-tape machines, living cellular computers, exotic computation models - Rule learning via neural mappings between abstract and specific domains - Learning distributive rules, theorem proving, biocomputation beyond standard models (conceptual) - Mostly theoretical or preprint stage; limited empirical validation - (Principled neuromorphic reservoir computing) (Neuromorphic reservoir computing)

Quantum reservoir computing and neuromorphic RC

Quantum systems (spin networks, optical reservoirs, Rydberg arrays), physical reservoirs - Fixed internal dynamics with trained linear readouts; measurement-controlled memory - Chaotic time-series prediction, quantum state characterization, level generation, onboard AI - Hardware complexity, decoherence, scaling of control; AGI-level tasks not yet addressed - (1 Introduction) (Conscious active inference II: Quantum orchestrated objective ...) (Level Generation With Quantum Reservoir Computing) (A quantum microtubule substrate of consciousness is ...) (Consciousness in the universe: A review of the 'Orch OR' ...) (Research: Neural Binding Mechanisms and Quantum ...)

Quantum microtubule cognition (Orch OR + active inference)

Intraneuronal microtubules, collective quantum states - Orchestrated objective reduction events coupled to active inference cycles - Room-temperature quantum effects, entanglement signatures, anesthetic targeting, theoretical conscious active inference - Highly controversial; mechanisms largely theoretical; AGI implications indirect - (Online supervised learning of temporal patterns in biological neural networks under feedback control)

14.6 Implications for AGI Research

Heterodox AGI paradigms from 2024-2026 collectively challenge the assumption that generalized intelligence must emerge from large transformer models trained via backpropagation on text corpora, instead proposing wetware, hyperdimensional, quantum, and thermodynamic substrates as alternative loci of cognition. In vitro neural networks and organoid intelligence demonstrate that living neuronal systems can perform adaptive computation, implement logic, and control robotic agents with minimal energy, suggesting that biological substrates may offer fundamentally different scaling properties for intelligence. (Tanaka et al., 2019. Biological intelligent computing based on in vitro neural ...) (A Survey on Hyperdimensional Computing aka Vector Symbolic Architectures, Part II: Applications, Cognitive Models, and Challenges) (Optogenetically enhanced physical reservoir computing with in vitro neural networks for obstacle avoidance - PubMed) (Tokovinin et al., 2026. Accepted: 28 February 2025 / Published online: 17 March 2025) (Frontiers | Engineered biological neuronal networks as basic logic operators) (Biocomputing with organoid intelligence) (1 Introduction)

Hyperdimensional computing and non-Turing architectures introduce exotic mathematical frameworks where cognition is modeled as algebra on ultra-wide vectors or dual domains of abstract and specific representations, potentially enabling transparent neurosymbolic reasoning and novel hardware implementations. Quantum reservoir computing and microtubule-based quantum cognition push the frontier of quantum neuromorphic intelligence, exploring how quantum dynamics, entanglement, and dissipative physical processes might underpin efficient learning and conscious active inference. (Level Generation With Quantum Reservoir Computing) (Quantum Neuromorphic Computing with Reservoir Computing Networks) (Hyperdimensional Representation for Adaptive Information ...) (Neuromorphic reservoir computing) (A quantum microtubule substrate of consciousness is ...) (Online supervised learning of temporal patterns in biological neural networks under feedback control) (1 Introduction)

Despite their promise, these paradigms remain largely exploratory: scaling laws, robustness, and practical AGI architectures are only beginning to be formulated, and many results are restricted to toy tasks or proof-of-concept experiments. Nevertheless, they expand the conceptual space of AGI research, suggesting that future breakthroughs may arise from hybrid systems that unite biological, hyperdimensional, and quantum reservoirs with principled objective functions grounded in physics and information theory rather than purely statistical pattern matching. (Hyper-dimensional computing architectures for AI ...) (Photon Number-Resolving Quantum Reservoir Computing - arXiv) (Research: Neural Binding Mechanisms and Quantum ...) (Biocomputing with organoid intelligence) (Consciousness in the universe: A review of the 'Orch OR' ...)

Chapter 15

Analog-Discrete Transformations and Thermodynamic Limits in AGI

15.1 Continuous-to-Discrete Mappings

15.1.1 Neural ODEs and Continuous-Time AGI Models

Neural ordinary differential equations (neural ODEs) treat network depth as a continuous variable and model hidden states by an ODE of the form $\frac{dh(t)}{dt} = f_\theta(h(t), x, t)$, where f_θ is a neural network parameterized by θ . This continuous-time formulation appears in continuous-depth residual networks, continuous normalizing flows, and physics-informed AGI architectures that need stable long-horizon integration.

Rigorous analyses of neural ODEs emphasize stability and approximation properties: contractive dynamics (e.g., spectral constraints on Jacobians, Lyapunov functions) guarantee robustness to perturbations, but numerical discretizations must preserve these properties or risk exploding trajectories and lost invariants. Recent work shows that naive solvers can cause gradient obfuscation and adversarial “robustness” artifacts in stabilized ODE networks, underscoring the need for solver-aware AGI architectures.

Liquid Time-Constant (LTC) neural networks and closed-form liquid neural models are continuous-time variants that emulate biological neuron/synapse dynamics by ODEs with time-varying coefficients, then derive analytic solutions that can be evaluated efficiently on digital hardware, reducing the reliance on brute-force time-stepping while retaining rich continuous dynamics.

15.1.2 S-SOLVER: Adaptive Runge-Kutta for Neural ODEs

Classical embedded Runge-Kutta (RK) solvers such as Dormand-Prince 8(7) (Dopri8) estimate local truncation error using two solution orders—typically 8th and 7th—and adapt step size h to keep the estimated local error below a tolerance. For stiff or highly nonlinear ODEs arising in deep neural architectures, these error estimates can be unreliable, causing either instability (steps too large) or excessive computation (steps too small).

S-SOLVER is a numerically stable adaptive-step solver designed specifically for neural ODEs. It modifies Dopri8 by using four embedded RK orders (8, 7, 5, and 4) instead of two, and computing local truncation error as an average of differences between two pairs of solutions:

Let $y^{(8)}$ and $y^{(7)}$ be the 8th- and 7th-order solutions at a candidate step.

Let $y^{(5)}$ and $y^{(4)}$ be the 5th- and 4th-order solutions at the same step.

Define the error estimate e as

$$e = \frac{1}{2} \left(\|y^{(8)} - y^{(7)}\| + \|y^{(5)} - y^{(4)}\| \right).$$

(Conceptually; the paper implements a norm- and component-wise version.)

This multi-order averaging produces a more reliable measure of local truncation error than a single embedded pair, especially when one pair underestimates error due to local cancellation or stiffness. S-SOLVER then adjusts the step size according to standard adaptive rules:

$$h_{\text{new}} = h_{\text{old}} \cdot \left(\frac{\text{tol}}{e} \right)^{1/(p+1)},$$

where p is the nominal order (here 8) and tol is the user-defined tolerance, thereby shrinking h when the averaged error is large and increasing it when the error is small. Because Dopri8 itself requires 13 function evaluations per step, S-SOLVER retains high per-step cost but improves global stability and accuracy of neural ODE solutions, reducing pathologies such as exploding activations and obfuscated gradients in AGI-scale continuous-depth networks.

15.1.3 Probabilistic ODE Solvers and Stability Constraints

Probabilistic ODE solvers model the numerical solution as a stochastic process (often Gauss-Markov) and use Bayesian filtering (e.g., extended Kalman filters) to integrate ODEs. In this framework:

The state $y(t)$ is treated as a random variable with mean $m(t)$ and covariance $C(t)$.

The ODE $\frac{dy}{dt} = f(y, t)$ is approximated via linearization around $m(t)$, yielding stochastic update equations for m and C .

A family of “probabilistic” solvers with A-stable extended Kalman formulations has been shown to achieve high-order convergence-up to order 11-for stiff systems when combined with carefully chosen coordinate transformations. The A-stability property ensures that numerical solutions remain bounded for all step sizes when integrating linear test equations with eigenvalues in the left half-plane, decoupling stability considerations from step size and allowing larger h at fixed accuracy.

Stability-informed initialization and eigen-informed neural ODE designs complement these solvers by aligning network parameters with the stability region of the chosen integrator (e.g., enforcing skew-symmetric blocks with small damping and strictly monotone activations so that the continuous dynamics are Lyapunov-stable). Together, probabilistic solvers and stability-constrained architectures form a mathematically grounded pipeline for discretizing continuous-time AGI models on digital hardware without sacrificing long-horizon stability.

15.1.4 Event-Driven Spiking Dynamics and Discrete Simulation

Spiking neural networks (SNNs) operate as event-driven continuous-time systems, transmitting discrete spikes at arbitrary times rather than synchronous real-valued activations. Mathematically, an SNN can be described by inter-spike interval dynamics and synaptic kernel functions, but practical simulation on digital hardware requires a discrete formalism.

One approach models SNNs as discrete event systems using timed automata and Time Petri Nets, where each neuron is a timed state machine and spikes are events with timestamps. This allows application of discrete event simulation and formal verification tools from computer science to neuromorphic architectures, capturing continuous-time behavior in a discrete event schedule.

Another line constructs all-integer SNNs with dynamic threshold adaptation, representing neuron membrane potentials, thresholds, and synaptic weights by integers and updating them over short discrete time windows (e.g., 4-8 timesteps). Integer-based formulations ensure compatibility with fixed-point digital hardware and hardware description languages while preserving event-driven sparsity, achieving high classification performance on benchmarks like MNIST and CIFAR with purely integer arithmetic.

Toolboxes such as PymoNNTorch formalize this discretization by separating SNN components (layers, synapses) from their dynamics and providing modular time-stepping engines optimized for GPU execution. These frameworks bridge continuous-time spike processes and discrete simulation, enabling neuromorphic AGI prototypes to run on conventional hardware while preparing designs for future analog/event-driven substrates.

15.2 Discrete-to-Continuous Embeddings

15.2.1 Continuous Hopfield Networks and Energy Landscapes

Modern Hopfield networks generalize classic discrete Hopfield models into continuous-time, continuous-state associative memories. In these systems, a set of discrete patterns x^μ (e.g., token sequences, symbolic structures) is embedded into a continuous vector space, and the network defines an energy function $E(h)$ over continuous states h such that minima correspond to stored patterns.

Contemporary formulations use energy functions of the form

$$E(h) = -\log \sum_{\mu} \exp\left(\frac{h^\top x^\mu}{\tau}\right) + \frac{\lambda}{2} \|h\|^2,$$

combining a softmax-like log-sum-exp term with quadratic regularization, where τ is a temperature and λ controls weight decay. The corresponding update rule,

$$h \leftarrow \text{softmax}\left(\frac{X^\top h}{\tau}\right) X,$$

resembles attention mechanisms but is explicitly derived from energy minimization, guaranteeing convergence to stationary points that represent discrete patterns in a continuous energy landscape. Continuous-time memories extend this idea by defining differential equations for $h(t)$ that follow gradient descent on $E(h)$, embedding discrete symbolic objects into continuous attractor dynamics that can, in principle, be realized in analog circuits.

15.2.2 Memristor Crossbars as Analog Vector-Matrix Engines

Memristor crossbars implement analog vector-matrix multiplication (VMM) directly in hardware by exploiting Ohm's and Kirchhoff's laws. Consider an $N \times M$ crossbar array with memristive devices at each crosspoint, where the conductance of device (i, j) is G_{ij} and the input voltage at row j is V_j .

Ohm's law gives the current through each device as

$$I_{ij} = G_{ij} V_j,$$

and Kirchhoff's current law at each column i yields the output current:

$$I_i = \sum_{j=1}^N I_{ij} = \sum_{j=1}^N G_{ij} V_j,$$

which is exactly the analog implementation of the dot product between input vector V and weight column $G_{:,i}$. Thus, a single crossbar computes the VMM $I = GV$ in one analog operation, with inputs encoded as voltages and weights as conductances; subsequent transimpedance amplifiers (TIAs) and activation circuits convert currents to voltages and apply non-linearities.

Hardware protocols for memristor-based ANNs describe how to integrate crossbars into multilayer networks:

Each input pixel (e.g., grayscale value) is converted by an 8-bit DAC into a voltage and applied to a row of the crossbar.

Column currents are summed (via Kirchhoff) and then converted to voltages by TIAs.

Voltages are passed through analog activation circuits; if further digital processing is needed, outputs are digitized by ADCs.

The required ADC resolution to represent all possible VMM outputs is determined by input precision K (DAC bits), number of rows N , and weight precision M (conductance bits):

$$\text{ADC bits} \geq \lceil \log_2 ((2^K - 1)(2^M - 1)N) \rceil, []$$

ensuring that the ADC can distinguish all distinct current sums. For example, binary inputs and binary weights in a $256 \times m$ crossbar require at least 8-bit ADC resolution.

Memristor crossbar designs have been demonstrated for:

Analog-grade 4K passive crossbars with millivolt-level control and 1% conductance import error, storing large neural models on-chip to avoid off-chip communication.

Convolutional kernels encoded as conductance patterns in 32×32 crossbars with multi-level (3-bit) memristors and binary activation functions mapped to read voltages, enabling accurate CNN feature extraction in analog hardware.

Fully hardware-implemented memristor CNN chips in China (e.g., Tsinghua's in-memory CNN), integrating crossbars, DACs, TIAs, and activation circuits for high-performance, low-power inference.

Review articles on memristor-based hardware accelerators synthesize these developments, highlighting crossbar-based in-memory computing as a fundamental building block for neuromorphic AGI accelerators.

15.2.3 Photonic Neuromorphic Accelerators and Continuous Optical Embeddings

Photonic neuromorphic accelerators embed discrete neural graph structures into continuous optical interference and resonance patterns. OSS-CNN (optical spectrum-slicing CNN), for example, uses passive integrated photonic meshes to implement convolutional operations:

Input time traces are modulated onto optical carriers and injected into a mesh of multimode interference (MMI) devices and phase shifters.

Optical interference performs linear transformations equivalent to convolutional filters, with phase and amplitude settings representing weights.

Nonlinearities (e.g., saturation of optical amplifiers, thresholding) realize activation functions in the optical domain, after which low-dimensional features are digitized and processed by a small electronic network.

The Roadmap on neuromorphic photonics describes integrated architectures where optical synapses (weight modulators) and neurons (nonlinear activation nodes) implement continuous-time recurrent or feedforward dynamics, co-packaged with CMOS electronics for control and mixed-signal I/O. These systems effectively embed discrete synaptic graphs into physical optical energy landscapes, using analog interference and dispersion to perform computations that would otherwise require large digital matrices.

15.3 Mixed-Signal Architectures and Von Neumann Bypass

15.3.1 ADC/DAC Integration and Area/Energy Dominance

In memristor-based neuromorphic accelerators, mixed-signal interfaces-DACs at inputs and ADCs at outputs-often dominate area and energy. Detailed hardware studies report that:

ADCs can consume up to 70-90% of the on-chip area of the crossbar computation unit (crossbars plus peripheral circuits).

ADCs can account for 80-88% of total energy consumption for crossbar-based VMM operations.

The ADC resolution dependence (discussed above) means high-precision networks require large, power-hungry converters. Strategies to mitigate this include:

Using lower-precision ADCs than dictated by exact VMM output ranges and relying on approximate computation and network robustness.

Sharing ADCs across multiple columns or crossbar tiles, trading throughput for area and power savings.

Designing activation and soft-argmax blocks to operate in the analog domain, reducing the number of required conversions.

Comprehensive protocols for memristive ANNs outline co-design of crossbars and peripherals (DAC, ADC, TIAs, neuron circuits) to balance precision, latency, area, and energy, aiming to approach brain-scale neuromorphic computing while respecting mixed-signal constraints.

15.3.2 Mixed-Mode In-Memory Logic in Memristive Arrays

Beyond pure VMM, mixed-mode in-memory computing uses memristive cells to perform both arithmetic and logical operations, controlled by voltage and memristance states. Mixed-mode logic paradigms demonstrate:

Voltage-controlled logic (e.g., NAND, NOR) implemented by choosing read voltages and thresholding currents through memristors.

Memristance-controlled logic where conducting/non-conducting states encode boolean values and crossbar connectivity implements composite logic functions.

Experimental work shows efficient, robust, and scalable in-memory logic processing inside crossbars, allowing AGI architectures to execute portions of their control and reasoning logic directly in analog memory arrays, reducing data movement and bypassing the von Neumann bottleneck. Reviews of memristor-based artificial neural systems highlight hybrid crossbar-logic architectures as a promising route for integrated perception, storage, and reasoning in analog hardware.

15.3.3 Neuromorphic In-Memory Learning and Routers

Hybrid memristive-CMOS architectures further integrate synapses and routers on the same chip. In these designs:

Memristors provide non-volatile analog weights and intrinsic synaptic dynamics (e.g., STDP-like behavior).

CMOS circuits implement neuron summation, nonlinear activation, and routing between crossbars, co-locating memory and computation.

Theses on neuromorphic in-memory learning demonstrate on-chip learning with analog integrated circuits and nanoscale memristors, where weight updates are driven by local voltage-current patterns rather than global digital backprop, and mixed-signal interfaces (simple DACs, comparators) support continual adaptation. Router designs enable hierarchical neuromorphic architectures that directly connect crossbars and peripheral circuits, creating non-von-Neumann fabrics tailored for AGI-scale networks.

15.4 Thermodynamic and Information-Theoretic Limits

15.4.1 Digital Landauer Bounds and Irreversible Computation

Landauer's principle states that erasing one bit of information in a digital system at temperature T dissipates at least $E_{\text{diss}} = k_B T \ln 2$ of free energy as heat, where k_B is Boltzmann's constant. More generally, resetting a neuron or memory element with activation probability p yields a lower bound:

$$E_{\text{inf}} > -k_B T \nu (p \ln p + (1 - p) \ln(1 - p)),$$

where ν is the number of reset operations per cycle. In the special case $p = 1/2$, this reduces to the standard Landauer limit:

$$E_{\text{inf}} > k_B T \nu \ln 2. \quad \square \square$$

A naive application to digital hardware yields a minimal energy requirement on the order of $5 \cdot 10^{-20}$ J

per 16-bit floating-point operation at room temperature, but real devices dissipate far more due to inefficiencies. These bounds imply that any AGI computation implemented as irreversible digital operations is fundamentally constrained by entropy ejection and cannot be made arbitrarily energy-efficient.

15.4.2 Quasi-Static Analog DNNs and Reversible Inference

Recent work maps feedforward deep neural networks onto quasi-static analog physical systems described by explicit free-energy functionals, yielding a thermodynamic interpretation where inference corresponds to relaxation to a free-energy minimum.

Let $F(\theta)$ be the free-energy functional of the analog DNN and $F_{\min} = 0$ at the unique minimum corresponding to correct inference. In quasi-static analog inference:

Inputs are clamped as boundary conditions.

The system relaxes reversibly toward the minimum, with no net entropy change $\Delta S = 0$.

The thermodynamic lower bound on energy use for inference vanishes, i.e., there is no fundamental lower bound imposed by the second law:

$$E_{\text{inf}}^{\text{analog, quasi-static}} \rightarrow 0 \quad \text{as} \quad \Delta S \rightarrow 0. \quad \square \square$$

This contrasts sharply with digital inference, which must eject entropy and therefore obey Landauer-type limits. The relation between free energy and information can be written as

$$J \equiv -\frac{\Delta S}{k_B \ln 2} = \frac{F}{k_B T \ln 2},$$

linking bits of information J to free-energy changes F . In quasi-static inference, $F_{\min} = 0$ implies no minimal free-energy dissipation, in principle allowing thermodynamically reversible AGI inference on analog substrates.

15.4.3 Lower Bounds on Training Energy

Training overconstrains the analog system by clamping both inputs and outputs, generating stresses that propagate backward through the architecture and reproduce backpropagation rules in purely physical terms. An analysis of these stresses yields a universal lower bound on training energy:

$$E_{\text{train}} \gtrsim 2NDk_BT \frac{\langle |\nabla_{\theta} F|^2 \rangle}{\text{var}(\nabla_{\theta} F)} \approx 2NDk_BT,$$

where N is the number of trainable parameters, D is dataset size, and the ratio of gradient-square mean to variance is approximated as unity in typical regimes. This bound has the same scaling in N and D as computational cost in digital training, showing that analog substrates can reduce constant factors but not circumvent fundamental scaling.

In other words, while inference can be made thermodynamically reversible in quasi-static analog DNNs, training necessarily dissipates energy proportional to the total information learned, and this cost grows linearly with both model size and data volume.

15.4.4 Energy-Time-Accuracy Tradeoffs in Thermodynamic Computing

Thermodynamic computing frameworks analyze energy-time-accuracy tradeoffs, often via energy-delay product (EDP) style bounds. For a computation mapping an initial distribution ρ_0 to a final distribution ρ_1 , one result establishes an EDP-like lower bound:

$$\text{EDDP} \geq \frac{W_2(\rho_0, \rho_1)^2}{\beta\gamma},$$

where W_2 is the Wasserstein-2 distance between distributions, $\beta = 1/(k_B T)$, and γ is a friction-like parameter. This connects the “distance” a computation travels in probability space to the minimal energy-time cost, implying that more complex transformations (e.g., AGI-level representation changes) have irreducible thermodynamic cost even on ideal analog hardware.

Information engine analyses show that overfitting in predictive memory can manifest as “thermodynamic overfitting”: engines with high predictive complexity dissipate extra energy in testing phase by overreacting to noise. This suggests that AGI systems with excessively complex internal models may incur thermodynamic penalties, hinting at an energy-regularized notion of generalization.

15.5 Comparative Table: Continuous, Discrete, and Mixed-Signal Approaches

Continuous-time models

Neural ODEs, liquid neural networks, continuous Hopfield memories - Continuous hidden states $h(t)$, energy landscapes - ODE integration via adaptive RK (Dopri8, S-SOLVER), probabilistic solvers, gradient flows on energy $E(h)$ - Stability via Lyapunov and A-stable solvers; quasi-static analog inference with $\Delta S = 0$ and no fundamental energy lower bound -

Discrete digital models

Integer SNNs, timed automata SNNs, standard digital ANNs - Discrete spikes, integer states, digital weights and activations - Time-stepped simulation, discrete event systems, fixed-point arithmetic - Constrained by Landauer bounds $E_{\text{diss}} \geq k_B T \ln 2$ per bit erase; numerical stability tied to step size and precision -

Mixed-signal analog-digital neuromorphic

Memristor crossbars, mixed-mode memristive logic, hybrid memristor-CMOS, photonic neuromorphic accelerators - Analog voltages/currents (weights as conductances), optical fields, digital control signals - VMM via Ohm’s and Kirchhoff’s laws $I_i = \sum_j G_{ij} V_j$; mixed-mode logic; optical interference and nonlinearity - In-memory computation bypasses von Neumann bottleneck; ADC/DAC overhead dominates area and energy; analog inference can approach reversible limit, while learning obeys $E_{\text{train}} \gtrsim 2NDk_B T$ -

This chapter formalizes how AGI-oriented neuromorphic research between 2024 and 2026 negotiates the boundary between analog and discrete computation: designing solvers that faithfully discretize continuous-time models, embedding discrete symbolic structures into continuous energy and hardware substrates, co-designing mixed-signal interfaces to overcome the von Neumann bottleneck, and respecting thermodynamic bounds on energy use in inference and learning.

Chapter 16

Entropic Physics, Synergetics, and Chaotic Resolution in AGI Architectures

16.1 Entropic Dynamics and Chaotic Resolution in Neural Systems

16.1.1 Entropy Production in Random Recurrent Neural Networks

Random recurrent neural networks (RRNNs) provide a tractable class of high-dimensional dynamical systems in which chaos and irreversibility can be analyzed using statistical physics. Consider a network of N units with state vector $x(t) \in \mathbb{R}^N$ evolving under

$$\dot{x}_i(t) = -x_i(t) + \sum_j J_{ij} \phi(x_j(t)) + \sigma \xi_i(t),$$

where J_{ij} is a random interaction matrix, ϕ is a nonlinearity, σ is noise strength, and $\xi_i(t)$ are independent Gaussian noise terms. Non-reciprocal interactions (i.e., $J_{ij} \neq J_{ji}$) introduce non-gradient components into the dynamics, making the system intrinsically irreversible.

Using dynamical mean-field theory (DMFT) in the thermodynamic limit $N \rightarrow \infty$, one can derive an exact expression for the averaged entropy production rate $\langle \dot{S}_{\text{prod}} \rangle$, which quantifies irreversibility and is directly related to chaotic behavior. In particular, the entropy production rate is expressed in terms of the non-reciprocal part of J_{ij} and the stationary two-time correlation and response functions of the network.

The model undergoes a transition from a fixed-point regime to a chaotic regime at a critical heterogeneity in the non-reciprocal interactions, and the entropy production rate behaves as an order parameter for irreversibility: it is zero in the quiescent regime and becomes positive in the chaotic regime. Near criticality, the entropy production rate scales as σ^{-2} with respect to noise strength σ , reflecting enhanced sensitivity of chaos to fluctuations, while deeper into the chaotic phase entropy production saturates. This provides a quantitative tool to measure and potentially control chaos in RRNNs relevant to AGI architectures.

16.1.2 Entropic Dynamics and Renormalization Group in Deep Networks

Entropic dynamics in deep feedforward networks analyzes information processing as a continuum-depth limit and draws analogies to renormalization group (RG) flows. In this framework, the association of concepts to patterns across layers is governed by entropy maximization subject to constraints derived from network weights and data statistics, leading to an “entropic dynamics” equation for layer-to-layer evolution.

Let $p(x, t)$ denote the probability distribution of internal representations at depth t . Entropic dynamics posits that p evolves according to a functional that maximizes Shannon entropy $S[p] = -\int p \log p \, dx$ subject to constraints representing conservation of relevant information (e.g., moments, invariants). This yields evolution equations similar to RG flows, with coarse-graining operations across depth playing the role of scale transformations.

The connection to Hamilton-Jacobi-Bellman (HJB) equations arises by interpreting the network’s training objective as a value function and its depth evolution as optimal control; entropic dynamics can be

rephrased in terms of HJB equations, where entropy constraints regularize the control policy and stabilize representations. Although primarily conceptual, this framework formalizes how maximizing entropy under structural constraints can drive self-organization in deep networks and suggests entropic phase transitions as a mechanism for abrupt representational changes.

16.1.3 Entropy Measures and Adaptive Control of Chaotic Systems

Entropy measures-such as approximate entropy, sample entropy, and fuzzy entropy-have been used to characterize chaos and guide neural control of non-equilibrium dynamical systems. For a four-dimensional chaotic system with hidden attractors, entropy analysis can detect transitions between regular and chaotic regimes as parameters vary, and these measures serve as diagnostic tools for the complexity of state trajectories.

Adaptive radial-basis-function neural networks (RBF-NNs) can then be designed to stabilize such chaotic systems. In one example, the uncertain nonlinear dynamics are approximated by an RBF-NN, and Lyapunov's direct method is used to derive adaptive learning laws for the network weights such that all closed-loop signals are semi-globally uniformly ultimately bounded. Entropy measures help identify the degree of chaos and evaluate the effectiveness of the control in reducing complexity, though the control scheme itself is Lyapunov-based rather than explicitly entropic.

16.1.4 Structured Chaos and Spike-Response Noise Entropy

Large networks of sparsely coupled excitatory and inhibitory neurons in the brain exhibit chaotic dynamics but still maintain reliable information transmission. To understand this, structured chaos has been analyzed via spike-response noise entropy, an information-theoretic quantity directly related to dynamical entropy production.

For a balanced spiking network, one can compute a bound on multi-cell spike-pattern entropy H by relating it to single-cell spike-response entropy and noise correlations. Surprisingly, multi-cell spike-response noise entropy is at least an order of magnitude lower than naive extrapolations from single-cell entropy, even though average noise correlations are small. For instance, in one model, the estimated spike-response noise entropy is $H_1^L \approx 1.12$ bits per time unit, compared to 3.67 bits per time unit for a homogeneous Poisson spike train with the same rate.

This reduction indicates that structured chaos-combined with network architecture and correlations-can shape variability in a way that stabilizes collective representations, suggesting that entropic bounds on spike entropy can act as constraints that prevent chaos from destroying neural codes.

16.2 Entropic Neural Kalman Filters and Non-Gaussian Noise

16.2.1 Maximum Correntropy Kalman Filters

Classical Kalman filters (KFs) assume Gaussian noise and minimize mean-squared error (MSE), but they remain MMSE-optimal based only on mean and covariance information and do not inherently require Gaussianity. For non-Gaussian noise, entropic and information-theoretic criteria can enhance robustness.

The maximum correntropy Kalman filter (MCKF) introduces the correntropy criterion as a robust similarity measure between state estimates and observations under non-Gaussian noise. Correntropy is defined as

$$V_\sigma(x, y) = \mathbb{E}[\kappa_\sigma(x - y)] = \mathbb{E} \left[\exp \left(-\frac{(x - y)^2}{2\sigma^2} \right) \right],$$

where κ_σ is a Gaussian kernel with bandwidth σ . Maximizing correntropy corresponds to minimizing a kernel-based generalized error measure that implicitly weights small errors more heavily and suppresses large deviations, making it robust to outliers.

In MCKF, the standard KF cost function is replaced by a maximum correntropy criterion (MCC), and the update equations are derived by optimizing this criterion, often via fixed-point iteration. The resulting filter achieves better performance in heavy-tailed noise environments than standard KFs, effectively

imposing an information-theoretic constraint on the error distribution. Multi-kernel maximum correntropy Kalman filters extend this approach by combining multiple kernels to capture different scales of non-Gaussianity and are used in distributed state estimation under deception attacks.

16.2.2 Minimum Error Entropy Kalman Filters

Minimum error entropy (MEE) Kalman filters replace the MSE criterion with minimization of error entropy, thereby directly controlling the information content of estimation errors. Error entropy H_e is defined via the probability density $p_e(e)$ of estimation error e :

$$H_e = - \int p_e(e) \log p_e(e) de,$$

or approximated via kernel density estimators. Minimizing H_e encourages error distributions that concentrate probability mass near zero, without assuming Gaussianity.

Numerically stable minimum error entropy Kalman filters derive update rules that minimize H_e while ensuring stability, often via approximations that avoid ill-conditioned Hessians and maintain bounded gains. Distributed variants such as MEEF-KF (minimum error entropy with fiducial points KF) incorporate fiducial points (representative error samples) to construct robust regression equations for multi-sensor fusion in non-Gaussian environments. In centralized and distributed MEEF-KF, consensus averaging across sensor nodes and entropy-based fusion yield robust state tracking for land vehicle navigation and power system monitoring under outliers and unusual data.

16.2.3 Max Entropy Moment Kalman Filter (MEM-KF)

The Max Entropy Moment Kalman Filter (MEM-KF) addresses nonlinear, non-Gaussian systems by representing beliefs using a moment-constrained maximum entropy distribution (MED). For polynomial systems with arbitrary noise, MEM-KF:

Represents the state distribution $p(x)$ by specifying a set of moments $m_k = \mathbb{E}[x^k]$ and solving a maximum entropy problem:

$$p(x) = \arg \max_p \left\{ - \int p(x) \log p(x) dx \right\} \quad \text{subject to} \quad \int x^k p(x) dx = m_k,$$

leading to an exponential-family distribution with parameters determined via convex optimization.

Propagates moments through the polynomial process model, avoiding symbolic integration, and reconstructs the MED after each prediction and update step.

Models process and observation noise via MEDs, enabling flexible non-Gaussian noise representations.

All steps, including point estimates and covariance extraction, are solved via convex optimization, making MEM-KF an entropic Kalman-type filter in which constraints on moments and maximum entropy principles govern state estimation under complex noise. This provides a mathematically rigorous entropic extension of KF suitable for chaotic and nonlinear AGI control tasks.

16.3 Synergetics, Order Parameters, and Deep/Multi-Agent Architectures

16.3.1 Haken's Synergetics and Maximum Entropy Principle

Synergetics, as introduced by Hermann Haken, is the study of self-organization in complex systems driven far from equilibrium. Its central concepts include:

Order parameters: collective variables that describe macroscopic patterns emerging from many degrees of freedom.

Slaving principle: once order parameters emerge, they “enslave” microscopic degrees of freedom, which rapidly adjust to the order parameter dynamics.

Phase transitions: qualitative changes in behavior at critical values of control parameters.

Haken extended Jaynes' maximum entropy principle by incorporating order-parameter-related constraints into the entropy maximization process. Instead of maximizing Gibbs-Jaynes entropy $S[p]$ subject only to usual constraints (e.g., normalization, energy), Haken added functions of order parameters as additional constraints, capturing nonequilibrium phase transitions and self-organized patterns.

In "Synergetic Computers and Cognition: A Top-Down Approach to Neural Nets," Haken applied these ideas to neural assemblies, suggesting that brain-like systems operate near instabilities where order parameters govern neural activities and enslave individual neuron states. Entropy maximization under synergetic constraints yields evolution equations for macroscopic variables that can be interpreted as control laws for pattern formation and cognition.

16.3.2 Order Parameter Dynamics in Complex Systems

Order parameter dynamics has been systematically studied across physical and biological complex systems. The synergetic theory posits that at the onset of macroscopic transitions, slow modes (order parameters) can be distinguished from fast modes, and the evolution of the slow modes obeys reduced equations derived via the slaving principle and center manifold reductions.

In model-based scenarios, order parameter dynamics appear in:

Synchronization of coupled phase oscillators, where the Kuramoto order parameter

$$R(t)e^{i\Phi(t)} = \frac{1}{N} \sum_{j=1}^N e^{i\phi_j(t)}$$

measures collective coherence and obeys reduced equations derived via Ott-Antonsen ansatz.

Chimera states and neuron network dynamics, where order parameters describe coexisting coherent and incoherent domains, with evolution governed by generalized mean-field equations.

In data-based scenarios, the eigen-microstate approach (EMP) reconstructs order-parameter dynamics from high-dimensional spatiotemporal data by decomposing evolution into eigenmodes and tracking Bose-Einstein-condensation-like transitions in dominant modes. This provides tools to infer macroscopic order parameters directly from neural or multi-agent data.

16.3.3 Synergetic Neural Networks and Entropy Monotonicity

Synergetic neural networks are competitive neural architectures inspired by Haken's synergetics, aimed at pattern recognition via self-organization. Their structure divides the network into matching and competition subnets, with order parameters measuring similarity between input patterns and prototype vectors (e.g., via distance, mean squared error, correlation).

A notable result shows that a suitably defined network entropy-quantifying the degree of order in the pattern recognition process-never decreases during operation, reflecting monotonic self-organization. Quick Haken networks further reduce computational time by using initial conditions and analytical order-parameter rules to determine competitive outcomes without iterative dynamics, preserving synergetic principles while achieving faster decisions.

Although these models are older and relatively small-scale, they demonstrate explicit use of entropy and order parameters in neural architectures, serving as conceptual precursors to modern synergetic interpretations of deep and multi-agent systems.

16.3.4 Synergetics in Multi-Agent AI Systems

Recent work adapts synergetic concepts to heterogeneous multi-agent AI systems by modeling agents as coupled oscillators in a Kuramoto-like framework. In one 2025 preprint, AI agents engaged in complex task execution are represented as oscillators with phase and amplitude dynamics; coupling strength, agent diversity, and network topology shape emergent coordination.

An order parameter analogous to Kuramoto’s $R(t)$ quantifies the degree of multi-agent synchronization, and simulations on all-to-all and scale-free networks show that increased coupling promotes robust synchronization despite heterogeneous agent capabilities. The authors draw a detailed correspondence between chain-of-thought prompting in AI reasoning and synchronization phenomena, treating chain-of-thought as an emergent pattern governed by collective modes (order parameters) that enslave local agent behaviors.

Synergetic control also appears in multi-agent consensus algorithms, where system errors are used as order parameters to construct finite-time synergetic evolution equations with exponential convergence properties in event-triggered synergetic control for multi-agent systems. These results extend synergetic theory to AI orchestration and suggest order-parameter-based control laws for agentic AGI.

16.3.5 Cybernetical Neuroscience and Speed-Gradient Entropy Models

A 2025 article on “information entropy dynamics, self-organization, and cybernetical neuroscience” develops speed-gradient evolution models for systems obeying Haken’s maximum information entropy principle. The speed-gradient method defines evolution laws by focusing on the rate of change of a goal functional (e.g., entropy or order parameter) and adjusting control variables proportional to the gradient of this rate.

By combining entropy maximization with synergetic order parameter constraints, the authors outline contours of “cybernetical neuroscience” dedicated to controlling neural systems via information-theoretic principles. This suggests a pathway toward entropic control of neural architectures, though explicit AGI implementations remain to be developed.

16.4 Empirical Attempts at Entropic Control of Neural Representations

16.4.1 Entropy-Based Guidance of Deep Networks

Entropy-based guidance has been investigated to accelerate convergence and improve generalization of deep networks. Methods such as entropy regularizing activations (ERA) constrain sampling entropy above thresholds by applying specially designed activations to model outputs in continuous control tasks. These activations shape output distributions to avoid collapse and encourage exploration, effectively using entropy as a control parameter in policy networks.

Entropy-based guidance for surrogate models in physical simulations uses entropy measures to steer training toward informative regions of parameter space, improving sample efficiency. While these studies introduce entropy into loss functions and activations, they typically operate at the level of scalar regularization rather than fundamental entropic dynamics or phase transitions.

16.4.2 Entropic Regularization in Reinforcement Learning

Maximum entropy reinforcement learning frameworks (e.g., soft Q-learning, SAC) incorporate entropy terms into objectives to encourage exploration, but recent work has shown that naive maximum entropy can mislead policy optimization. For instance, maximizing entropy without careful constraint can drive policies toward unsafe or uninformative behaviors, revealing tension between entropic objectives and control goals.

This suggests that entropy must be coupled to appropriate order parameters or constraints to yield meaningful control in AGI systems, aligning with Haken’s view that entropy maximization should incorporate physical and structural conditions via order parameter functions.

16.4.3 Entropic Bounds and Chaotic Neural Circuits

Empirical analysis of chaotic neural circuits shows that chaos can be “boosted” or suppressed near the onset of collective activity, with noise entropy and spike-pattern variability controlled by structural parameters. For example, boosting chaos at the onset of collective spiking can increase sensitivity to inputs while maintaining bounded noise entropy through structured connectivity and balanced excitation-inhibition.

However, the practical translation of these entropic bounds and structured chaos into AGI architectures is incomplete: most work focuses on biological circuits or simplified models, and there is no consensus framework for entropic control of representations in large-scale artificial networks.

16.4.4 Limits of Entropic Control and Cognitive Unification

Across entropic dynamics, entropic Kalman filters, and synergetic applications, current research falls short of a unified theory of entropy-based cognitive control in AGI:

Entropic dynamics and RRNN entropy production provide tools to quantify and analyze chaos but do not yet yield constructive algorithms for stabilizing latent spaces in high-dimensional deep networks.

Entropy-based Kalman filters (MCKF, MEE-KF, MEM-KF) successfully handle non-Gaussian noise and complex belief spaces in state estimation but are applied mainly to robotics, sensor networks, and tracking, not to internal representational dynamics of AGI models.

Synergetics and order parameter dynamics furnish a rich theoretical language for self-organization, but direct synergetic design of deep architectures remains mostly conceptual, with a few older synergetic neural networks and recent multi-agent models as prototypes.

There is, as yet, no consensus on how to couple entropy and order parameters to provide real-time cognitive control-i.e., mechanisms by which AGI systems can dynamically adjust representational complexity, stabilize chaotic modes, and orchestrate multi-agent reasoning in response to entropy-based diagnostics. Entropic physics and synergetics offer powerful frameworks, but their integration into operational AGI architectures remains a frontier for 2024-2026 and beyond.